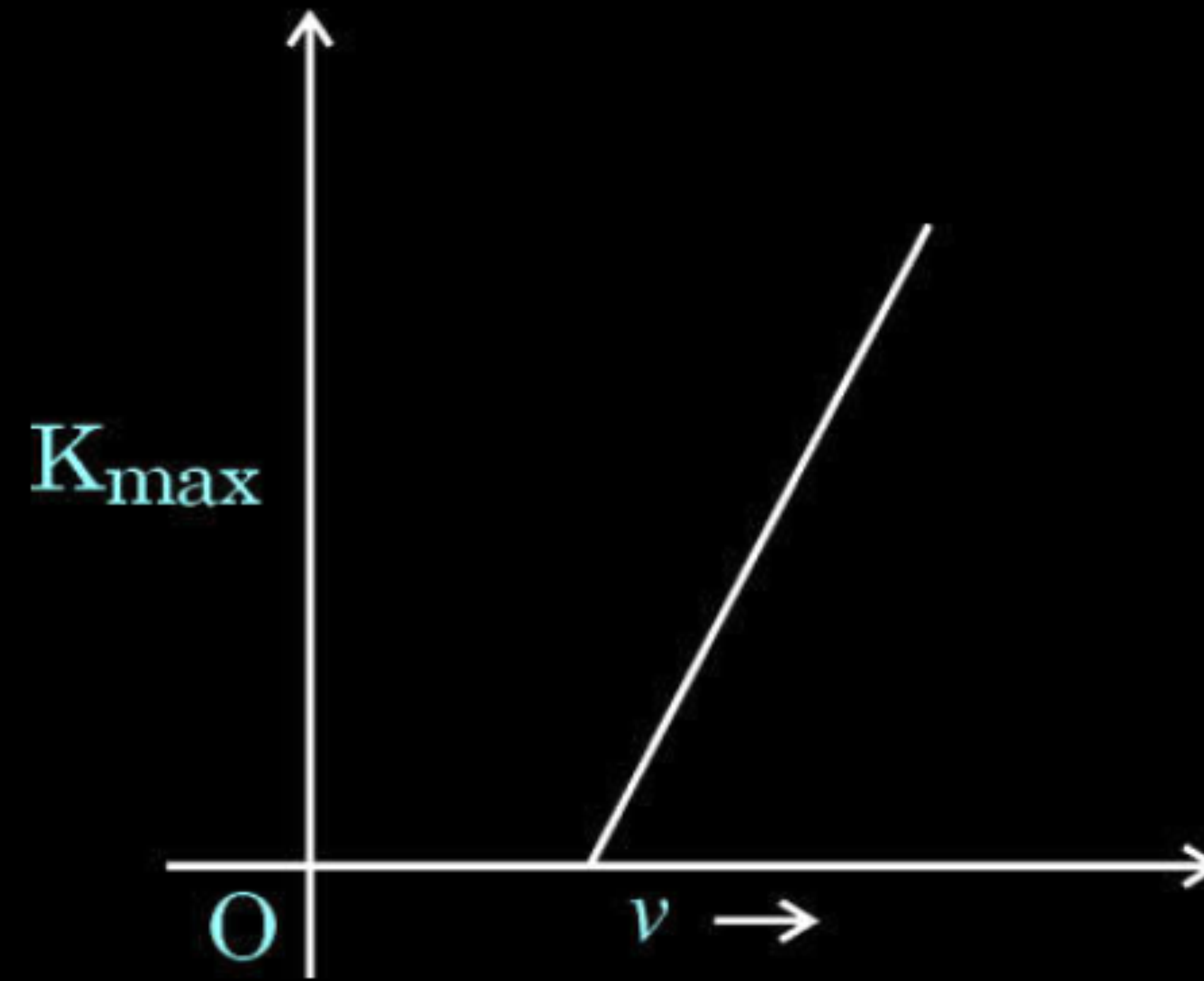


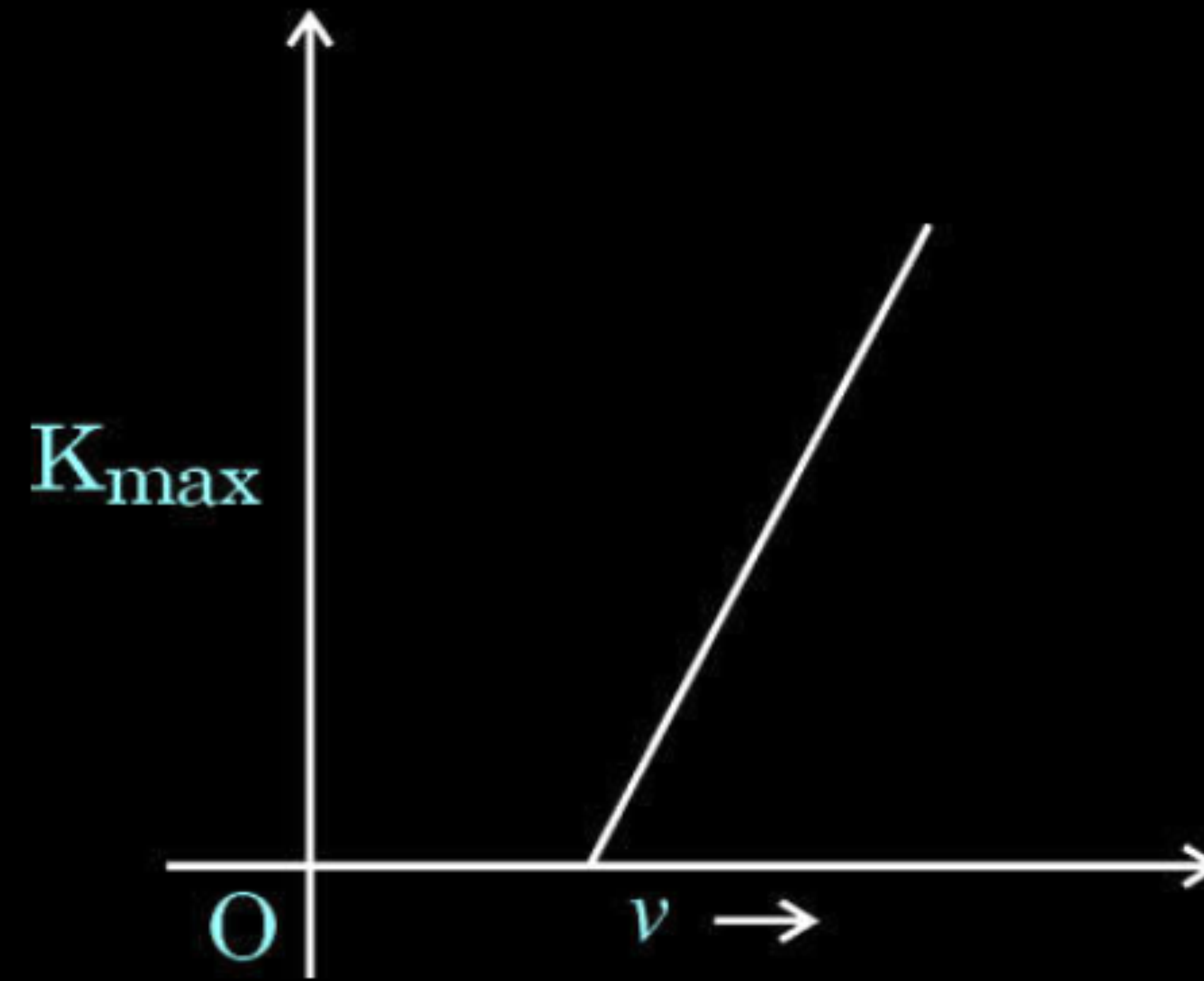
8. The figure shows the variation of maximum kinetic energy ($K_{\max}$) of emitted electrons as a function of the frequency ν of radiation incident on a photosensitive surface. The slope of the curve is :



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- | | |
|-------------------|---------------------|
| (A) h | (B) $\frac{h}{\nu}$ |
| (C) $\frac{h}{e}$ | (D) he |

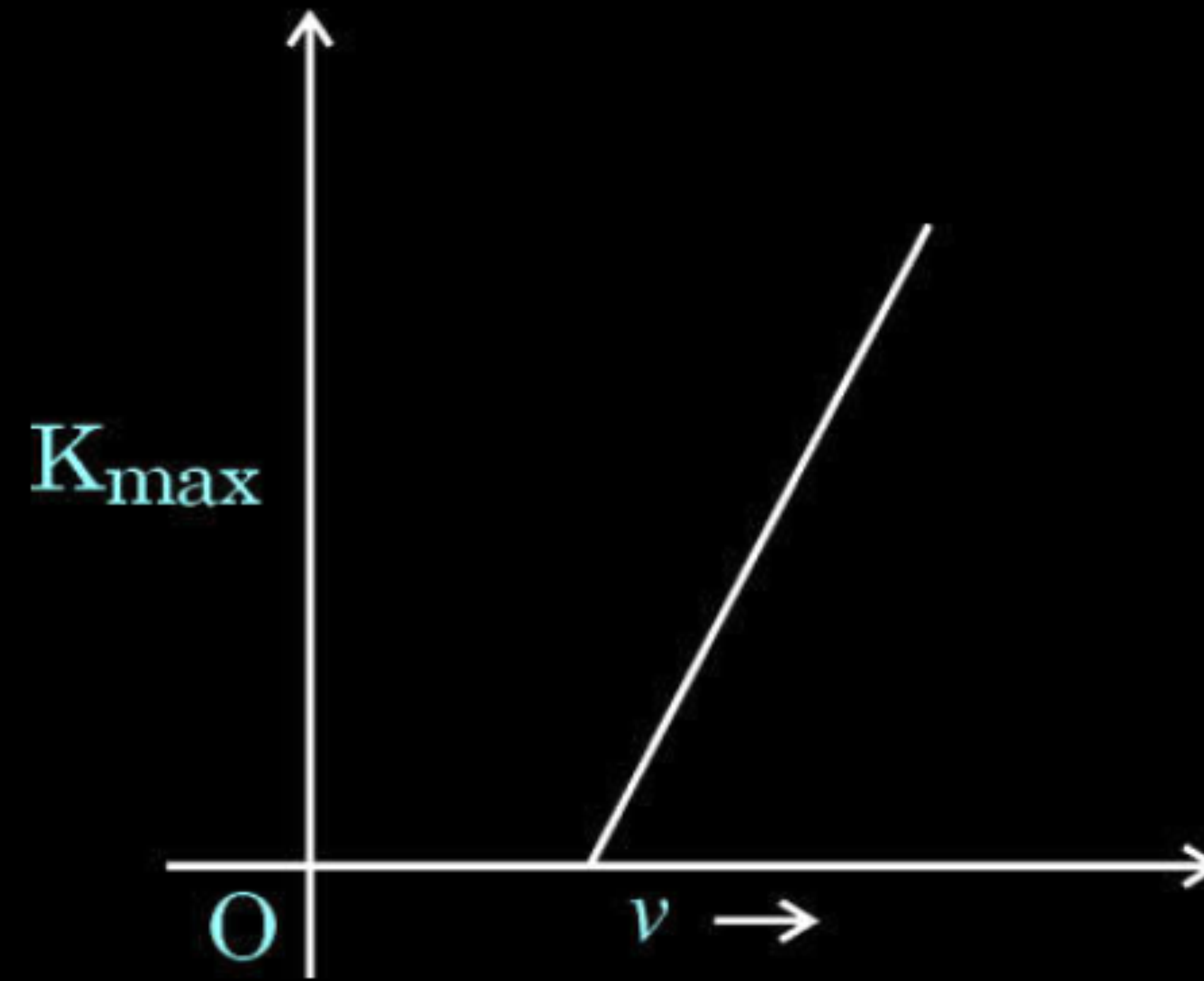
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- (A) h (B) $\frac{h}{v}$

(C) $\frac{h}{e}$ (D) he

(A) h

13. *Assertion (A) :* In process of photoelectric emission using a monochromatic light, all emitted electrons do not have the same kinetic energy.

Reason (R) : If radiation falling on a photosensitive surface consists of different wavelengths, the energy of emitted photoelectrons by absorbing photons of different wavelengths, shall be different.

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13. *Assertion (A) :* In process of photoelectric emission using a monochromatic light, all emitted electrons do not have the same kinetic energy.

Reason (R) : If radiation falling on a photosensitive surface consists of different wavelengths, the energy of emitted photoelectrons by absorbing photons of different wavelengths, shall be different.

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(B) Both Assertion (A)and Reason (R)are true ,but Reason (R) is not the correct explanation of the Assertion (A).

- 17.** An electron of mass m and charge e is accelerated from rest through a potential V . Prove that, after acceleration, the de Broglie wavelength associated with the electron is proportional to $\frac{1}{\sqrt{V}}$.

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2

Momentum and kinetic energy are related as

$$p = \sqrt{2mK}$$

$$p = \sqrt{2meV}$$

De Broglie wavelength

$$\lambda = \frac{h}{p}$$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\lambda \propto \frac{1}{\sqrt{V}}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

- 17.** Monochromatic light of frequency 7.0×10^{14} Hz is produced by a laser. The power emitted is 21 mW. Calculate (i) the energy of a photon in the light beam, and (ii) the number of photons per second, on an average, emitted by the source.

17. Monochromatic light of frequency 7.0×10^{14} Hz is produced by a laser. The power emitted is 21 mW. Calculate (i) the energy of a photon in the light beam, and (ii) the number of photons per second, on an average, emitted by the source.

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2

(i) $E = h\nu$

$$= 6.63 \times 10^{-34} \text{ J.s} \times 7.0 \times 10^{14} \text{ s}^{-1}$$

$$= 4.641 \times 10^{-19} \text{ J}$$

(ii) $P = \frac{nh\nu}{t}$

$$21 \times 10^{-3} = \frac{n}{t} \times 4.641 \times 10^{-19}$$

$$\text{number of photons emitted per second} = 4.5 \times 10^{16}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

17. The work function of Caesium is 2.14 eV .

- (a) Find the threshold frequency for Caesium.
- (b) Find the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.86 V .

17. The work function of Caesium is 2.14 eV.

- (a) Find the threshold frequency for Caesium.
- (b) Find the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.86 V.

2

$$\begin{aligned}
 \text{(a)} \quad \frac{\phi_0}{h} &= \nu_0 \\
 &= \frac{2.14 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}} \\
 &= 5.16 \times 10^{14} \text{ Hz}
 \end{aligned}$$

 $\frac{1}{2}$ $\frac{1}{2}$

$$\begin{aligned}
 \text{(b)} \quad \frac{hc}{\lambda} &= K_{\max} + \phi_0 \\
 \lambda &= \frac{hc}{K_{\max} + \phi_0} \\
 \lambda &= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{(2.14 + 0.86) \times 1.6 \times 10^{-19}} \\
 \lambda &= 4.14 \times 10^{-7} \text{ m}
 \end{aligned}$$

 $\frac{1}{2}$ $\frac{1}{2}$

- 9.** A beam of red light and a beam of blue light have equal intensities. It implies that :
- (A) Red light beam has higher number of photons than that in blue light beam.
 - (B) Blue light beam has higher number of photons than that in red light beam.
 - (C) Both beams contain the same number of photons.
 - (D) The momentum of a photon of red light beam is more than that of a photon of blue light beam.

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9. A beam of red light and a beam of blue light have equal intensities. It implies that :
- (A) Red light beam has higher number of photons than that in blue light beam.
 - (B) Blue light beam has higher number of photons than that in red light beam.
 - (C) Both beams contain the same number of photons.
 - (D) The momentum of a photon of red light beam is more than that of a photon of blue light beam.

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(A) Red light beam has higher number of photons than that in blue light beam.

14. *Assertion (A) :* If the maximum kinetic energy of an electron in a photoelectric experiment is one-fourth the energy of a photon of wavelength λ incident on a surface, the work function of the surface is $\frac{3}{4} \frac{hc}{\lambda}$.

Reason (R) : Photoelectric emission can be explained by Einstein's photoelectric equation.

14. *Assertion (A) :* If the maximum kinetic energy of an electron in a photoelectric experiment is one-fourth the energy of a photon of wavelength λ incident on a surface, the work function of the surface is $\frac{3}{4} \frac{hc}{\lambda}$.

Reason (R) : Photoelectric emission can be explained by Einstein's photoelectric equation.

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(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

- 27.** (a) Use Einstein's photoelectron equation to explain the experimental features of photoelectric emissions.
- (b) Light of wavelength 450 nm is incident on three metals A, B and C of work functions 3.2 eV, 2.7 eV and 1.75 eV respectively. Which metal(s) will exhibit photoelectric effect ?

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- 27.** (a) Use Einstein's photoelectron equation to explain the experimental features of photoelectric emissions.
- (b) Light of wavelength 450 nm is incident on three metals A, B and C of work functions 3.2 eV, 2.7 eV and 1.75 eV respectively. Which metal(s) will exhibit photoelectric effect ?

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3

(a) Einstein's photoelectric equation

$$K_{\max} = h\nu - h\nu_0$$

1. If $\nu > \nu_0$, $K_{\max}$ of ejected electron increases with frequency of incident radiation.

2. For $\nu < \nu_0$, $K_{\max}$ is negative which is not possible. So, no photoelectric emission takes place below threshold frequency.
(Give full credit to the student for explaining any other feature)

$$(b) E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{450 \times 10^{-9} \times 1.6 \times 10^{-19}} eV$$

$$= 2.76 eV$$

$\therefore$ Metal B & C will exhibit photo-electric effect.

1/2

1/2

1/2

1/2

1/2

1/2

7. The frequency of a photon of energy 1.326 eV is :

(A) 1.18×10^{14} Hz

(B) 3.20×10^{14} Hz

(C) 4.20×10^{15} Hz

(D) 4.80×10^{15} Hz

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7. The frequency of a photon of energy 1.326 eV is :

(A) 1.18×10^{14} Hz

(B) 3.20×10^{14} Hz

(C) 4.20×10^{15} Hz

(D) 4.80×10^{15} Hz

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(B) 3.20×10^{14} Hz

15. *Assertion (A) :* The minimum negative potential applied to the anode in a photoelectric experiment at which photoelectric current becomes zero, is called cut-off voltage.

Reason (R) : The threshold frequency for a metal is the minimum frequency of incident radiation below which emission of photoelectrons does not take place.

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15. *Assertion (A) :* The minimum negative potential applied to the anode in a photoelectric experiment at which photoelectric current becomes zero, is called cut-off voltage.

Reason (R) : The threshold frequency for a metal is the minimum frequency of incident radiation below which emission of photoelectrons does not take place.

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(B) Both Assertion(A) and Reason (R) are true but Reason(R) is **not** the correct explanation of Assertion(A).

24. Answer the following, giving reasons :

3

- (a) The maximum kinetic energy of the photoelectrons is independent of the intensity of incident radiation.
- (b) Photoelectric current increases with the increase in the intensity of the incident radiation.
- (c) The stopping potential V_0 varies linearly with the frequency ν of the incident radiation for a given photosensitive surface.

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(a) With the variation in the intensity of incident radiations, there is a change in the number of photons not on the energies of the photons so the maximum kinetic energy of photoelectrons remains same.

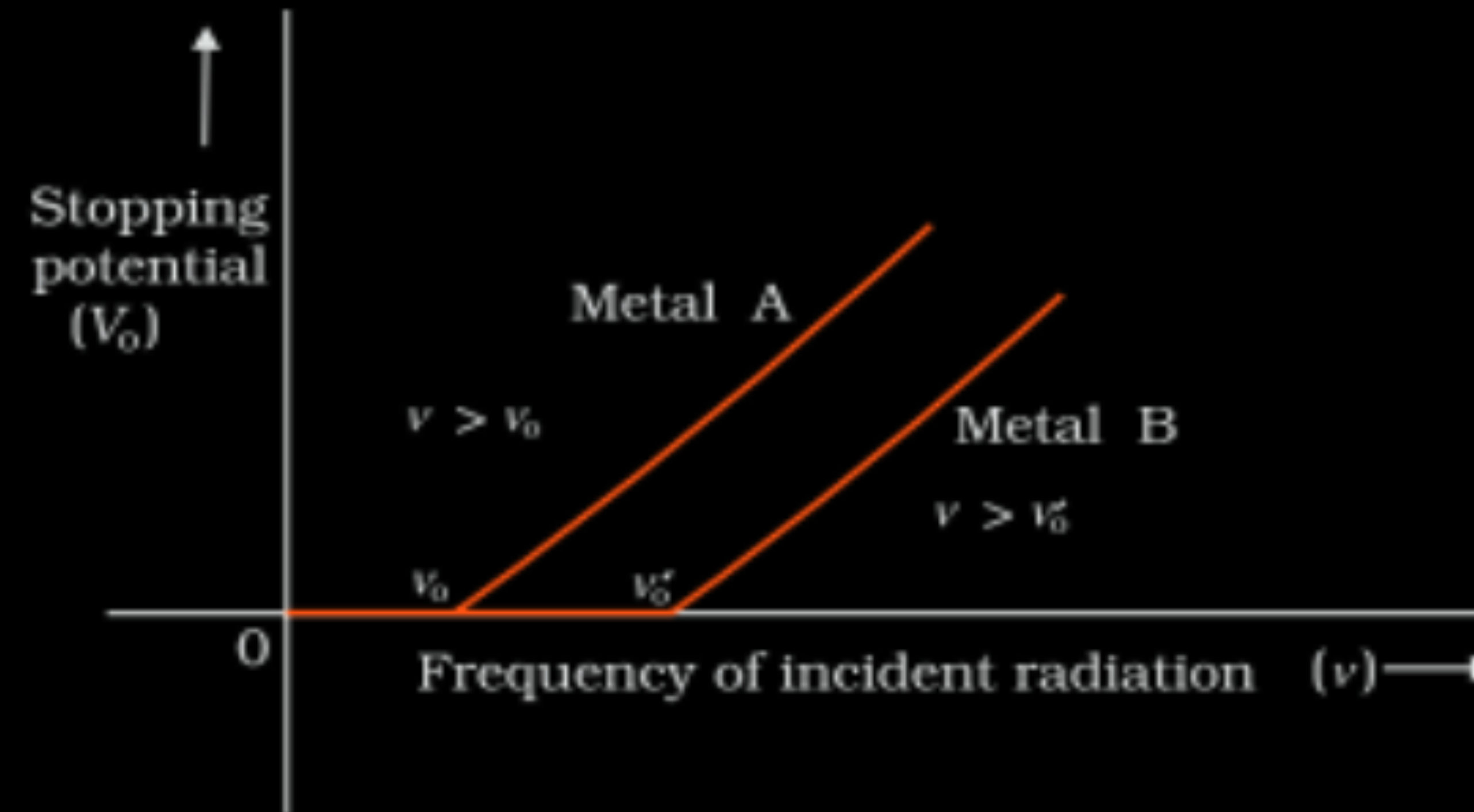
1

(b) With the increase in intensity of incident radiation number of photons increases as a result photoelectric current increase.

1

(c) With the increase in frequency of incident radiation, the maximum kinetic energy of photoelectrons increases. Hence higher value of stopping potential (retarding potential) is required.

1



Alternatively:

$$K_{\max} = eV_o = h\nu - h\nu_0$$

23. (a) Briefly explain Einstein's photoelectric equation.

(b) Four metals with their work functions are listed below :

$K = 2.3 \text{ eV}$, $Na = 2.75 \text{ eV}$, $Mo = 4.17 \text{ eV}$ and $Ni = 5.15 \text{ eV}$.

The radiation of wavelength 330 nm from a laser source placed 1 m away, falls on these metals. Which of these metals will not show photoelectric emission ? What will happen if the laser source is brought closer to a distance of 50 cm ?

23.
- (a)

Briefly explain Einstein’s photoelectric equation.
- (b)

Four metals with their work functions are listed below :

$K = 2.3 \text{ eV}$, $Na = 2.75 \text{ eV}$, $Mo = 4.17 \text{ eV}$ and $Ni = 5.15 \text{ eV}$.

The radiation of wavelength 330 nm from a laser source placed 1 m away, falls on these metals. Which of these metals will not show photoelectric emission ? What will happen if the laser source is brought closer to a distance of 50 cm ?

3

(a) $h\nu = \phi_0 + K_{\text{max}}$ $h\nu$ = Energy of incident radiation ϕ_0 = Work function or minimum energy required to emit an electron from metal surface K_{max} = maximum kinetic energy of emitted electron Alternatively: The energy of incident radiation $E (> \phi_0)$ incident on a metal surface, a part of it is used to overcome the work function & remaining energy provides maximum kinetic energy to the electrons.	
(b) $\lambda = 330 \text{ nm}$ $E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{330 \times 10^{-9} \times 1.6 \times 10^{-19}}$ $= 3.77 \text{ eV}$	
Mo and Ni will not show photoelectric emission.	
No change.	

26. (a) Define 'work function' of a metal. How can its value be determined from a graph between stopping potential and frequency of the incident radiation ?

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(b) The work function of a metal is 2.4 eV . A stopping potential of 0.6 V is required to reduce the photocurrent to zero, in a photoelectric experiment. Calculate the wavelength of light used.

26. (a) Define 'work function' of a metal. How can its value be determined from a graph between stopping potential and frequency of the incident radiation ?

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(b) The work function of a metal is 2.4 eV. A stopping potential of 0.6 V is required to reduce the photocurrent to zero, in a photoelectric experiment. Calculate the wavelength of light used.

3

Minimum energy required by an electron to escape from metal surface.	1/2
The intercept on the y axis for a graph between stopping potential & frequency gives $\frac{\phi_0}{e}$.	
$\therefore \phi_0 = e \times \text{intercept on y-axis.}$	1
<u>Alternatively:</u>	
Work function $\phi_0 = h \times \text{intercept on x-axis.}$	
(b) $eV_0 = \frac{hc}{\lambda} - \phi_0$	1/2
$0.6 = \frac{hc}{\lambda} - 2.4$	
$3 = \frac{hc}{\lambda} \Rightarrow 3 \times 1.6 \times 10^{-19} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{\lambda}$	1/2
$\lambda = \frac{1241}{3} = 413.6 \text{ nm}$	1/2

10. The wavelength of a photon is equal to the wavelength associated with an electron. Both will have the same value of :

1

(A) energy

(B) angular momentum

(C) velocity

(D) linear momentum

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10. The wavelength of a photon is equal to the wavelength associated with an electron. Both will have the same value of : *1*

(A) energy

(B) angular momentum

(C) velocity

(D) linear momentum

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(D) linear momentum

12. The momentum (in kg m/s) of a photon of frequency 6.0×10^{14} Hz is :

- (A) 6.63×10^{-25}
- (B) 1.326×10^{-27}
- (C) 2.652×10^{-26}
- (D) 3.978×10^{-24}

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12. The momentum (in kg m/s) of a photon of frequency 6.0×10^{14} Hz is :

- (A) 6.63×10^{-25}
- (B) 1.326×10^{-27}
- (C) 2.652×10^{-26}
- (D) 3.978×10^{-24}

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(B) 1.326×10^{-27}

10. The de Broglie wavelength associated with an electron moving with energy 5 eV is :

(A) 0.75 nm

(B) 1.2 nm

(C) 2.4 nm

(D) 0.55 nm

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10. The de Broglie wavelength associated with an electron moving with energy 5 eV is :

(A) 0.75 nm

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(D) 0.55 nm

- 19.** A laser beam of wavelength 500 nm and power 5 mW strikes normally on a perfectly reflecting surface of area 1 mm^2 of a body. It rebounds back from the surface. Find the force exerted by the laser beam on the body.

2

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- 19.** A laser beam of wavelength 500 nm and power 5 mW strikes normally on a perfectly reflecting surface of area 1 mm² of a body. It rebounds back from the surface. Find the force exerted by the laser beam on the body. 2

$$\text{No. of photons ejected per second} = \frac{P}{E}$$

$$= \frac{P}{\left(\frac{hc}{\lambda}\right)}$$

1/2

$$\text{Momentum of photon, } p = \frac{h}{\lambda}$$

1/2

Force exerted by laser beam

$$F = 2 \times \frac{P}{\left(\frac{hc}{\lambda}\right)} \times \frac{h}{\lambda}$$

1/2

$$= \frac{2P}{c}$$

$$= \frac{2 \times 5 \times 10^{-3}}{3 \times 10^8}$$

$$F = 3.33 \times 10^{-11} \text{ N}$$

1/2

10. The work function of a material is 2.21 eV. Which of the following *cannot* produce photoelectrons from it ?

(A) Red light

(B) Blue light

(C) Violet light

(D) Green light

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10. The work function of a material is 2.21 eV. Which of the following *cannot* produce photoelectrons from it ?

(A) Red light

(B) Blue light

(C) Violet light

(D) Green light

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(A) Red Light

- 19.** A laser beam of frequency 3.0×10^{14} Hz produces average power of 9 mW. Find (i) the energy of photon of the beam, and (ii) the number of photons emitted per second on an average by the source.

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- 19.** A laser beam of frequency 3.0×10^{14} Hz produces average power of 9 mW. Find (i) the energy of photon of the beam, and (ii) the number of photons emitted per second on an average by the source.

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2

$$\begin{aligned} \text{(i) } E &= h\nu \\ &= 6.63 \times 10^{-34} \times 3.0 \times 10^{14} \\ &= 1.99 \times 10^{-19} \text{ J} \end{aligned}$$

$$\begin{aligned} \text{(ii) } N &= \frac{P}{E} \\ &= \frac{9 \times 10^{-3}}{1.99 \times 10^{-19}} \\ &= 4.5 \times 10^{16} \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

9. The de-Broglie wavelengths associated with an electron (mass m_e) and a proton (mass m_p) moving with the same velocity are λ_e and λ_p respectively. The value of (λ_e/λ_p) is –

1

(A) $\sqrt{\frac{m_p}{m_e}}$

(B) $\frac{m_p}{m_e}$

(C) $\frac{m_e}{m_p}$

(D) $\sqrt{\frac{m_e}{m_p}}$

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9. The de-Broglie wavelengths associated with an electron (mass m_e) and a proton (mass m_p) moving with the same velocity are λ_e and λ_p respectively. The value of (λ_e/λ_p) is – 1

(A) $\sqrt{\frac{m_p}{m_e}}$

(B) $\frac{m_p}{m_e}$

(C) $\frac{m_e}{m_p}$

(D) $\sqrt{\frac{m_e}{m_p}}$

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(B) $\frac{m_p}{m_e}$

16. **Assertion (A)** : A photon behaves as a particle. **1**
Reason (R) : A photon possesses both energy and momentum.

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16. **Assertion (A)** : A photon behaves as a particle. **1**
Reason (R) : A photon possesses both energy and momentum.

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(A) Both assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the assertion (A).

30. A photon is a quantum of energy. The energy of a photon depends on its frequency (ν). When light of suitable frequency is incident on a metal surface, photo-electrons are emitted from the surface. The minimum frequency is called threshold frequency (ν_0) and the minimum energy required to emit an electron is called the work function (ϕ_0) of that metal. The energy of an incident photon is utilised in two ways (i) liberating the electron and (ii) the remaining energy is given to the ejected electron as its kinetic energy i.e. $h\nu = \phi_0 + \text{K.E.}$ $4 \times 1 = 4$

- (i) The packet of electromagnetic energy is called – **1**
- | | |
|--------------|-------------|
| (A) Photon | (B) Neutron |
| (C) Molecule | (D) Quark |

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- (i) The packet of electromagnetic energy is called – **1**
- | | |
|--------------|-------------|
| (A) Photon | (B) Neutron |
| (C) Molecule | (D) Quark |

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(A) Photon

- (ii) A photon has energy 3.3×10^{-19} J. Its momentum is – **1**
- (A) 9.9×10^{-11} kg.m/s (B) 9.9×10^{-26} kg.m/s
- (C) 1.1×10^{-27} kg.m/s (D) 3.3×10^{-25} kg.m/s

- (ii) A photon has energy 3.3×10^{-19} J. Its momentum is – **1**
- (A) 9.9×10^{-11} kg.m/s (B) 9.9×10^{-26} kg.m/s
- (C) 1.1×10^{-27} kg.m/s (D) 3.3×10^{-25} kg.m/s

(C) 1.1×10^{-27} kg m/s

- (iii) Monochromatic light of frequency 5.0×10^{14} Hz is produced by a source. The power emitted is 3.315 mW. The number of photons emitted per second, on an average, by the source is – **1**
- (A) 5.0×10^{14} (B) 6.0×10^{15}
(C) 1.0×10^{16} (D) 2.0×10^{17}

(iii) Monochromatic light of frequency 5.0×10^{14} Hz is produced by a source. The power emitted is 3.315 mW. The number of photons emitted per second, on an average, by the source is – **1**

(A) 5.0×10^{14}

(B) 6.0×10^{15}

(C) 1.0×10^{16}

(D) 2.0×10^{17}

(C) 1.0×10^{16}

(iv) (a) The work function of a metal is 2.14 eV. When light of frequency 6×10^{14} Hz is incident on the metal surface, photoemission of electrons occurs. The maximum kinetic energy of the emitted electrons is –

1

(A) 0.34×10^{-18} J

(B) 0.40×10^{-17} J

(C) 0.51×10^{-18} J

(D) 0.55×10^{-19} J

(iv) (a) The work function of a metal is 2.14 eV. When light of frequency 6×10^{14} Hz is incident on the metal surface, photoemission of electrons occurs. The maximum kinetic energy of the emitted electrons is – **1**

- | | |
|------------------------------|------------------------------|
| (A) 0.34×10^{-18} J | (B) 0.40×10^{-17} J |
| (C) 0.51×10^{-18} J | (D) 0.55×10^{-19} J |

(D) 0.55×10^{-19} J

(b) The work function of a metal is 2.21 eV. The threshold frequency for the metal is

1

(A) $2.1 \times 10^{13} \text{ Hz}$

(B) $3.2 \times 10^{13} \text{ Hz}$

(C) $5.3 \times 10^{14} \text{ Hz}$

(D) $4.1 \times 10^{15} \text{ Hz}$

(b) The work function of a metal is 2.21 eV. The threshold frequency for the metal is

1

(A) $2.1 \times 10^{13} \text{ Hz}$

(B) $3.2 \times 10^{13} \text{ Hz}$

(C) $5.3 \times 10^{14} \text{ Hz}$

(D) $4.1 \times 10^{15} \text{ Hz}$

(C) $5.3 \times 10^{14} \text{ Hz}$

10. Which of the following electromagnetic waves has photons of largest momentum ?

(A) X-rays

(B) AM radio waves

(C) Microwaves

(D) TV waves

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(A) X-rays

(B) AM radio waves

(C) Microwaves

(D) TV waves

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(A) X- rays

11. The kinetic energy of an alpha particle is four times the kinetic energy of a proton. The ratio $\left(\frac{\lambda_{\alpha}}{\lambda_p}\right)$ of de Broglie wavelengths associated with them will be :

(A) $\frac{1}{16}$

(B) $\frac{1}{8}$

(C) $\frac{1}{4}$

(D) $\frac{1}{2}$

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11. The kinetic energy of an alpha particle is four times the kinetic energy of a proton. The ratio $\left(\frac{\lambda_{\alpha}}{\lambda_p}\right)$ of de Broglie wavelengths associated with them will be :

(A) $\frac{1}{16}$

(B) $\frac{1}{8}$

(C) $\frac{1}{4}$

(D) $\frac{1}{2}$

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(C) $\frac{1}{4}$

- 21.** The threshold frequency for a given metal is 3.6×10^{14} Hz. If monochromatic radiations of frequency 6.8×10^{14} Hz are incident on this metal, find the cut-off potential for the photoelectrons.

2

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- 21.** The threshold frequency for a given metal is 3.6×10^{14} Hz. If monochromatic radiations of frequency 6.8×10^{14} Hz are incident on this metal, find the cut-off potential for the photoelectrons.

2

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$eV = h(\nu - \nu_0)$	$\frac{1}{2}$
$V = \frac{6.63 \times 10^{-34} \times (6.8 - 3.6) \times 10^{14}}{1.6 \times 10^{-19}}$	1
$= 1.33 \text{ V}$	$\frac{1}{2}$

30. Einstein explained photoelectric effect on the basis of Planck's quantum theory, where light travels in the form of small bundles of energy called photons. The energy of each photon is $h\nu$, where ν is the frequency of incident light and h is Planck's constant. The number of photons in a beam of light determines the intensity of the incident light. A photon incident on a metal surface transfers its total energy $h\nu$ to a free electron in the metal. A part of this energy is used in ejecting the electron from the metal and is called its work function. The rest of the energy is carried by the ejected electron as its kinetic energy.

(i) Which of the following graphs can be used to obtain the value of Planck's constant ?

- (A) Photocurrent versus Intensity of incident light
- (B) Photocurrent versus Frequency of incident light
- (C) Cut-off potential versus Frequency of incident light
- (D) Cut-off potential versus Intensity of incident light

1

30. Einstein explained photoelectric effect on the basis of Planck's quantum theory, where light travels in the form of small bundles of energy called photons. The energy of each photon is $h\nu$, where ν is the frequency of incident light and h is Planck's constant. The number of photons in a beam of light determines the intensity of the incident light. A photon incident on a metal surface transfers its total energy $h\nu$ to a free electron in the metal. A part of this energy is used in ejecting the electron from the metal and is called its work function. The rest of the energy is carried by the ejected electron as its kinetic energy.

(i) Which of the following graphs can be used to obtain the value of Planck's constant ?

- (A) Photocurrent versus Intensity of incident light
- (B) Photocurrent versus Frequency of incident light
- (C) Cut-off potential versus Frequency of incident light
- (D) Cut-off potential versus Intensity of incident light

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(C) cut– off potential versus frequency of incident light

- (ii) (a) Red light, yellow light and blue light of the same intensity are incident on a metal surface successively. K_R , K_Y and K_B represent the maximum kinetic energy of photoelectrons respectively, then :

- (A) $K_R > K_Y > K_B$ (B) $K_Y > K_B > K_R$
(C) $K_B > K_Y > K_R$ (D) $K_R > K_B > K_Y$

- (ii) (a) Red light, yellow light and blue light of the same intensity are incident on a metal surface successively. K_R , K_Y and K_B represent the maximum kinetic energy of photoelectrons respectively, then :

- (A) $K_R > K_Y > K_B$ (B) $K_Y > K_B > K_R$
(C) $K_B > K_Y > K_R$ (D) $K_R > K_B > K_Y$

1

(C) $K_B > K_Y > K_R$

(b) Which of the following metals exhibits photoelectric effect with visible light ?

1

(A) Caesium

(B) Zinc

(C) Cadmium

(D) Magnesium

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(b) Which of the following metals exhibits photoelectric effect with visible light ? 1

(A) Caesium

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(A) Caesium

(iii) When the frequency of the incident light is increased without changing its intensity, the saturation current : *1*

- (A) increases linearly
- (B) decreases
- (C) increases non-linearly
- (D) remains the same

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(iii) When the frequency of the incident light is increased without changing its intensity, the saturation current : *1*

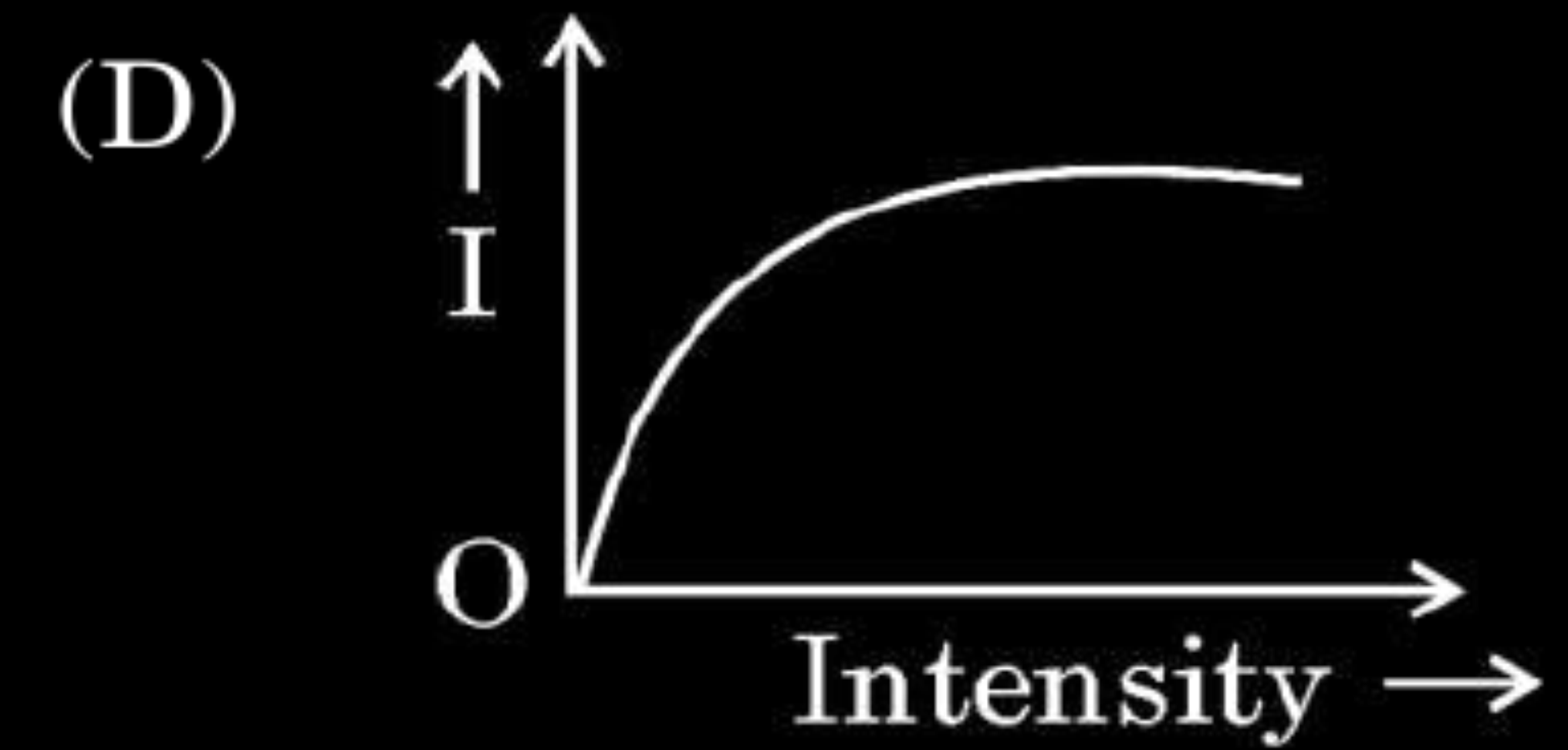
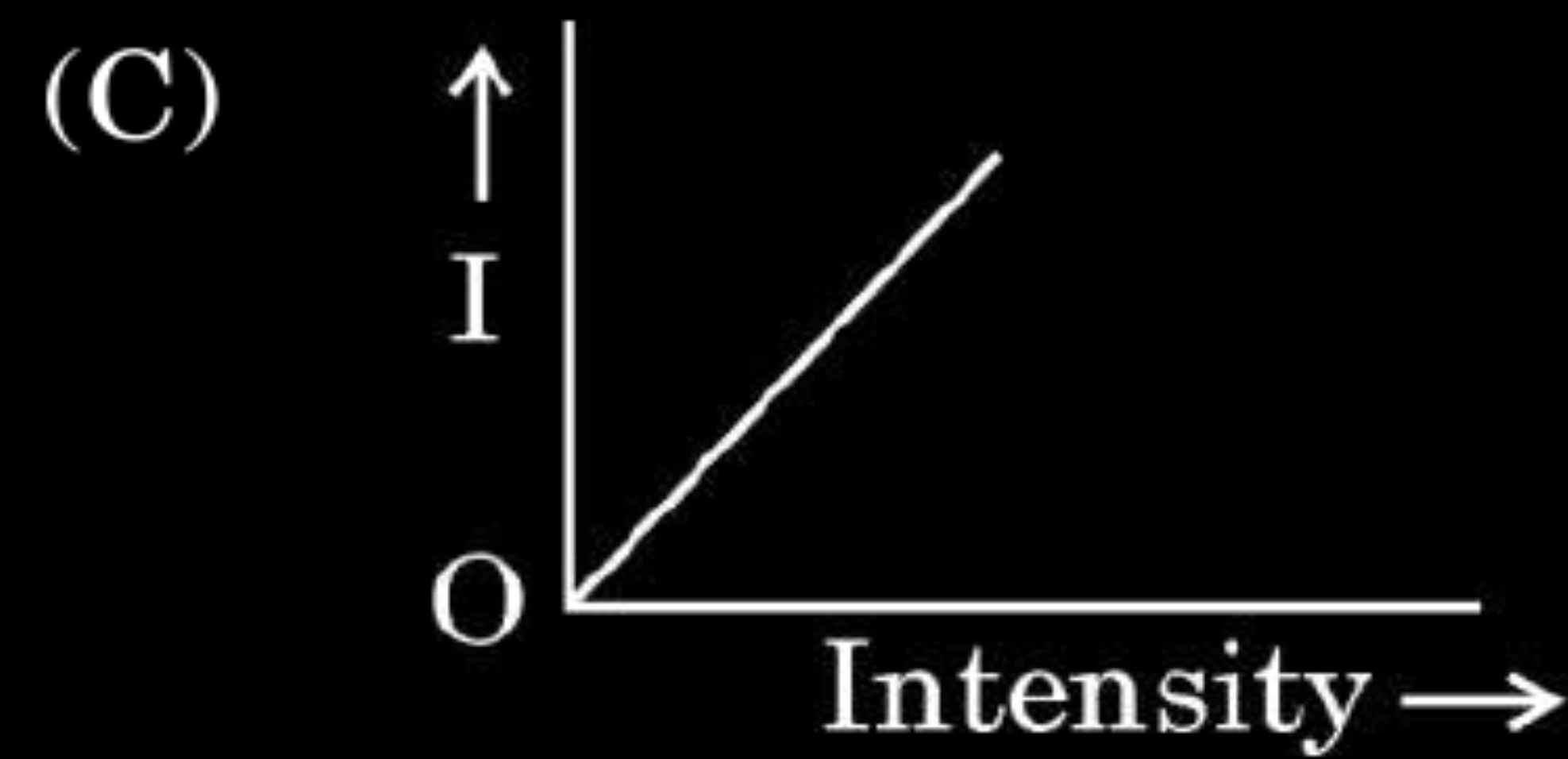
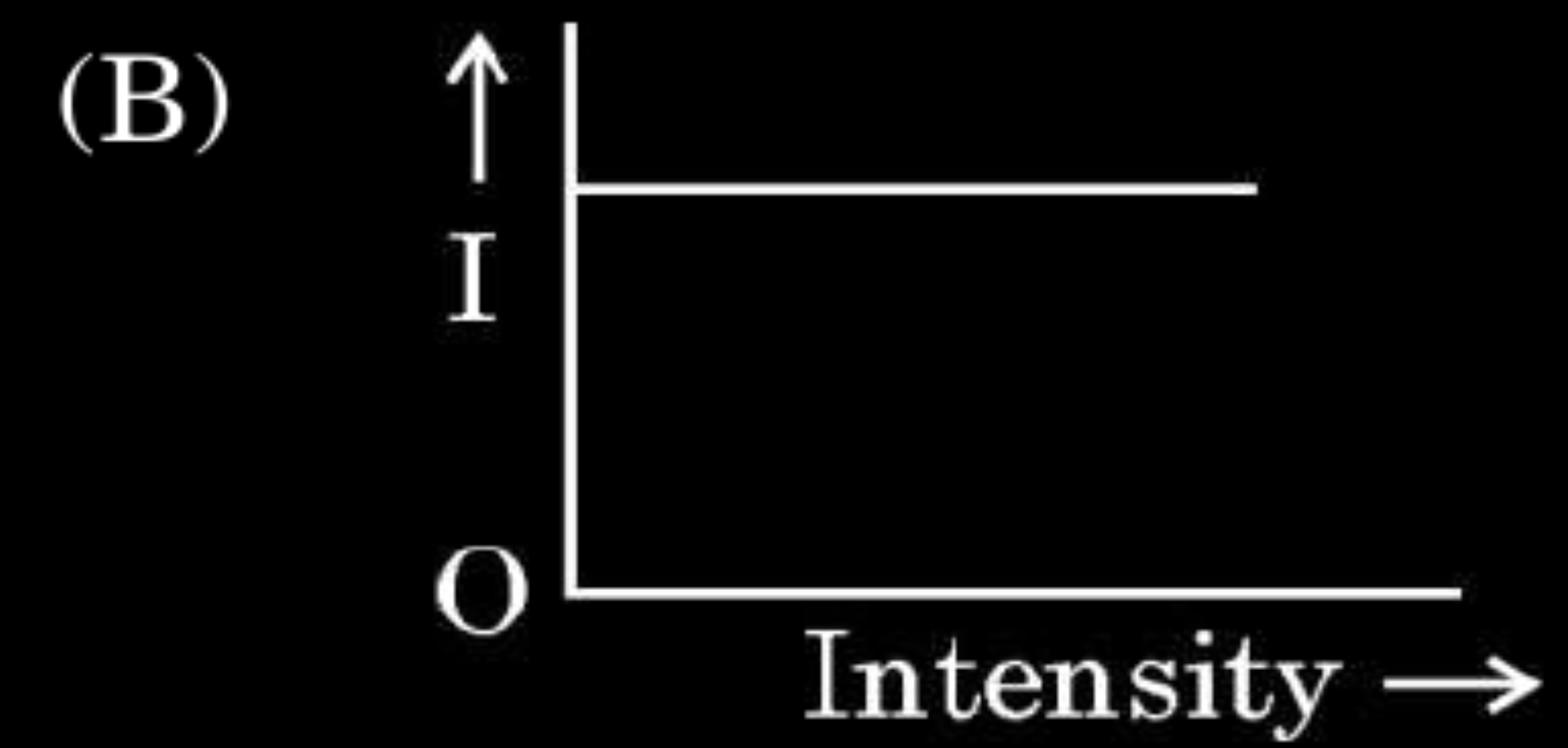
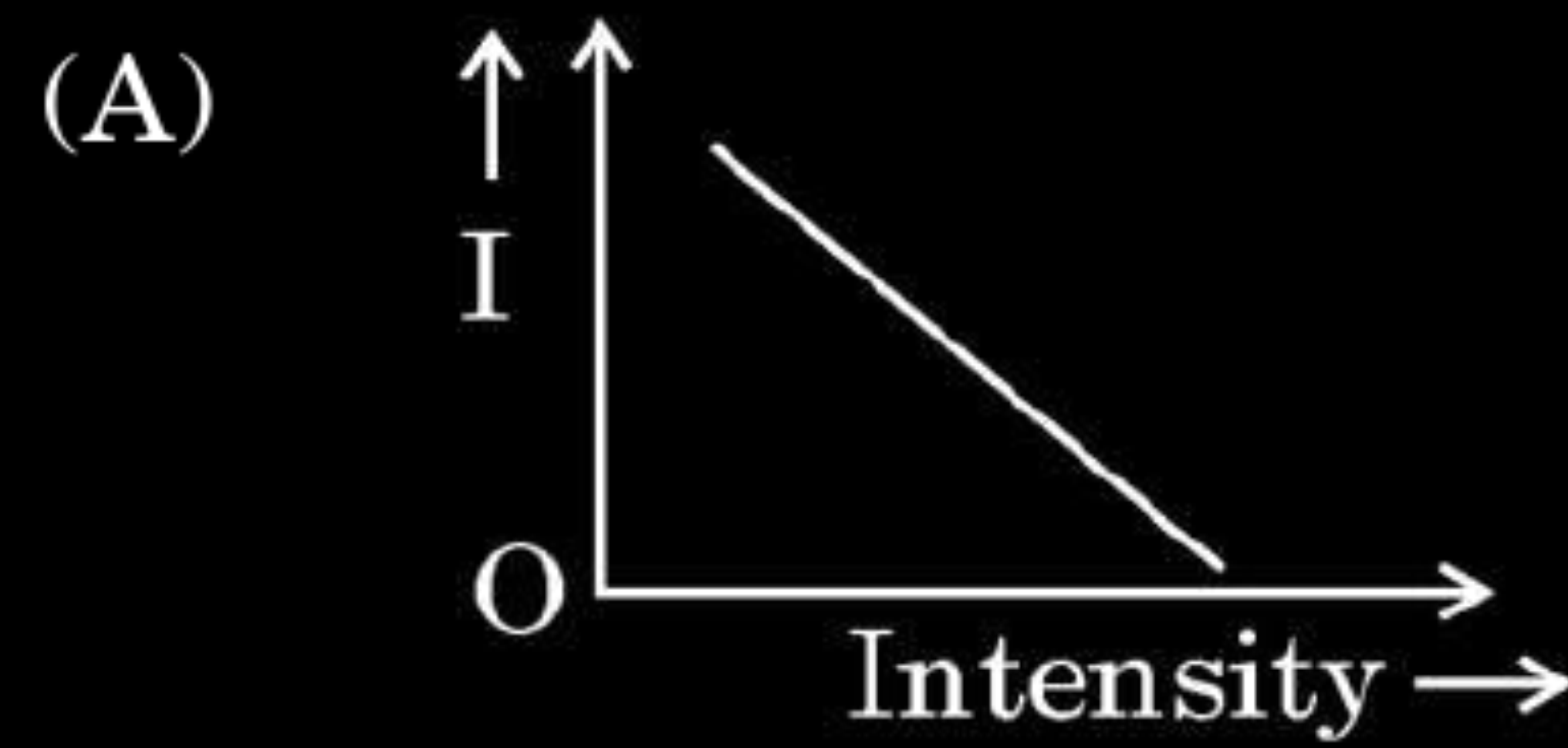
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(D) Remains the same

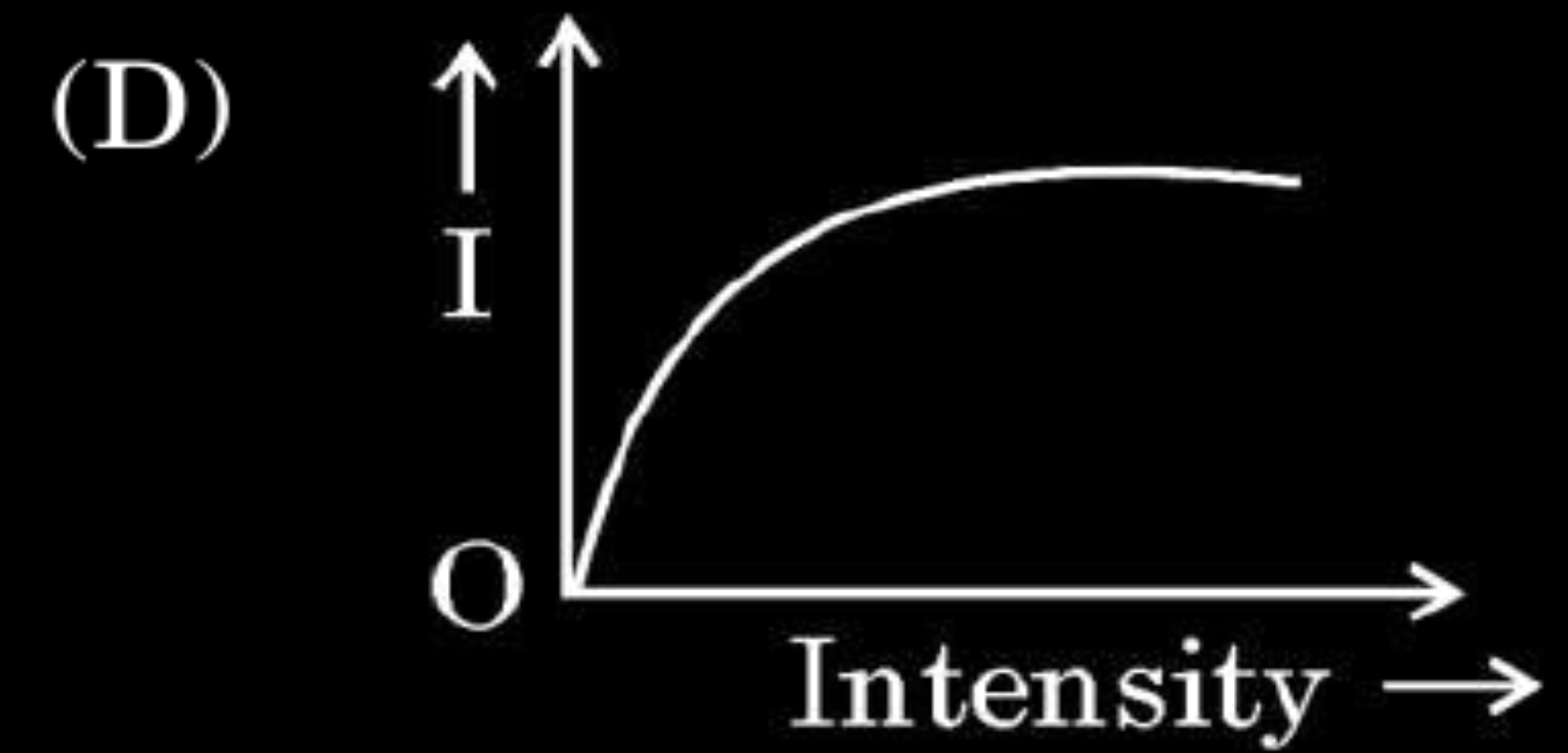
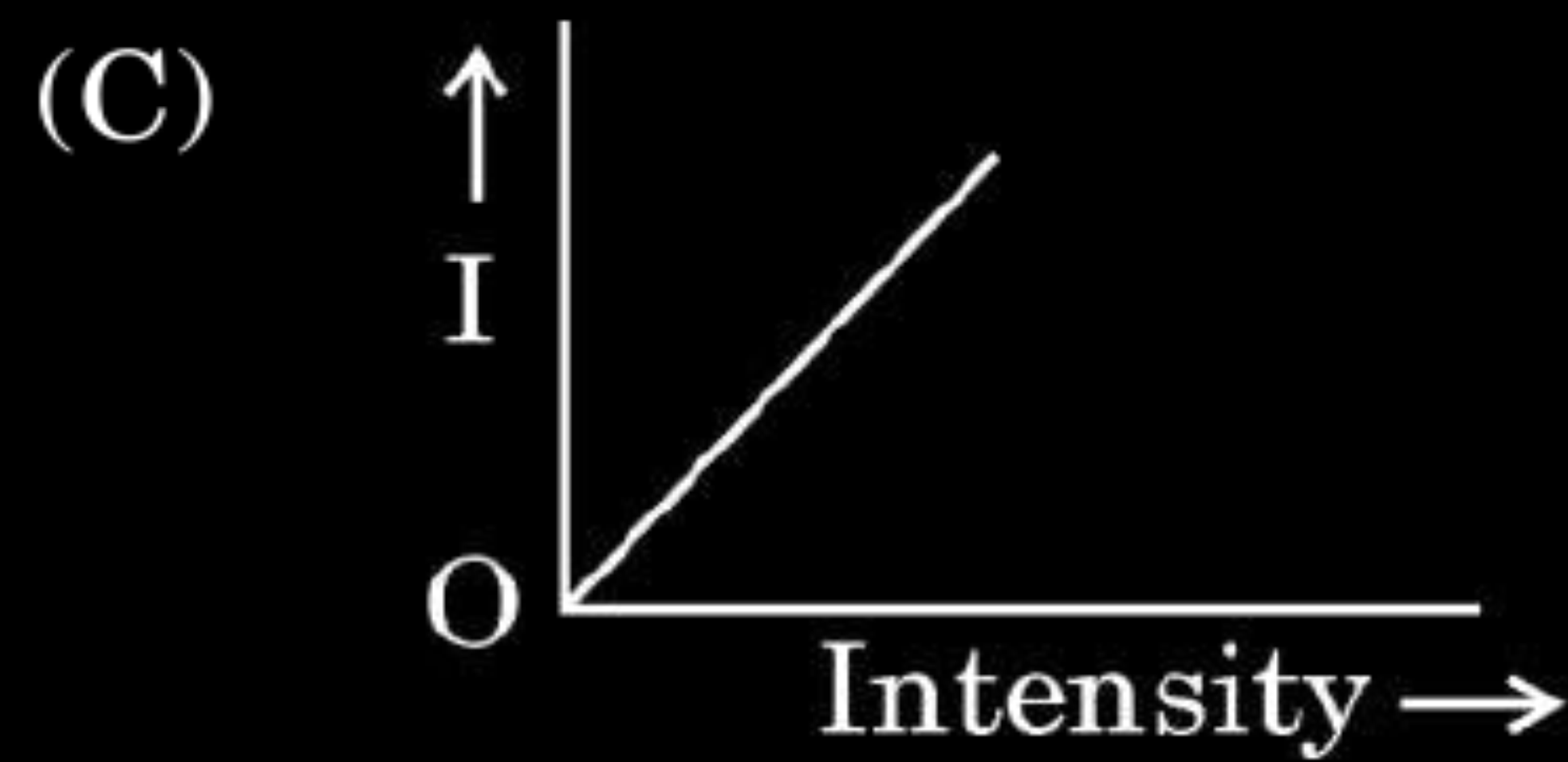
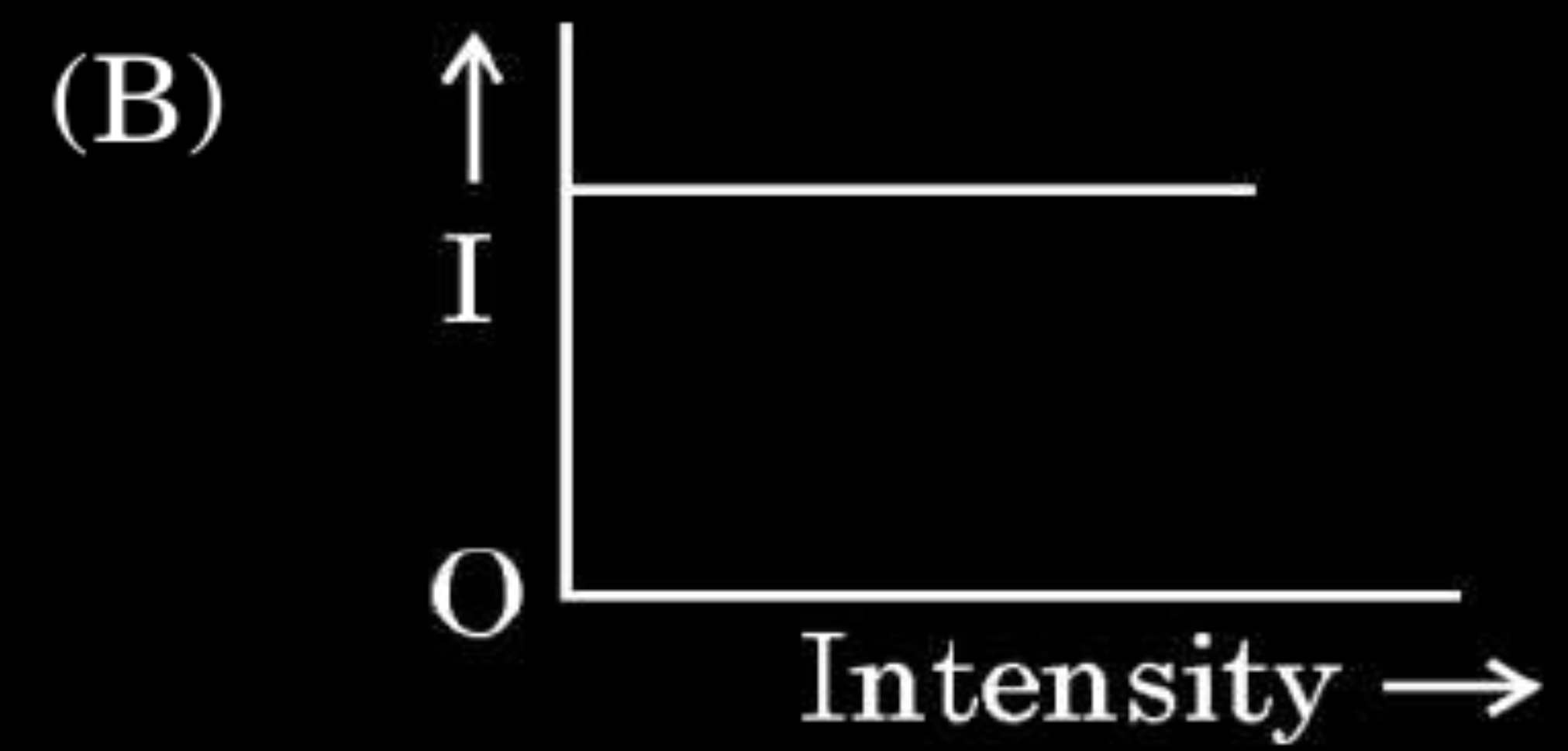
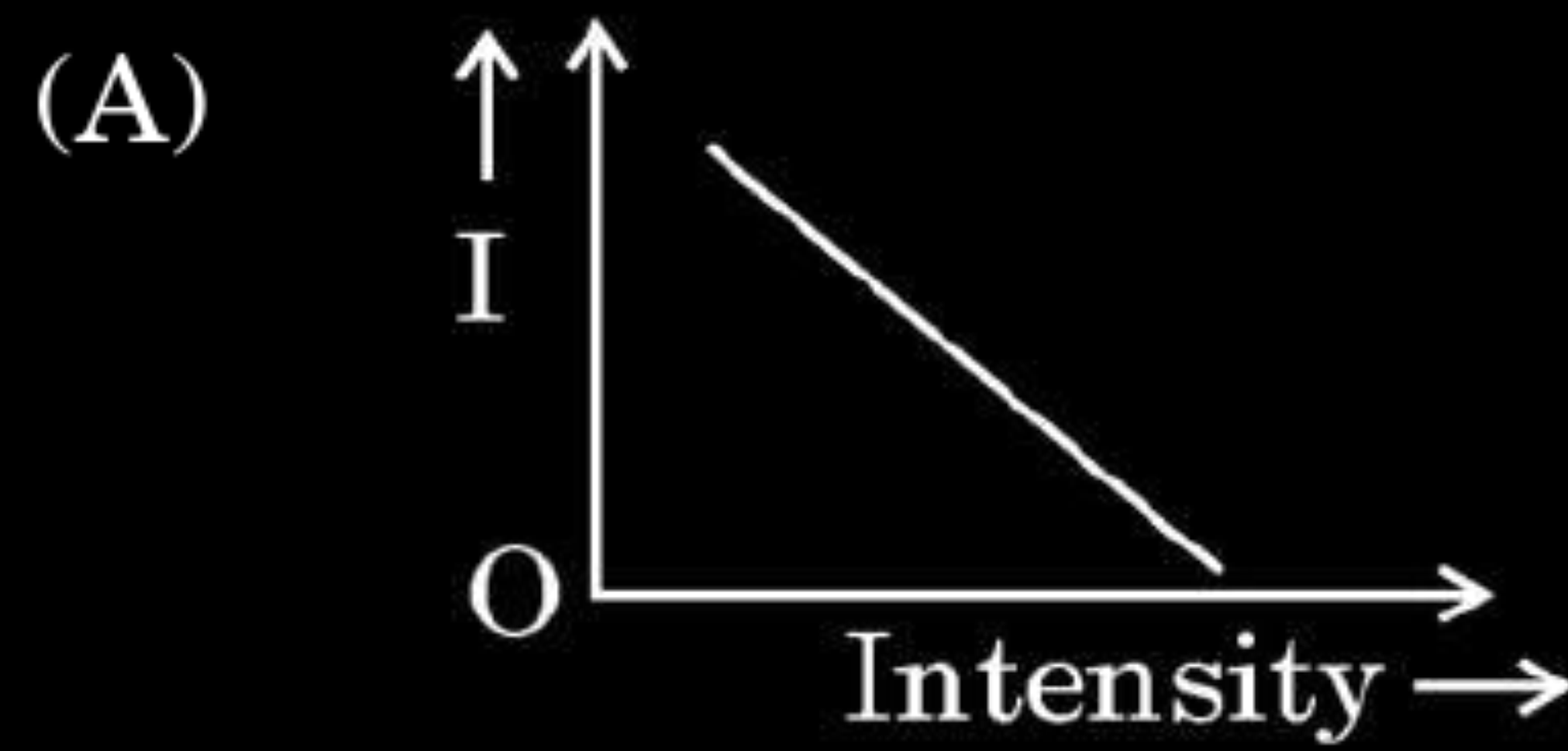
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1

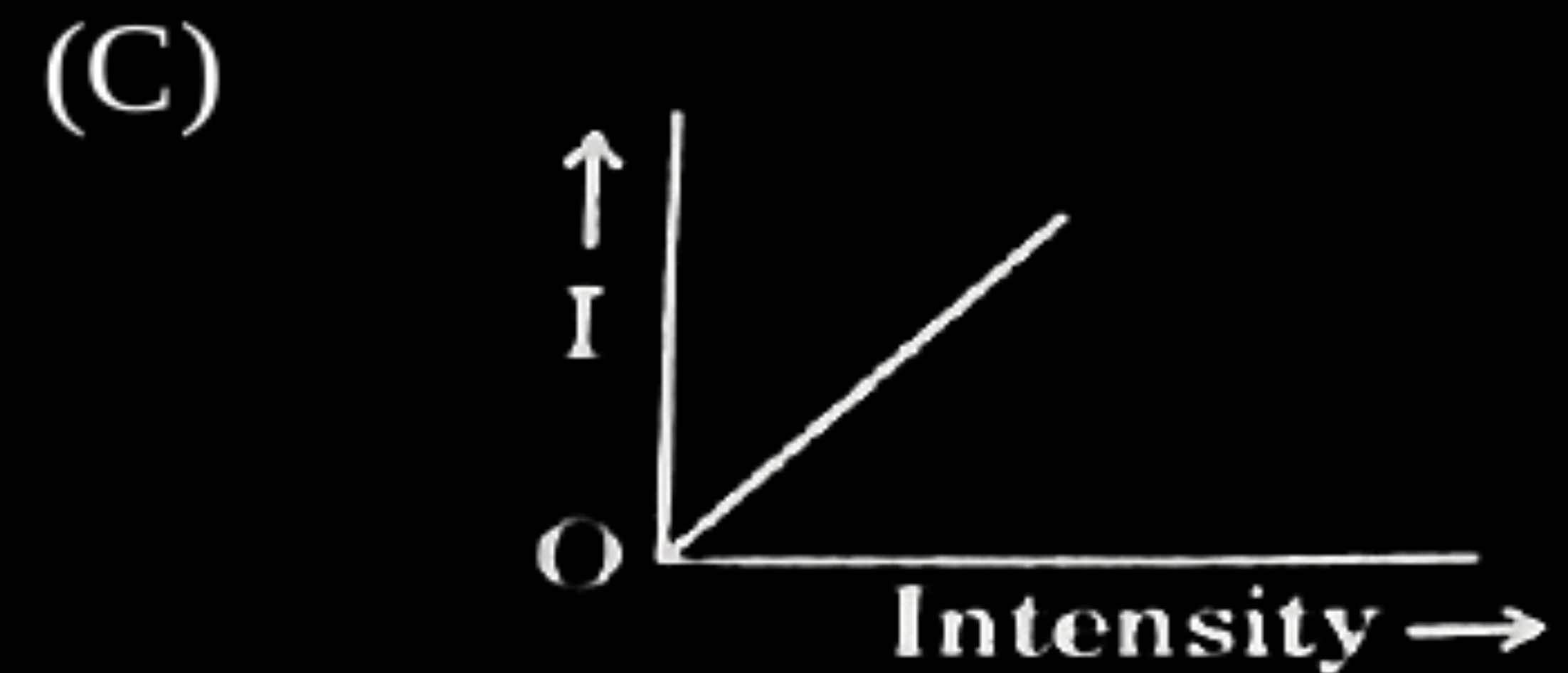


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- (iv) Which of the following graphs shows the variation of photoelectric current I with the intensity of light ? 1



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10. Choose the correct statement :

- (A) Photons of light show diffraction whereas electrons do not show diffraction.
- (B) Electrons have momentum whereas photons do not have momentum.
- (C) Photons of light and electrons both exhibit dual nature.
- (D) All electromagnetic radiations do not have photons.

10. Choose the correct statement :

- (A) Photons of light show diffraction whereas electrons do not show diffraction.
- (B) Electrons have momentum whereas photons do not have momentum.
- (C) Photons of light and electrons both exhibit dual nature.
- (D) All electromagnetic radiations do not have photons.

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(C) Photons of light and electrons both exhibit dual nature

- 11.** A beam of red light and a beam of blue light have equal intensities. Which of the following statements is true ?
- (A) The blue beam has more number of photons than the red beam.
 - (B) The red beam has more number of photons than the blue beam.
 - (C) Wavelength of red light is lesser than wavelength of blue light.
 - (D) The blue light beam has lesser energy per photon than that in the red light beam.

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 - (C) Wavelength of red light is lesser than wavelength of blue light.
 - (D) The blue light beam has lesser energy per photon than that in the red light beam.

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(B) The red beam has more numbers of photons than the blue beam

14. *Assertion (A) :* For monochromatic incident radiation, the emitted photoelectrons from a given metal have speed ranging from zero to a certain maximum value.

Reason (R) : Each metal has a definite work function.

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14. *Assertion (A) :* For monochromatic incident radiation, the emitted photoelectrons from a given metal have speed ranging from zero to a certain maximum value.

Reason (R) : Each metal has a definite work function.

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(B) Both Assertion (A) and Reason (R) are true but (R) is not the correct explanation of the Assertion (A)

25. Explain the following observations using Einstein's photoelectric equation : 3

- (a) Photoelectric emission does not occur from a surface when the frequency of the light incident on it is less than a certain minimum value.
- (b) It is the frequency, and not the intensity of the incident light which affects the maximum kinetic energy of the photoelectrons.
- (c) The cut-off voltage (V_0) versus frequency (ν) of the incident light curve is a straight line with a slope $\frac{h}{e}$.

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(a) Einstein Photo electric equation

$$h\nu = h\nu_o + K_{\max}$$

$$K_{\max} = h(\nu - \nu_o)$$

For $\nu < \nu_o$, $K_{\max}$ will be negative

Hence, Photoelectric emission is not possible.

(b) According to Einstein Photoelectric equation

$$K_{\max} = h(\nu - \nu_o)$$

Hence $K_{\max} \propto \nu$

It shows $K_{\max}$ depends upon frequency only and not depends upon intensity.

(c) $eV_o = h\nu - h\nu_o$

$$V_o = \frac{h}{e}\nu - \frac{h}{e}\nu_o$$

This equation represents the equation of straight line ($y = mx + c$) with the

slope $\frac{h}{e}$.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

20. The threshold wavelength of a metal is 450 nm. Calculate (i) the work function of the metal in eV and (ii) the maximum energy of the ejected photoelectrons in eV by incident radiation of 250 nm.

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20. Radiations of two frequencies are incident on a metal surface of work function 2.0 eV one by one. The energies of their photons are 2.5 eV and 4.5 eV respectively. Find the ratio of the maximum speed of the electrons emitted in the two cases.

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2

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2

$$h\nu = \phi_0 + K_{\max}$$

For radiation having photons of energy 2.5 eV

$$2.5 \text{ eV} = 2 \text{ eV} + \frac{1}{2}mv_1^2$$

$$0.5 \text{ eV} = \frac{1}{2}mv_1^2 \dots\dots\dots(i)$$

For radiation having photons of energy 4.5 eV

$$4.5 \text{ eV} = 2 \text{ eV} + \frac{1}{2}mv_2^2$$

$$2.5 \text{ eV} = \frac{1}{2}mv_2^2 \dots\dots\dots(ii)$$

Dividing eq. (i) by (ii): -

$$\frac{v_1}{v_2} = \sqrt{\frac{1}{5}}$$

OR

$$\frac{v_2}{v_1} = \sqrt{5}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

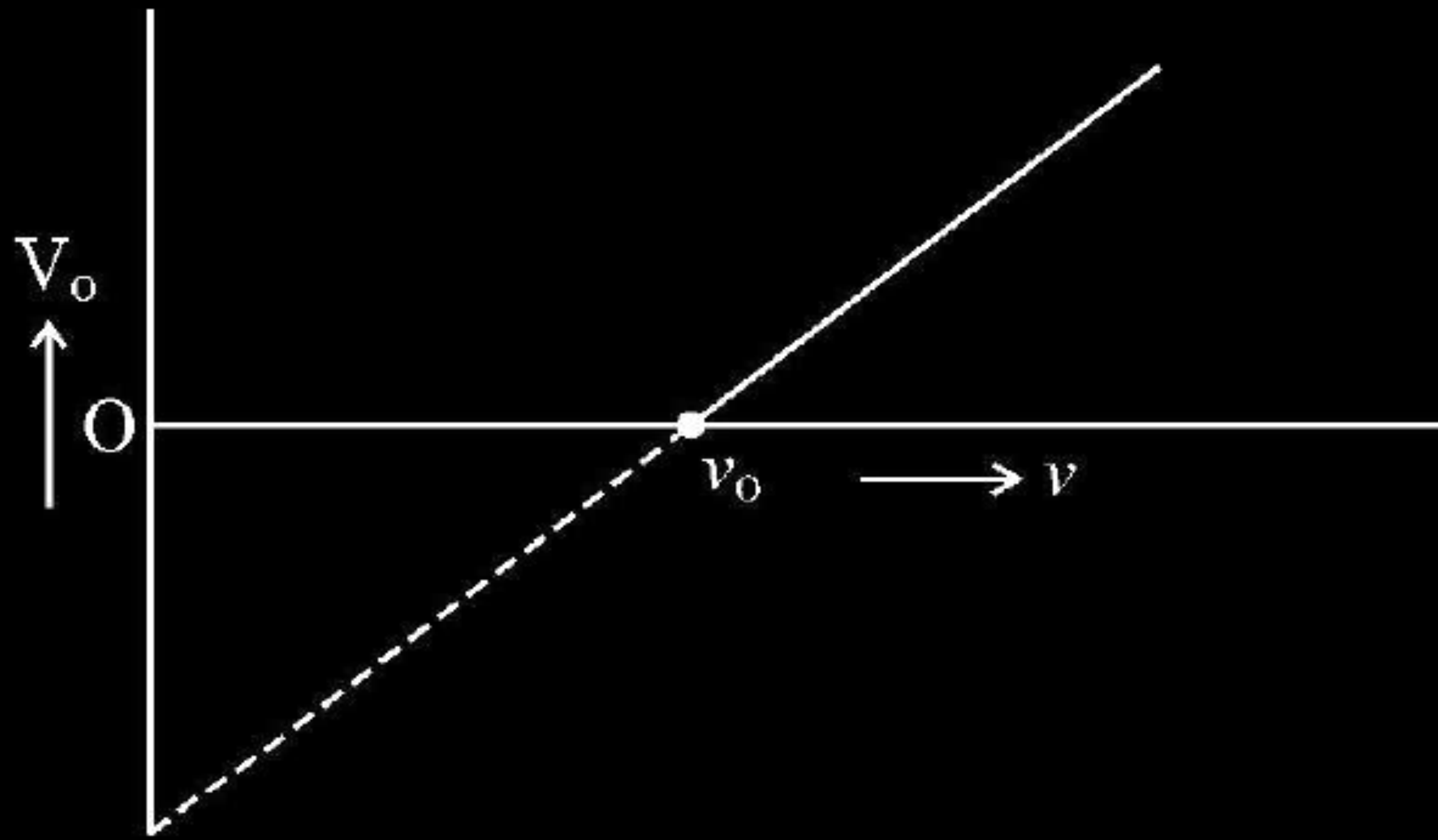
$\frac{1}{2}$

29. When a photon of suitable frequency is incident on a metal surface, photoelectron is emitted from it. If the frequency is below a threshold frequency (ν_0) for the surface, no photoelectron is emitted. For a photon of frequency ν ($\nu > \nu_0$), the kinetic energy of the emitted photoelectrons is $h(\nu - \nu_0)$. The photocurrent can be stopped by applying a potential V_0 called 'stopping potential' on the anode. Thus maximum kinetic energy of photoelectrons $K_m = eV_0 = h(\nu - \nu_0)$. The experimental graph between V_0 and ν for a metal is shown in figure. This is a straight line of slope m . $4 \times 1 = 4$

(i) The straight line graphs obtained for two metals

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- (A) coincide each other.
- (B) are parallel to each other.
- (C) are not parallel to each other and cross at a point on ν -axis.
- (D) are not parallel to each other and do not cross at a point on ν -axis.



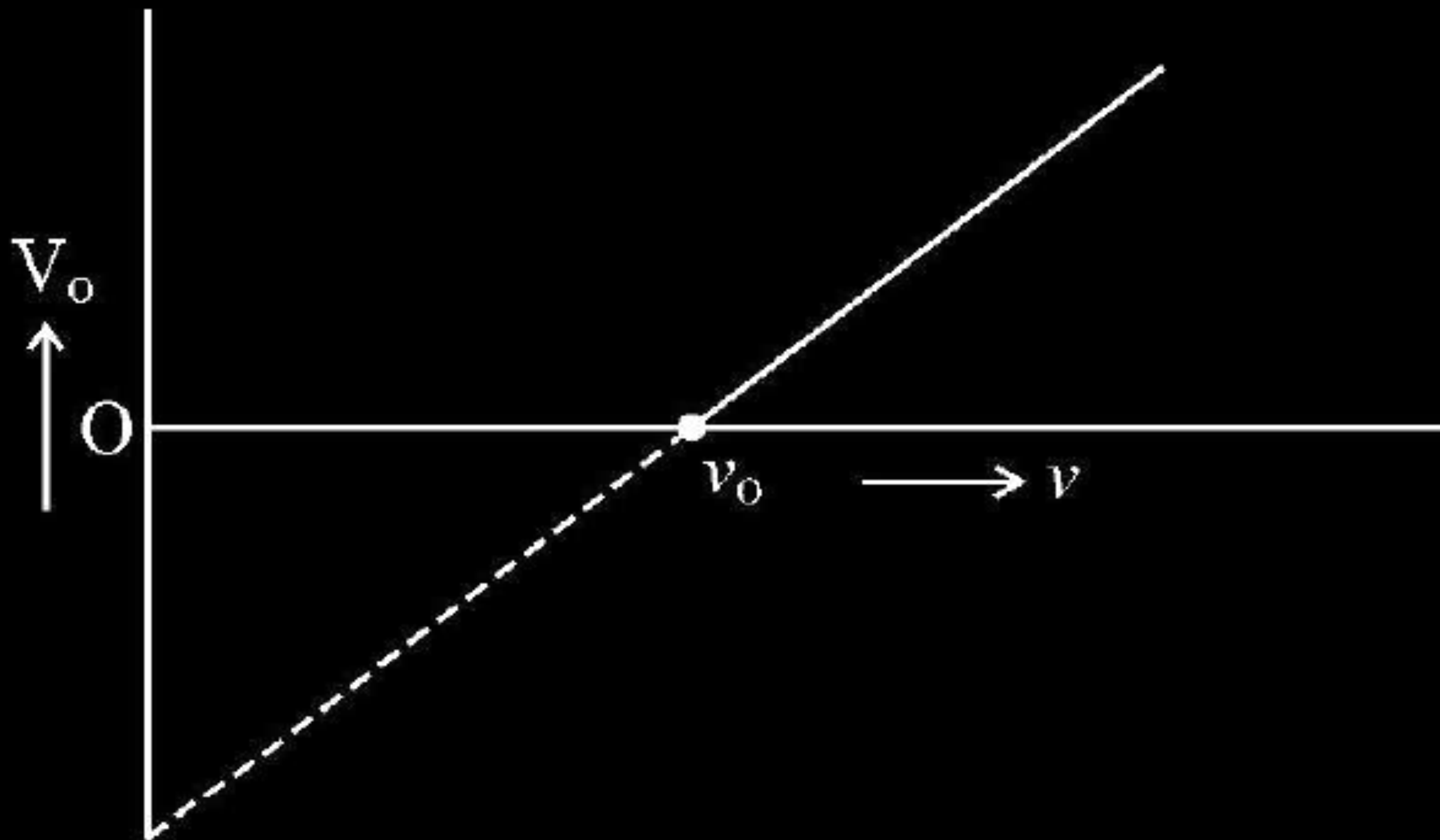
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CBSE 2025

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- (D) are not parallel to each other and do not cross at a point on ν -axis.

(B) are parallel to each other.



(ii) The value of Planck's constant for this metal is

(A) $\frac{e}{m}$

(B) $\frac{1}{me}$

(C) me

(D) $\frac{m}{e}$

(ii) The value of Planck's constant for this metal is

(A) $\frac{e}{m}$

(B) $\frac{1}{me}$

(C) me

(D) $\frac{m}{e}$

(C) me

(iii) The intercepts on ν -axis and V_o -axis of the graph are respectively :

(A) $\nu_o, \frac{h\nu_o}{e}$

(B) $\nu_o, h\nu_o$

(C) $\frac{h\nu_o}{e}, \nu_o$

(D) $h\nu_o, \nu_o$

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(C) $\frac{h\nu_o}{e}, \nu_o$

(D) $h\nu_o, \nu_o$

(iii) Full 1 mark to be awarded to all the students who have attempted this part of the question.

(iii) When the wavelength of a photon is doubled, how many times its wave number and frequency become, respectively ?

(A) $2, \frac{1}{2}$

(B) $\frac{1}{2}, \frac{1}{2}$

(C) $\frac{1}{2}, 2$

(D) $2, 2$

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(B) $\frac{1}{2}, \frac{1}{2}$

(C) $\frac{1}{2}, 2$

(D) $2, 2$

(B) $\frac{1}{2}, \frac{1}{2}$

(iv) The momentum of a photon is 5.0×10^{-29} kg. m/s. Ignoring relativistic effects (if any), the wavelength of the photon is

(A) $1.33 \mu\text{m}$

(B) $3.3 \mu\text{m}$

(C) $16.6 \mu\text{m}$

(D) $13.3 \mu\text{m}$

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20. A light of wavelength 400 nm is incident on metal surface whose work function is 3.0×10^{-19} J. Calculate the speed of the fastest photoelectrons emitted. **2**

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20. A light of wavelength 400 nm is incident on metal surface whose work function is 3.0×10^{-19} J. Calculate the speed of the fastest photoelectrons emitted.

2

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$$h\nu = \phi_0 + K_{\max}$$

$$\frac{hc}{\lambda} = \phi_0 + \frac{1}{2}mv_{\max}^2$$

$$\frac{6.63 \times 10^{-34} \times 3 \times 10^8}{4 \times 10^{-7}} = 3 \times 10^{-19} + \frac{1}{2}mv_{\max}^2$$

$$\frac{19.89}{4} \times 10^{-19} = 3 \times 10^{-19} + \frac{1}{2}mv_{\max}^2$$

$$\frac{4 \times 10^{-19}}{9 \times 10^{-31}} = v_{\max}^2$$

$$v_{\max} = \frac{2}{3} \times 10^6 \text{ m/s}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

11. Let λ_e , λ_p and λ_d be the wavelengths associated with an electron, a proton and a deuteron, all moving with the same speed. Then the correct relation between them is

(A) $\lambda_d > \lambda_p > \lambda_e$

(B) $\lambda_e > \lambda_p > \lambda_d$

(C) $\lambda_p > \lambda_e > \lambda_d$

(D) $\lambda_e = \lambda_p = \lambda_d$

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(D) $\lambda_e = \lambda_p = \lambda_d$

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1

(B) $\lambda_e > \lambda_p > \lambda_d$

26. (a) Mention any three features of results of experiment on photoelectric effect which cannot be explained using the wave theory of light. 3
- (b) In his experiment on photoelectric effect, Robert A. Millikan found the slope of the cut-off voltage versus frequency of incident light plot to be 4.12×10^{-15} Vs. Calculate the value of Planck's constant from it.

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a) Three features	
i) The existence of threshold frequency (ν_0)	$\frac{1}{2}$
ii) Maximum Kinetic energy of photoelectrons is independent of intensity of incident radiation.	$\frac{1}{2}$
iii) Instantaneous nature of photoelectric effect.	$\frac{1}{2}$
b) slope = $\frac{h}{e}$	1
$\therefore h = e \times \text{slope}$ $= 1.6 \times 10^{-19} \times 4.12 \times 10^{-15}$ $= 6.6 \times 10^{-34} \text{ Js}$	$\frac{1}{2}$

9. A source produces monochromatic light of frequency 5.0×10^{14} Hz and the power emitted is 3.31 mW. The number of photons emitted per second by the source, on an average is

(A) 10^{16}

(B) 10^{24}

(C) 10^{10}

(D) 10^{20}

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CBSE 2025

1

(A) 10^{16}

26. (a) Draw a plot of frequency ν of incident radiations as a function of stopping potential V_0 for a given photo emissive material. What information can be obtained from the value of the intercept on the stopping potential axis ?

3

(b) Calculate : (i) the momentum and (ii) de Broglie wavelength, of an electron with kinetic energy of 80 eV.

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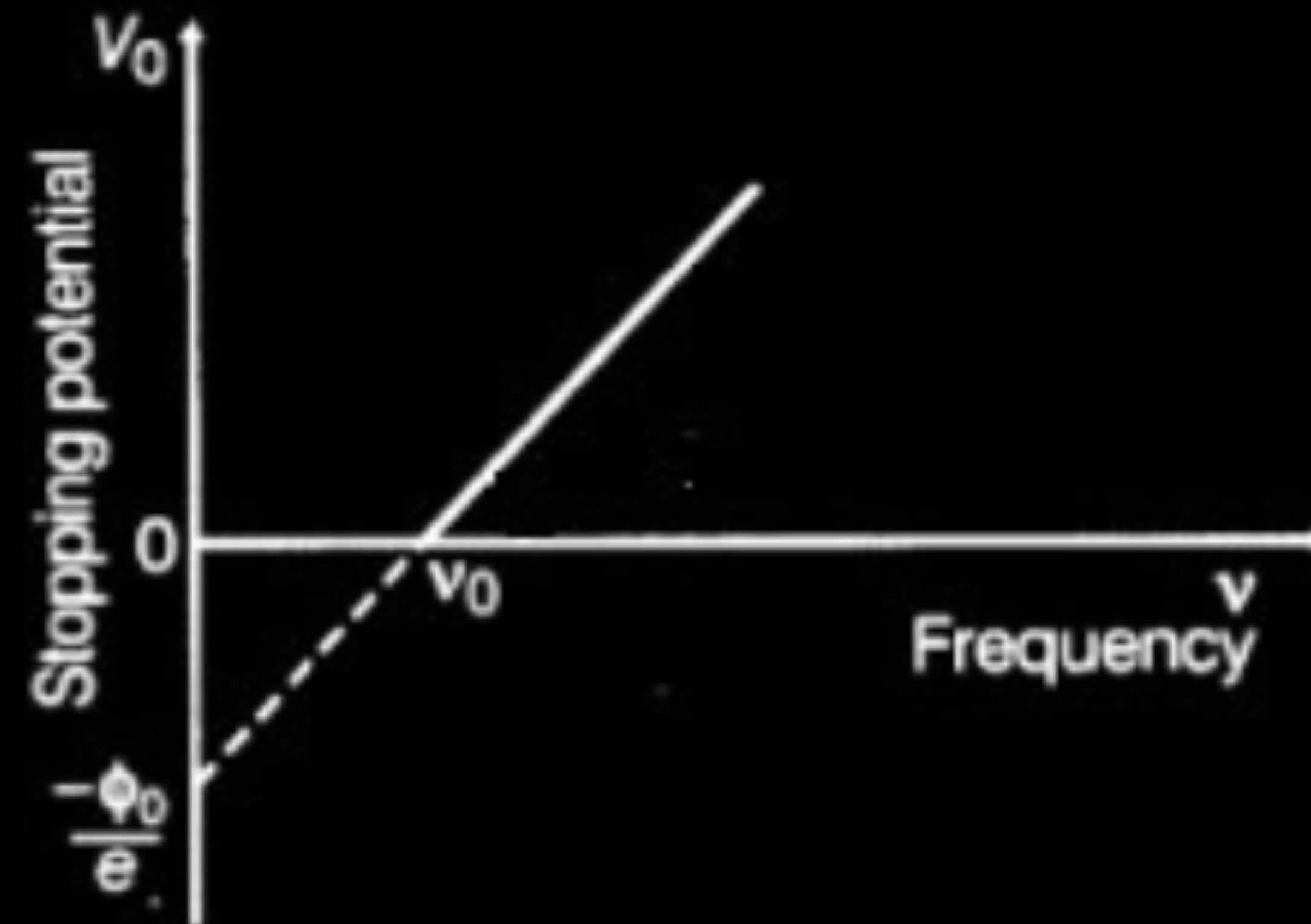
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a)



Value of work function can be obtained from intercept.

b) i) $p = \sqrt{2mK}$

$$= \sqrt{2 \times 9.1 \times 10^{-31} \times 80 \times 1.6 \times 10^{-19}}$$

$$= 4.8 \times 10^{-24} \text{ kg m/s}$$

ii) $\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{4.8 \times 10^{-25}} = 1.38 \times 10^{-9} \text{ m}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

9. The stopping potential V_0 measured in a photoelectric experiment for a metal surface is plotted against frequency ν of the incident radiation. Let m be the slope of the straight line so obtained. Then the value of charge of an electron is given by (h is the Planck's constant.)

(A) mh

(B) $\frac{m}{h}$

(C) $\frac{h}{m}$

(D) $\frac{1}{mh}$

CBSE 2025

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(A) mh

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(C) $\frac{h}{m}$

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CBSE 2025

(C) $\frac{h}{m}$

26. Answer the following giving reason :

- (a) All the photo electrons do not eject with the same kinetic energy when monochromatic light is incident on a metal surface.
- (b) The saturation current in case (a) is different for different intensity.
- (c) If one goes on increasing the wavelength of light incident on a metal surface, keeping its intensity constant, emission of photo electrons stop at a certain wavelength for this metal.

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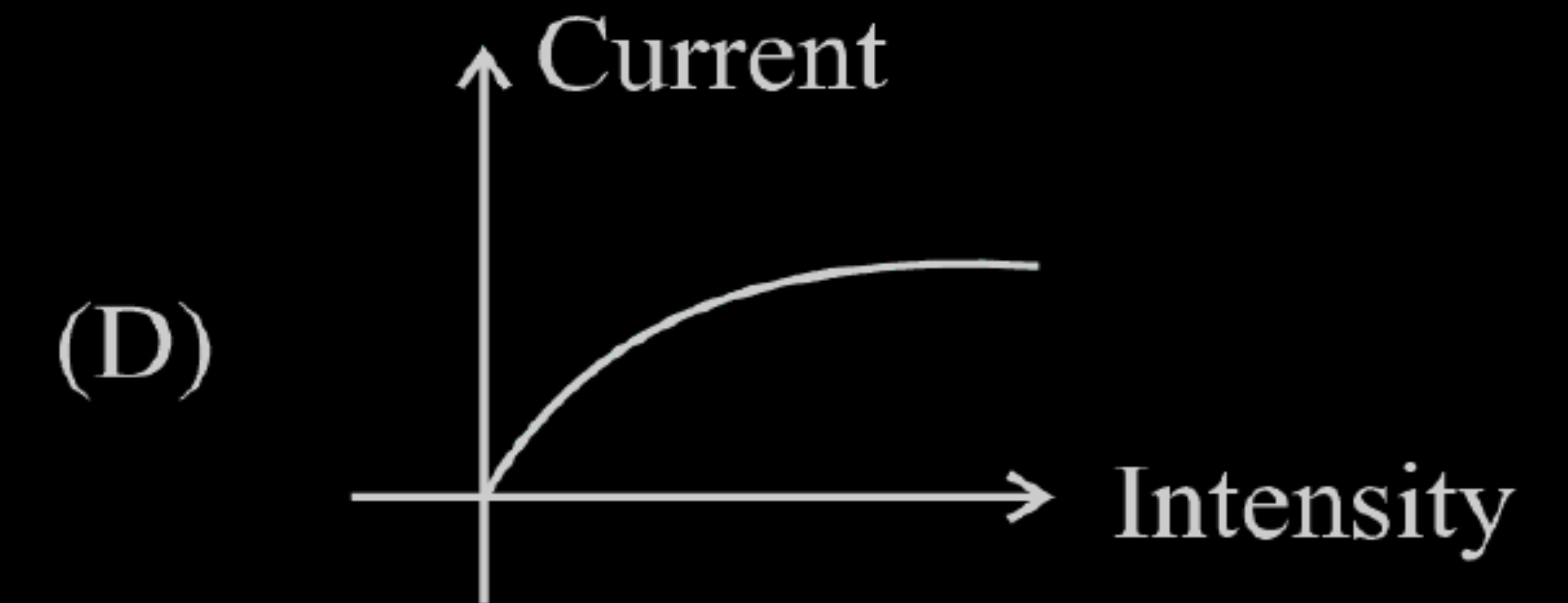
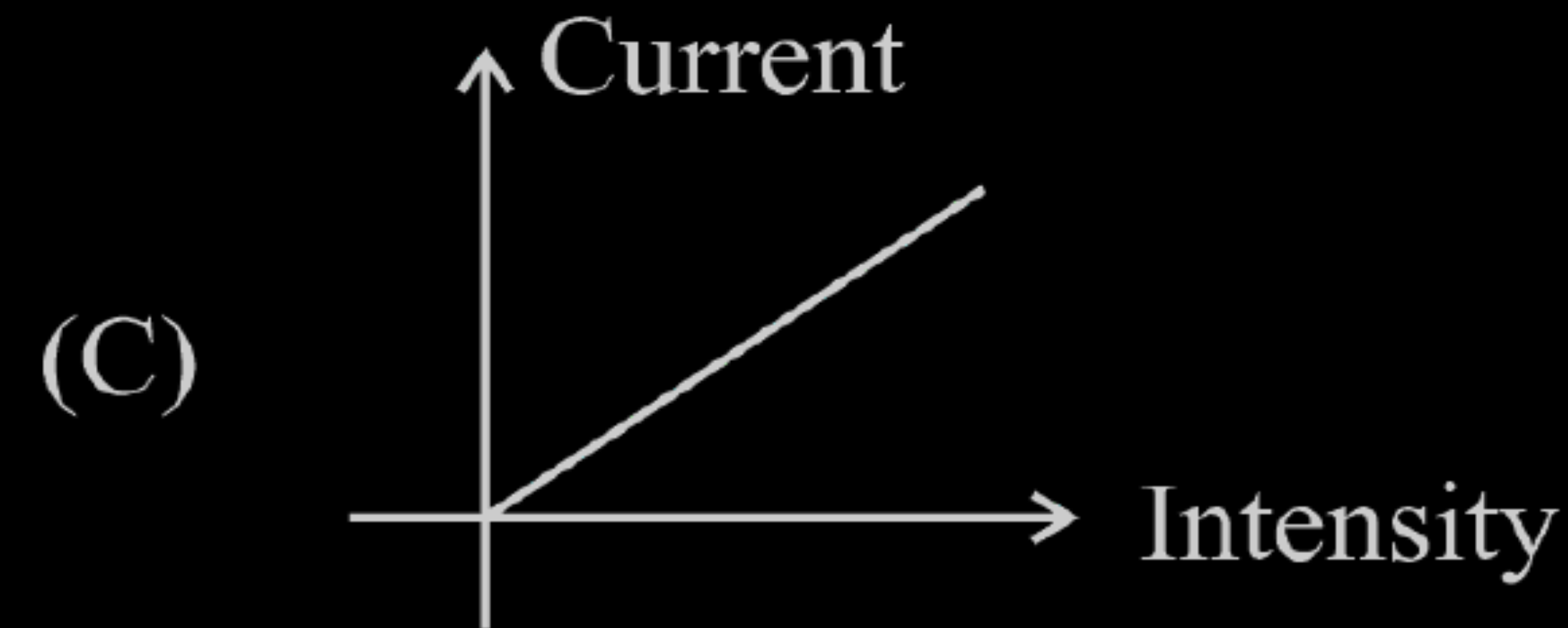
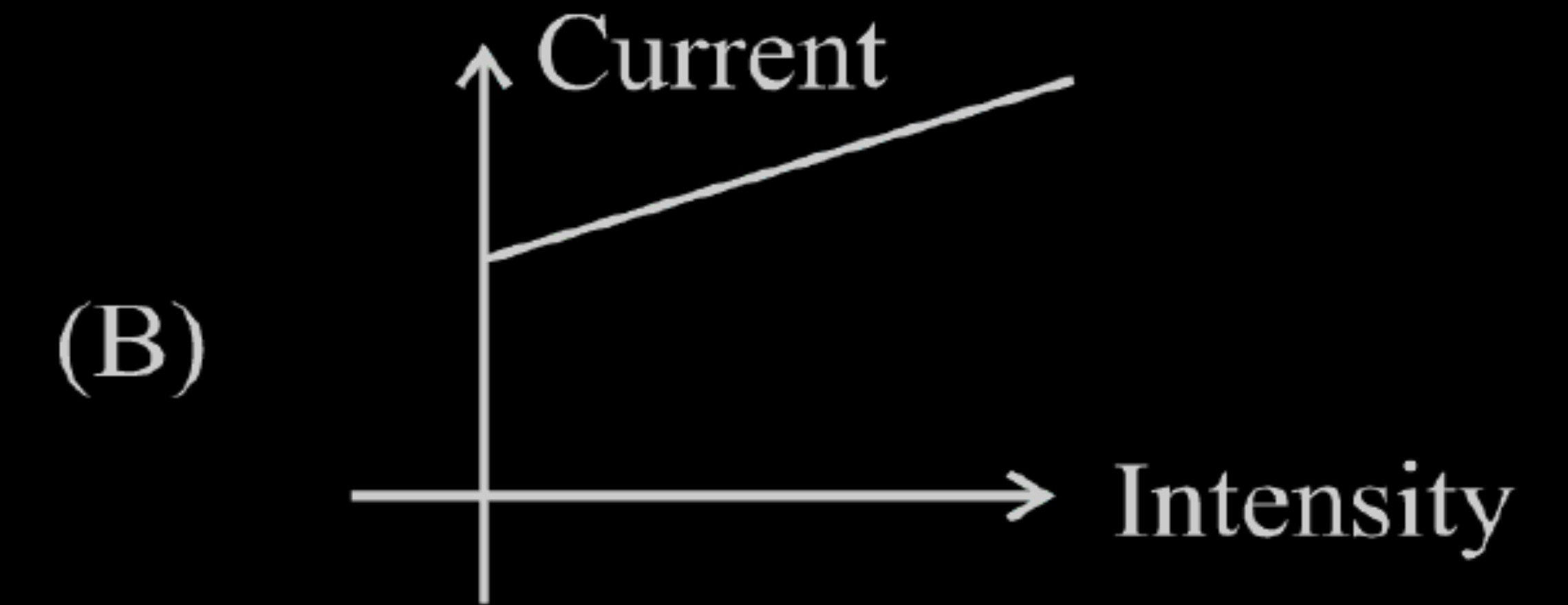
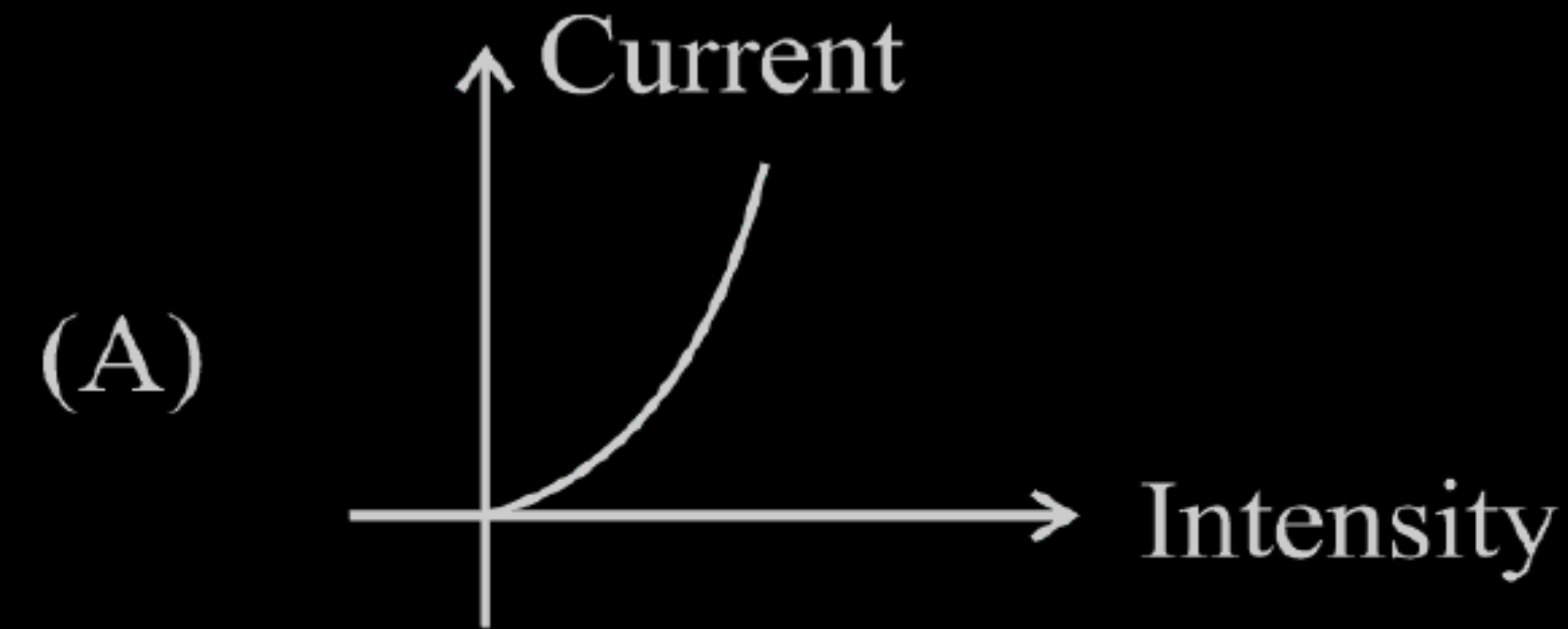
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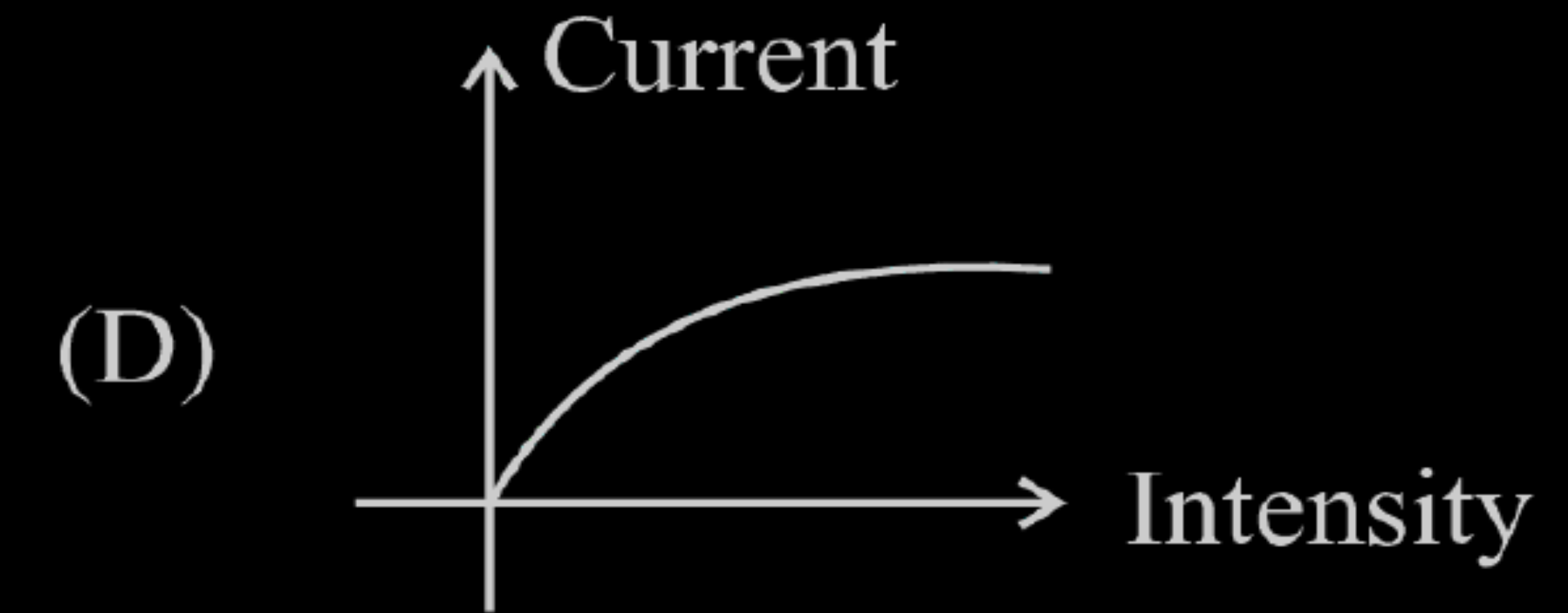
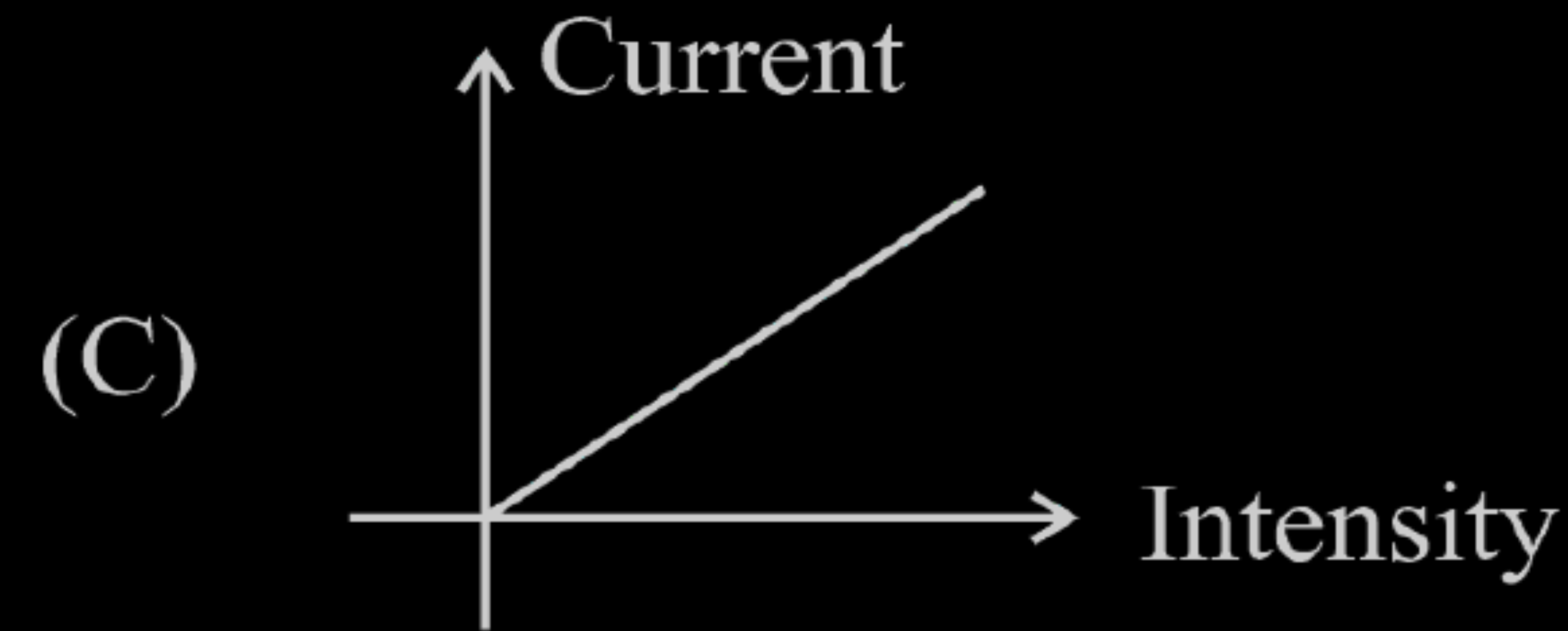
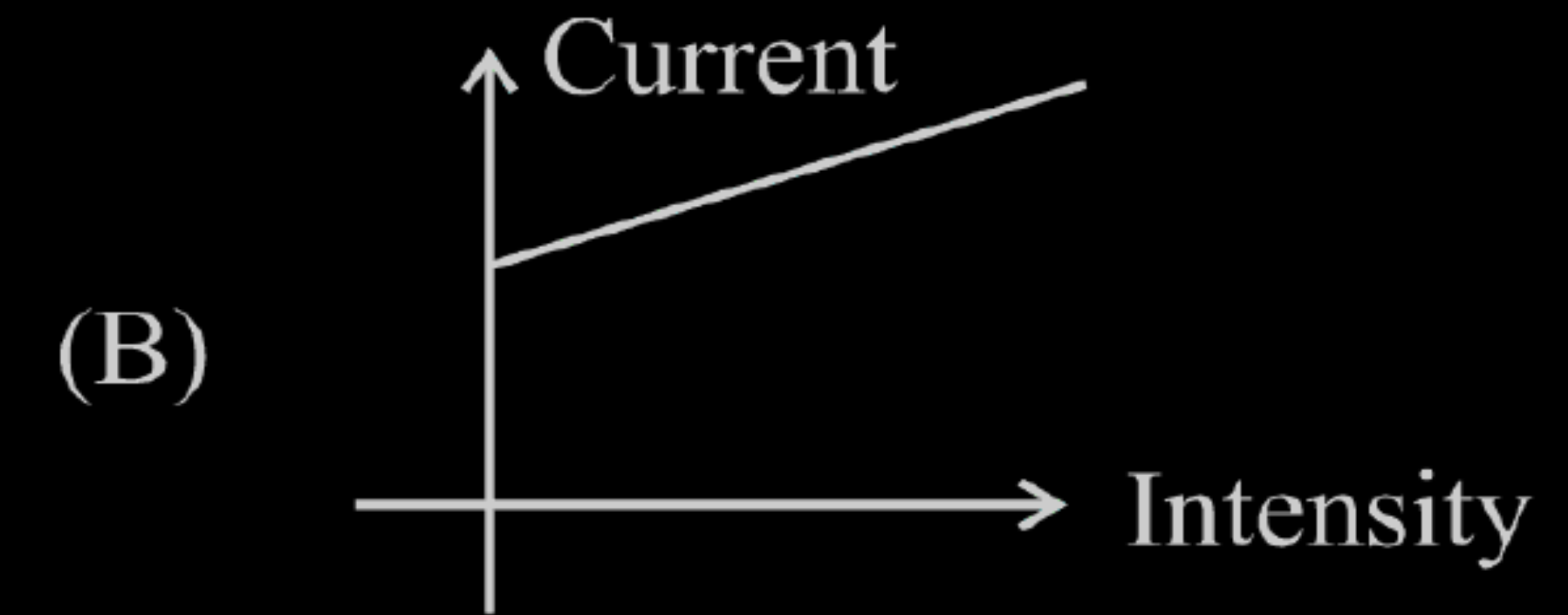
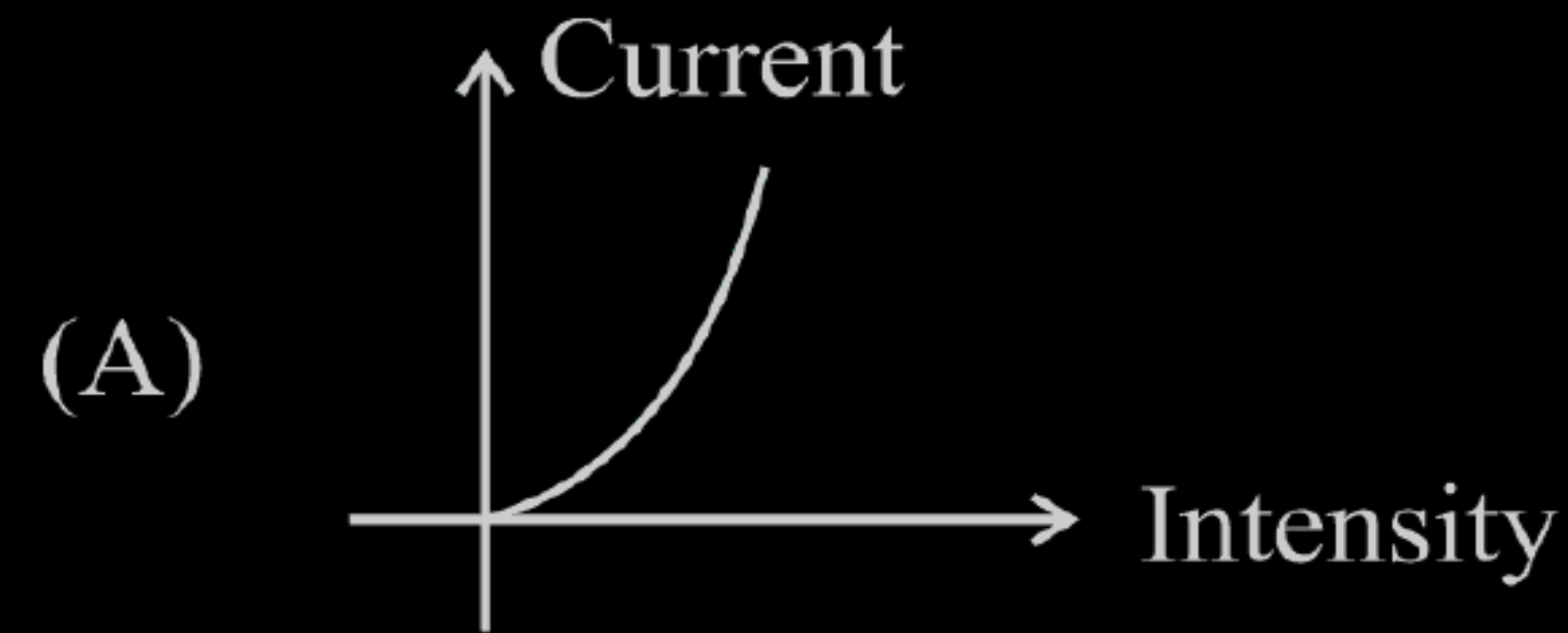
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a) When monochromatic light is incident on a metal surface then more/less tightly bound electrons will emerge with less/more kinetic energy. So all the photoelectrons do not eject with same kinetic energy.	1
b) Maximum number of photoelectrons ejected per second (saturation current) is directly proportional to the Intensity of incident radiation Hence saturation current is different for different intensities.	1
c) when λ increases , ν decreases and energy of incident photon ($h\nu$) also decreases. When $\lambda > \lambda_0$, $\nu < \nu_0$ (threshold frequency) , no photoelectron is ejected. Emission of photoelectrons stop at $\lambda > \lambda_0$.	1

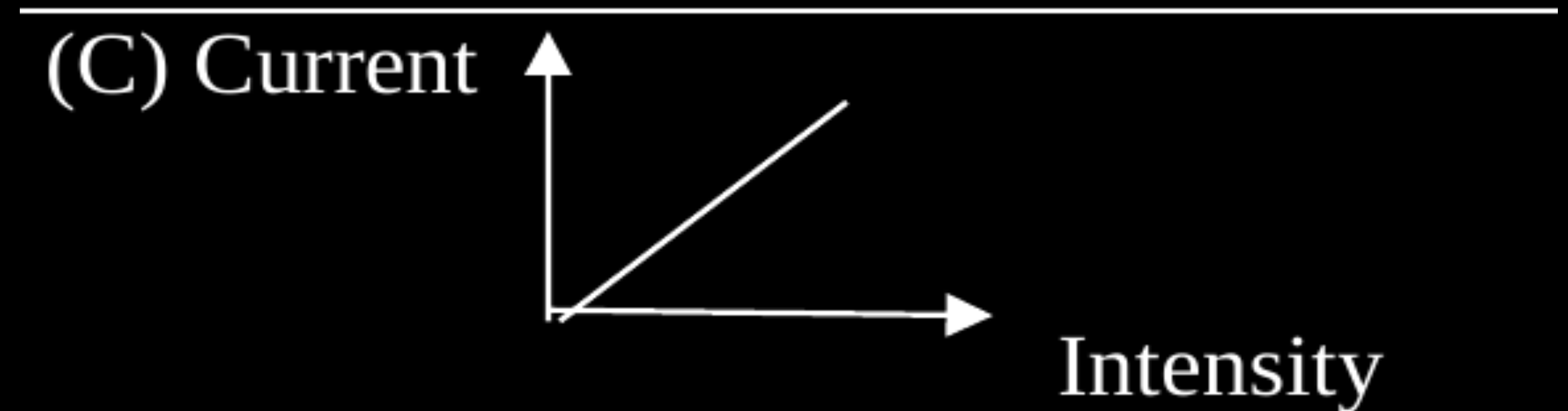
10. Which one of the given graphs correctly represents the variation of photoelectric current with the intensity of incident radiation, keeping other parameters fixed ?



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CBSE 2024 C



25. Radiations of frequency 3.0×10^{15} Hz are incident on a photosensitive surface of threshold frequency 1.0×10^{15} Hz. Calculate :

3

- (a) work function of the surface
- (b) maximum kinetic energy of photoelectrons

CBSE 2024 C

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CBSE 2024 C

(a) $\phi_0 = h\nu_0$

$$= 6.63 \times 10^{-34} \times 1.0 \times 10^{15}$$

$$= 6.63 \times 10^{-19} \text{ J}$$

$$= 4.14 \text{ eV}$$

(b) $K.E._{\text{max}} = h(\nu - \nu_0)$

$$= 6.63 \times 10^{-34} [3.0 \times 10^{15} - 1.0 \times 10^{15}]$$

$$= 13.26 \times 10^{-19} \text{ J}$$

$$= 8.29 \text{ eV}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 33.** (a) (i) What are matter waves ? A particle of mass m and charge q is accelerated from rest through a potential difference V . Obtain an expression for de Broglie wavelength associated with the particle.
- (ii) Monochromatic light of frequency 5.0×10^{14} Hz is produced by a source of power output 3.315 mW. Calculate :
- (1) energy of the photon in the beam
 - (2) number of photons emitted per second by the source

5

(i) Wave associated with a mass in motion is called matter wave.
Particle of mass m and charge q gains energy in the form of kinetic energy.

$$\frac{1}{2}mv^2 = qV$$

$$(mv)^2 = 2qVm$$

$$mv = \sqrt{2mqV}$$

Accordingly to de-Broglie relation

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

(ii) (1)

$$E = h\nu$$

$$= 6.63 \times 10^{-34} \times 5 \times 10^{14}$$

$$= 3.315 \times 10^{-19} \text{ J}$$

(2)

$$n = \frac{P}{E}$$

$$= \frac{3.315 \times 10^{-3}}{3.315 \times 10^{-19}} = 10^{16} \text{ s}^{-1}$$

24. In photoelectric effect experiment, show the variation of
- (a) photocurrent with collector plate potential for a given surface for different intensities of incident radiation. Do the curves meet at any point ? If so, why ?
 - (b) photocurrent with intensity of radiation incident on a surface keeping the frequency and plate potential fixed.

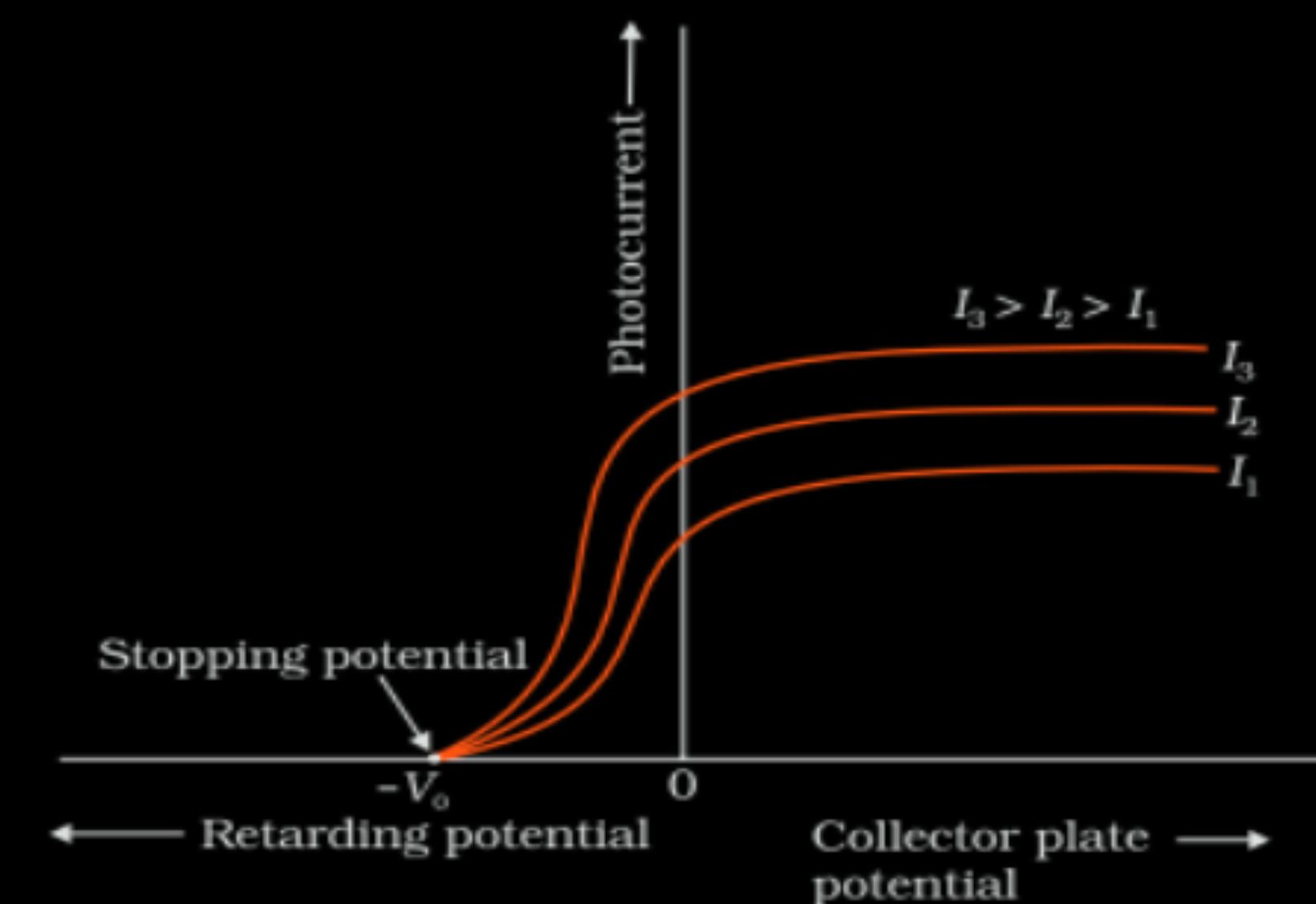
CBSE 2024 C

24. In photoelectric effect experiment, show the variation of

- (a) photocurrent with collector plate potential for a given surface for different intensities of incident radiation. Do the curves meet at any point ? If so, why ?
- (b) photocurrent with intensity of radiation incident on a surface keeping the frequency and plate potential fixed.

CBSE 2024 C

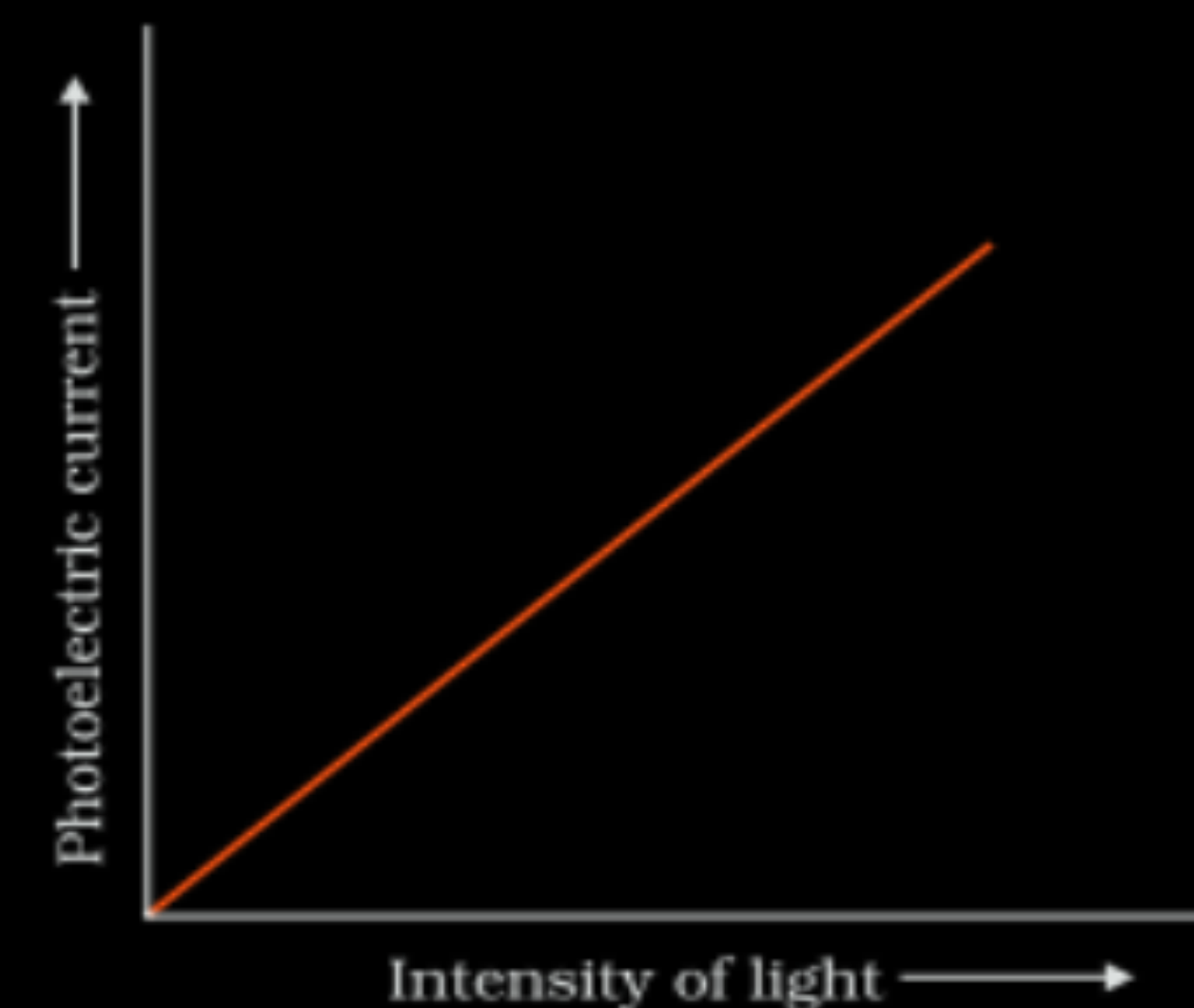
3



Yes, these curves meet at stopping potential

For any surface , as the energy of photons does not depend upon intensity of light, at the stopping potential current reduces to zero.

(b)



1

$\frac{1}{2}$

$\frac{1}{2}$

1

9. The ratio of the de Broglie wavelengths associated with an electron of mass m and of proton of mass M when both are accelerated through a potential difference V is :

(A) $\left(\frac{M}{m}\right)^{3/2}$

(B) $\frac{m}{M}$

(C) $\sqrt{\frac{M}{m}}$

(D) $\sqrt{\frac{m}{M}}$

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(D) $\sqrt{\frac{m}{M}}$

CBSE 2024 C

(C) $\sqrt{\frac{M}{m}}$

- 13.** *Assertion (A) :* The kinetic energy of the emitted photoelectrons depends upon the work function of the surface.
- Reason (R) :* The maximum value of kinetic energy of photoelectrons is the difference of energy of incident photons and work function of the surface.

13. *Assertion (A) :* The kinetic energy of the emitted photoelectrons depends upon the work function of the surface.

Reason (R) : The maximum value of kinetic energy of photoelectrons is the difference of energy of incident photons and work function of the surface.

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(A) Both Assertion (A) and Reason (R) are true and
 (A) Reason (R) is the correct explanation of the Assertion (A).

27. The threshold frequency for a metal surface for photoelectric emission is 4×10^{14} Hz. When a light of frequency ν is incident upon it, the emitted photoelectrons are stopped by a stopping potential of 4.98 V. Calculate : 3
- (a) work function of the metal, and
- (b) the value of ν .

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- (a) work function of the metal, and
- (b) the value of ν .

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$$\text{Work Function } (\phi_0) = h\nu_0$$

$$= 6.63 \times 10^{-34} \times 4 \times 10^{14}$$

$$= 2.65 \times 10^{-19} \text{ J}$$

From the Photoelectric equation:-

$$eV_0 = h\nu - h\nu_0$$

$$1.6 \times 10^{-19} \times 4.98 = 6.63 \times 10^{-34} [\nu - 4 \times 10^{14}]$$

On solving:-

$$\nu = 1.6 \times 10^{15} \text{ Hz}$$

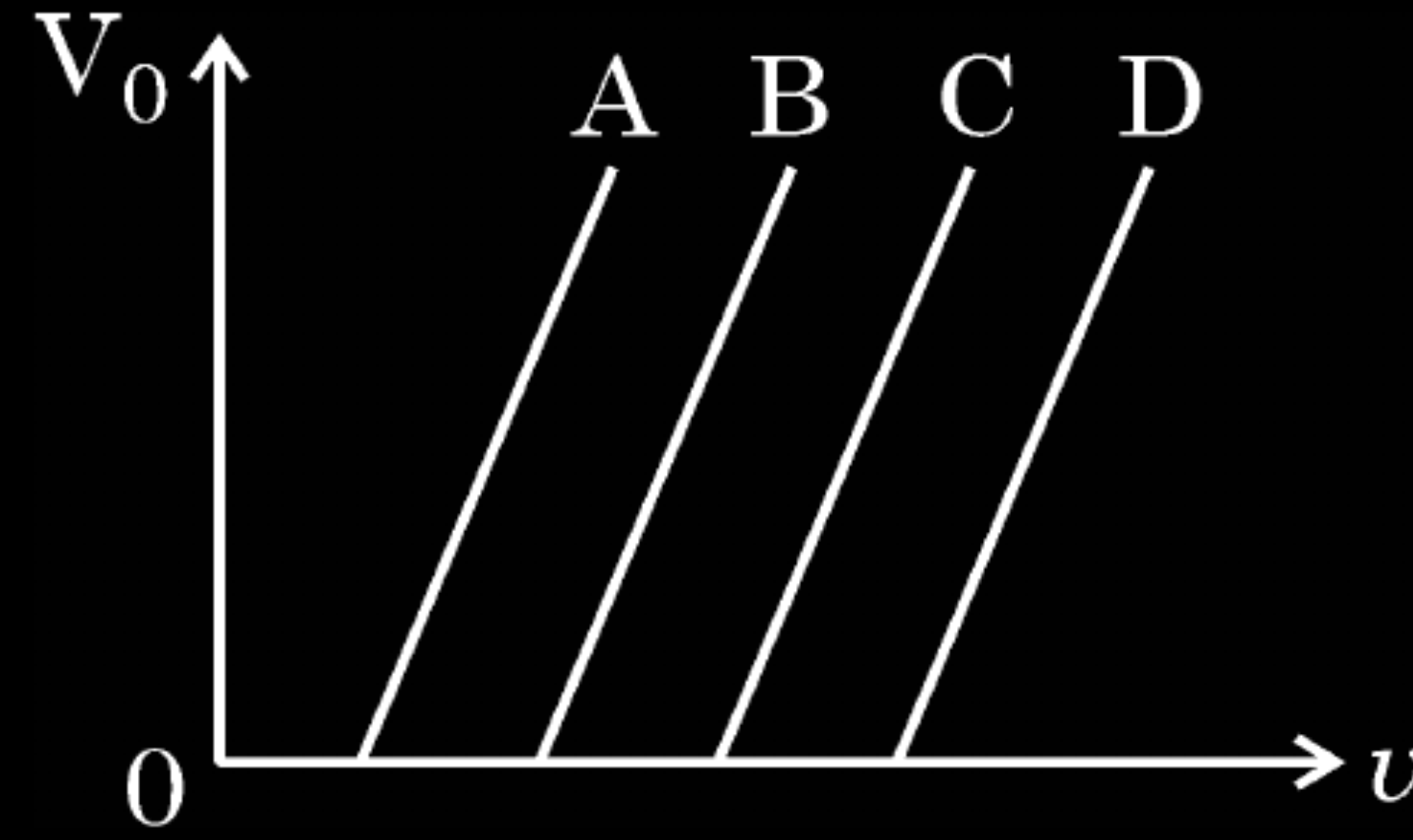
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1

1/2

1

10. The variation of the stopping potential (V_0) with the frequency (ν) of the incident radiation for four metals A, B, C and D is shown in the figure. For the same frequency of incident radiation producing photo-electrons in all metals, the kinetic energy of photo-electrons will be maximum for metal

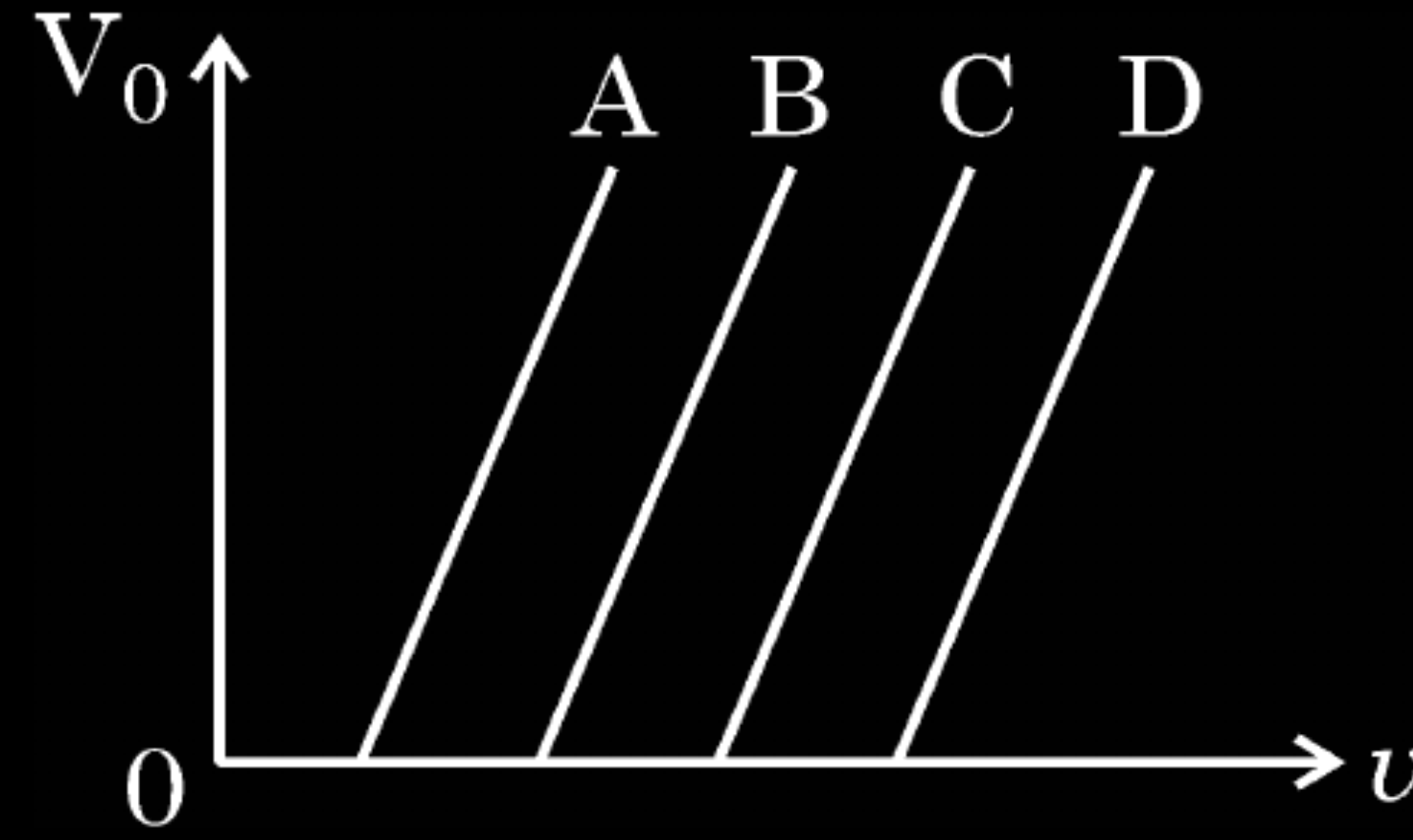


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(A) A
(C) C

(B) B
(D) D

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(A) A
(C) C

(B) B
(D) D

(A) A

23. The threshold frequency for a metal is 3.0×10^{14} Hz. A beam of frequency 9.0×10^{14} Hz is incident on the metal. Calculate (i) the work function (in eV) of the metal and (ii) the maximum speed of photoelectrons.

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$$\begin{aligned} \text{(i)} \quad \phi_0 = h\nu_0 &= \frac{6.63 \times 10^{-34} \times 3.0 \times 10^{14}}{1.6 \times 10^{-19}} \\ &= 1.24 \text{ eV} \end{aligned}$$

(ii)

$$K_{\max} = h\nu - h\nu_0$$

$$\frac{1}{2} m V_{\max}^2 = h(\nu - \nu_0)$$

$$V_{\max} = \left[\frac{2h(\nu - \nu_0)}{m} \right]^{\frac{1}{2}}$$

$$V_{\max} = \left[\frac{2 \times 6.63 \times 10^{-34} (9 - 3) \times 10^{14}}{9.1 \times 10^{-31}} \right]^{\frac{1}{2}}$$

$$= 9.35 \times 10^5 \text{ m/s}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

22. A photosensitive surface of work function 2.1 eV is irradiated by radiation of wavelength 150 nm. Calculate (i) the threshold wavelength, (ii) energy (in eV) of an incident photon, and (iii) maximum kinetic energy of emitted photoelectron.

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22. A photosensitive surface of work function 2.1 eV is irradiated by radiation of wavelength 150 nm. Calculate (i) the threshold wavelength, (ii) energy (in eV) of an incident photon, and (iii) maximum kinetic energy of emitted photoelectron.

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(a)

$$\phi_0 = \frac{hc}{\lambda_0}$$

$\frac{1}{2}$

$$\begin{aligned} \text{(i) } \lambda_0 &= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2.1 \times 1.6 \times 10^{-19}} \\ &= 5.92 \times 10^{-7} \text{ m} \end{aligned}$$

$\frac{1}{2}$

$$\text{(ii) Energy of incident photon} = \frac{hc}{\lambda}$$

$\frac{1}{2}$

$$\begin{aligned} &= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{150 \times 10^{-9} \times 1.6 \times 10^{-19}} \\ &= 8.29 \text{ eV} \end{aligned}$$

$\frac{1}{2}$

(iii) Using Einstein equation

$$\frac{hc}{\lambda} = \phi_0 + K_{\max}$$

$\frac{1}{2}$

$$\begin{aligned} K_{\max} &= (8.29 - 2.1) \text{ eV} \\ &= 6.2 \text{ eV} \end{aligned}$$

$\frac{1}{2}$

22. (a) Write Einstein's photoelectric equation. How did Millikan prove the validity of this equation ?
- (b) Explain the existence of threshold frequency of incident radiation for photoelectric emission from a given surface.

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22. (a) Write Einstein's photoelectric equation. How did Millikan prove the validity of this equation ?
- (b) Explain the existence of threshold frequency of incident radiation for photoelectric emission from a given surface.

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(a) $h\nu = h\nu_0 + K_{\max} = h\nu_0 + eV_0$

By finding the value of Planck's constant using V_0 versus ν straight line plot for sodium.

(b) Since $K_{\max}$ must be non-negative therefore photo-electric emission is possible only when $h\nu > h\nu_0$, which implies the existence of ν_0 .

1½

1½

10. The quantum nature of light explains the observations on photoelectric effect as –
- (A) there is a minimum frequency of incident radiation below which no electrons are emitted.
 - (B) the maximum kinetic energy of photoelectrons depends only on the frequency of incident radiation.
 - (C) when the metal surface is illuminated, electrons are ejected from the surface after sometime.
 - (D) the photoelectric current is independent of the intensity of incident radiation.

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1

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(A) There is a minimum frequency of incident radiation below which no electrons are emitted.

16. **Assertion (A)** : Electrons are ejected from the surface of zinc when it is irradiated by yellow light.

Reason (R) : Energy associated with a photon of yellow light is more than the work function of zinc.

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Reason (R) : Energy associated with a photon of yellow light is more than the work function of zinc.

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1

(D) Both Assertion (A) and Reason (R) are false.

18. The ratio of de Broglie wavelengths of a proton and a deuteron accelerated by potential V_p and V_d respectively, $\left(\frac{\lambda_p}{\lambda_d}\right)$ is $\frac{1}{2}$. Find $\frac{V_p}{V_d}$.

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2

18. The ratio of de Broglie wavelengths of a proton and a deuteron accelerated by potential V_p and V_d respectively, $\left(\frac{\lambda_p}{\lambda_d}\right)$ is $\frac{1}{2}$. Find $\frac{V_p}{V_d}$.

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2

De Broglie wavelength of proton

$$\lambda_p = \frac{h}{\sqrt{2meV_p}}$$

De Broglie wavelength of deuteron

$$\lambda_d = \frac{h}{\sqrt{2(2m)eV_d}}$$

$$\frac{\lambda_p}{\lambda_d} = \frac{1}{2} = \frac{\sqrt{2(2m)eV_d}}{\sqrt{2meV_p}}$$

On solving

$$\frac{V_p}{V_d} = 8$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

18. What are matter waves ? A proton, an electron and an α -particle have the same kinetic energy. Write the de Broglie wavelengths associated with them in increasing order.

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2

18. What are matter waves ? A proton, an electron and an α -particle have the same kinetic energy. Write the de Broglie wavelengths associated with them in increasing order.

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2

The waves associated with every moving particle are called matter waves.

$$\lambda = \frac{h}{\sqrt{2mK}}$$

For same kinetic energy, $\lambda \propto \frac{1}{\sqrt{m}}$

$$m_{\alpha} > m_p > m_e$$

$$\therefore \lambda_{\alpha} < \lambda_p < \lambda_e$$

1

$\frac{1}{2}$

$\frac{1}{2}$

18. A proton and α -particle are accelerated through different potentials V_1 and V_2 respectively so that they have the same de Broglie wavelengths.

Find $\frac{V_1}{V_2}$.

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Find $\frac{V_1}{V_2}$.

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2

De Broglie wavelength of a proton

$$\lambda_p = \frac{h}{\sqrt{2m_p q_p V_1}} = \frac{h}{\sqrt{2meV_1}}$$

$\frac{1}{2}$

De Broglie wavelength of an α particle

$$\lambda_\alpha = \frac{h}{\sqrt{2m_\alpha q_\alpha V_2}} = \frac{h}{\sqrt{2(4m)(2e)V_2}}$$

$\frac{1}{2}$

$$\lambda_p = \lambda_\alpha$$

$$\frac{h}{\sqrt{2meV_1}} = \frac{h}{\sqrt{16meV_2}}$$

$\frac{1}{2}$

$$\frac{V_1}{V_2} = 8$$

$\frac{1}{2}$

- 10.** The wavelength of matter wave associated with an electron of kinetic energy K is λ . If the kinetic energy of the electron is doubled, the associated wavelength becomes :

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- (A) $\frac{\lambda}{\sqrt{2}}$
- (B) $\frac{\lambda}{2}$
- (C) $\sqrt{2} \lambda$
- (D) 2λ

14. *Assertion (A) :* Photoelectric effect demonstrates the particle nature of light.

Reason (R) : Photoelectric current is proportional to frequency of incident radiation.

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14. *Assertion (A) :* Photoelectric effect demonstrates the particle nature of light.

Reason (R) : Photoelectric current is proportional to frequency of incident radiation.

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(C) Assertion (A) is true but Reason (R) is false.

20. Light of wavelength 600 nm is incident on potassium (work function 2.3 eV). Will photoemission of electrons occur ? What is the longest wavelength that will cause photoemission of electrons ?

2

(Take $hc = 1240 \text{ eV} \cdot \text{nm}$)

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- 20.** Light of wavelength 600 nm is incident on potassium (work function 2.3 eV). Will photoemission of electrons occur ? What is the longest wavelength that will cause photoemission of electrons ?

2

(Take $hc = 1240 \text{ eV} \cdot \text{nm}$)

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$$\begin{aligned}
 E &= \frac{hc}{\lambda} \\
 &= \frac{1240 \text{ eV} \cdot \text{nm}}{600 \text{ nm}} \\
 &= 2.06 \text{ eV}
 \end{aligned}$$

$\therefore$ Work function $\phi_0 = 2.3 \text{ eV}$

$\therefore E < \phi_0$ No emission will take place.

$$\begin{aligned}
 \lambda_{\text{max}} &= \frac{hc}{\phi} \\
 &= \frac{1240 \text{ eV} \cdot \text{nm}}{2.3 \text{ eV}}
 \end{aligned}$$

$$\lambda_{\text{max}} = 539.13 \text{ nm}$$

$\frac{1}{2}$

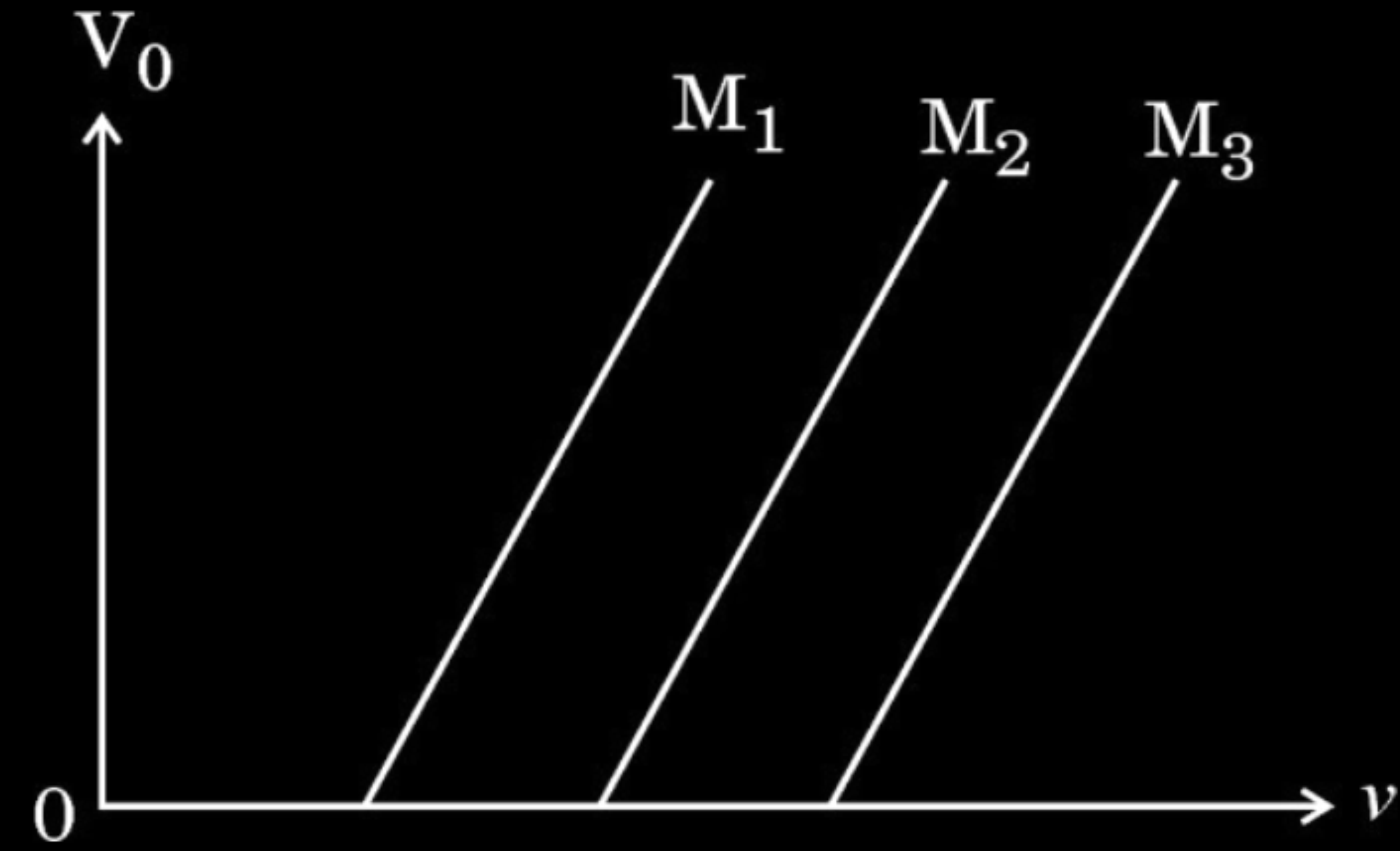
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

10. The figure shows the variation of stopping potential V_0 with frequency ν of incident radiation, for three materials M_1 , M_2 and M_3 with work functions ϕ_1 , ϕ_2 and ϕ_3 respectively. Then :

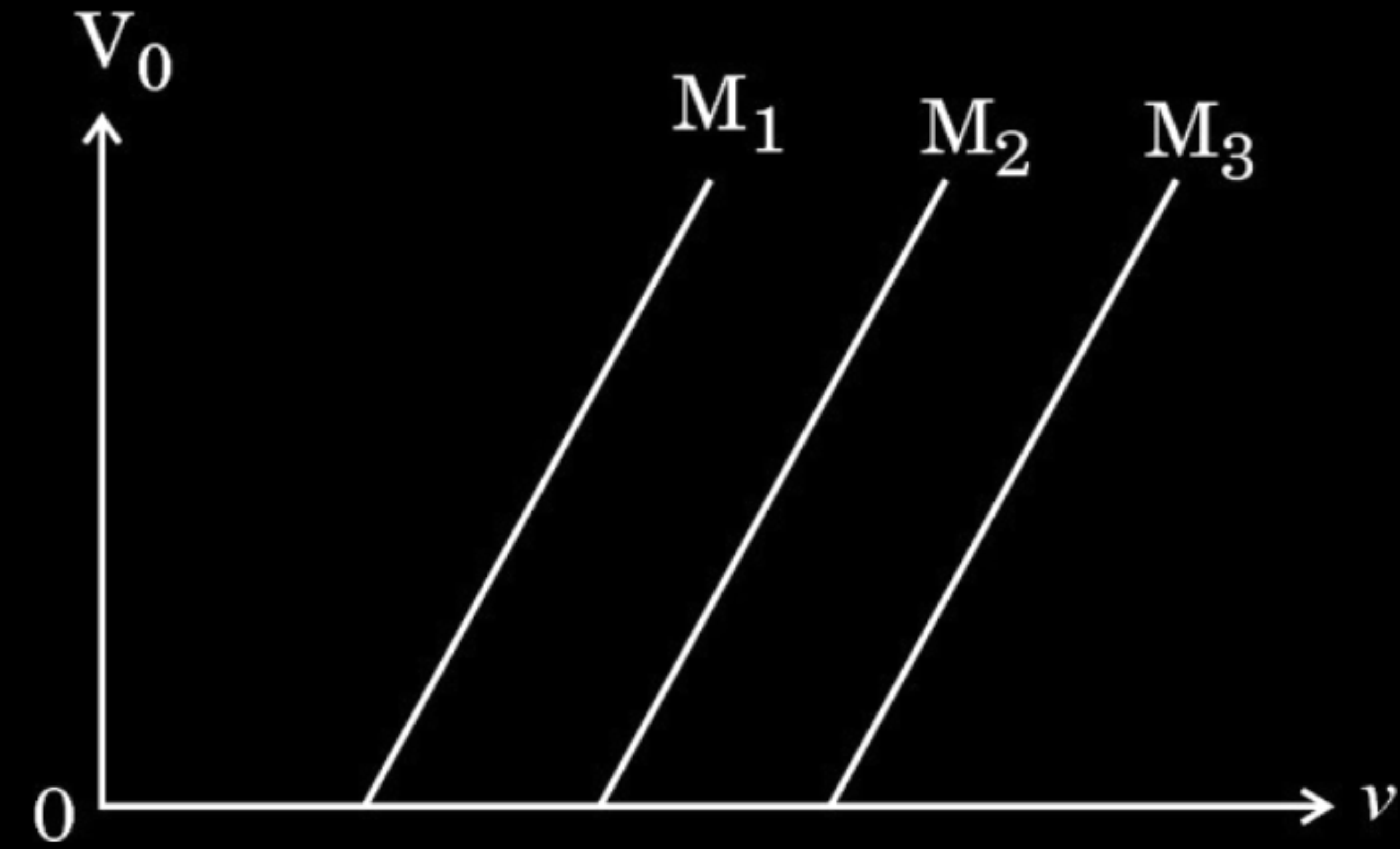
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- (A) $\phi_1 > \phi_2 > \phi_3$
(B) $\phi_2 > \phi_3 > \phi_1$
(C) $\phi_3 > \phi_2 > \phi_1$
(D) $\phi_2 > \phi_1 > \phi_3$

10. The figure shows the variation of stopping potential V_0 with frequency ν of incident radiation, for three materials M_1 , M_2 and M_3 with work functions ϕ_1 , ϕ_2 and ϕ_3 respectively. Then :

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- (A) $\phi_1 > \phi_2 > \phi_3$
- (B) $\phi_2 > \phi_3 > \phi_1$
- (C) $\phi_3 > \phi_2 > \phi_1$
- (D) $\phi_2 > \phi_1 > \phi_3$

(C) $\phi_3 > \phi_2 > \phi_1$

- 20.** Light of wavelength 500 nm is incident on caesium metal (work function 2.14 eV) and photoemission of electrons occurs. Calculate the (i) kinetic energy (in eV) of the fastest electrons and (ii) stopping potential for this situation. (Take $hc = 1240 \text{ eV}\cdot\text{nm}$)

- 20.** Light of wavelength 500 nm is incident on caesium metal (work function 2.14 eV) and photoemission of electrons occurs. Calculate the (i) kinetic energy (in eV) of the fastest electrons and (ii) stopping potential for this situation. (Take $hc = 1240 \text{ eV}\cdot\text{nm}$)

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2

Using Einstein Photoelectric equation

$$\frac{hc}{\lambda} = K.E_{\text{max}} + \phi_0$$

$$K.E_{\text{max}} = \frac{hc}{\lambda} - \phi_0$$

$$= \frac{1240 \text{ eV}\cdot\text{nm}}{500 \text{ nm}} - 2.14 \text{ eV}$$

$$K.E_{\text{max}} = 0.34 \text{ eV}$$

$$K.E_{\text{max}} = eV_0$$

$$\therefore V_0 = 0.34 \text{ V}$$

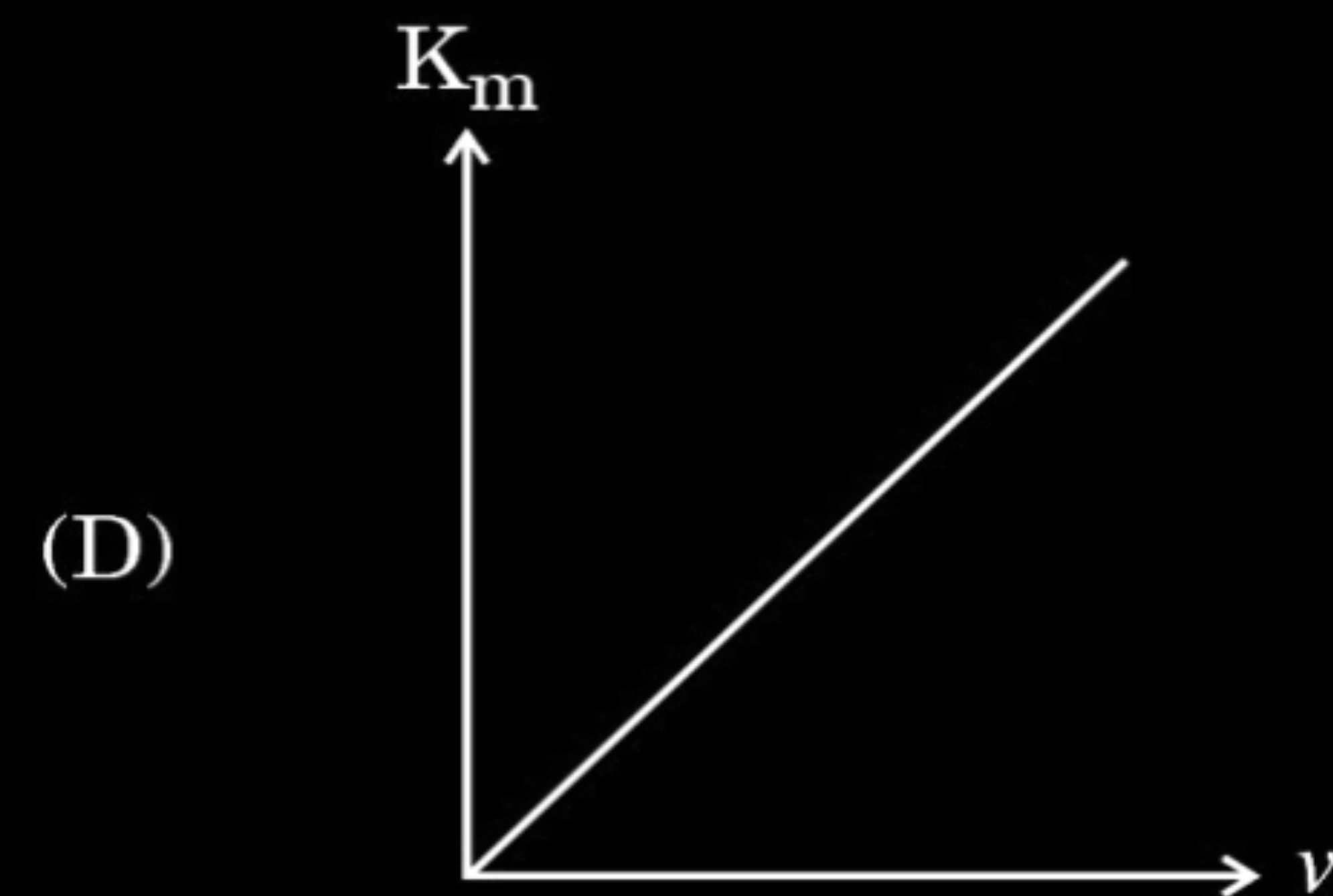
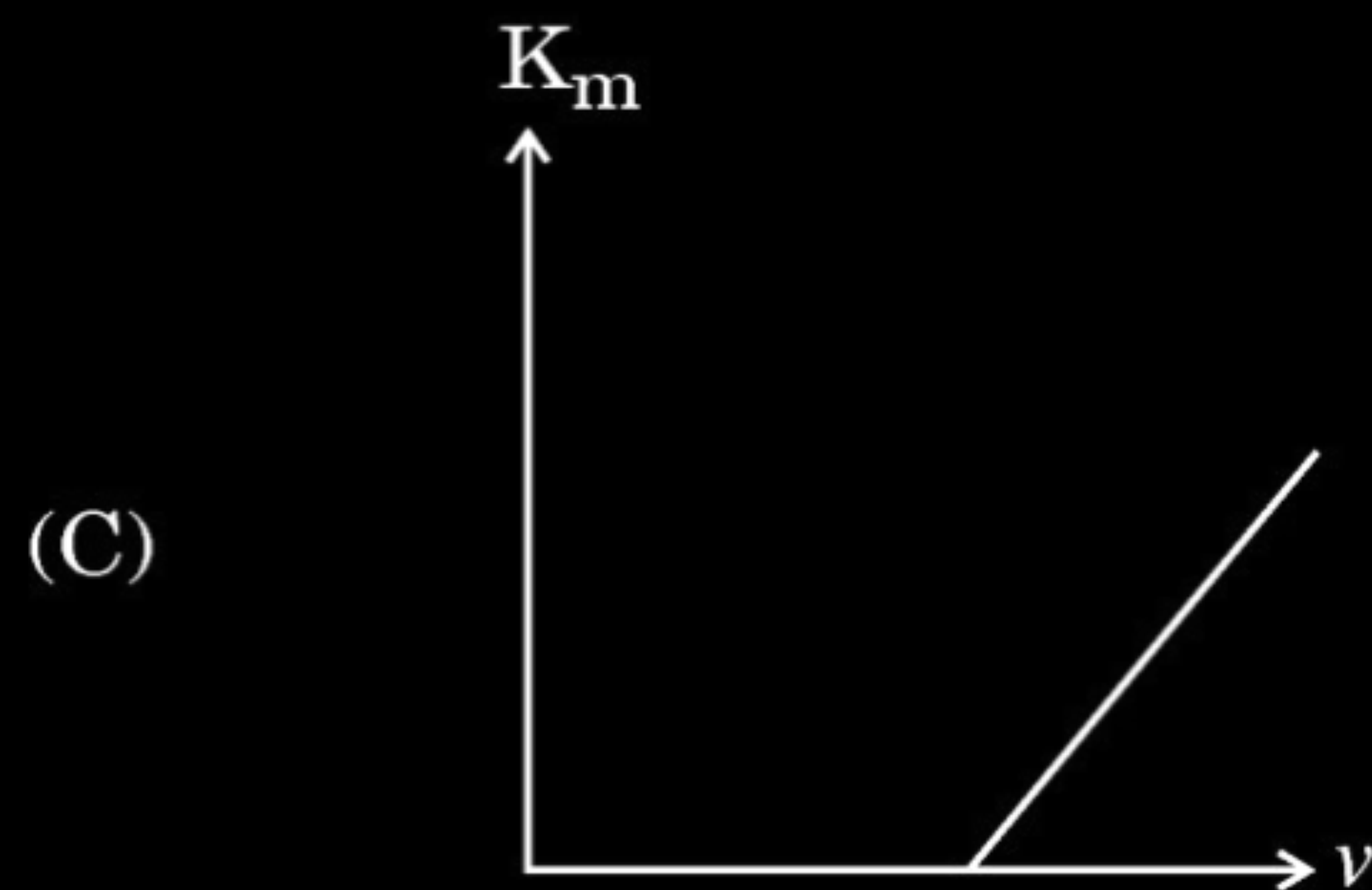
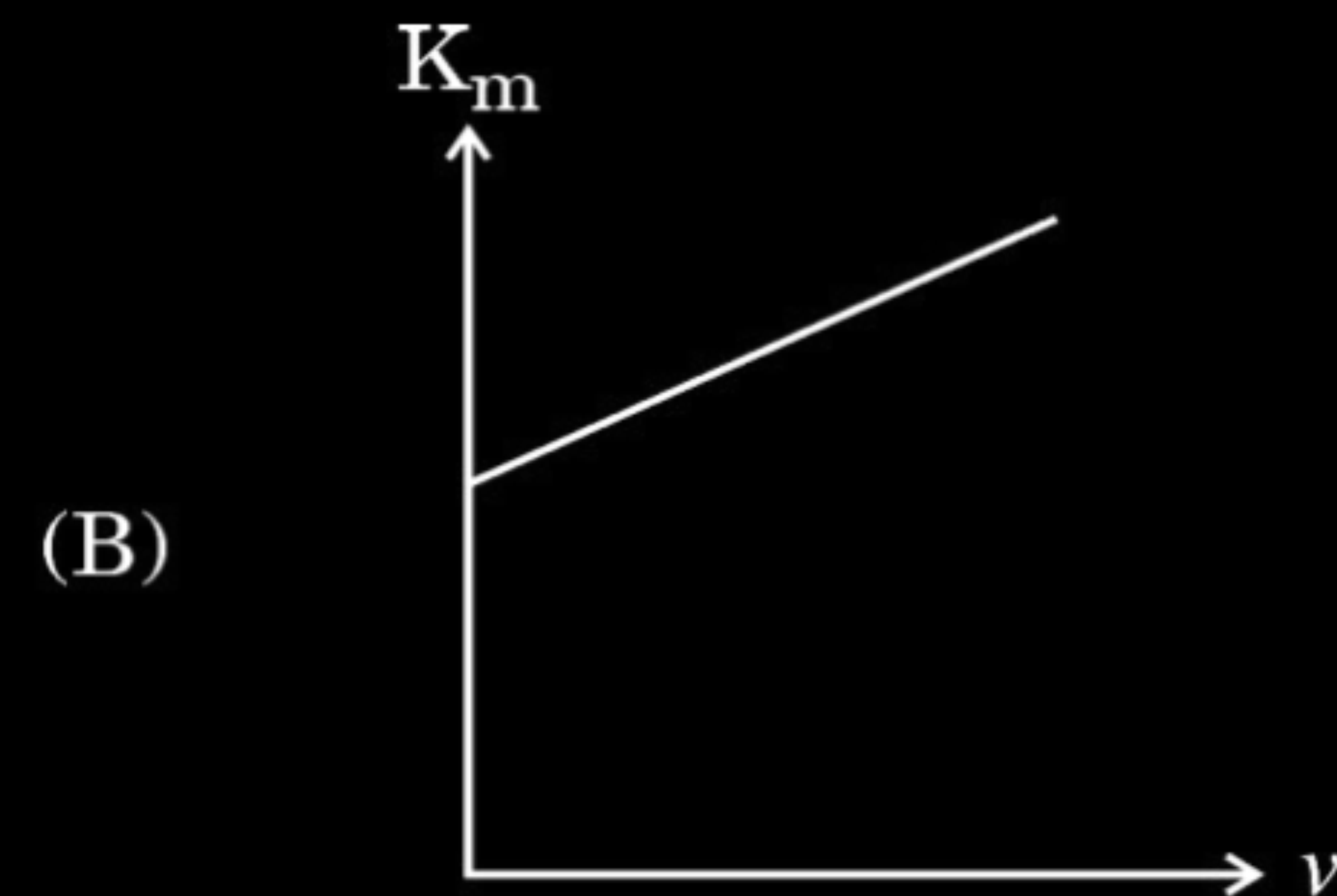
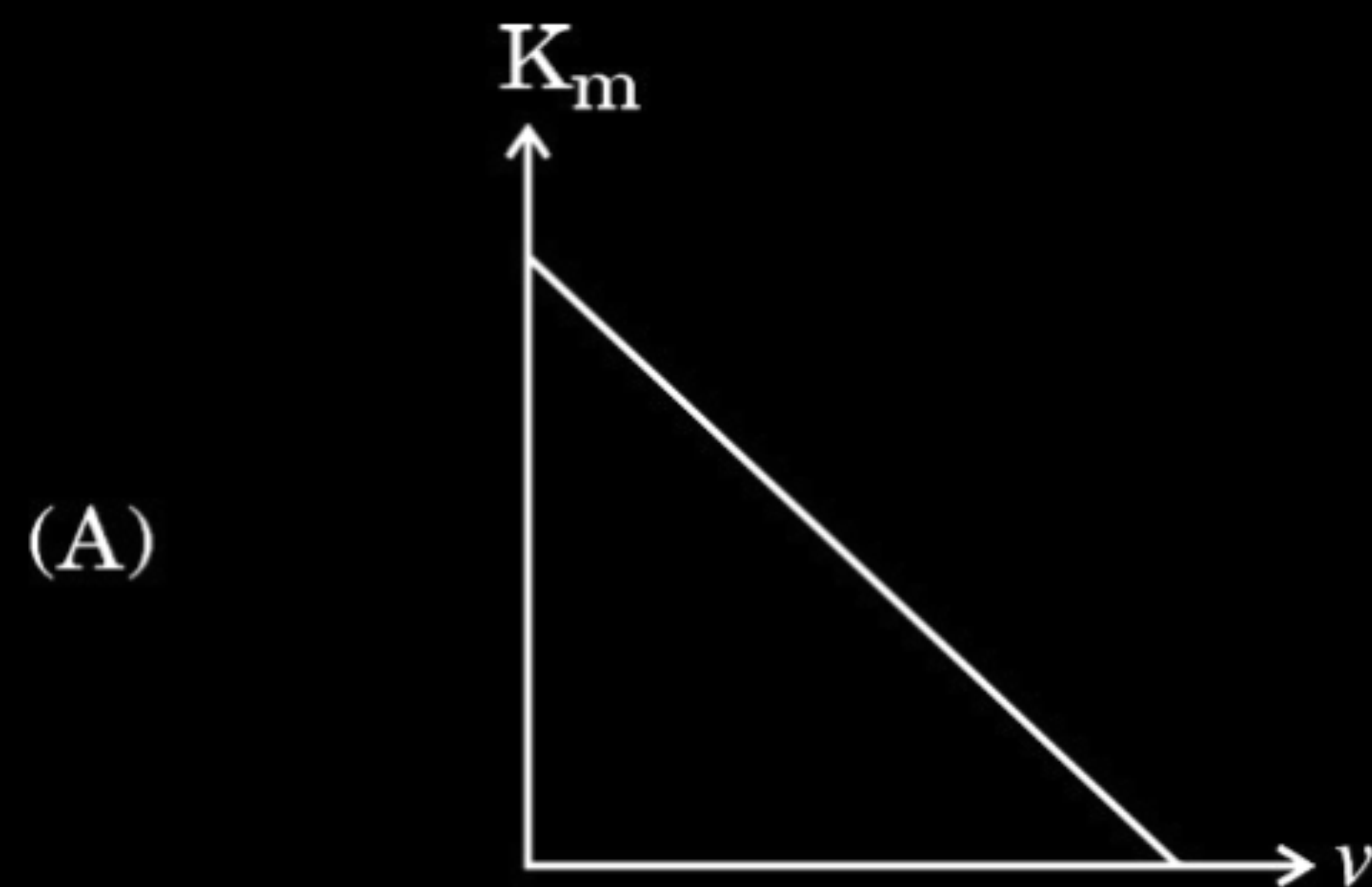
$\frac{1}{2}$

$\frac{1}{2}$

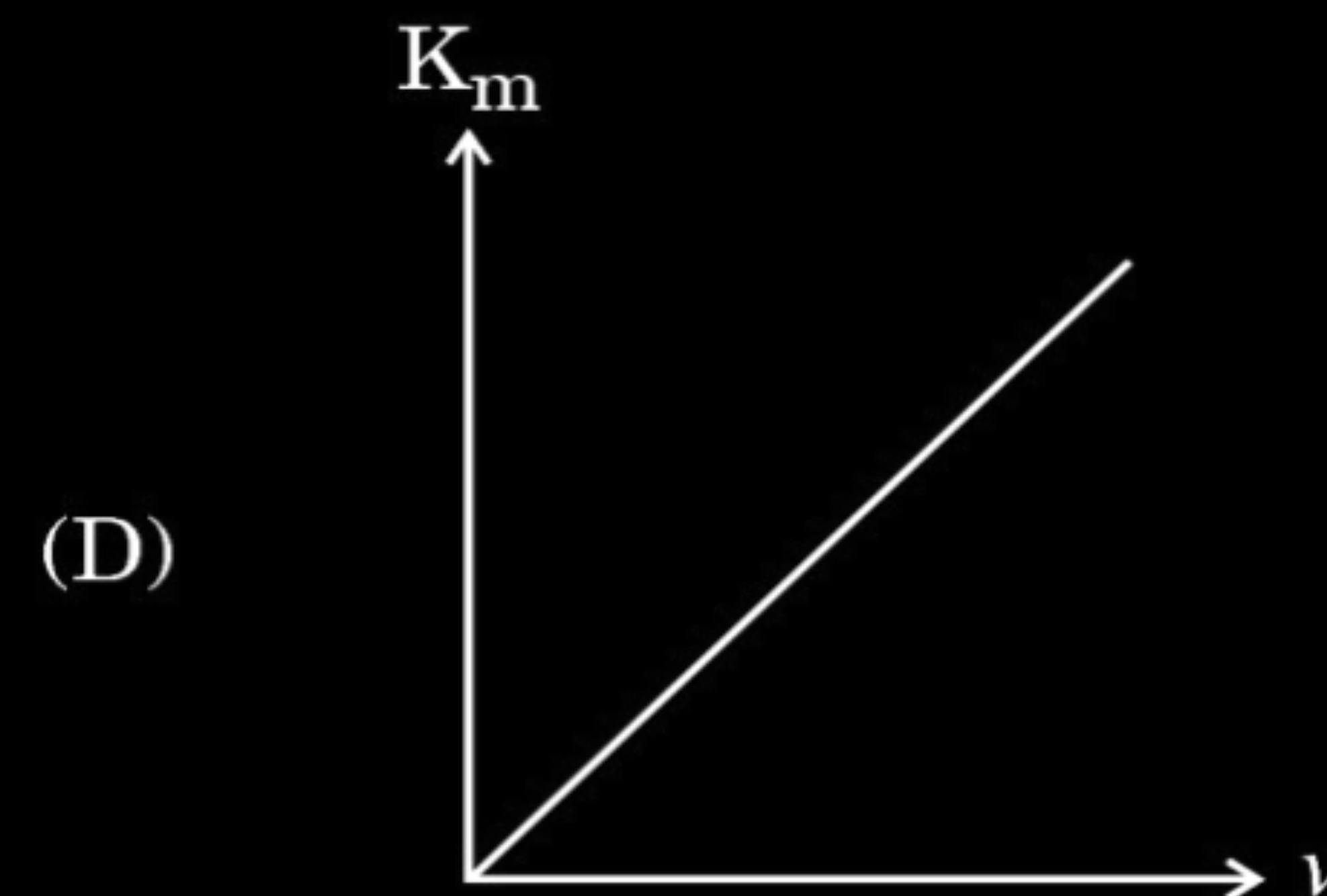
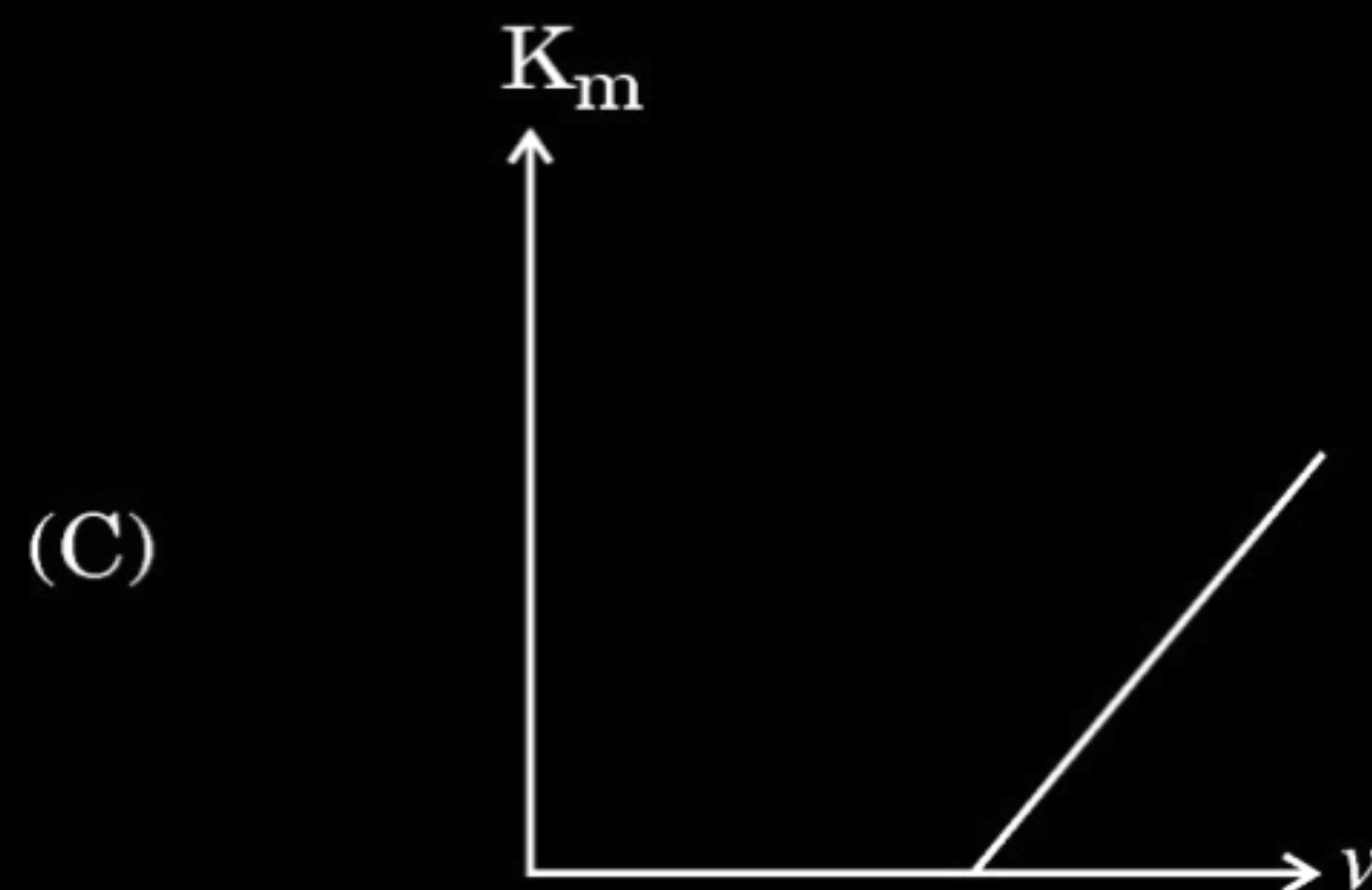
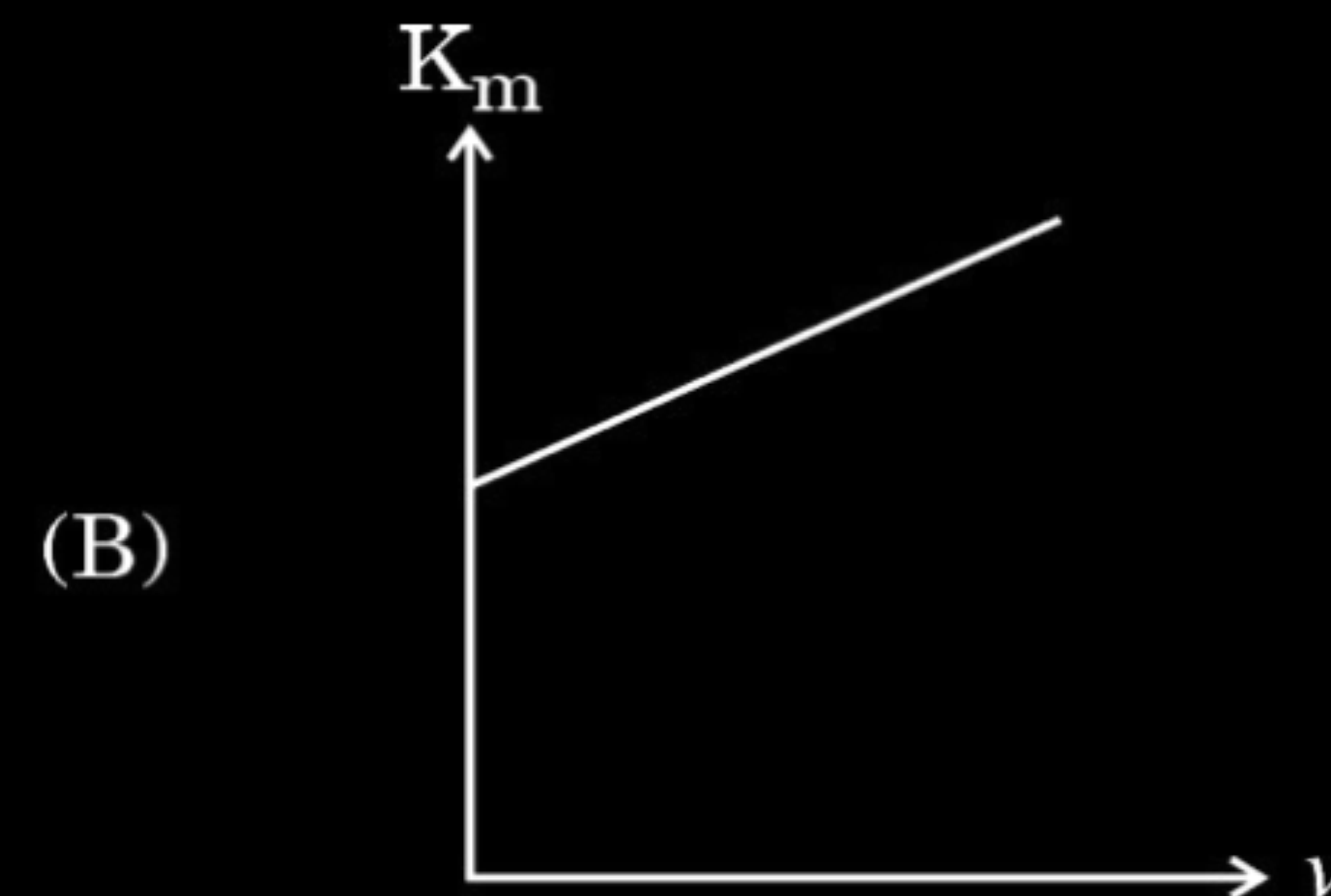
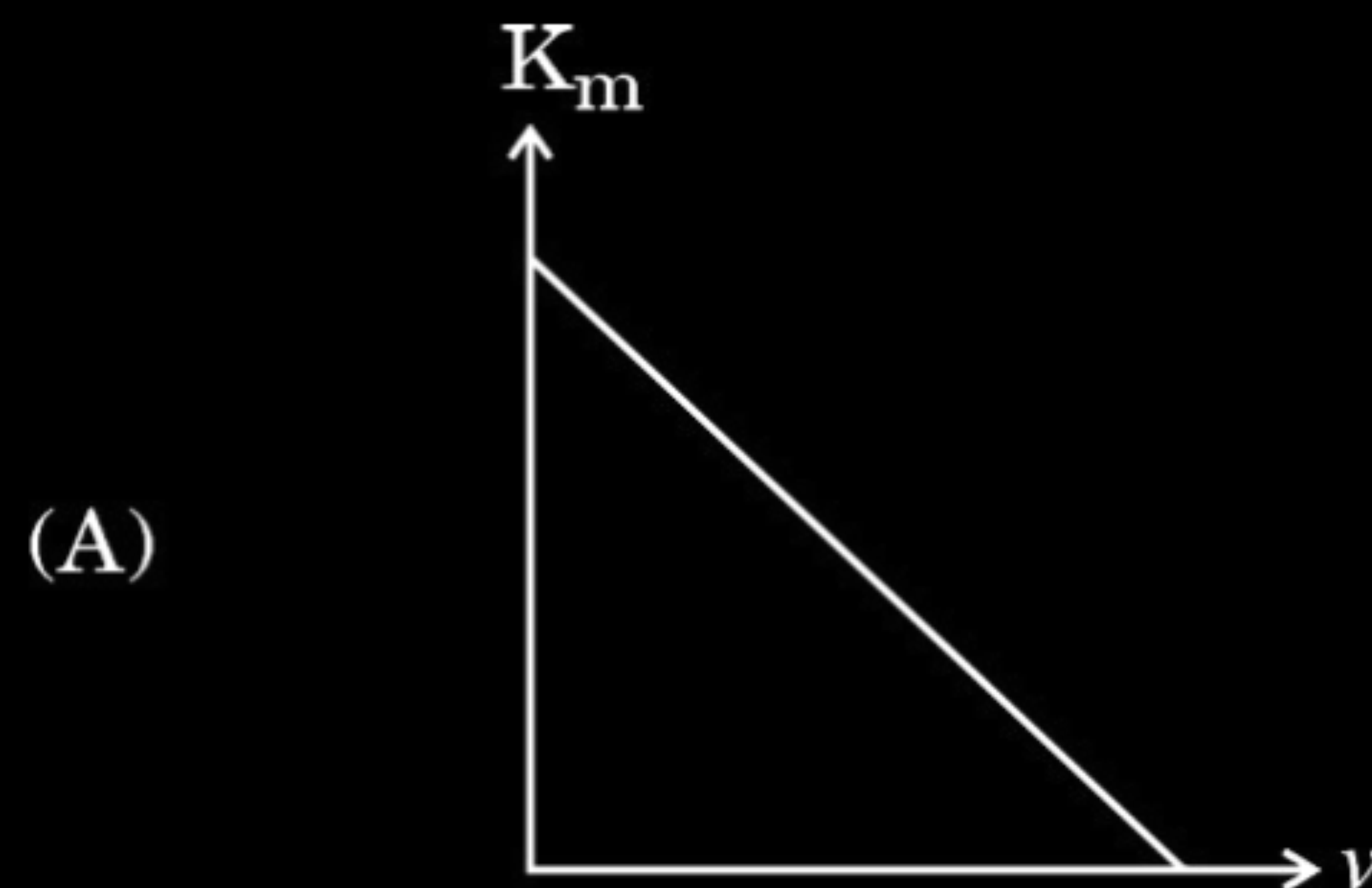
$\frac{1}{2}$

$\frac{1}{2}$

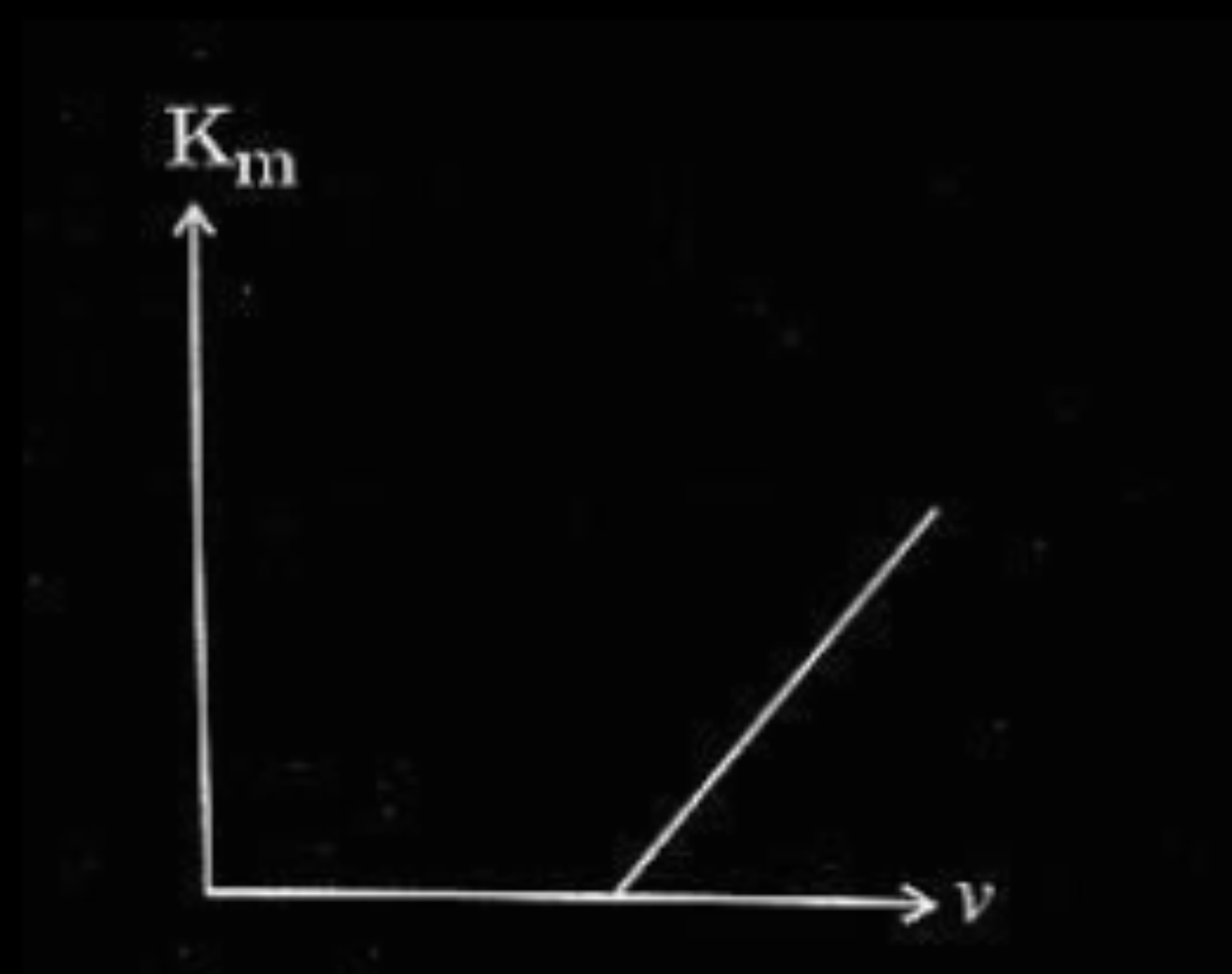
10. Which one of the following is the correct graph between the maximum kinetic energy (K_m) of the emitted photoelectrons and the frequency of incident radiation (ν) for a given photosensitive surface ?



10. Which one of the following is the correct graph between the maximum kinetic energy (K_m) of the emitted photoelectrons and the frequency of incident radiation (ν) for a given photosensitive surface ?



(C)



- 20.** The minimum intensity of white light that our eyes can perceive is about 0.1 nWm^{-2} . Calculate the number of photons of this light entering our pupil (area 0.4 cm^2) per second.

2

(Take average wavelength of white light = 500 nm and Planck's constant = $6.6 \times 10^{-34} \text{ Js}$)

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2

(Take average wavelength of white light = 500 nm and Planck's constant = $6.6 \times 10^{-34} \text{ Js}$)

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Total Energy gained per second from photon = IA

$$E = N h \nu$$

$$IA = N \times \frac{hc}{\lambda}$$

$$N = \frac{[IA]\lambda}{hc}$$

$$N = \frac{[0.1 \times 10^{-9} \times 0.4 \times 10^{-4}] \times 500 \times 10^{-9}}{6.6 \times 10^{-34} \times 3 \times 10^8}$$

$$N = 1.01 \times 10^4$$

$\frac{1}{2}$

1

$\frac{1}{2}$

8. The momentum (in SI units) associated with a photon of energy 1.5 eV is :

(A) 4×10^{-27}

(B) 8×10^{-28}

(C) 2×10^{-30}

(D) 6×10^{-29}

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(A) 4×10^{-27}

(B) 8×10^{-28}

(C) 2×10^{-30}

(D) 6×10^{-29}

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(B) 8×10^{-28}

16. *Assertion (A) :* Photoelectric current increases with an increase in intensity of incident radiation, for a given frequency of incident radiation and the accelerating potential.

Reason (R) : Increase in the intensity of incident radiation results in an increase in the number of photoelectrons emitted per second and hence an increase in the photocurrent.

16. *Assertion (A) :* Photoelectric current increases with an increase in intensity of incident radiation, for a given frequency of incident radiation and the accelerating potential.

Reason (R) : Increase in the intensity of incident radiation results in an increase in the number of photoelectrons emitted per second and hence an increase in the photocurrent.

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(A) Both Assertion(A) and Reason (R) are true and Reason(R) is the correct explanation of the Assertion (A)

8. A photosensitive surface has a work function of 2.00 eV. The maximum kinetic energy of electrons ejected from this surface by radiation of wavelength 300 nm is :

(A) 0.54 eV

(B) 1.07 eV

(C) 1.61 eV

(D) 2.14 eV

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8. A photosensitive surface has a work function of 2.00 eV. The maximum kinetic energy of electrons ejected from this surface by radiation of wavelength 300 nm is :

(A) 0.54 eV

(B) 1.07 eV

(C) 1.61 eV

(D) 2.14 eV

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(D) 2.14 e V

- 26.** (a) Briefly explain de Broglie hypothesis for wave nature of matter.
- (b) Find the ratio of de Broglie wavelength associated with a proton and an alpha particle when both are
- (i) accelerated from rest through the same potential difference, and
 - (ii) moving with the same kinetic energy.

3

(a) Moving particles of matter display wave like properties under suitable conditions.

The wave length λ associated with a particle of momentum p is given as

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

1

λ is the attribute of a wave while momentum is a typical attribute of particle.

(b) (i) $\lambda = \frac{h}{\sqrt{2meV}}$

$\frac{1}{2}$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{\sqrt{2 \times 4m_p \times 2e \times V}}{\sqrt{2 \times m_p \times e \times V}}$$

$$= 2\sqrt{2}$$

$\frac{1}{2}$

(ii) $\lambda = \frac{h}{\sqrt{2mK}}$

$\frac{1}{2}$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{\sqrt{2 \times 4m_p \times K}}{\sqrt{2 \times m_p \times K}}$$

$$= 2$$

$\frac{1}{2}$

8. The work function for a photosensitive surface is 3.315 eV. The cut-off wavelength for photoemission of electrons from this surface is :

(A) 150 nm

(B) 200 nm

(C) 375 nm

(D) 500 nm

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8. The work function for a photosensitive surface is 3.315 eV . The cut-off wavelength for photoemission of electrons from this surface is :

(A) 150 nm

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(C) 375 nm

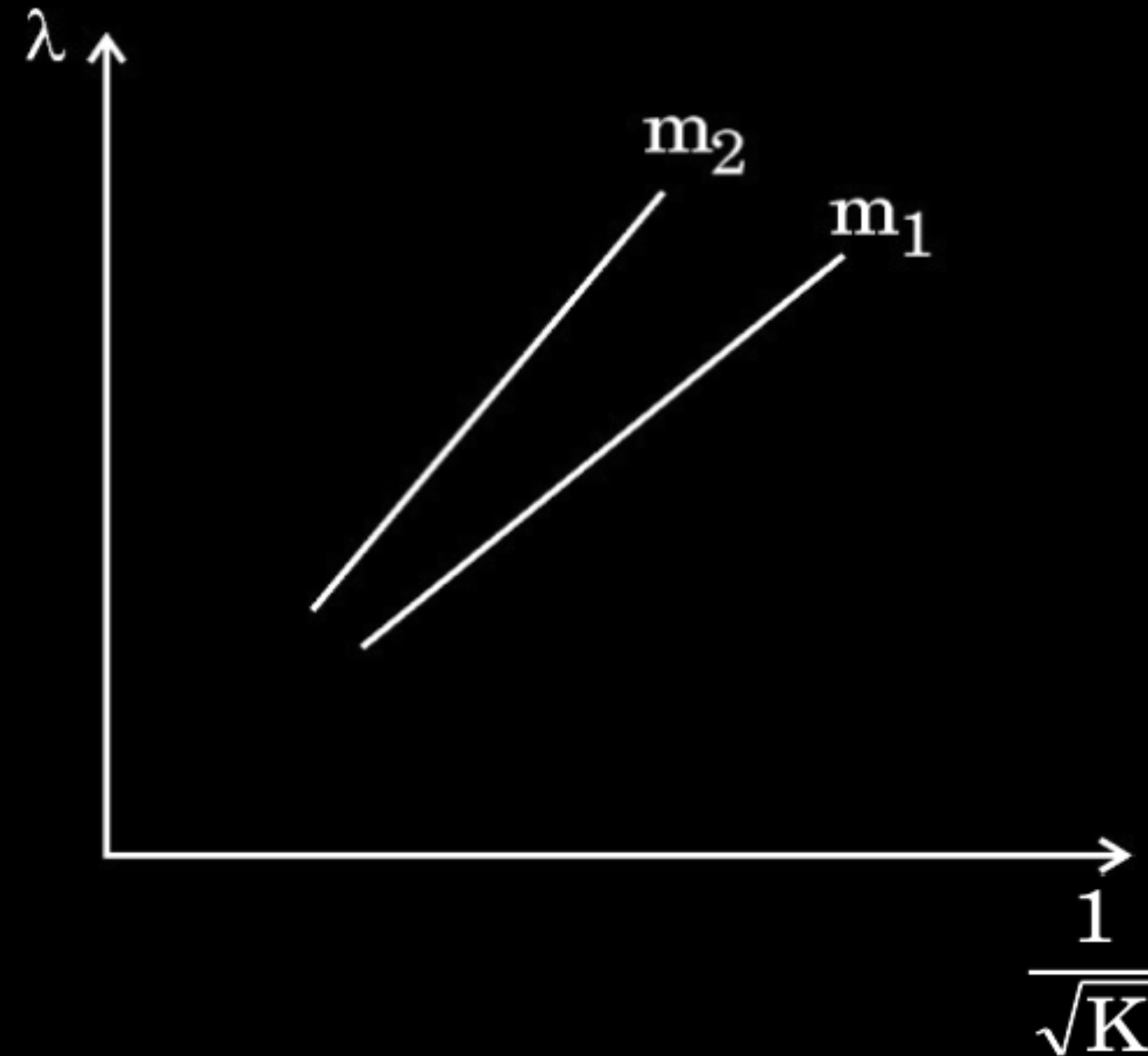
(D) 500 nm

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(C) 375 nm

27. de Broglie wavelength λ as a function of $\frac{1}{\sqrt{K}}$, for two particles of masses m_1 and m_2 are shown in the figure. Here, K is the energy of the moving particles.

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- (a) What does the slope of a line represent ?
 - (b) Which of the two particles is heavier ?
 - (c) Is this graph also valid for a photon ?
- Justify your answer in each case.

(a) $\lambda = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2m}} \times \frac{1}{\sqrt{K}}$	$\frac{1}{2}$
$slope = \frac{h}{\sqrt{2m}}$	$\frac{1}{2}$
(b) $slope \propto \frac{1}{\sqrt{m}}$	$\frac{1}{2}$
Slope of m_2 is more than that of m_1 . Therefore, m_1 is heavier.	$\frac{1}{2}$
(c) No	$\frac{1}{2}$
Momentum $(p) = \sqrt{2mK}$ is not valid for a photon	$\frac{1}{2}$

9. The waves associated with a moving electron and a moving proton have the same wavelength λ . It implies that they have the same :

(A) momentum

(B) angular momentum

(C) speed

(D) energy

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(A) momentum

(B) angular momentum

(C) speed

(D) energy

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(A) momentum

10. Two beams, A and B whose photon energies are 3.3 eV and 11.3 eV respectively, illuminate a metallic surface (work function 2.3 eV) successively. The ratio of maximum speed of electrons emitted due to beam A to that due to beam B is :

(A) 3

(B) 9

(C) $\frac{1}{3}$

(D) $\frac{1}{9}$

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- (A) 3 (B) 9 (C) $\frac{1}{3}$ (D) $\frac{1}{9}$

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(C) $\frac{1}{3}$

- 19.** Consider a neutron (mass m) of kinetic energy E and a photon of the same energy. Let λ_n and λ_p be the de Broglie wavelength of neutron and the wavelength of photon respectively. Obtain an expression for $\frac{\lambda_n}{\lambda_p}$. 2

- 19.** Consider a neutron (mass m) of kinetic energy E and a photon of the same energy. Let λ_n and λ_p be the de Broglie wavelength of neutron and the wavelength of photon respectively. Obtain an expression for $\frac{\lambda_n}{\lambda_p}$. 2

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$$E = \frac{hc}{\lambda_p} \Rightarrow \lambda_p = \frac{hc}{E}$$

$$\lambda_n = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

$$\frac{\lambda_n}{\lambda_p} = \frac{h}{\sqrt{2mE}} \times \frac{E}{hc}$$

$$\frac{\lambda_n}{\lambda_p} = \sqrt{\frac{E}{2mc^2}}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

15. *Assertion (A) :* In photoelectric effect, the kinetic energy of the emitted photoelectrons increases with increase in the intensity of the incident light.

Reason (R) : Photoelectric current depends on the wavelength of the incident light.

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15. *Assertion (A) :* In photoelectric effect, the kinetic energy of the emitted photoelectrons increases with increase in the intensity of the incident light.

Reason (R) : Photoelectric current depends on the wavelength of the incident light.

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(D) Assertion (A) is false and Reason (R) is also false.

8. In the photoelectric effect, electrons are emitted from a given metal surface when the incident radiation has a certain minimum value of :

(A) frequency

(B) intensity

(C) wavelength

(D) angle of incidence

8. In the photoelectric effect, electrons are emitted from a given metal surface when the incident radiation has a certain minimum value of :

(A) frequency

(B) intensity

(C) wavelength

(D) angle of incidence

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(A) frequency

13. *Assertion (A) :* Work functions of different metals are different.

Reason (R) : Metals differ in their conductivity.

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13. *Assertion (A) :* Work functions of different metals are different.

Reason (R) : Metals differ in their conductivity.

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(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

26. Answer the following giving reasons :

3

- (a) The speed of emitted electrons depends on the work function of the photosensitive surface, for the incident radiation of a given frequency.
- (b) The stopping potential (V_0) increases linearly with the frequency of incident radiation, for a given photosensitive surface.
- (c) The photoelectric current increases linearly with the intensity of incident radiation, for a given surface.

(a)The kinetic energy of the emitted electron $\text{KE} = \frac{1}{2}mv^2 = h\nu - \phi_0$ From the above relation it is clear that kinetic energy of the emitted electrons depends upon the work function (ϕ_0).Hence the speed of the emitted electrons depends upon the work function.	1
(b)According to Einstein's photoelectric equation, $eV_0 = \text{KE} = h\nu - \phi_0 \quad V_0 = \text{Stopping potential}$ For constant ϕ_0, $V_0 \propto \nu$	1
(c)As intensity of incident radiation increases, the number of photons incident on the metal surface increases. As a result number of electrons ejected from the metal surface increases. More photoelectrons ejected result in more photoelectric current. Hence photoelectric current increases linearly with increase in intensity.	1

10. The work function of potassium is 2.3 eV and threshold wavelength for it is 540 nm . The threshold wavelength for a metal whose work function is 4.6 eV will be :

(a) 135 nm

(b) 270 nm

(c) 540 nm

(d) 1080 nm

CBSE 2023 C

10. The work function of potassium is 2.3 eV and threshold wavelength for it is 540 nm . The threshold wavelength for a metal whose work function is 4.6 eV will be :

(a) 135 nm

(b) 270 nm

(c) 540 nm

(d) 1080 nm

CBSE 2023 C

(b) 270 nm

18. *Assertion (A) :* The photoelectric current increases with increase in intensity of incident radiation, keeping the frequency of the incident radiation and the accelerating potential fixed.

Reason (R) : The number of photoelectrons emitted per second is directly proportional to the intensity of incident radiation.

CBSE 2023 C

18. *Assertion (A) :* The photoelectric current increases with increase in intensity of incident radiation, keeping the frequency of the incident radiation and the accelerating potential fixed.

Reason (R) : The number of photoelectrons emitted per second is directly proportional to the intensity of incident radiation.

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(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

- 30.** (a) Write Einstein's photoelectric equation. Use it to :
- (i) explain the existence of threshold frequency for a given photosensitive surface.
 - (ii) show that the stopping potential, V_0 varies linearly with the frequency ν of incident radiation.

Explain how the value of Planck's constant can be found from the V_0 versus ν curve.

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(a) Einstein's photoelectric equation

$$h\nu = \phi_o + E_k^{\max}$$

$\frac{1}{2}$

(i) $E_k^{\max} = h\nu - \phi_o$

$$E_k^{\max} = h(\nu - \nu_o)$$

In case $\nu < \nu_o$, $E_k^{\max} < 0$

Which is not possible

So, no frequency is possible below threshold frequency

1

(ii)

$$h\nu = \phi_o + eV_o$$

$$eV_o = h\nu - \phi_o$$

$$V_o = \frac{h}{e}\nu - \frac{\phi_o}{e}$$

Which is equation of a straight line

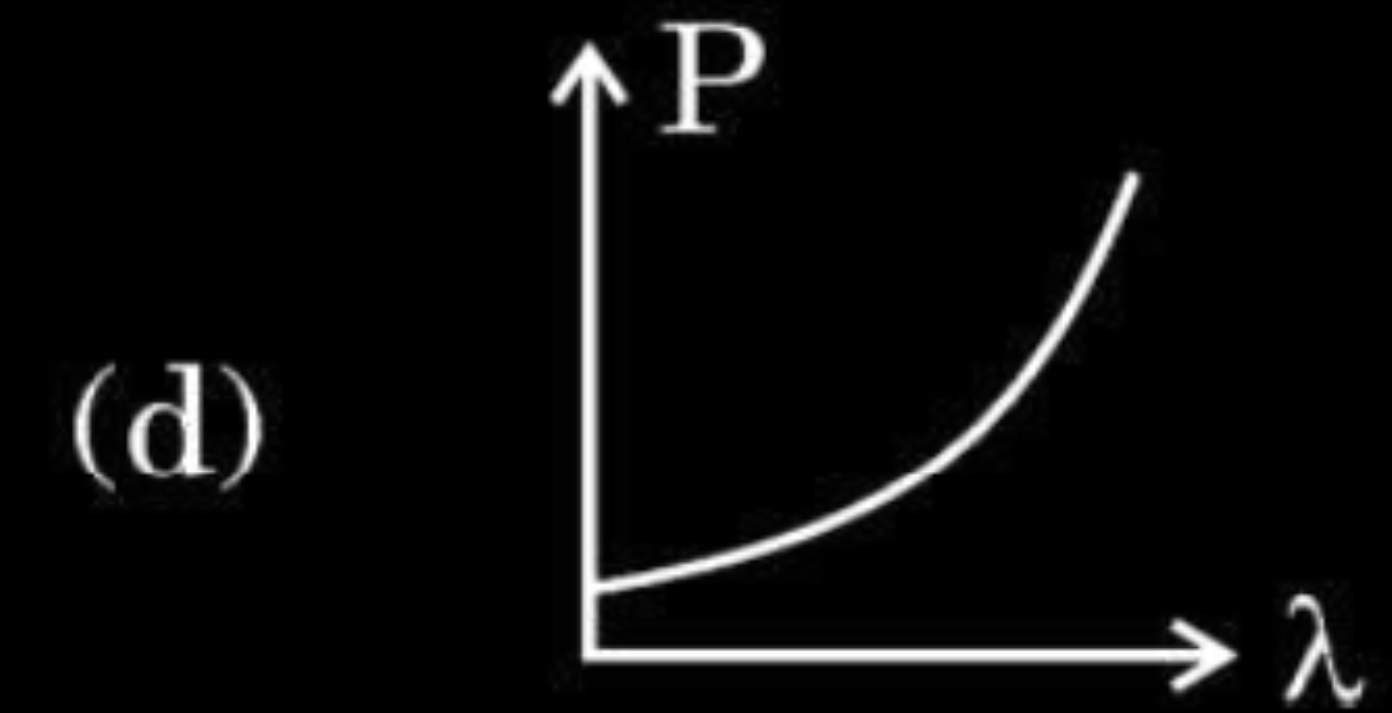
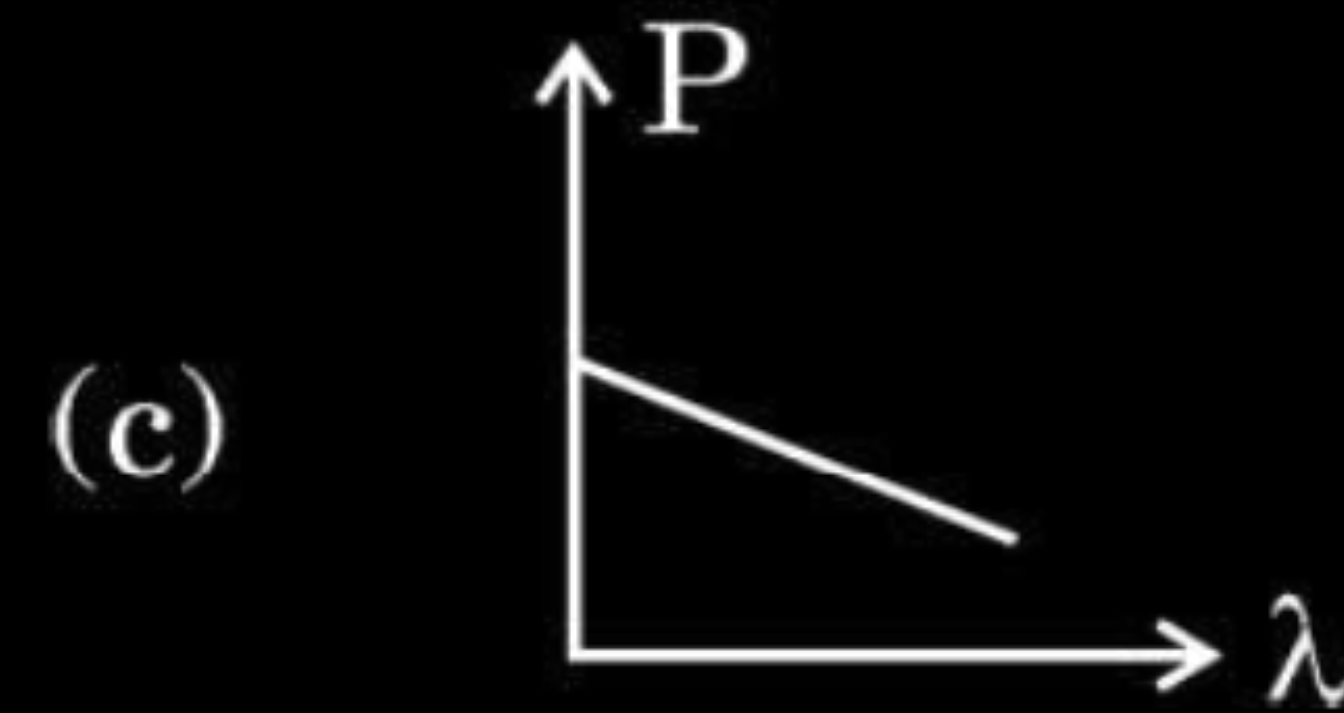
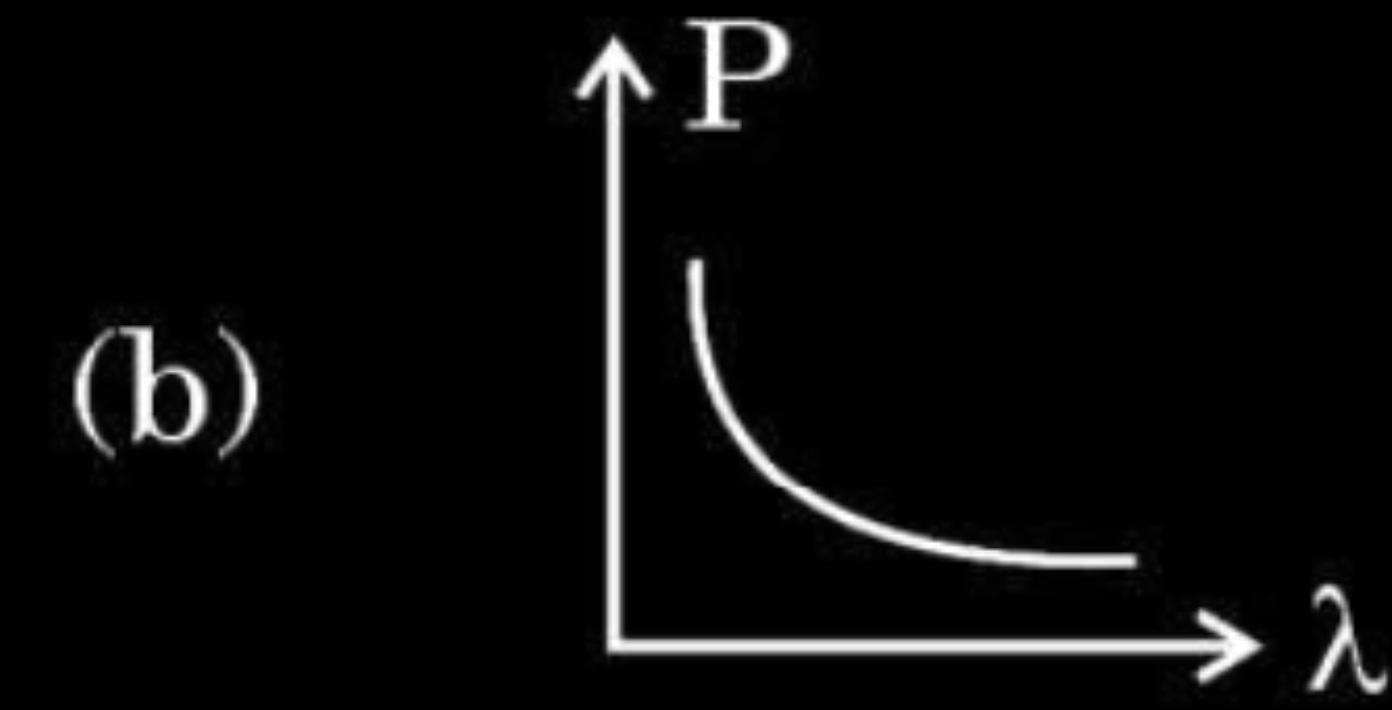
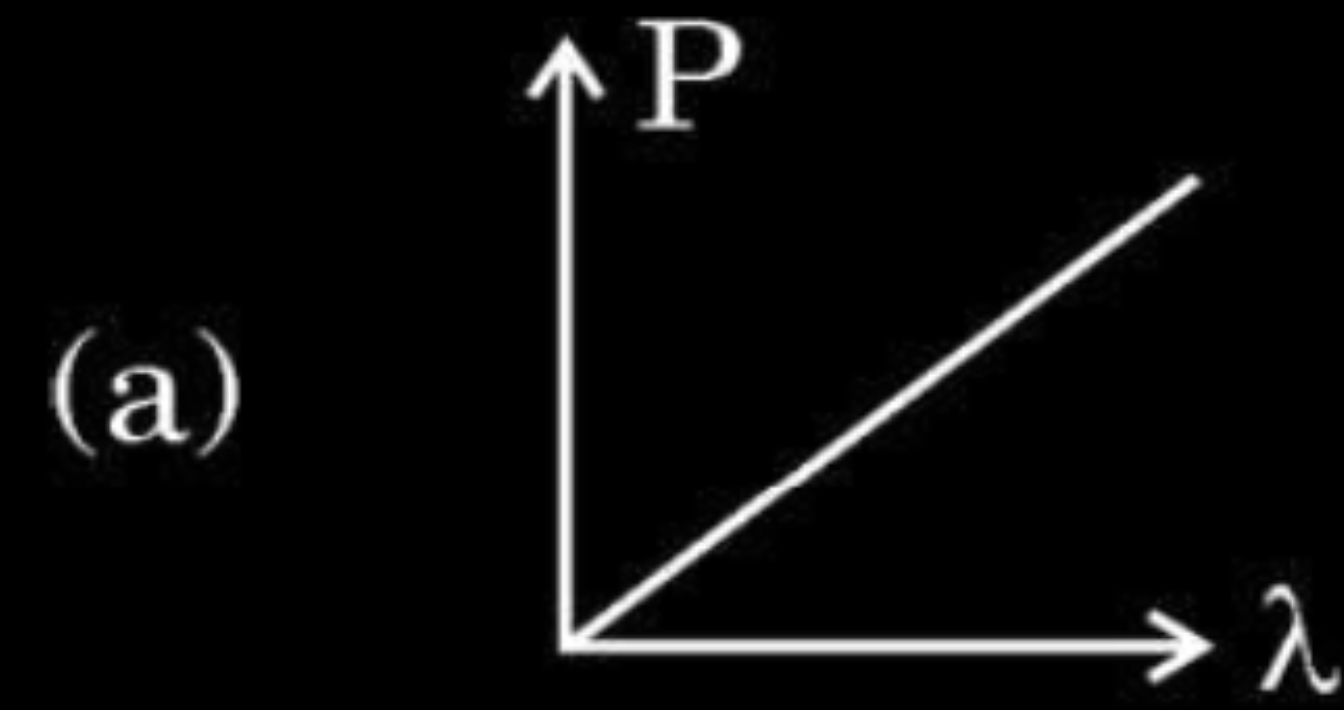
as $\frac{h}{e} = m$ (slope of the line)

So by calculating slope of the line Planck's constant can be calculated.

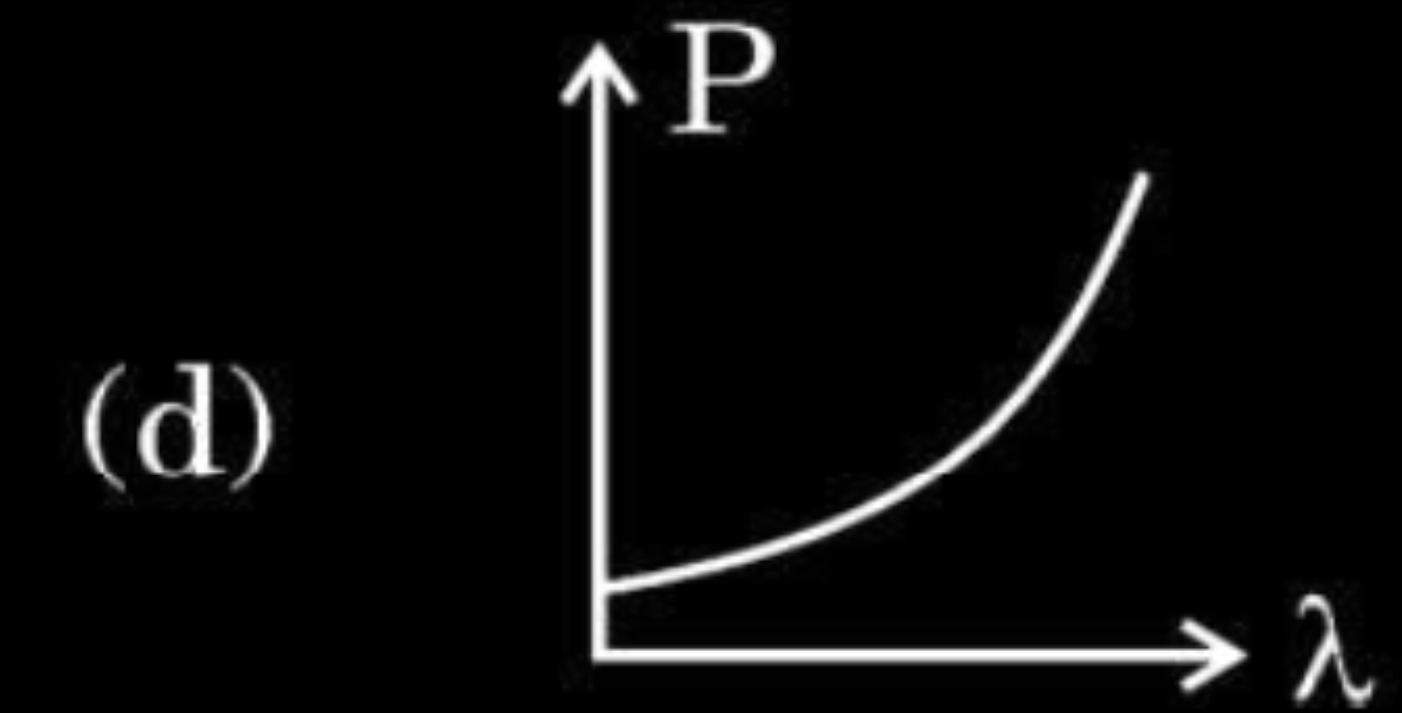
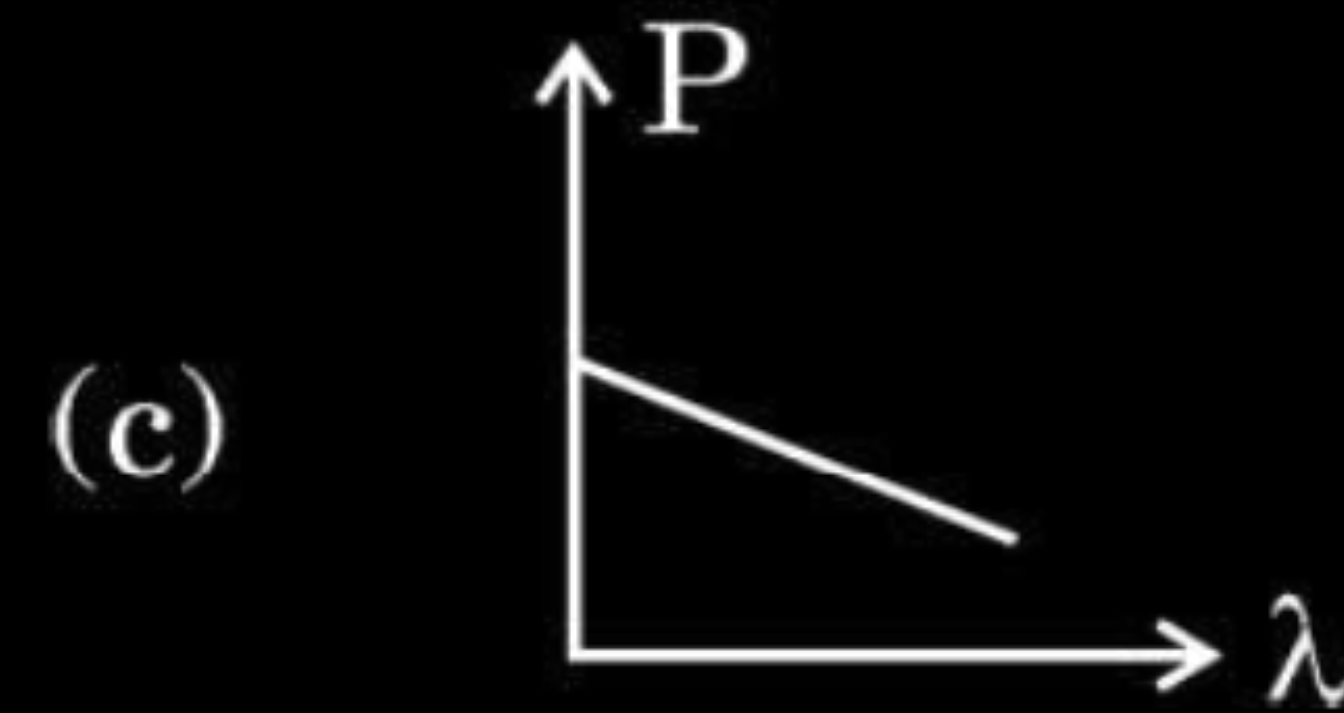
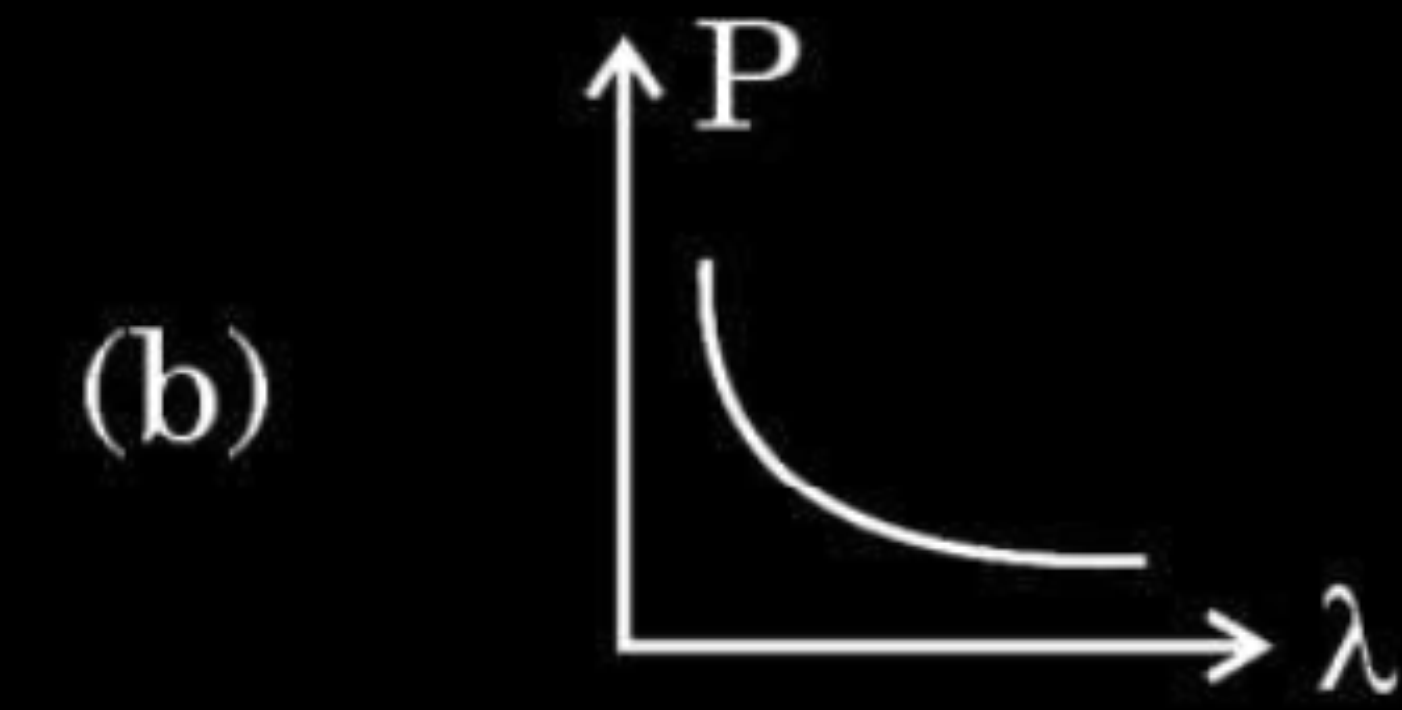
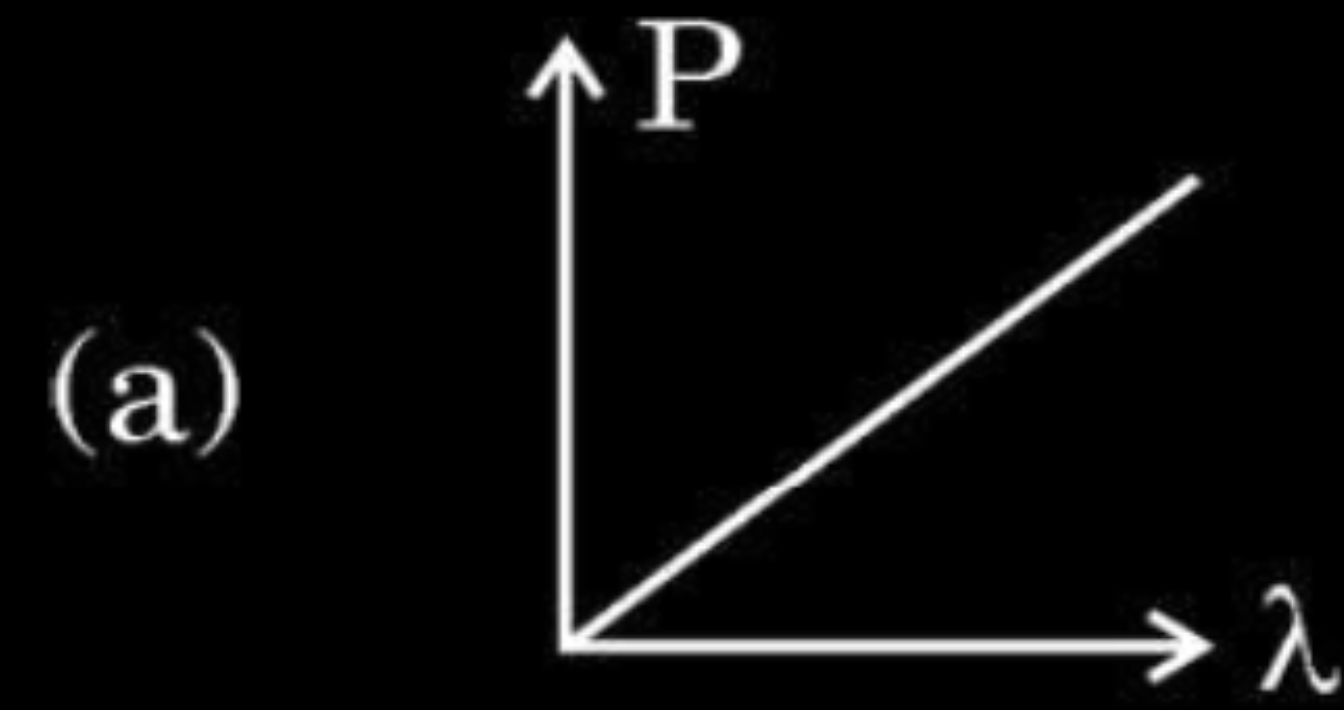
1

$\frac{1}{2}$

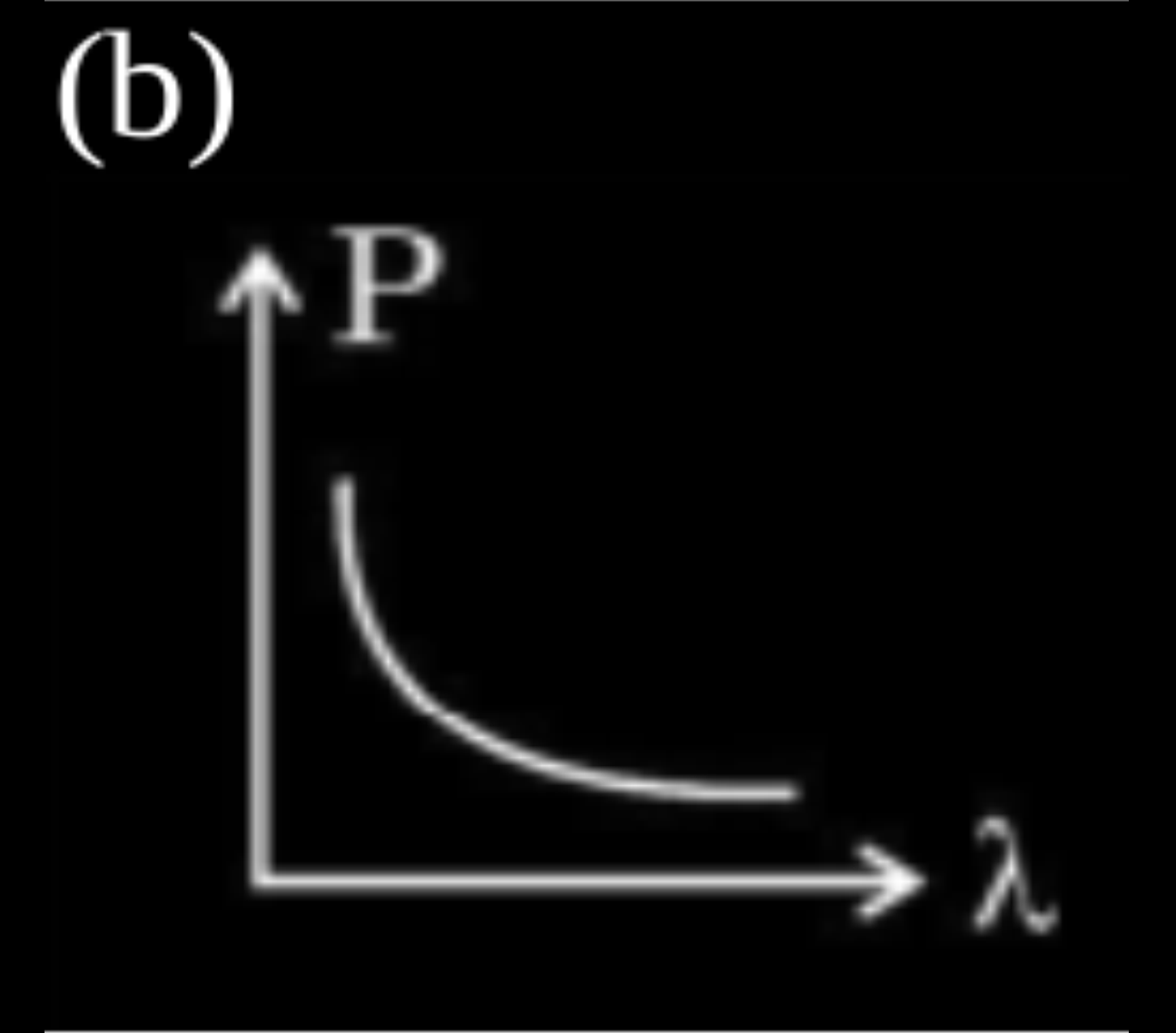
7. Which of the following figures represents the variation of a particle's momentum with the de Broglie wavelength associated with it ?



7. Which of the following figures represents the variation of a particle's momentum with the de Broglie wavelength associated with it ?



CBSE 2023 C



9. Light of frequency $1.5 \nu_0$ is incident on a photosensitive material of threshold frequency ν_0 . If the frequency of the incident radiation is kept constant and intensity is increased, the photo current will :
- (a) increase
 - (b) decrease
 - (c) not change
 - (d) first decrease and then become zero

CBSE 2023 C

9. Light of frequency $1.5 \nu_0$ is incident on a photosensitive material of threshold frequency ν_0 . If the frequency of the incident radiation is kept constant and intensity is increased, the photo current will :
- (a) increase
 - (b) decrease
 - (c) not change
 - (d) first decrease and then become zero

CBSE 2023 C

(a) increase

- 21.** Light of wavelength $3500 \text{ \AA}$ is incident on two metals A and B. Which of them will yield photoelectrons, if their work functions are 4.2 eV and 1.9 eV respectively ? Make the necessary calculations to justify your answer.

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21. Light of wavelength 3500 Å is incident on two metals A and B. Which of them will yield photoelectrons, if their work functions are 4.2 eV and 1.9 eV respectively ? Make the necessary calculations to justify your answer.

CBSE 2023 C

2

$$E = \frac{hc}{\lambda}$$
$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{3500 \times 10^{-10} \times 1.6 \times 10^{-19}} \text{ eV}$$
$$E = 3.55 \text{ eV}$$

Metal B

Since energy of incident light is more than the work function of metal 'B'.

(Note: Give full credit of one mark if student writes 'Metal B' only.)

1/2

1/2

1/2

1/2

- 23.** A beam of high energy photons, each of energy of E is incident on a metal surface and electrons are ejected from the surface. Assuming that the work function of the metal, ϕ is negligible, find the relation between the de Broglie wavelength λ associated with photoelectrons and the energy of photons E . Show in a figure, the variation of λ as a function of E .

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2

- 23.** A beam of high energy photons, each of energy of E is incident on a metal surface and electrons are ejected from the surface. Assuming that the work function of the metal, ϕ is negligible, find the relation between the de Broglie wavelength λ associated with photoelectrons and the energy of photons E . Show in a figure, the variation of λ as a function of E . CBSE 2023 C²

$$h\nu = \Phi_o + E_v^{\max}$$

As Φ_o is negligible

$$E_v^{\max} = h\nu$$

$$E = h\nu$$

$$E = \frac{p^2}{2m}$$

$$p = \sqrt{2mE}$$

$$\lambda = \frac{h}{p}$$

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$\frac{1}{2}$

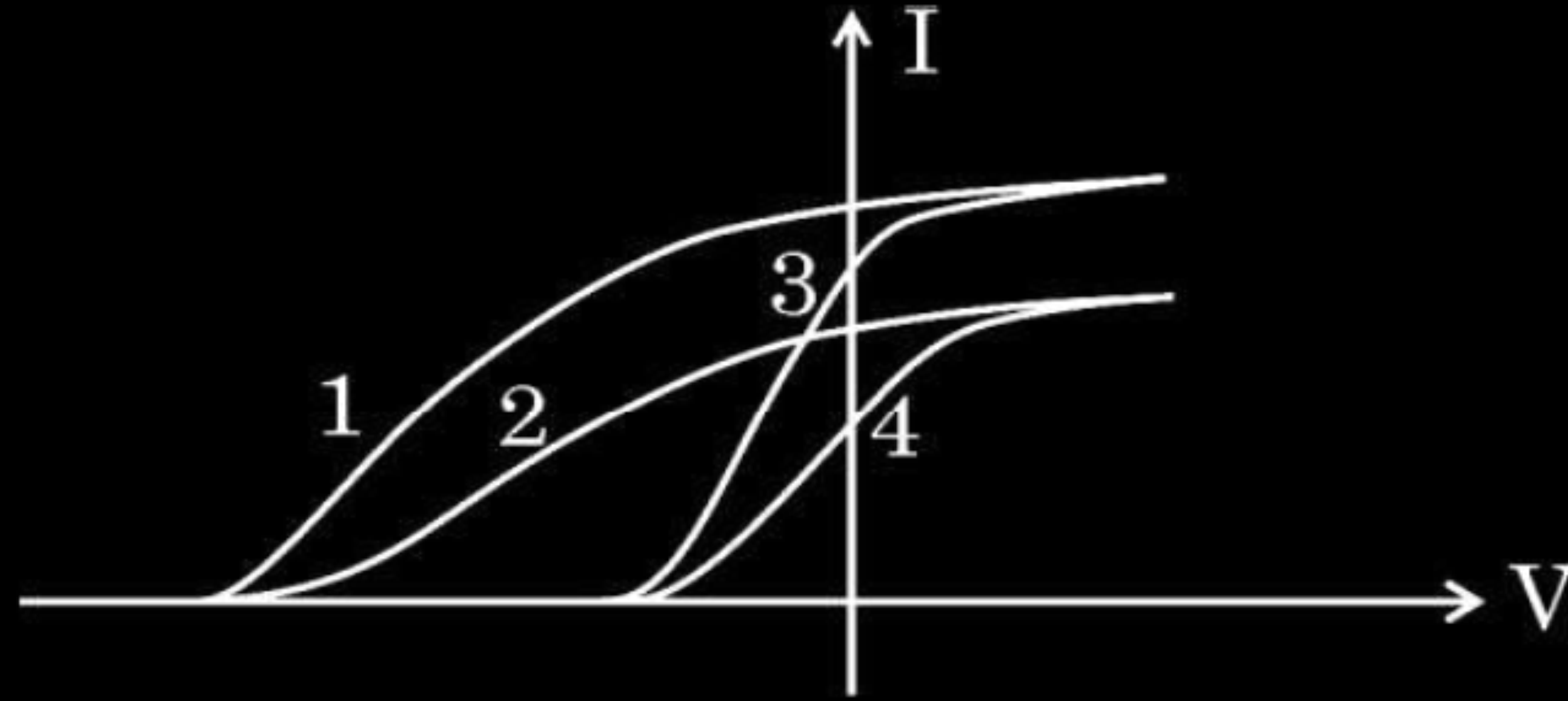
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 25.** The figure shows the variation of photoelectric current (I) versus applied voltage (V) for two photosensitive materials and for two different intensities of the incident radiation. Identify the pairs of curves that correspond to different materials for same intensity of incident radiation. Justify your answers in brief.

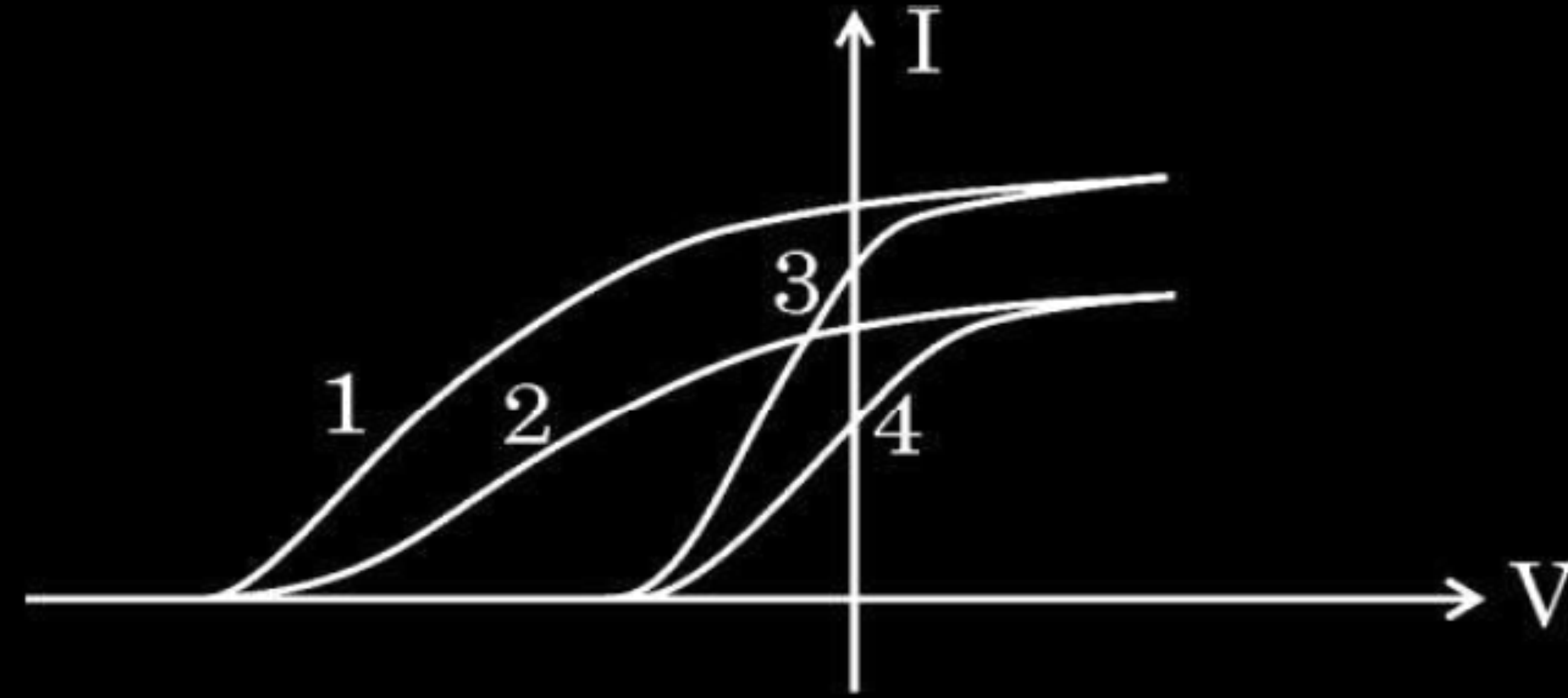
2



CBSE 2023 C

- 25.** The figure shows the variation of photoelectric current (I) versus applied voltage (V) for two photosensitive materials and for two different intensities of the incident radiation. Identify the pairs of curves that correspond to different materials for same intensity of incident radiation. Justify your answers in brief.

2



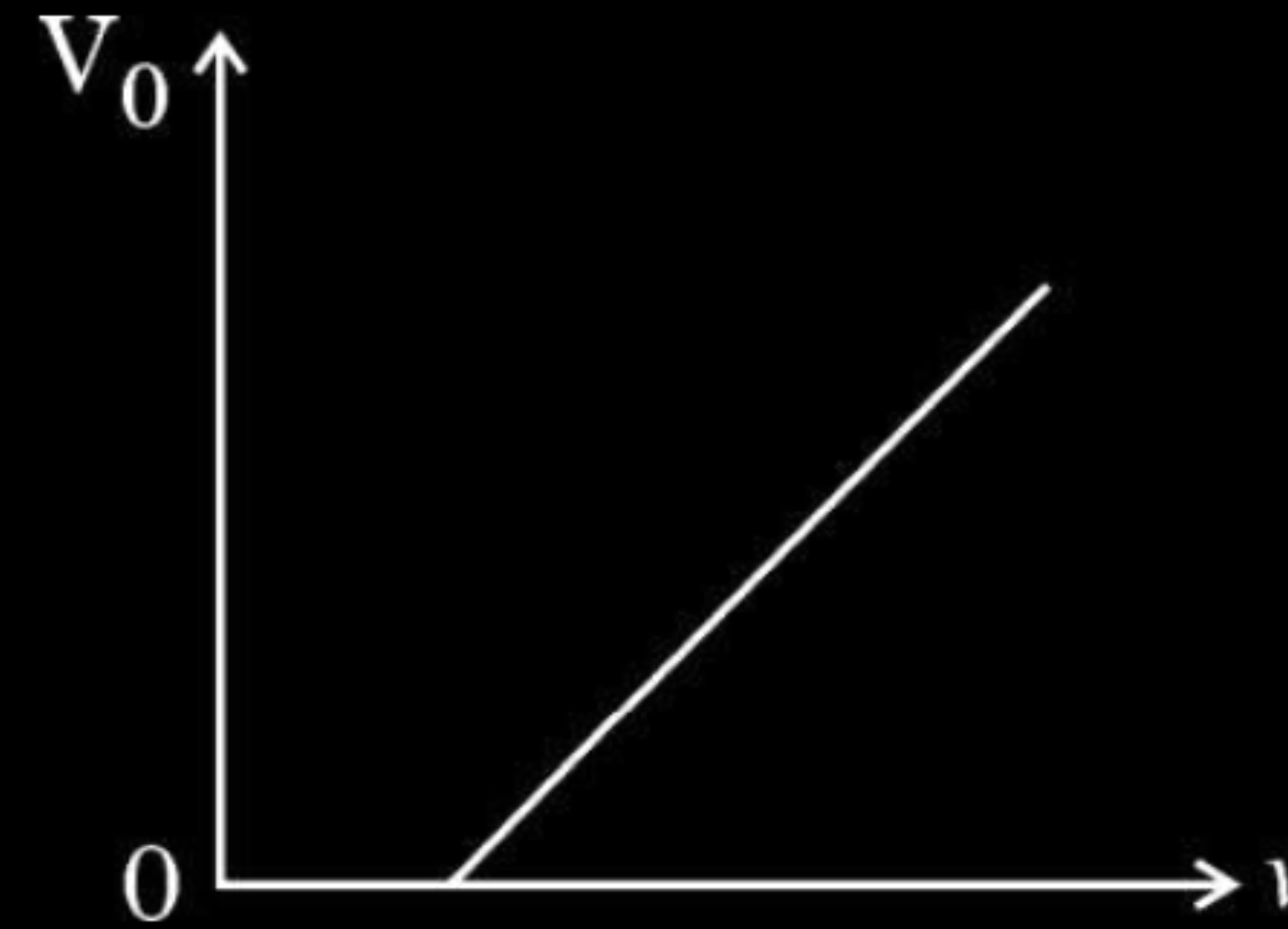
CBSE 2023 C

Curves 1 and 2, 3 and 4 meet at the same point on the voltage axis (same cut off voltage) therefore they belong to same material.
For same intensity and different materials curves 1 and 3, 2 and 4 because they have same saturation current and correspond to different materials.

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2} + \frac{1}{2}$

24. The figure shows the variation of stopping potential V_0 with frequencies of incident monochromatic beam for a metal. Such a graph was first obtained by R.A. Millikan in 1916 for sodium.



CBSE 2023 C

Explain how, using Einstein's photoelectric equation and the graph, you can obtain the value of (i) Planck's constant, and (ii) work function of the metal, given the value of charge on electron.

According to Einstein's equation

$$h\nu = \Phi_o + eV_o$$

$$eV_o = h\nu - \Phi_o$$

$$V_o = \frac{h}{e}\nu - \frac{\Phi_o}{e}$$

This is equation of straight line as shown in figure, where slope is $\frac{h}{e}$.

So from the value of slope from the graph and value of charge on electron value of Planck's constant can be calculated. Extrapolating the graph we get y - intercept $\frac{\Phi_o}{e}$. From this we can find work function given value of e.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

11. The energy of a photon in a beam of red light with a wavelength of 6.60×10^{-7} m is close to :

(a) 4.42×10^{-19} J

(b) 4.0×10^{-19} J

(c) 2.19×10^{-19} J

(d) 3.0×10^{-19} J

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11. The energy of a photon in a beam of red light with a wavelength of 6.60×10^{-7} m is close to :

(a) 4.42×10^{-19} J

(b) 4.0×10^{-19} J

(c) 2.19×10^{-19} J

(d) 3.0×10^{-19} J

CBSE 2023

(d) 3.0×10^{-19} J

18. *Assertion (A) :* The energy (E) and momentum (p) of a photon are related as $p = \frac{h}{\lambda}$.

Reason (R) : Photons behave as a wave.

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18. *Assertion (A) :* The energy (E) and momentum (p) of a photon are related as $p = \frac{h}{\lambda}$.

Reason (R) : Photons behave as a wave.

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(d) [A]- False [R]- False

29. Briefly explain how Einstein's photoelectric equation accounts for all observations on photoelectric effect.

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3

29. Briefly explain how Einstein's photoelectric equation accounts for all observations on photoelectric effect.

CBSE 2023

3

Einstein's Photoelectric equation

$$E_{K_{\max}} = h\nu - \phi_o = h(\nu - \nu_o)$$

Observations:

(i) Kinetic energy of emitted electron depends upon frequency and is independent of intensity of incident radiation.

(ii) there exists a threshold energy/cut off frequency below which no electron emission is possible As $E_{K_{\max}} > 0$

$$\therefore \nu > \nu_o$$

(iii) The photoelectric current depends upon intensity of incident radiation as more the intensity, more is the quanta incident and hence more photoelectrons emitted, thus more is photoelectric current.

(iv) Photoelectric emission is instantaneous, as absorption of quantum by an electron is instantaneous and therefore the emission of electrons is also instantaneous.

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

2. A light of frequency ν is incident on a metal surface whose work function is W_0 . The kinetic energy of emitted electron is K . If the frequency of the incident light is doubled then the kinetic energy of emitted electron will be :

(a) $2K$

(b) more than $2K$

(c) between K and $2K$

(d) less than K

CBSE 2023

2. A light of frequency ν is incident on a metal surface whose work function is W_0 . The kinetic energy of emitted electron is K . If the frequency of the incident light is doubled then the kinetic energy of emitted electron will be :

(a) $2K$

(b) more than $2K$

(c) between K and $2K$

(d) less than K

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(b) more than $2 K$

5. Photons of energy 3.2 eV are incident on a photosensitive surface. If the stopping potential for the emitted electrons is 1.5 V , the work function for the surface is :

- (a) 1.5 eV (b) 1.7 eV (c) 3.2 eV (d) 4.7 eV

5. Photons of energy 3.2 eV are incident on a photosensitive surface. If the stopping potential for the emitted electrons is 1.5 V , the work function for the surface is :

- (a) 1.5 eV (b) 1.7 eV (c) 3.2 eV (d) 4.7 eV

5. Photons of energy 3.2 eV are incident on a photosensitive surface. If the stopping potential for the emitted electrons is 1.5 V , the work function for the surface is :

(a) 1.5 eV (b) 1.7 eV (c) 3.2 eV (d) 4.7 eV

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(b) 1.7 eV

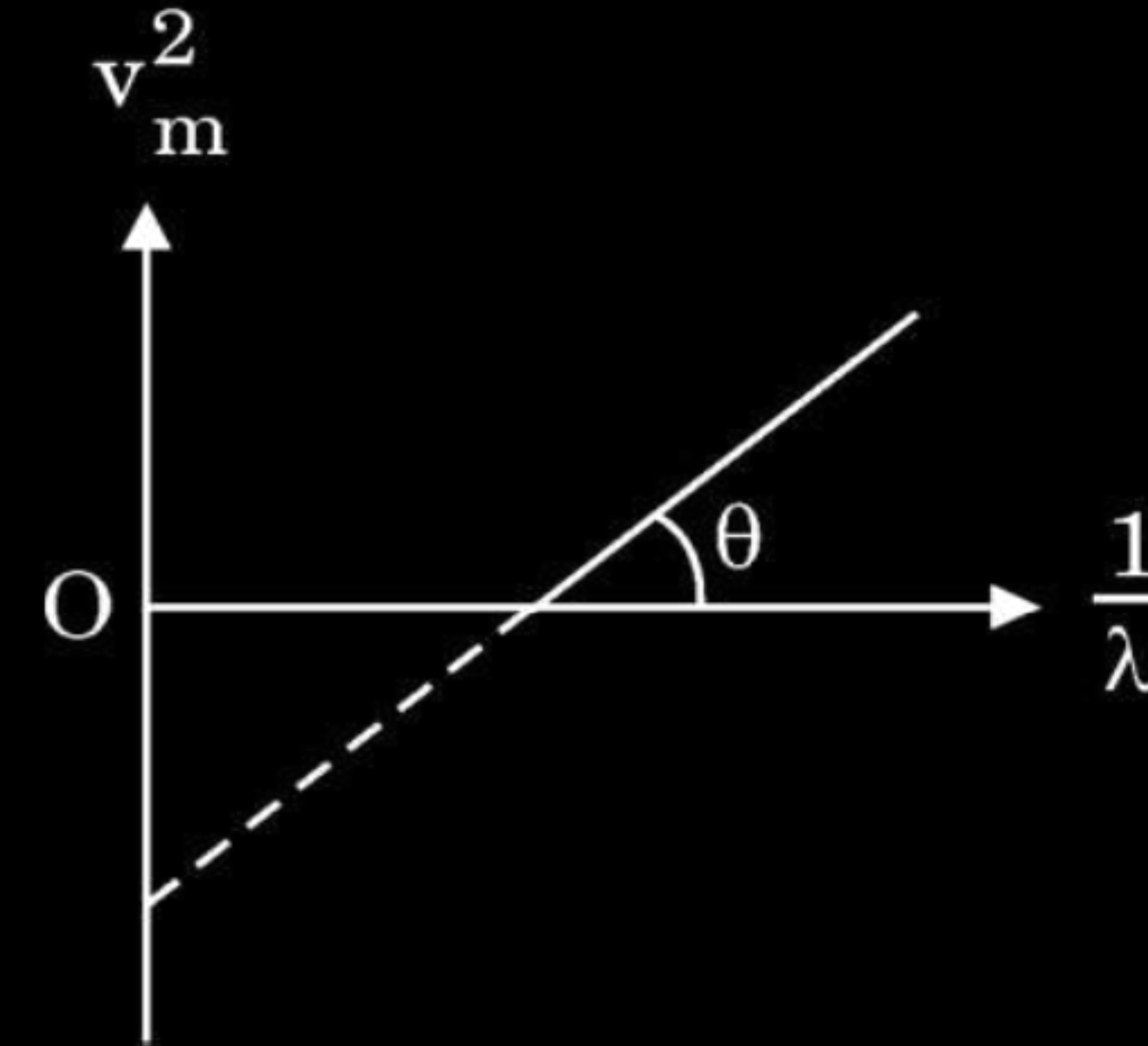
19. The figure shows v_m^2 versus $\frac{1}{\lambda}$ graph for photoelectrons emitted from a surface where v_m is the maximum speed of electrons and λ is the wavelength of incident radiation. Using this graph and Einstein's photoelectric equation, obtain the expression for Planck's constant and work function of the surface.

2

$$\frac{1}{2}mv_m^2 = \frac{hc}{\lambda} - \phi_0$$

$$v_m^2 = \left(\frac{2hc}{m} \right) \frac{1}{\lambda} - \frac{2}{m} \phi_0$$

$\frac{1}{2}$



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According to this equation a plot of v_m^2 versus $(1/\lambda)$ is a straight line.

Slope of the graph = $\frac{2hc}{m}$

Intercept = $\frac{2}{m} \phi_0$

Slope and intercept can be found from the graph

$h = \frac{m}{2c} \times \text{slope}$

$\phi_0 = \frac{m}{2} \times \text{intercept}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

4. Photons of energy 4.3 eV are incident on a photosensitive surface of work function 2.3 eV . The stopping potential for photoelectrons is :

(a) 2.0 eV

(b) 2.3 eV

(c) 4.3 eV

(d) 6.6 eV

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4. Photons of energy 4.3 eV are incident on a photosensitive surface of work function 2.3 eV . The stopping potential for photoelectrons is :

(a) 2.0 eV

(b) 2.3 eV

(c) 4.3 eV

(d) 6.6 eV

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**None of the option is correct.
Award 1 mark to each student.**

14. A proton and an alpha particle have the same kinetic energy. The ratio of de Broglie wavelengths associated with the proton to that with the alpha particle is :

- (a) 1 (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$

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14. A proton and an alpha particle have the same kinetic energy. The ratio of de Broglie wavelengths associated with the proton to that with the alpha particle is :

- (a) 1 (b) 2 (c) $2\sqrt{2}$ (d) $\frac{1}{2}$

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(b) 2

2. The momentum of a photon of wavelength λ is :

(a) $\frac{h\lambda}{c^2}$

(b) $\frac{h\lambda}{c}$

(c) $\frac{h}{\lambda}$

(d) zero

CBSE 2023

2. The momentum of a photon of wavelength λ is :

(a) $\frac{h\lambda}{c^2}$

(b) $\frac{h\lambda}{c}$

(c) $\frac{h}{\lambda}$

(d) zero

CBSE 2023

(c) $\frac{h}{\lambda}$

9. Light of frequency 6.4×10^{14} Hz is incident on a metal of work function 2.14 eV. The maximum kinetic energy of the emitted electrons is about :

(a) 0.25 eV

(b) 0.51 eV

(c) 1.02 eV

(d) 0.10 eV

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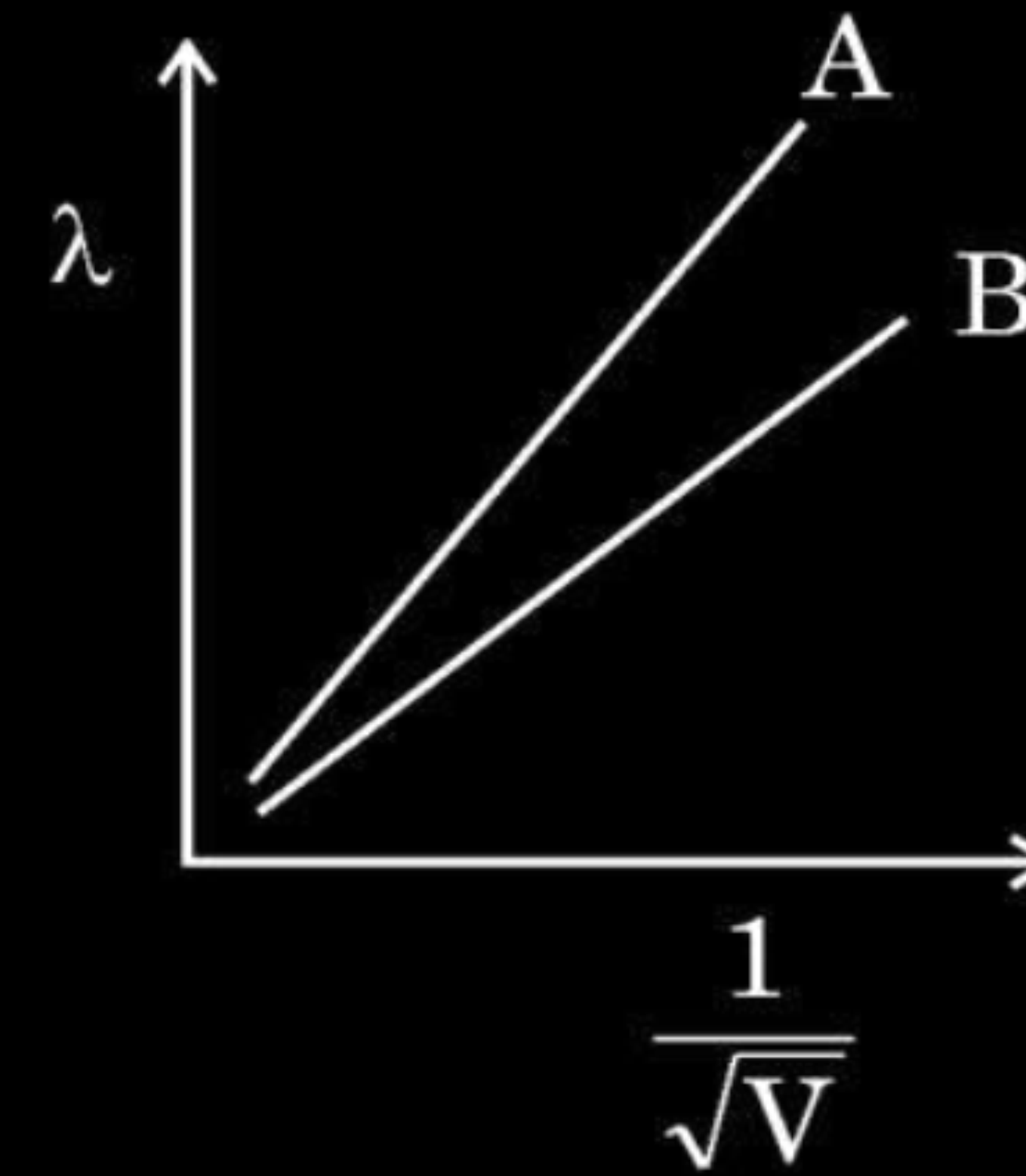
9. Light of frequency 6.4×10^{14} Hz is incident on a metal of work function 2.14 eV. The maximum kinetic energy of the emitted electrons is about :

- (a) 0.25 eV
- (b) 0.51 eV
- (c) 1.02 eV
- (d) 0.10 eV

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(b) 0.51 eV

- 30.** (a) The figure shows de Broglie wavelength (λ) for two particles A and B having same charges but different masses as a function of $\frac{1}{\sqrt{V}}$, where V is the potential through which the particles are accelerated.



- (i) Which of them is heavier and why ?
- (ii) What does the slope of the line represent ?
- (b) Calculate the momentum of an electron having de Broglie wavelength of $3.0 \text{ \AA}$.

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(a)(i) B is heavier than A

Reason- Slope $\propto \frac{1}{\sqrt{m}}$

Slope of A > slope of B;

Hence $m_B > m_A$

(ii) $\lambda = \frac{h}{\sqrt{2mqV}}$

$$\lambda = \frac{h}{\sqrt{2mq}} \times \frac{1}{\sqrt{V}}$$

So, slope = $\frac{h}{\sqrt{2mq}}$

(b) Momentum, $p = \frac{h}{\lambda}$

$$= \frac{6.63 \times 10^{-34}}{3 \times 10^{-10}}$$

$$p = 2.21 \times 10^{-24} \text{ kgm/s}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

7. In an experiment on photoelectric effect, the intensity of incident radiation is increased, keeping the frequency ν the same. The number of photoelectrons emitted will :

- (a) increase (b) decrease
(c) remain same (d) depend on frequency

7. In an experiment on photoelectric effect, the intensity of incident radiation is increased, keeping the frequency ν the same. The number of photoelectrons emitted will :

(a) increase

(b) decrease

(c) remain same

(d) depend on frequency

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(b) 0.51 eV

- 26.** (a) A particle of mass m and charge q is accelerated through a potential difference V . Plot a graph of de Broglie wavelength λ associated with it as a function of V .
- (b) Calculate the energy acquired by and de Broglie wavelength associated with, an electron accelerated through a potential difference of 400 V.

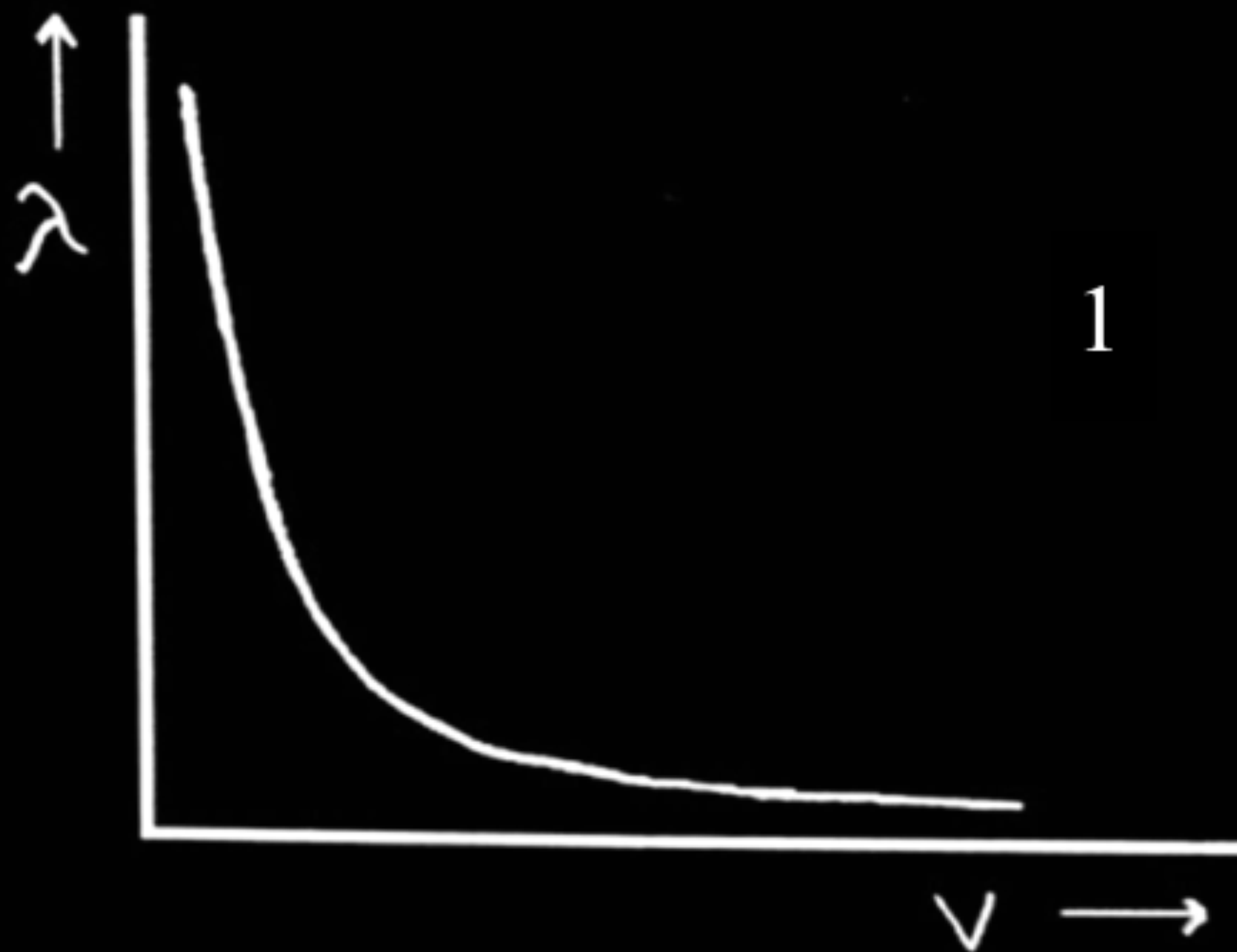
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26. (a) A particle of mass m and charge q is accelerated through a potential difference V . Plot a graph of de Broglie wavelength λ associated with it as a function of V .
- (b) Calculate the energy acquired by and de Broglie wavelength associated with, an electron accelerated through a potential difference of 400 V.

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3

(a)



(b) Energy acquired = qV_o

$$= 1.6 \times 10^{-19} \times 400$$

$$= 6.4 \times 10^{-17} \text{ J} = 400 \text{ eV}$$

$$\lambda = \frac{h}{\sqrt{2mK}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 6.4 \times 10^{-17}}}$$

$$\lambda = 0.6145 \text{ nm} = 0.6145 \text{ Å}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

10. A graph is plotted between the stopping potential (on y-axis) and the frequency of incident radiation (on x-axis) for a metal. The product of the slope of the straight line obtained and the magnitude of charge on an electron is equal to :

(a) h

(b) $\frac{h}{c}$

(c) $\frac{2h}{c}$

(d) $\frac{h}{2c}$

10. A graph is plotted between the stopping potential (on y-axis) and the frequency of incident radiation (on x-axis) for a metal. The product of the slope of the straight line obtained and the magnitude of charge on an electron is equal to :

(a) h

(b) $\frac{h}{c}$

(c) $\frac{2h}{c}$

(d) $\frac{h}{2c}$

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(a) h

- 29.** Calculate the wavelength of de Broglie waves associated with a proton having $\left(\frac{500}{1.673}\right)$ eV energy. How will the wavelength be affected for an alpha particle having the same energy ?

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- 29.** Calculate the wavelength of de Broglie waves associated with a proton having $\left(\frac{500}{1.673}\right)$ eV energy. How will the wavelength be affected for an alpha particle having the same energy ?

CBSE 2023

3

$$\lambda = \frac{h}{\sqrt{2mK}}$$

$\frac{1}{2}$

$$K = \left(\frac{500}{1.673}\right) eV$$

$$= \frac{500 \times 1.6 \times 10^{-19}}{1.673} J$$

$\frac{1}{2}$

$$\lambda_p = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 1.673 \times 10^{-27} \times \left(\frac{500 \times 1.6 \times 10^{-19}}{1.673}\right)}}$$

$\frac{1}{2}$

$$= 1.65 \times 10^{-12} m$$

$\frac{1}{2}$

$$\lambda \propto \frac{1}{\sqrt{m}}$$

$\frac{1}{2}$

For α particle; $m_\alpha = 4m_p$

$$\Rightarrow \lambda_\alpha = \frac{\lambda_p}{2}$$

$\frac{1}{2}$

6. In an experiment, photons of energy 7.5 eV are incident on a metal surface. Electrons emitted from the metal surface are stopped by an electrode at a potential of -4.5 V w.r.t. the metal. The work function of the metal is :

(a) 3.0 eV

(b) 4.5 eV

(c) 7.5 eV

(d) 12.0 eV

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6. In an experiment, photons of energy 7.5 eV are incident on a metal surface. Electrons emitted from the metal surface are stopped by an electrode at a potential of -4.5 V w.r.t. the metal. The work function of the metal is :

(a) 3.0 eV

(b) 4.5 eV

(c) 7.5 eV

(d) 12.0 eV

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(a) 3.0 eV

16. *Assertion (A) :* Photoelectric effect demonstrates the particle nature of light.

Reason (R) : Photoelectric current is proportional to intensity of incident radiation for frequencies more than the threshold frequency.

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16. *Assertion (A) :* Photoelectric effect demonstrates the particle nature of light.

Reason (R) : Photoelectric current is proportional to intensity of incident radiation for frequencies more than the threshold frequency.

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(b) Both the assertion (A) and Reason (R) are true , but Reason (R) is **not** the correct explanation of the Assertion (A).

- 30.** A particle is associated with a de Broglie wave of wavelength 1.2 nm. The kinetic energy of the particle is made four times. Calculate the new value of the wavelength of the wave.

CBSE 2023

- 30.** A particle is associated with a de Broglie wave of wavelength 1.2 nm. The kinetic energy of the particle is made four times. Calculate the new value of the wavelength of the wave.

CBSE 2023

3

$$\lambda = \frac{h}{\sqrt{2mK}}$$

$$\lambda' = \frac{h}{\sqrt{2m \times 4K}}$$

$$= \frac{1}{2} \frac{h}{\sqrt{2mK}}$$

$$= \frac{\lambda}{2}$$

$$= \frac{1.2 \text{ nm}}{2}$$

$$= 0.6 \text{ nm}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

4. A photon of energy 7 eV is incident on a metal surface having the work function of 3.75 eV. The stopping potential is :

(a) 1.75 eV

(b) 2.45 eV

(c) 3.25 eV

(d) 3.75 eV

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4. A photon of energy 7 eV is incident on a metal surface having the work function of 3.75 eV. The stopping potential is :

(a) 1.75 eV

(b) 2.45 eV

(c) 3.25 eV

(d) 3.75 eV

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Since no option is correct award 1 mark even if student does not attempt.

- 27.** The threshold wavelength for a metal is $3315 \text{ \AA}$. What should be the wavelength of light incident on the metal surface so that the maximum kinetic energy of the emitted photoelectrons be 1.25 eV ?

CBSE 2023

3

- 27.** The threshold wavelength for a metal is 3315 Å. What should be the wavelength of light incident on the metal surface so that the maximum kinetic energy of the emitted photoelectrons be 1.25 eV ?

CBSE 2023

3

$$\lambda_o = 3315 \text{ Å}$$

$$\begin{aligned} K_{\max} &= 1.25 \text{ eV} = 1.25 \times 1.6 \times 10^{-19} \text{ J} \\ &= 2 \times 10^{-19} \text{ J} \\ &= 0.2 \times 10^{-18} \text{ J} \end{aligned}$$

$$\begin{aligned} \text{Work function} &= \phi_o = \frac{hc}{\lambda_o} \\ &= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{3315 \times 10^{-10}} = 0.6 \times 10^{-18} \text{ J} \end{aligned}$$

$$\text{Using } h\nu = \phi_o + K_{\max}$$

$$\begin{aligned} \frac{hc}{\lambda} &= \phi_o + K_{\max} \\ &= (0.6 \times 10^{-18} + 0.2 \times 10^{-18}) \text{ J} \\ &= 0.8 \times 10^{-18} \text{ J} \end{aligned}$$

$$\begin{aligned} \lambda &= \frac{hc}{0.8 \times 10^{-18}} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{0.8 \times 10^{-18}} \\ &= 24.86 \times 10^{-8} \text{ m} \\ &= 2486 \times 10^{-10} \text{ m} = 2486 \text{ Å} \end{aligned}$$

1/2

1/2

1/2

1/2

1/2

1/2

29. How does Einstein's photoelectric equation explain the emission of electrons from a metal surface ? Explain briefly.

Plot the variation of photocurrent with :

- (a) collector plate potential for different intensity of incident radiation, and
- (b) intensity of incident radiation.

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Plot the variation of photocurrent with :

- (a) collector plate potential for different intensity of incident radiation, and
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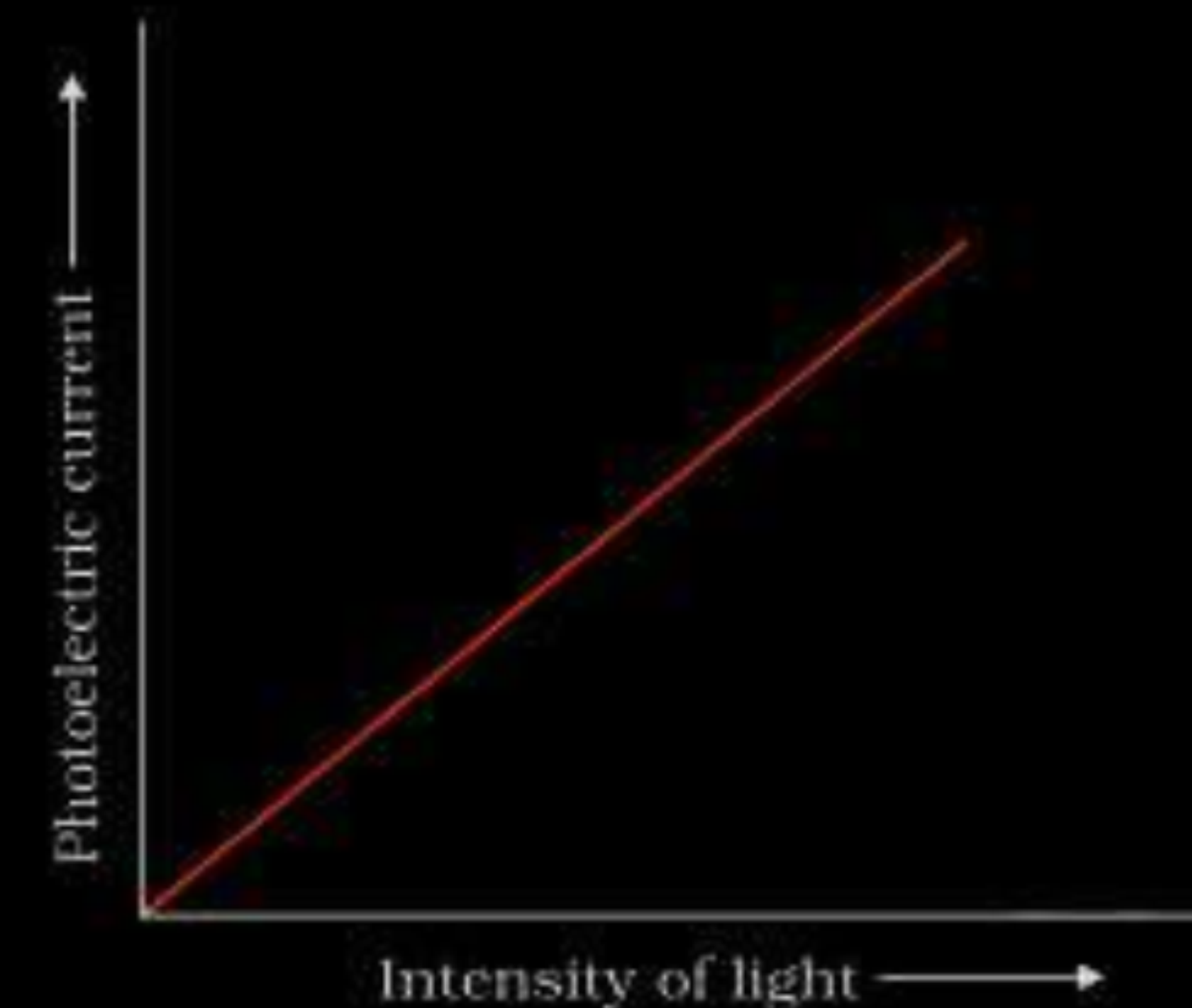
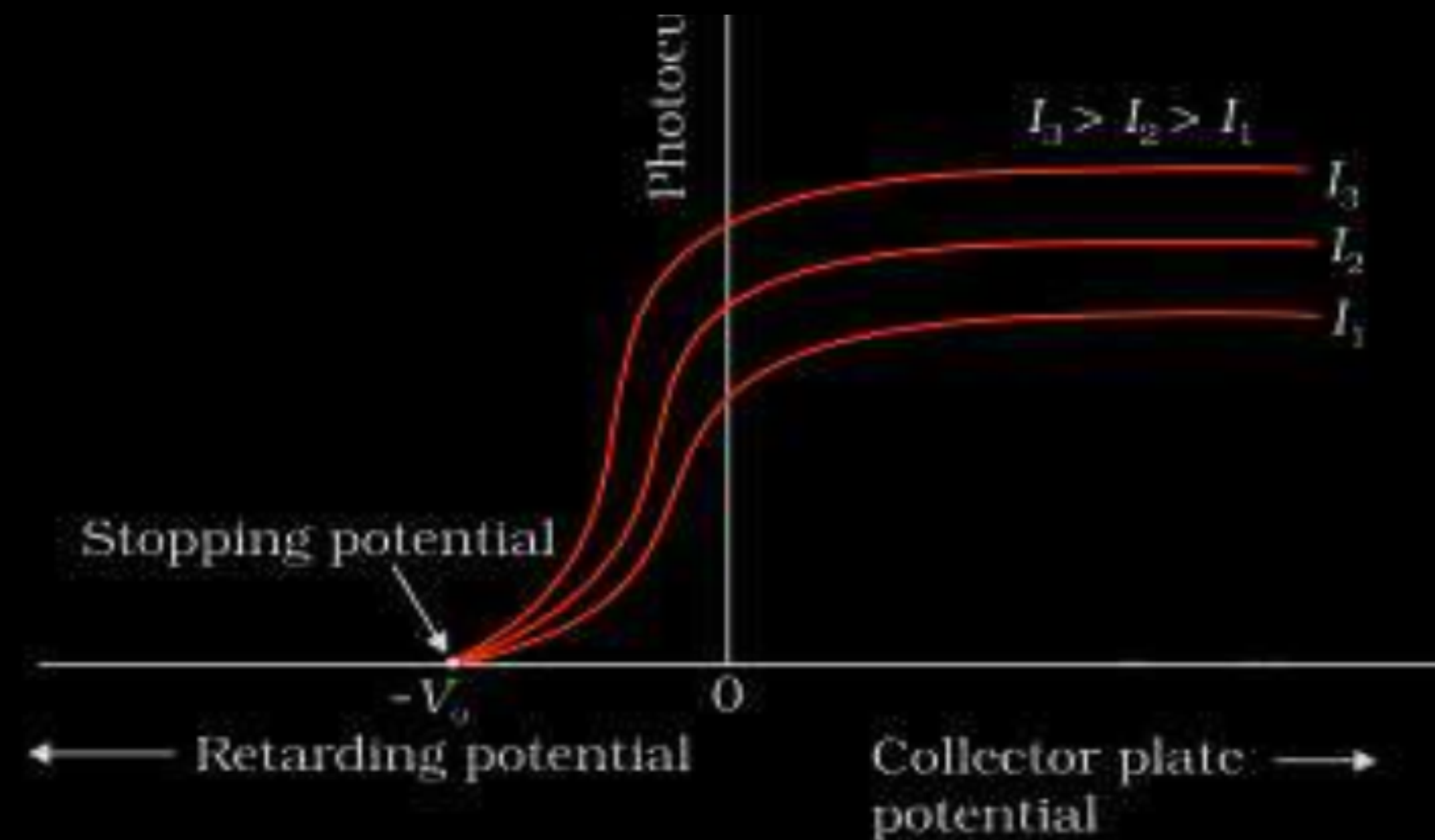
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According to Einstein's photoelectric equation

An electron absorbs a quantum of energy ' $h\nu$ ' of incident radiation. If the energy of absorbed quantum exceeds the minimum energy needed by the electron to escape from the metal surface (work function ϕ_0), the electron is emitted.

$$K_{\max} = h\nu - \phi_0$$

b)



1

2. The energy of a photon of wavelength 663 nm is 1

(a) $6.64 \times 10^{-20} \text{ J}$

(b) $5.18 \times 10^{-19} \text{ J}$

(c) $3.0 \times 10^{-19} \text{ J}$

(d) $2.0 \times 10^{-20} \text{ J}$

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2. The energy of a photon of wavelength 663 nm is 1

(a) $6.64 \times 10^{-20} \text{ J}$

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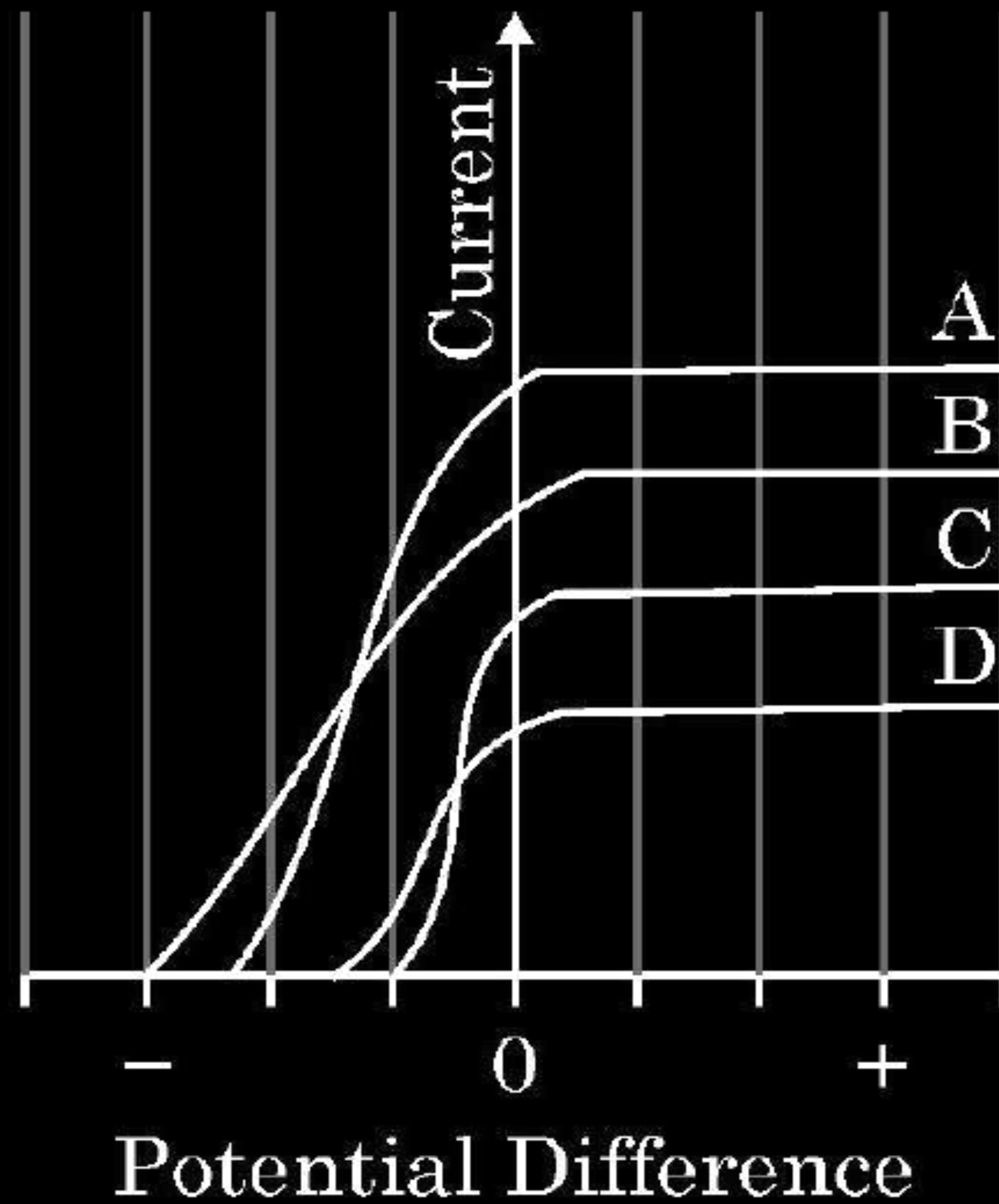
(c) $3.0 \times 10^{-19} \text{ J}$

(d) $2.0 \times 10^{-20} \text{ J}$

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(c) $3.0 \times 10^{-19} \text{ J}$

35. (a) Figure shows the variation of photoelectric current measured in a photo cell circuit as a function of the potential difference between the plates of the photo cell when light beams A, B, C and D of different wavelengths are incident on the photo cell. Examine the given figure and answer the following questions :



- (i) Which light beam has the highest frequency and why ?
- (ii) Which light beam has the longest wavelength and why ?
- (iii) Which light beam ejects photoelectrons with maximum momentum and why ?

4

OR

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- (b) What is the effect on threshold frequency and stopping potential on increasing the frequency of incident beam of light ? Justify your answer.

(i) The light beam B because it requires maximum retarding potential to reduce the photoelectric current to zero.	$\frac{1}{2} + \frac{1}{2}$
(ii) The light beam C because it requires minimum retarding potential to reduce photoelectric current to zero.	$\frac{1}{2} + \frac{1}{2}$
(iii) The light beam B ejects photoelectrons with maximum momentum. because highest frequency light beam ejects photoelectrons with highest kinetic energy and hence highest momentum.	1
OR	
There is no effect on threshold frequency since it is characteristic of the metal.	$\frac{1}{2} + \frac{1}{2}$
With increase in frequency of incident beam of light, stopping potential increases because to stop the photoelectrons of higher kinetic energy, larger retarding potential is required.	$\frac{1}{2} + \frac{1}{2}$
Alternatively Give full credit if a candidate explains the effect of frequency on stopping potential using the following formula. $eV_0 = h(\nu - \nu_0)$	

12. A photon of wavelength 663 nm is incident on a metal surface. The work function of the metal is 1.50 eV. The maximum kinetic energy of the emitted photo electrons is

1

- (a) 3.0×10^{-20} J
- (b) 6.0×10^{-20} J
- (c) 4.5×10^{-20} J
- (d) 9.0×10^{-20} J

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12. A photon of wavelength 663 nm is incident on a metal surface. The work function of the metal is 1.50 eV. The maximum kinetic energy of the emitted photo electrons is

1

(a) $3.0 \times 10^{-20} \text{ J}$

(b) $6.0 \times 10^{-20} \text{ J}$

(c) $4.5 \times 10^{-20} \text{ J}$

(d) $9.0 \times 10^{-20} \text{ J}$

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(b) $6.0 \times 10^{-20} \text{ J}$

10. Which one of the following metals does not exhibit emission of electrons from its surface when irradiated by visible light ?

1

(a) Rubidium

(b) Sodium

(c) Cadmium

(d) Caesium

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10. Which one of the following metals does not exhibit emission of electrons from its surface when irradiated by visible light ?

1

(a) Rubidium

(b) Sodium

(c) Cadmium

(d) Caesium

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(c)
Cadmium

7. E, c and ν represent the energy, velocity and frequency of a photon. Which of the following represents its wavelength ?

1

(A) $\frac{h\nu}{c^2}$

(B) $h\nu$

(C) $\frac{hc}{E}$

(D) $\frac{h\nu}{c}$

7. E, c and ν represent the energy, velocity and frequency of a photon. Which of the following represents its wavelength ?

1

(A) $\frac{h\nu}{c^2}$

(B) $h\nu$

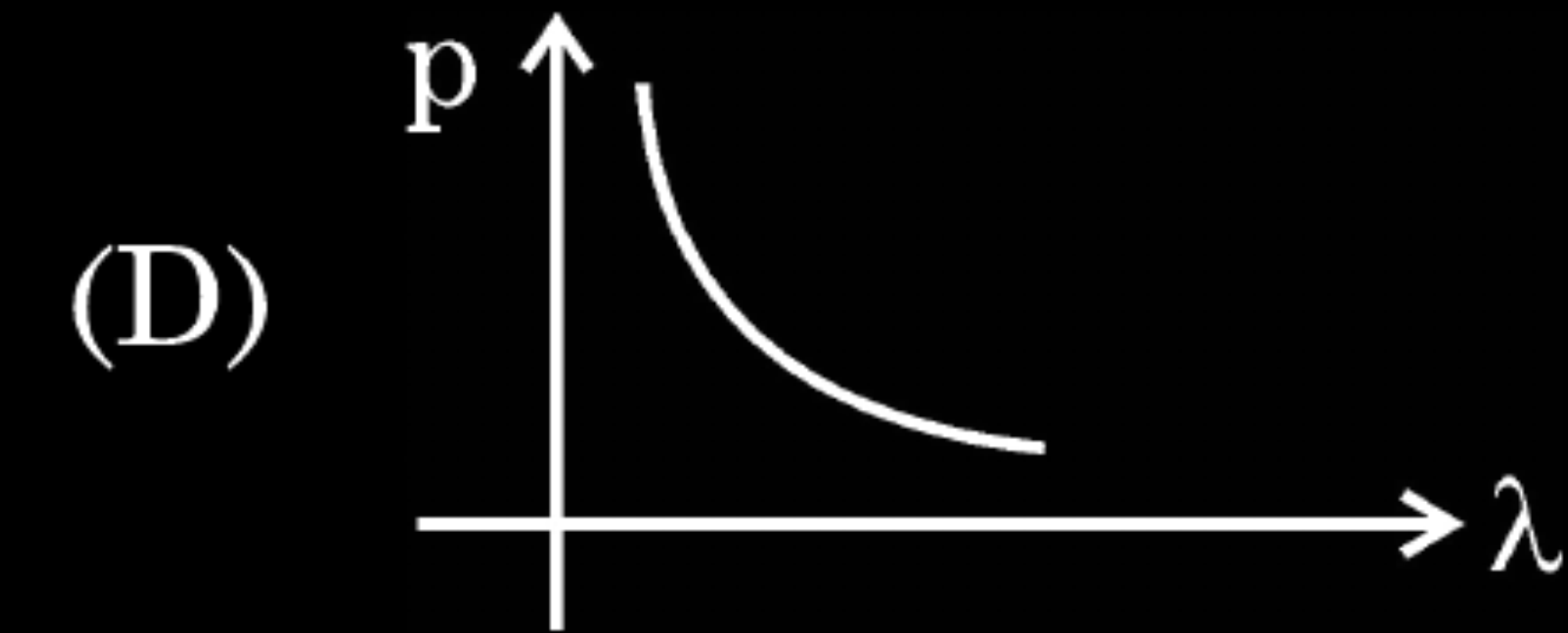
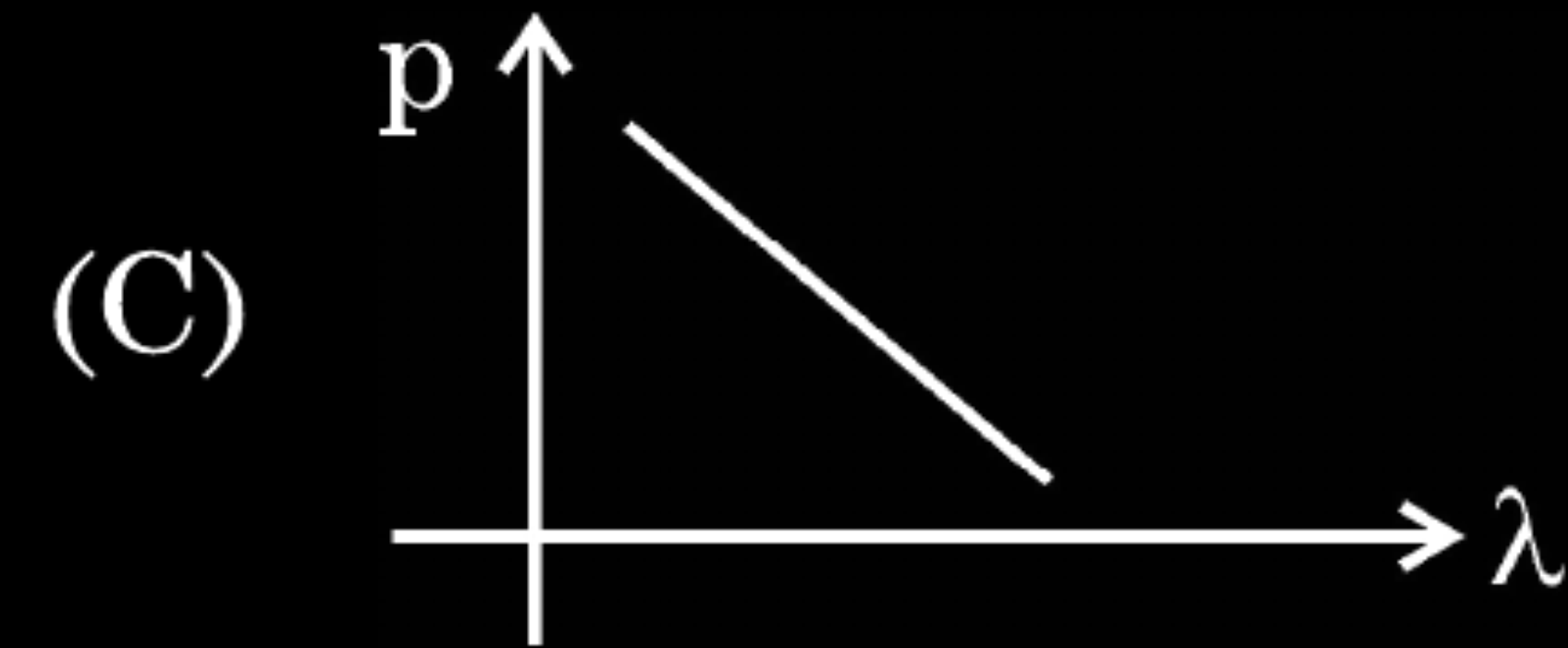
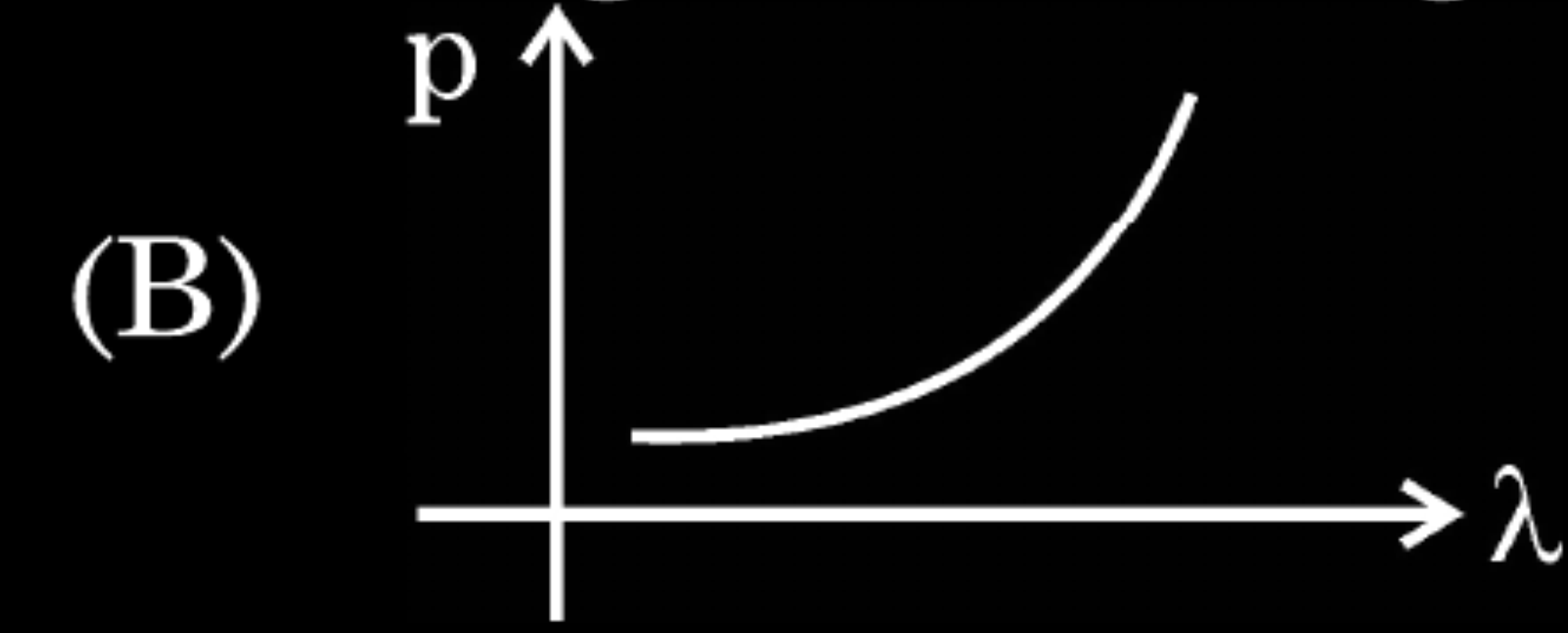
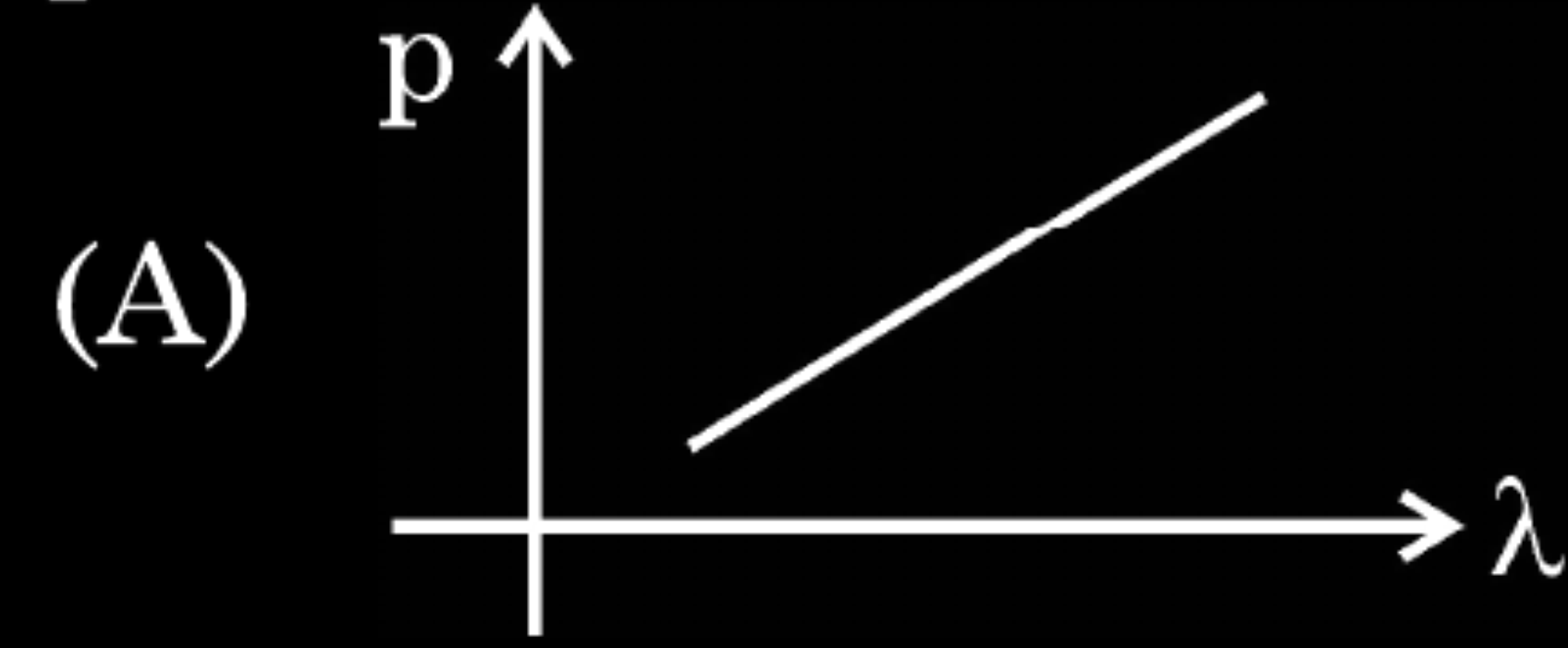
(C) $\frac{hc}{E}$

(D) $\frac{h\nu}{c}$

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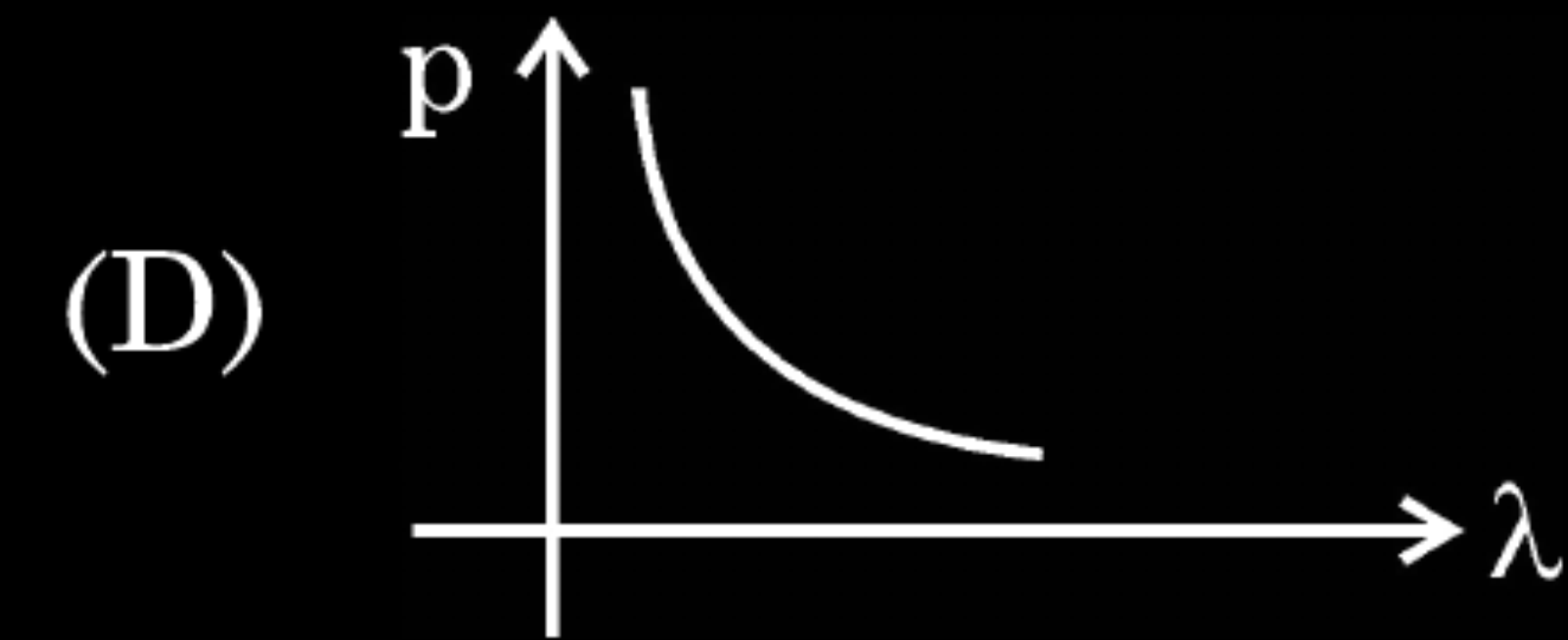
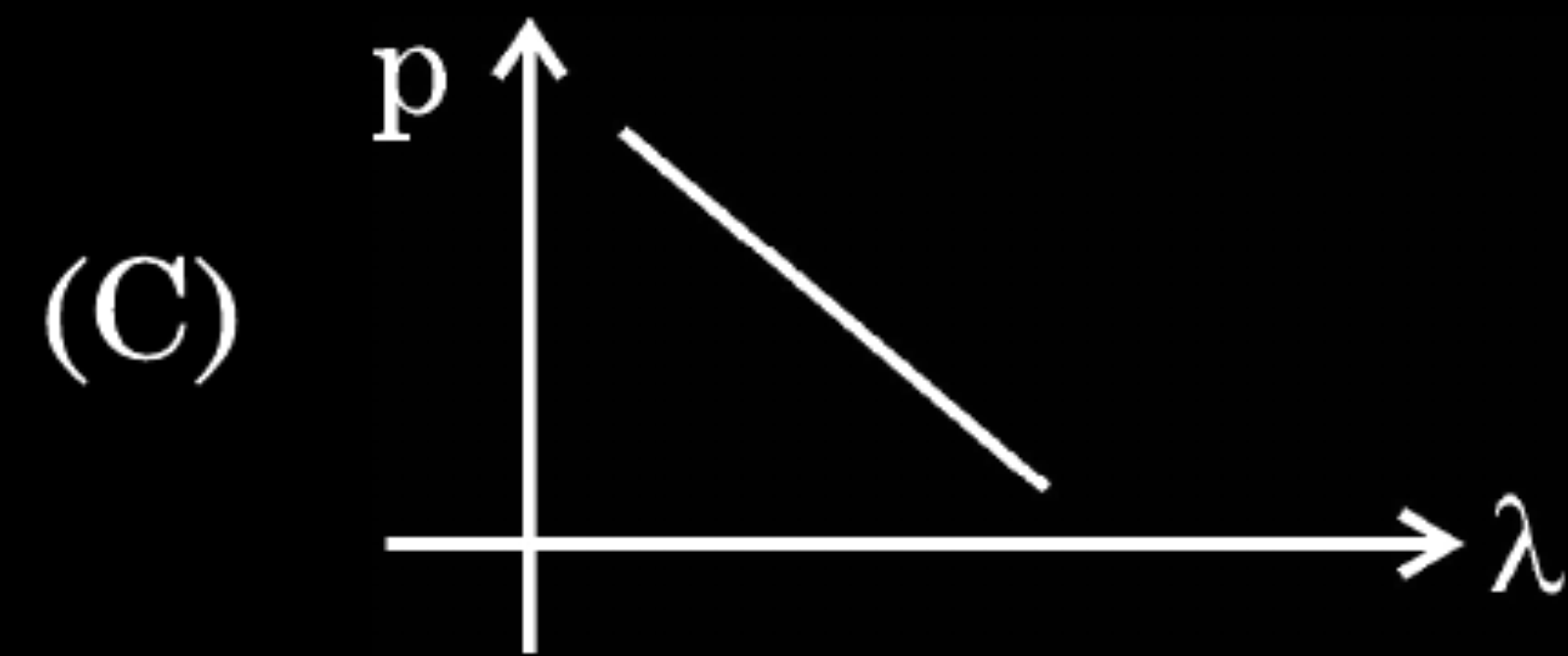
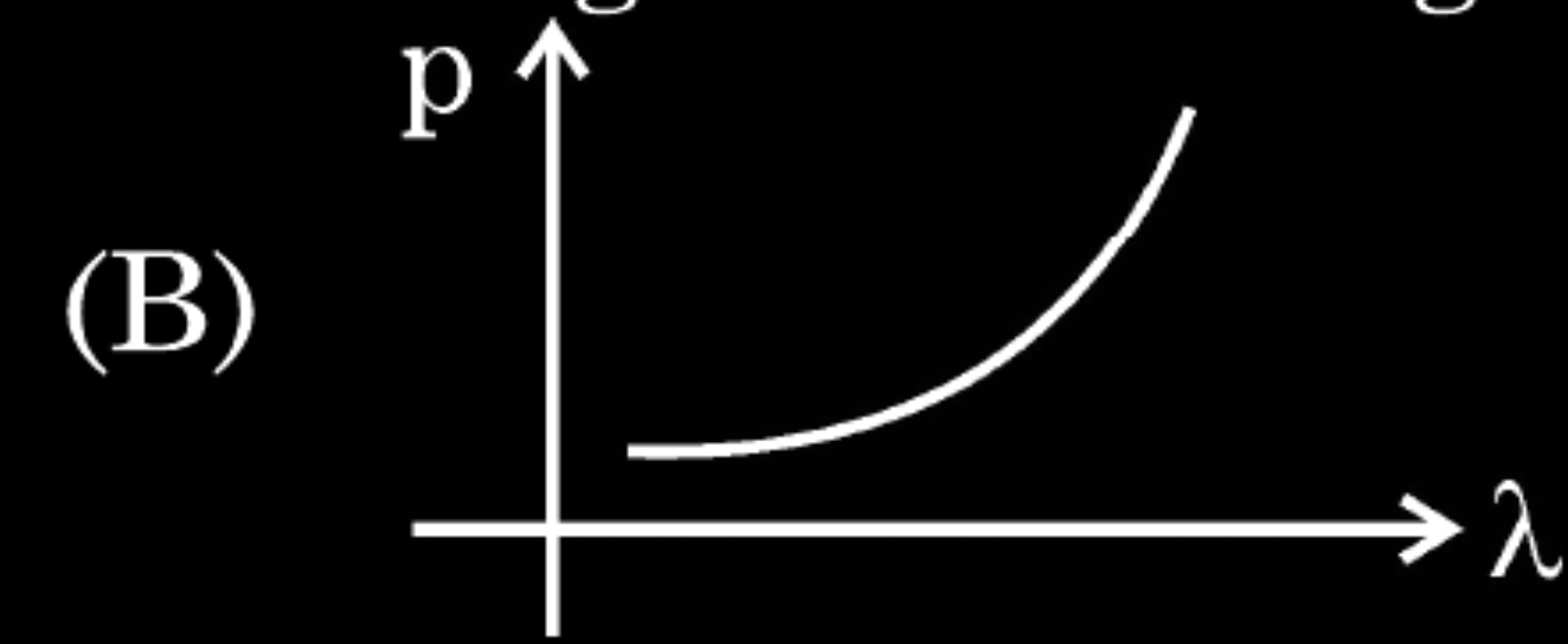
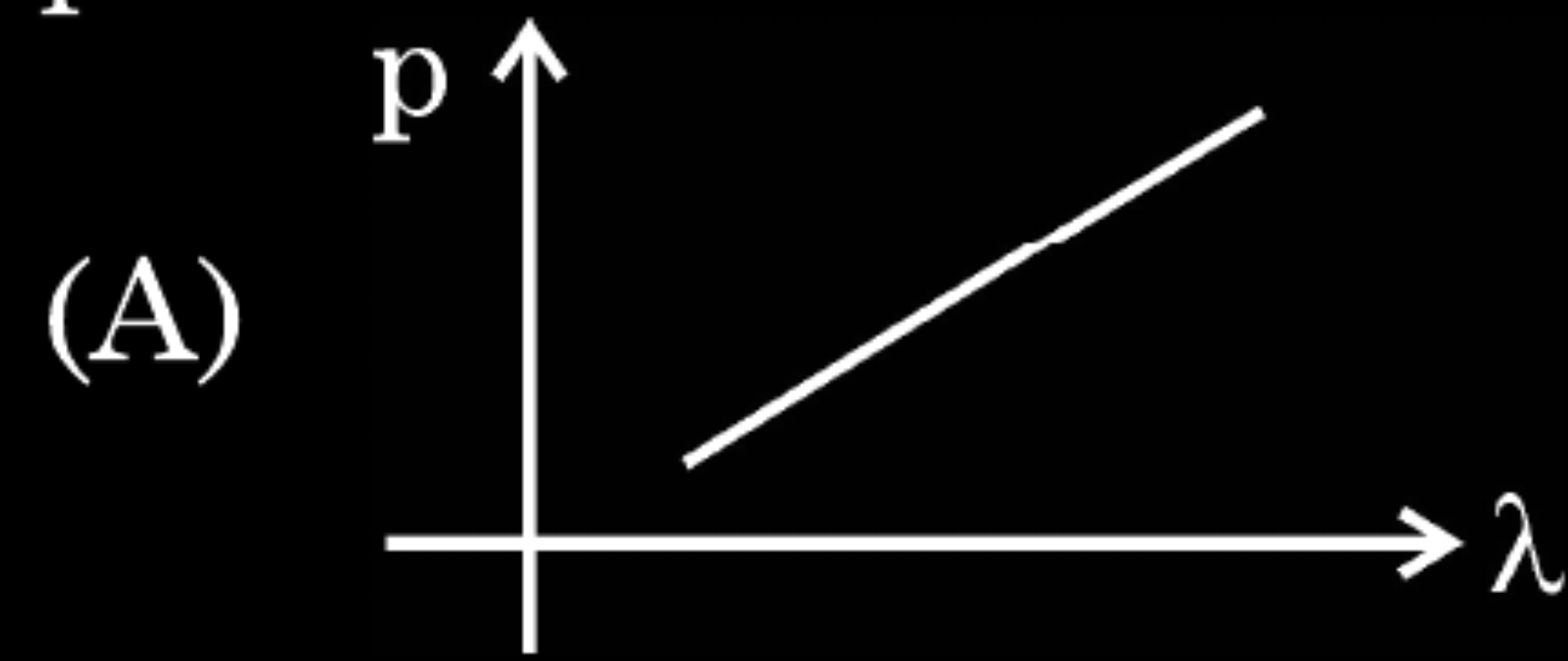
(C) hc/E

11. Which of the following graphs correctly represents the variation of a particle momentum with its associated de-Broglie wavelength ? 1



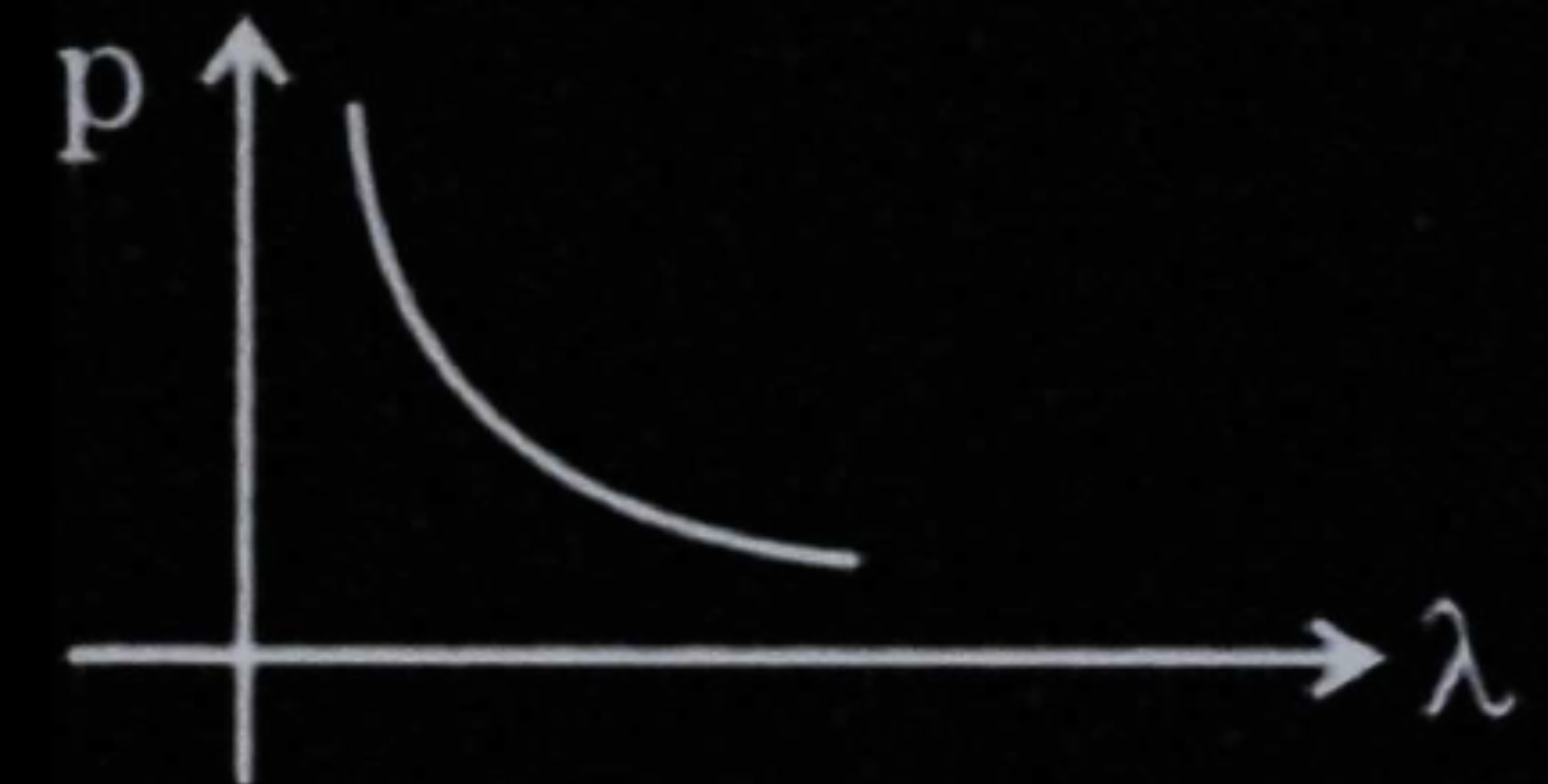
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11. Which of the following graphs correctly represents the variation of a particle momentum with its associated de-Broglie wavelength ? 1



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(D)



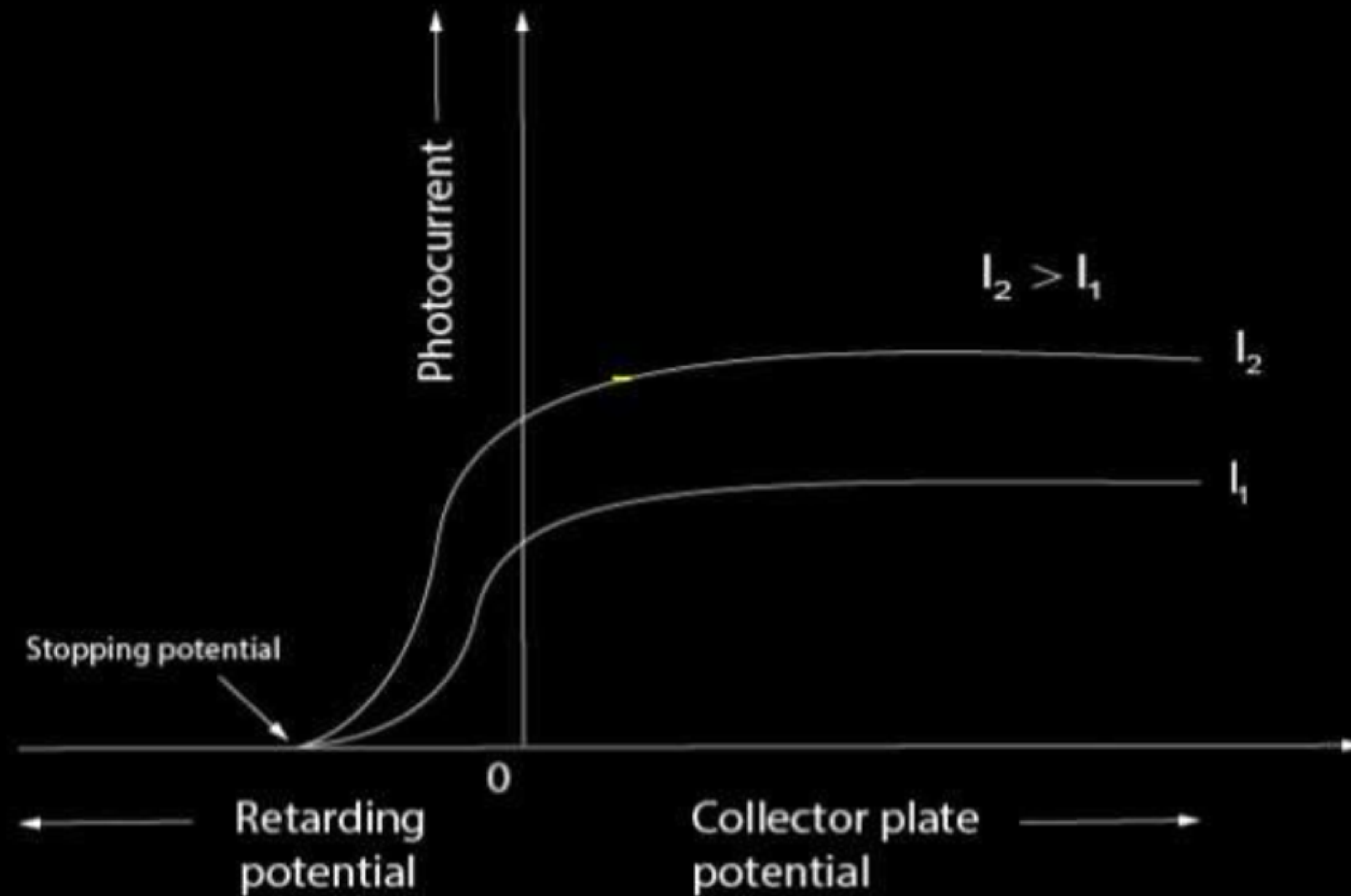
19. Plot a graph showing the variation of photo electric current, as a function of anode potential for two light beams having the same frequency but different intensities I_1 and I_2 ($I_1 > I_2$). Mention its important features.

2

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19. Plot a graph showing the variation of photo electric current, as a function of anode potential for two light beams having the same frequency but different intensities I_1 and I_2 ($I_1 > I_2$). Mention its important features.

2



1

Increase in intensity does not change the cut-off potential.
The increase in intensity increases the value of saturation current.

$\frac{1}{2} + \frac{1}{2}$

20. (a) How will the De Broglie wavelength associated with an electron be affected when the (i) velocity of the electron decreases ? and (ii) accelerating potential is increased ? Justify your answer.

2

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20. (a) How will the De Broglie wavelength associated with an electron be affected when the (i) velocity of the electron decreases ? and (ii) accelerating potential is increased ? Justify your answer. 2

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(i) $\lambda = h/mv$

$\frac{1}{2}$

With decrease in velocity, de- Broglie wavelength increases

$\frac{1}{2}$

(ii) $\lambda = \frac{h}{\sqrt{2meV}}$

$\frac{1}{2}$

With increase in accelerating potential, de- Broglie wavelength decreases

$\frac{1}{2}$

- (b) How would the stopping potential for a given photosensitive surface change if (i) the frequency of the incident radiation were increased ? and (ii) the intensity of incident radiation were decreased ? Justify your answer.

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- (b) How would the stopping potential for a given photosensitive surface change if (i) the frequency of the incident radiation were increased ? and (ii) the intensity of incident radiation were decreased ? Justify your answer.

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2

(i) With increase in frequency stopping potential will increase.	$\frac{1}{2}$
$V_0 = \frac{h}{e} \nu - \frac{\phi_0}{e}$	$\frac{1}{2}$
(ii) Stopping potential remains same.	$\frac{1}{2}$
Energy of photoelectron ejected is independent of intensity of radiation	$\frac{1}{2}$

7. The energy of a photon of wavelength λ is
- (A) $hc \lambda$ (B) hc/λ
(C) λ/hc (D) $\lambda h/c$

1

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- (A) $hc \lambda$ (B) hc/λ
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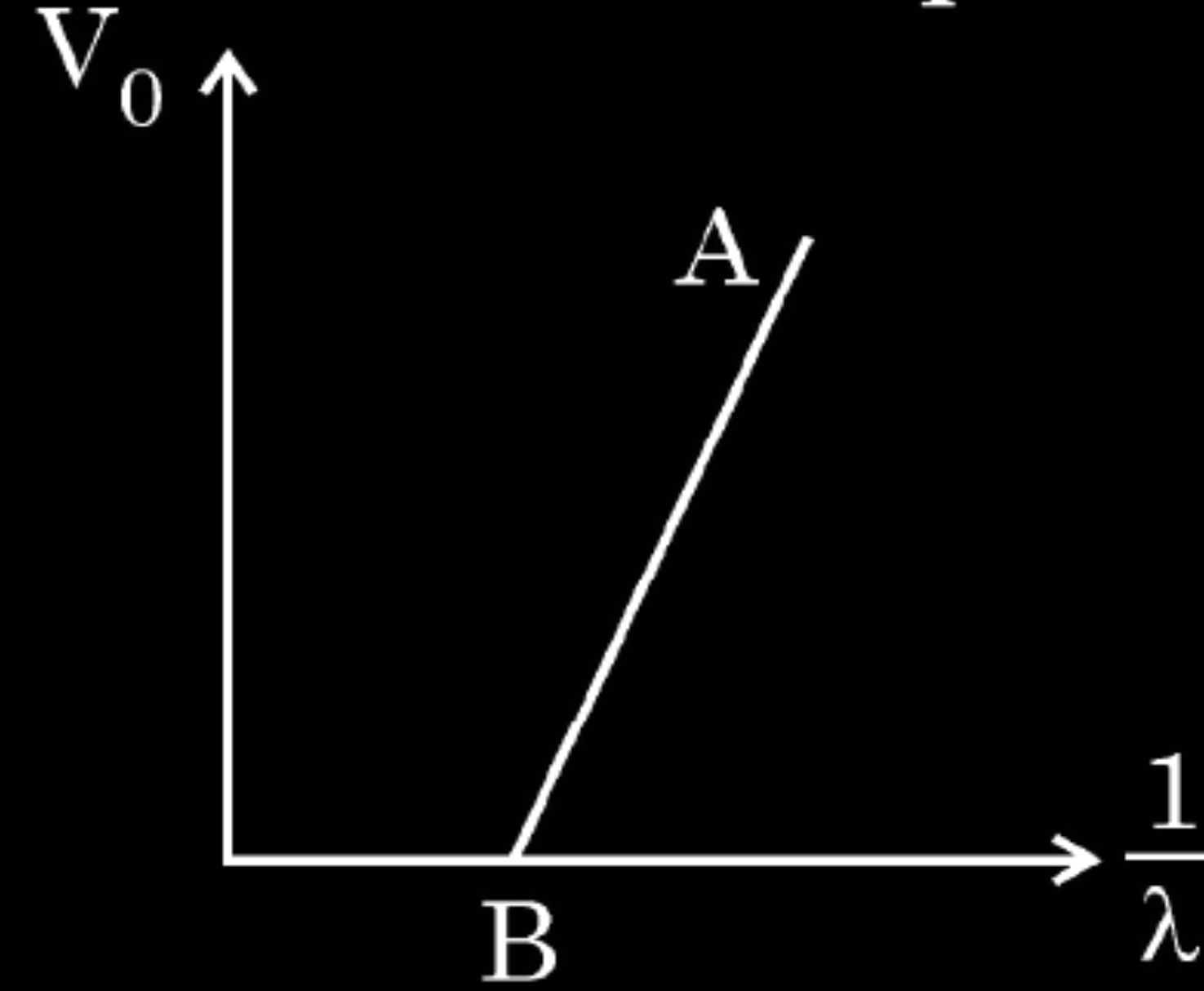
1

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(B) hc / λ

11. Figure shows a plot of stopping potential (V_0) versus $\frac{1}{\lambda}$, where λ is the wavelength of the radiation causing photoelectric emission from a surface. The slope of the line is equal to

1



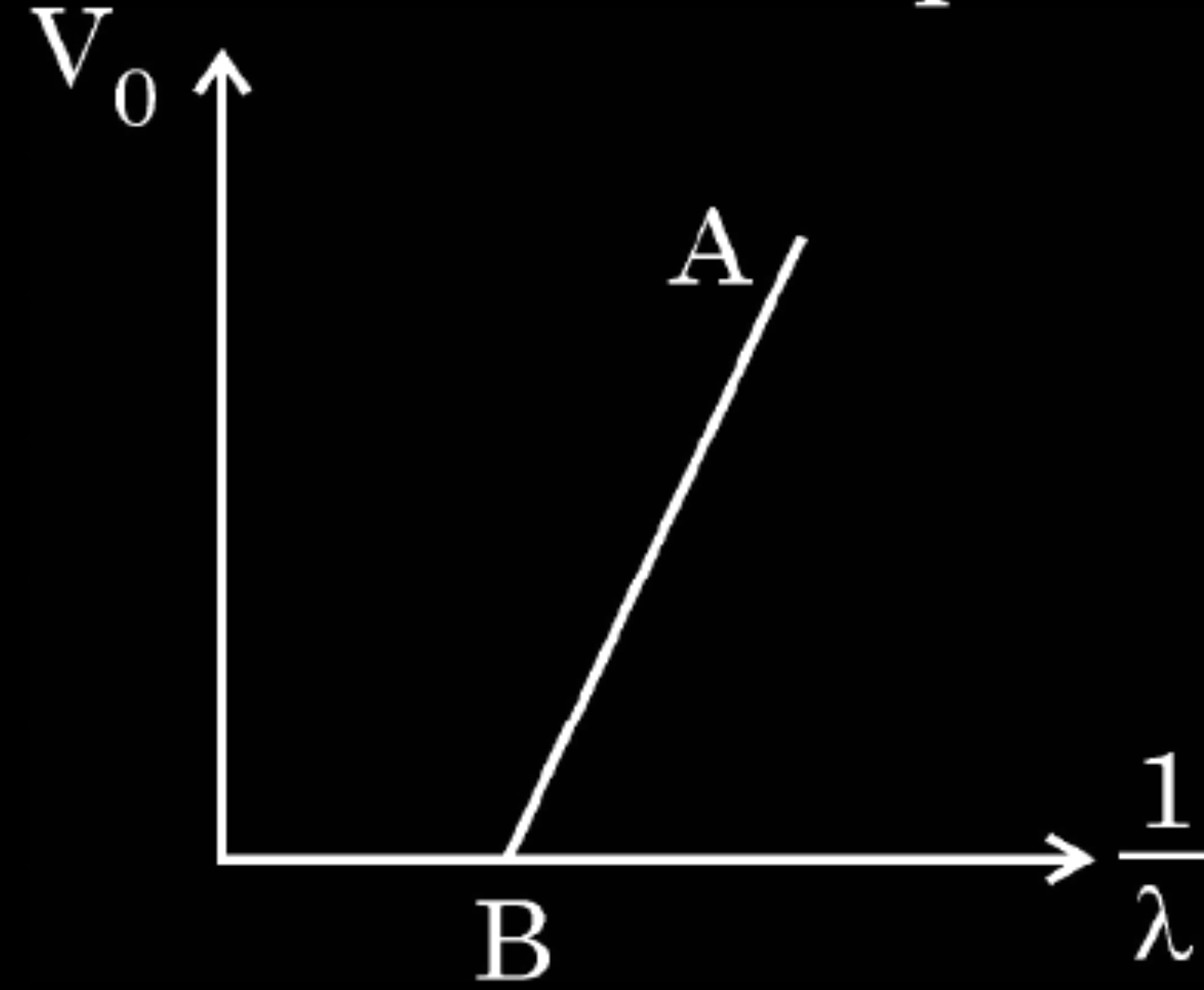
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- (A) ϕ_0 (B) $\frac{h}{e}$ (C) $\frac{hc}{e}$ (D) $\frac{h^2c}{e^2}$

11. Figure shows a plot of stopping potential (V_0) versus $\frac{1}{\lambda}$, where λ is the wavelength of the radiation causing photoelectric emission from a surface. The slope of the line is equal to

1

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(A) ϕ_0

(B) $\frac{h}{e}$

(C) $\frac{hc}{e}$

(D) $\frac{h^2c}{e^2}$

(C) $\frac{hc}{e}$

- (b) How is the photoelectric current affected with
- (i) the increase of the intensity of the incident radiation, keeping its frequency same ?
 - (ii) the decrease of the frequency of the incident radiation, keeping its intensity same ?

Justify your answer.

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- (i) the increase of the intensity of the incident radiation, keeping its frequency same ?
 - (ii) the decrease of the frequency of the incident radiation, keeping its intensity same ?

Justify your answer.

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2

(i) Increases :

As the intensity of the incident radiation increases, the number of photons in the beam increases implying the emission of more number of photoelectrons.

$\frac{1}{2} + \frac{1}{2}$

(ii) No Effect :

As the decrease in frequency will effect the energy of the photon and not the number of photoelectrons.

$\frac{1}{2} + \frac{1}{2}$

- 9.** (a) Write two differences between matter waves and electromagnetic waves.
- (b) Find the ratio of de Broglie wavelengths associated with deuterium and tritium if both have the same kinetic energies.

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3

(a)

Matter Waves	Electromagnetic Waves
(1) Matter waves are associated with every moving mass.	(1) EM waves can only be produced by accelerating charged particles.
(2) Matter waves associated with moving particle cannot achieve a speed of 3×10^8 m/s.	(2) Speed of EM wave in free space is 3×10^8 m/s.

(Any other two difference)

(a) $\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2m \times E}}$ where E is the kinetic energy of the particle

$$\frac{\lambda_D}{\lambda_T} = \frac{h}{\sqrt{2m_D E}} \times \frac{\sqrt{2m_T E}}{h}$$

$$\frac{\lambda_D}{\lambda_T} = \sqrt{\frac{3}{2}}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

5. (a) A proton and an α particle are moving with the same speed. Which one of them has greater value of de Broglie wavelength associated with it ? Justify your answer. 1
- (b) Lights of wavelengths 430 nm, 450 nm and 660 nm are incident on a metal surface whose threshold wavelength is 600 nm. In which case/cases will photoemission take place and why ? Calculate the threshold frequency of the metal surface. 2

CBSE 2022 C

5. (a) A proton and an α particle are moving with the same speed. Which one of them has greater value of de Broglie wavelength associated with it ? Justify your answer. 1

(b) Lights of wavelengths 430 nm, 450 nm and 660 nm are incident on a metal surface whose threshold wavelength is 600 nm. In which case/cases will photoemission take place and why ? Calculate the threshold frequency of the metal surface. 2

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a) de- Broglie wavelength , $\lambda = \frac{h}{mv}$

$$\frac{\lambda_{\alpha}}{\lambda_p} = \frac{m_p}{m_{\alpha}}$$

$\frac{1}{2}$

As, $m_{\alpha} > m_p \Rightarrow \lambda_p > \lambda_{\alpha}$

$\frac{1}{2}$

b) For photoelectric emission wavelength of radiation must be lesser than the threshold wavelength.

$\frac{1}{2}$

Thus lights of wavelength 430 nm and 450 nm can cause photoelectric emission

$\frac{1}{2}$

Threshold frequency (ν_o)

$\frac{1}{2} + \frac{1}{2}$

$$\nu_o = \frac{c}{\lambda_o} = \frac{3 \times 10^8}{600 \times 10^{-9}} = 5 \times 10^{14} \text{ Hz}$$

- (b) Why it is the frequency and not the intensity of light source that determines whether emission of photoelectrons will occur or not ?

Explain

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- (b) Why it is the frequency and not the intensity of light source that determines whether emission of photoelectrons will occur or not ?

2

Explain

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According to the photon picture of light the emission of photoelectrons depend on the energy of photon incident on the metal surface which is determined by the frequency not by the intensity.

2

7. (a) Give an example each of a metal from which photoelectric emission takes place when irradiated by (i) UV light (ii) visible light.
- (b) The work function of a metal is 4.50 eV. Find the frequency of light to be used to eject electrons from the metal surface with a maximum kinetic energy of $6.06 \times 10^{-19} \text{ J}$.

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(a)

(i) UV light --- Zinc, Magnesium, Cadmium

$\frac{1}{2}$

(ii) Visible light --- Alkali metals like Sodium, Potassium, Lithium

$\frac{1}{2}$

(Any one example of each)

(b) From Einstein for equation of photo electric effect

$$h\nu = W_o + KE_{\max}$$

$\frac{1}{2}$

$$= 4.5 \times 1.6 \times 10^{-19} + 6.06 \times 10^{-19}$$

$\frac{1}{2}$

$$= (7.2 + 6.06) \times 10^{-19} \text{ J}$$

$$= 13.26 \times 10^{-19} \text{ J}$$

$\frac{1}{2}$

$$\nu = \frac{13.26 \times 10^{-19}}{6.6 \times 10^{-34}}$$

$$= 2 \times 10^{15} \text{ Hz}$$

$\frac{1}{2}$

5. An alpha particle is accelerated through a potential difference of 100 V. Calculate : 3

- (i) The speed acquired by the alpha particle, and
- (ii) The de-Broglie wavelength associated with it.
(Take mass of alpha particle = 6.4×10^{-27} kg)

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5. An alpha particle is accelerated through a potential difference of 100 V. Calculate : 3

- (i) The speed acquired by the alpha particle, and
- (ii) The de-Broglie wavelength associated with it.
(Take mass of alpha particle = 6.4×10^{-27} kg)

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$$\begin{aligned} \text{i) } v &= \sqrt{\frac{2qV}{m}} \\ &= \sqrt{\frac{2 \times 2 \times 1.6 \times 10^{-19} \times 100}{6.4 \times 10^{-27}}} \\ &= 1 \times 10^5 \text{ m/s} \end{aligned}$$

$$\text{ii) } \lambda = \frac{h}{p} = \frac{h}{mv}$$

$$= \frac{6.63 \times 10^{-34}}{6.4 \times 10^{-27} \times 10^5}$$

$$\lambda = 10^{-12} \text{ m}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

9. (a) Calculate the energy and momentum of a photon in a monochromatic beam of wavelength 331.5 nm.
- (b) How fast should a hydrogen atom travel in order to have the same momentum as that of the photon in part (a) ?

3

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(a) Energy of photon, $E = h\nu$

$$E = h \frac{c}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{331.5 \times 10^{-9}}$$

$\frac{1}{2}$

$E = 6 \times 10^{-19} \text{ J}$
momentum(p)

$\frac{1}{2}$

$$p = \frac{h}{\lambda}$$

$\frac{1}{2}$

$$= \frac{6.63 \times 10^{-34}}{331.5 \times 10^{-9}}$$

$\frac{1}{2}$

$$p = 2 \times 10^{-27} \text{ kg ms}^{-1}$$

(b) Momentum of H atom = momentum of the photon = $2 \times 10^{-27} \text{ kg ms}^{-1}$

$\frac{1}{2}$

$$p = mu$$

$$u = \frac{p}{m} = \frac{2 \times 10^{-27}}{1.67 \times 10^{-27}} \text{ ms}^{-1}$$

$$u = 1.20 \text{ ms}^{-1}$$

$\frac{1}{2}$

5. The work function of a metal is 2.31 eV. Photoelectric emission occurs when light of frequency 6.4×10^{14} Hz is incident on the metal surface. Calculate :
(i) the energy of the incident radiation, (ii) the maximum kinetic energy of the emitted electron and (iii) the stopping potential of the surface. **3**

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 (i) the energy of the incident radiation, (ii) the maximum kinetic energy of the emitted electron and (iii) the stopping potential of the surface. 3

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i) $E = h \nu$ $= 6.63 \times 10^{-34} \times 6.4 \times 10^{14}$ $= 4.24 \times 10^{-19} \text{ J}$ $= 2.65 \text{ eV}$	$\frac{1}{2}$ $\frac{1}{2}$
ii) $K_{\text{max}} = E - \phi_0$ $= 2.65 - 2.31$ $= 0.34 \text{ eV}$	$\frac{1}{2}$ $\frac{1}{2}$
iii) $eV_0 = K_{\text{max}}$ $V_0 = 0.34 \text{ V}$	$\frac{1}{2}$ $\frac{1}{2}$

4. (a) Calculate the frequency of a photon of energy $6.5 \times 10^{-19} \text{ J}$.
- (b) Can this photon cause emission of an electron from the surface of Cs of work function 2.14 eV ? If yes, what will be maximum kinetic energy of the photoelectron ?

3

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4. (a) Calculate the frequency of a photon of energy $6.5 \times 10^{-19} \text{ J}$.
 (b) Can this photon cause emission of an electron from the surface of Cs of work function 2.14 eV ? If yes, what will be maximum kinetic energy of the photoelectron ?

3

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$$E = h\nu$$

$$\nu = \frac{E}{h} = \frac{6.5 \times 10^{-19}}{6.63 \times 10^{-34}}$$

$$= 9.8 \times 10^{14} \text{ Hz}$$

b) Energy of photon in eV

$$E = 6.5 \times 10^{-19} \text{ J} = 4.06 \text{ eV}$$

$\therefore E > \phi_0$ There will be photoelectric emission.

K.E of photoelectron $E_k = E - \phi_0$

$$= 4.06 - 2.14$$

$$= 1.92 \text{ eV}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) (i) Name the factors on which photoelectric emission from a surface depends.
- (ii) Define the term ‘threshold frequency’ for a photosensitive material.

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(i) Factors	
(a) Frequency of incident radiation	$\frac{1}{2}$
(b) Work function of the surface	$\frac{1}{2}$
(ii) The minimum frequency of the incident radiation below which photoelectric emission does not take place.	1

9. An electron is accelerated from rest through a potential difference of 100 V.
Find :
- (i) the wavelength associated with
 - (ii) the momentum of and
 - (iii) the velocity required by, the electron.

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Find :

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- (i) the wavelength associated with
(ii) the momentum of and
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3

(i)	$\lambda = \frac{1.23}{\sqrt{V}} \text{ nm} = \frac{1.23}{\sqrt{100}}$ $= 0.123 \text{ nm}$	$\frac{1}{2}$ $\frac{1}{2}$
(ii)	$p = \frac{h}{\lambda}$ $= \frac{6.63 \times 10^{-34}}{0.123 \times 10^{-9}}$ $= 5.4 \times 10^{-24} \text{ kg m s}^{-1}$	$\frac{1}{2}$ $\frac{1}{2}$
(iii)	$v = \frac{p}{m}$ $v = \frac{5.4 \times 10^{-24}}{9.1 \times 10^{-31}}$ $= 5.9 \times 10^6 \text{ m/s}$	Alternative method $v = \sqrt{\frac{2eV}{m}}$ $v = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 100}{9.1 \times 10^{-31}}}$ $= 5.9 \times 10^6 \text{ m/s}$ $\frac{1}{2}$ $\frac{1}{2}$

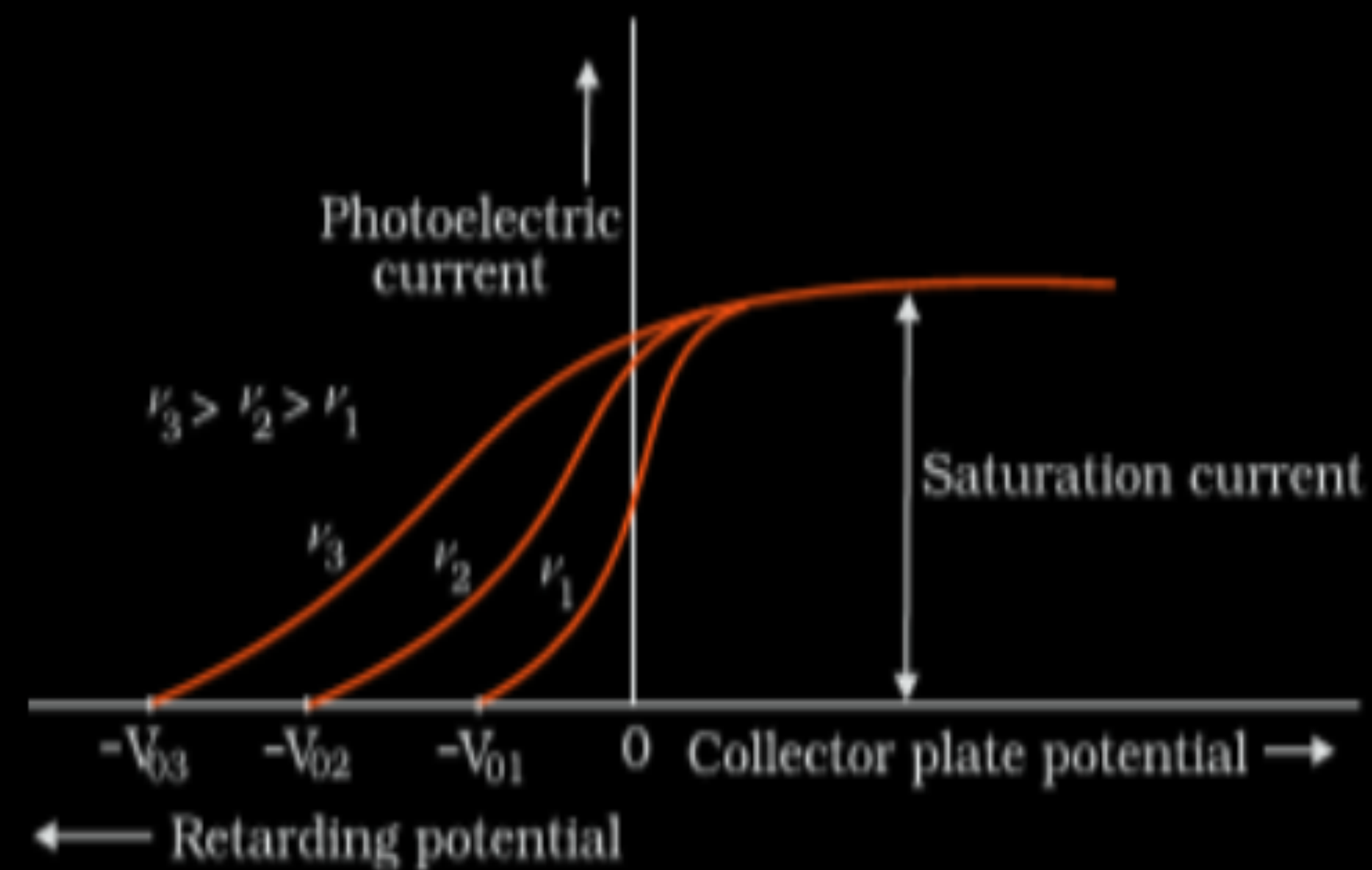
- (b) Plot suitable graphs to show the variation of photoelectric current with the collector plate potential for the incident radiation of
- (i) the same intensity but different frequencies ν_1 , ν_2 and ν_3 ($\nu_1 < \nu_2 < \nu_3$)
 - (ii) the same frequency but different intensities I_1 , I_2 and I_3 ($I_1 < I_2 < I_3$)

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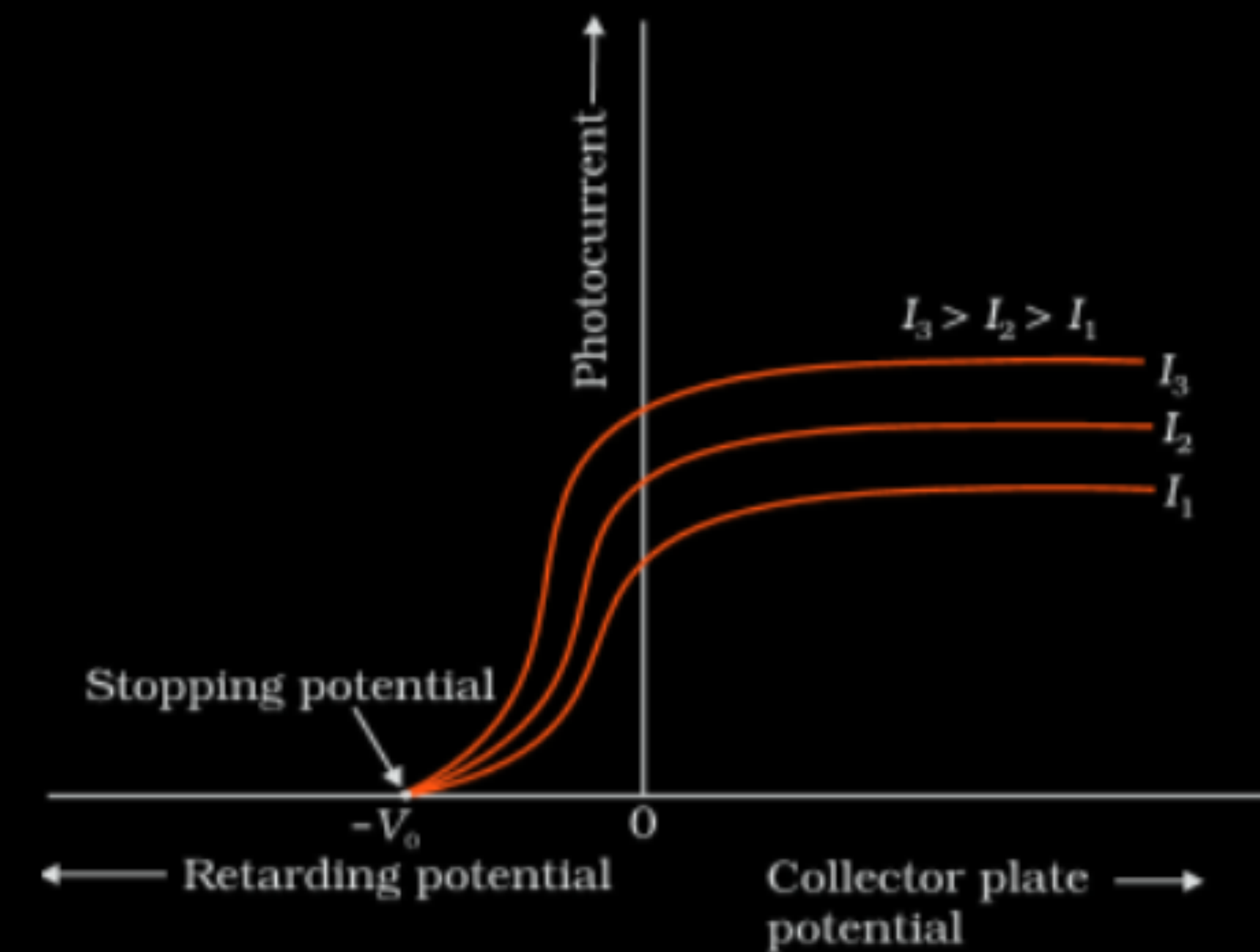
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 - (ii) the same frequency but different intensities I_1 , I_2 and I_3 ($I_1 < I_2 < I_3$)

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i)



ii)



1+1

7. The de-Broglie wavelength associated with an electron is 0.30 nm.
Calculate : (a) the speed, and (b) the kinetic energy (in eV), of the electron. **3**

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7. The de-Broglie wavelength associated with an electron is 0.30 nm. Calculate : (a) the speed, and (b) the kinetic energy (in eV), of the electron. 3

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$$\text{a) } \lambda = \frac{h}{mv}$$

1/2

$$v = \frac{h}{m\lambda} = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 0.30 \times 10^{-9}}$$

1/2

$$= 2.41 \times 10^6 \text{ m/s}$$

1/2

$$\text{b) K.E.} = \frac{1}{2} mv^2$$

1/2

$$= \frac{1}{2} \times 9.1 \times 10^{-31} \times (2.41 \times 10^6)^2 \text{ J}$$

$$= 26.42 \times 10^{-19} \text{ J}$$

1/2

K.E in eV

$$= \frac{26.42 \times 10^{-19}}{1.6 \times 10^{-19}} = 16.5 \text{ eV}$$

1/2

5. Ultra-violet light of wavelength 200 nm from a source is incident on a metal surface. If the stopping potential is -2.5 V, (a) calculate the work function of the metal, and (b) How would the surface respond to a high intensity red light of wavelength $6328 \text{ \AA}$ produced by a laser ?

3

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5. Ultra-violet light of wavelength 200 nm from a source is incident on a metal surface. If the stopping potential is -2.5 V , (a) calculate the work function of the metal, and (b) How would the surface respond to a high intensity red light of wavelength $6328\text{ \AA}$ produced by a laser ?

3

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a) $\lambda = 200\text{ nm}$, stopping potential = -2.5 V

$$\text{K.E.} = 2.5 \times 1.6 \times 10^{-19}\text{ J} = 4 \times 10^{-19}\text{ J}$$

$$E = h\nu = h \frac{c}{\lambda}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{200 \times 10^{-9}}$$

$$= 9.945 \times 10^{-19}\text{ J}$$

Work function = $E - eV$

$$= (9.945 \times 10^{-19} - 4 \times 10^{-19})\text{ J}$$

$$= 5.595 \times 10^{-19}\text{ J}$$

b) Wavelength of red light = $6328\text{ \AA}$

$$E = h\nu = h \frac{c}{\lambda}$$

$$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{6328 \times 10^{-10}}$$

$$= 0.00314 \times 10^{-16}\text{ J}$$

$$= 3.14 \times 10^{-19}\text{ J}$$

Energy of red light < work function of metal surface

$\therefore$ No emission of photoelectrons takes place.

Alternatively (b)

$$\text{work function} = h\nu_0 = h \frac{c}{\lambda'}$$

$$\lambda' = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{5.595 \times 10^{-19}}$$

$$= 3.55 \times 10^{-7}\text{ J}$$

$$= 3550\text{ \AA}$$

Wavelength required for emission < wavelength of red light

$\therefore$ no emission of photoelectron

9. Photoelectrons are emitted from a metal surface when illuminated with UV light of wavelength 330 nm. The minimum amount of energy required to emit the electrons from the surface is 3.5×10^{-19} J. Calculate : **3**
- (i) the energy of the incident radiation, and
 - (ii) the kinetic energy of the photoelectron.

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9. Photoelectrons are emitted from a metal surface when illuminated with UV light of wavelength 330 nm. The minimum amount of energy required to emit the electrons from the surface is 3.5×10^{-19} J. Calculate :

3

- (i) the energy of the incident radiation, and
(ii) the kinetic energy of the photoelectron.

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i) Energy of incident radiation

$$\begin{aligned} E &= h\nu = h \frac{c}{\lambda} \\ &= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{330 \times 10^{-9}} \\ &= 6.027 \times 10^{-19} \text{ J} \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

ii) Kinetic energy of photoelectron

$$\begin{aligned} \text{K.E.} &= E - \phi_0 \\ &= (6.027 \times 10^{-19} - 3.5 \times 10^{-19}) \text{ J} \\ &= 2.527 \times 10^{-19} \text{ J} \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(b) Explain briefly any two features of photoelectric effect. 2

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(b) Explain briefly any two features of photoelectric effect.

2

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The photoelectric current is directly proportional to the intensity of radiation incident on the photoelectric surface/for the radiation of particular frequency incident on the surfaces, photo electrons of different kinetic energies are emitted simultaneously/ for a photosensitive surface there is a minimum frequency of the incident radiation below which the photoemission is not possible/ there is no time lag between the incidence of radiations and the emission of the electron from the surface.

(Any two)

1+1

9. A beam of light of wavelength 500 nm is incident on a photosensitive surface. If the threshold wavelength for the surface is 663 nm, then calculate :

(a) the work function for the surface, and

(b) the maximum kinetic energy (in eV) of the photoelectrons.

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3

9. A beam of light of wavelength 500 nm is incident on a photosensitive surface. If the threshold wavelength for the surface is 663 nm, then calculate :

(a) the work function for the surface, and

(b) the maximum kinetic energy (in eV) of the photoelectrons.

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3

(a)	$E = h\nu = h \frac{c}{\lambda}$	$\frac{1}{2}$
	$= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{500 \times 10^{-9}} \text{ J}$	$\frac{1}{2}$
	$= 3.99 \times 10^{-19} \text{ J} = 2.49 \text{ eV}$	$\frac{1}{2}$
(b)	$\therefore E_k = h\nu - W_0 = h\nu - \frac{hc}{\lambda_0}$	$\frac{1}{2}$
	$= 2.49 \text{ eV} - \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{663 \times 10^{-9} \times 1.6 \times 10^{-19}} \text{ eV}$	$\frac{1}{2}$
	$= 2.49 \text{ eV} - 1.89 = 0.6 \text{ eV}$	$\frac{1}{2}$

- (b) What are matter waves ? A proton and an alpha particle are accelerated through the same potential difference. Find the ratio of the de Broglie wavelength associated with the proton to that with the alpha particle.

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2

- (b) What are matter waves ? A proton and an alpha particle are accelerated through the same potential difference. Find the ratio of the de Broglie wavelength associated with the proton to that with the alpha particle.

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2

Wave associated with moving material particle

$$\lambda = \frac{h}{\sqrt{2qmV}}$$

$$\therefore \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{q_\alpha m_\alpha}{q_p m_p}}$$

$$\therefore \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{2e \cdot 4m}{e \cdot m}}$$

$$= \frac{2\sqrt{2}}{1}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

8. A proton is accelerated through a potential difference V . After acceleration, the de Broglie wavelength associated with it is λ . If the proton is replaced by an alpha particle, then find the de Broglie wavelength associated with it if it were accelerated through the same potential difference V . What will be the momentum of the alpha particle ?

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8. A proton is accelerated through a potential difference V . After acceleration, the de Broglie wavelength associated with it is λ . If the proton is replaced by an alpha particle, then find the de Broglie wavelength associated with it if it were accelerated through the same potential difference V . What will be the momentum of the alpha particle ?

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3

$$\lambda_p = \frac{h}{\sqrt{2q_p m_p V}},$$

$$\lambda_\alpha = \frac{h}{\sqrt{2 \cdot 2q_p \cdot 4m_p V}}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \frac{1}{2\sqrt{2}}$$

$$\lambda_\alpha = \frac{h}{p_p}, \quad \lambda_\alpha = \frac{h}{p_\alpha}$$

$$\frac{p_\alpha}{p_p} = \frac{1}{2\sqrt{2}},$$

$$p_\alpha = 2\sqrt{2} p_p$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

8. The wavelength of the waves associated with a particle having kinetic energy E is λ . How and by what factor should its kinetic energy be changed so that the wavelength becomes $\left(\frac{\lambda}{2}\right)$? Also, find the ratio of the final to the initial velocity of the particle.

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3

8. The wavelength of the waves associated with a particle having kinetic energy E is λ . How and by what factor should its kinetic energy be changed so that the wavelength becomes $\left(\frac{\lambda}{2}\right)$? Also, find the ratio of the final to the initial velocity of the particle.

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3

$$\lambda = \frac{h}{\sqrt{2mE_1}} ,$$

$$\frac{\lambda}{2} = \frac{h}{\sqrt{2mE_2}}$$

$$E_2 = 4E_1$$

$$\lambda = \frac{h}{mv} ,$$

$$\lambda \propto \frac{1}{v}$$

$$\frac{v_f}{v_i} = \frac{2}{1}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

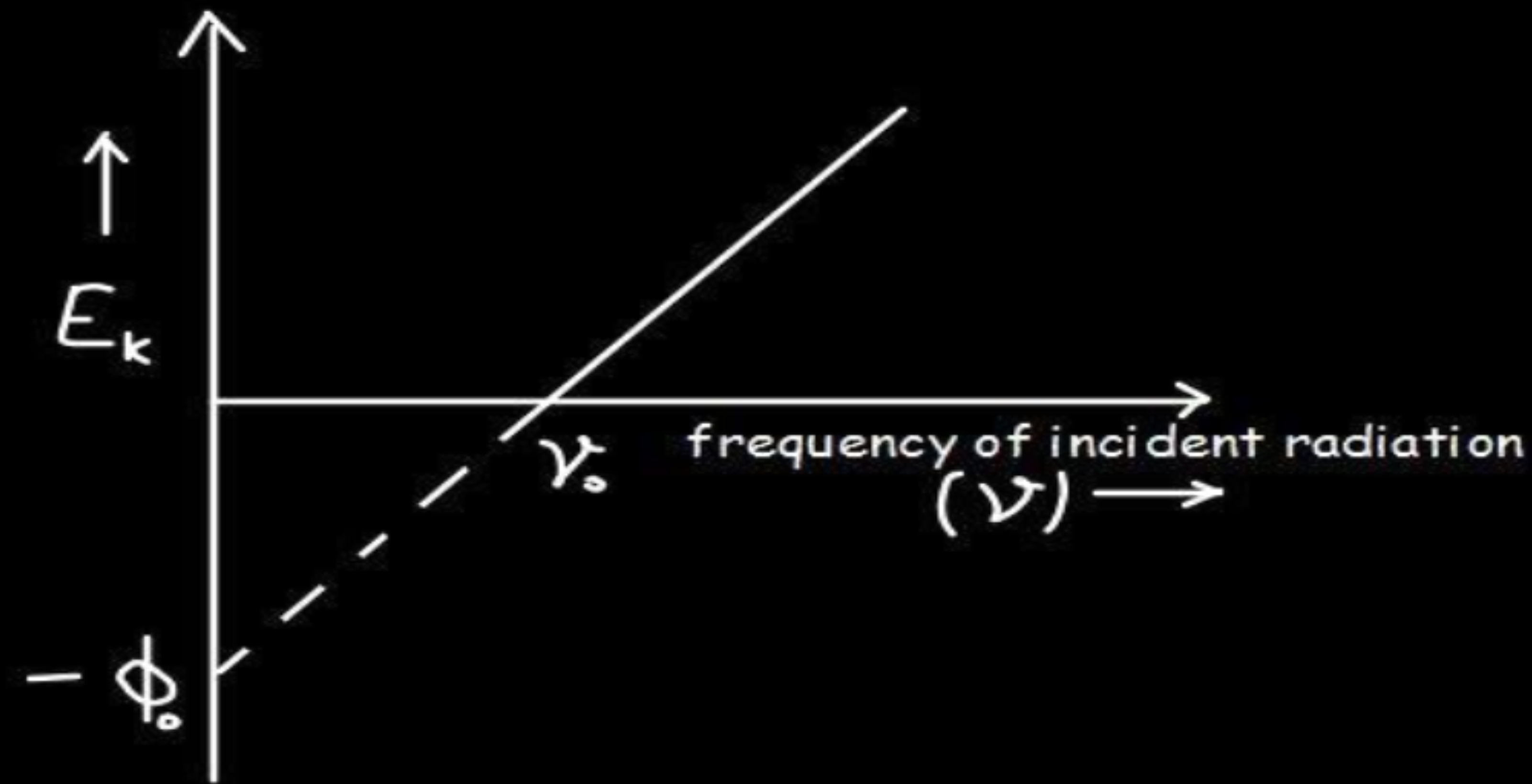
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 9.** (a) Use Einstein's photoelectric equation to depict the variation of the maximum kinetic energy (E_k) of electrons emitted, with the frequency (ν) of the incident radiation.
- (b) A photosensitive surface is illuminated with a beam of (i) yellow light, and (ii) red light, both of the same intensity.
- In which case will
- (I) photoelectrons have more E_k ?
- (II) more numbers of electrons be emitted ?
- Justify your answer in each case.

(a)



(b) (I)

$$E_K = h\nu - \phi_0$$

$$\text{as } \nu_Y > \nu_R$$

$$(E_K)_Y > (E_K)_R$$

For a given photosensitive surface, K.E. of photo-electrons will be more for yellow light.

(II) Since the number of photons incident on unit area per unit time for same intensity of two lights will be more in Red light ($N_R > N_Y$) hence more number of photoelectrons will be ejected for Red light.

1

$\frac{1}{2}$

$\frac{1}{2}$

1

- (b) Photoelectric emission occurs when a surface is irradiated with the radiation of frequency (i) ν_1 , and (ii) ν_2 . The maximum kinetic energy of the electrons emitted in the two cases are K and $2K$ respectively. Obtain the expression for the threshold frequency for the surface. 2

- (b) Photoelectric emission occurs when a surface is irradiated with the radiation of frequency (i) ν_1 , and (ii) ν_2 . The maximum kinetic energy of the electrons emitted in the two cases are K and $2K$ respectively. Obtain the expression for the threshold frequency for the surface. 2

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$$K = h\nu_1 - \phi_0 \text{ and } 2K = h\nu_2 - \phi_0$$

$$\Rightarrow 2(h\nu_1 - \phi_0) = h\nu_2 - \phi_0$$

$$\Rightarrow 2h\nu_1 - 2\phi_0 = h\nu_2 - \phi_0$$

$$\Rightarrow h(2\nu_1 - \nu_2) = \phi_0 = h\nu_0$$

$$\Rightarrow (2\nu_1 - \nu_2) = \phi_0 = \nu_0$$

$$\Rightarrow \nu_0 = 2\nu_1 - \nu_2$$

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

10. Find the ratio of the de Broglie wavelengths associated with an alpha particle and a proton, if both

CBSE 2022

- (a) have the same speeds,
- (b) have the same kinetic energy,
- (c) are accelerated through the same potential difference.

10. Find the ratio of the de Broglie wavelengths associated with an alpha particle and a proton, if both

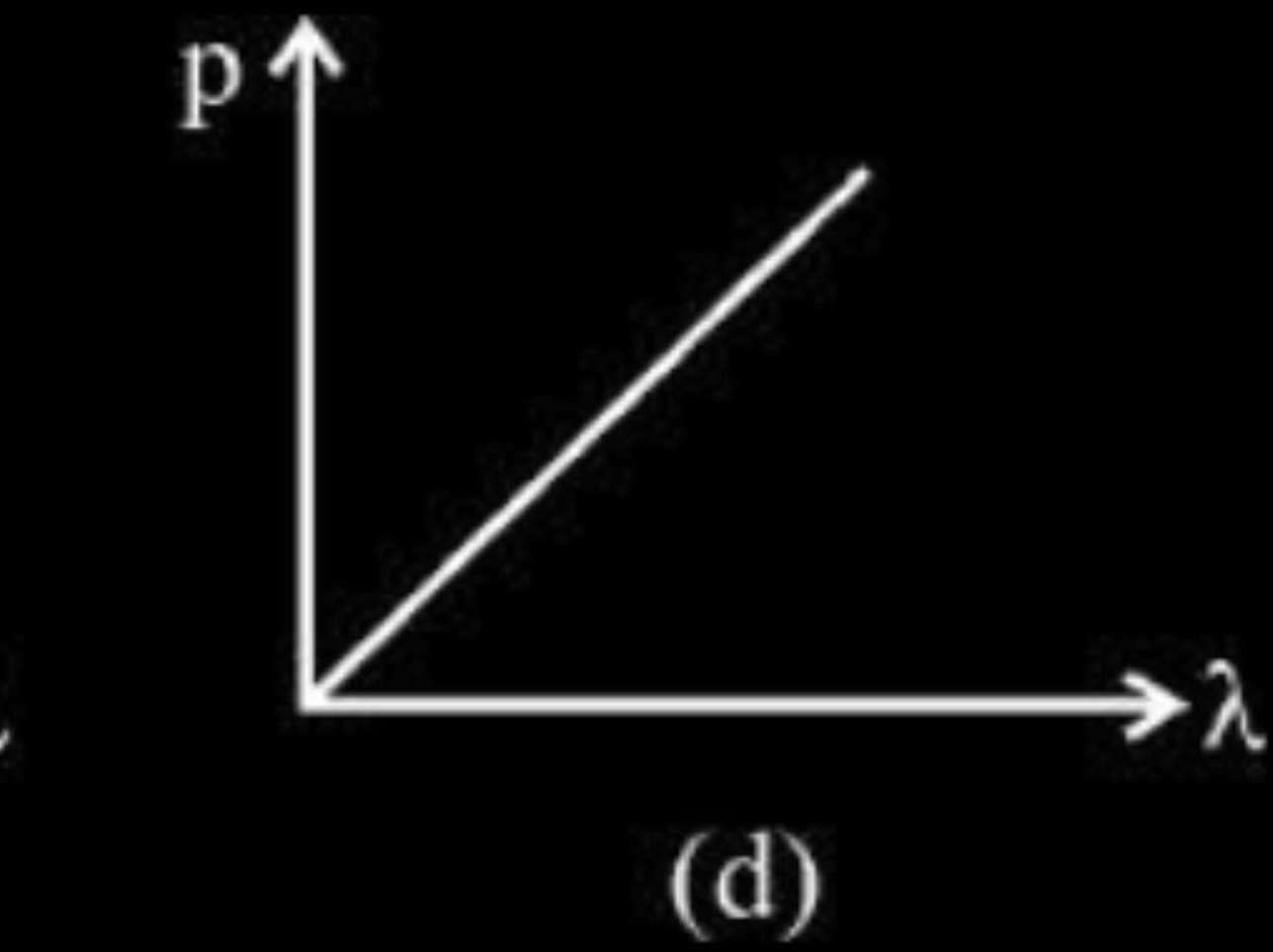
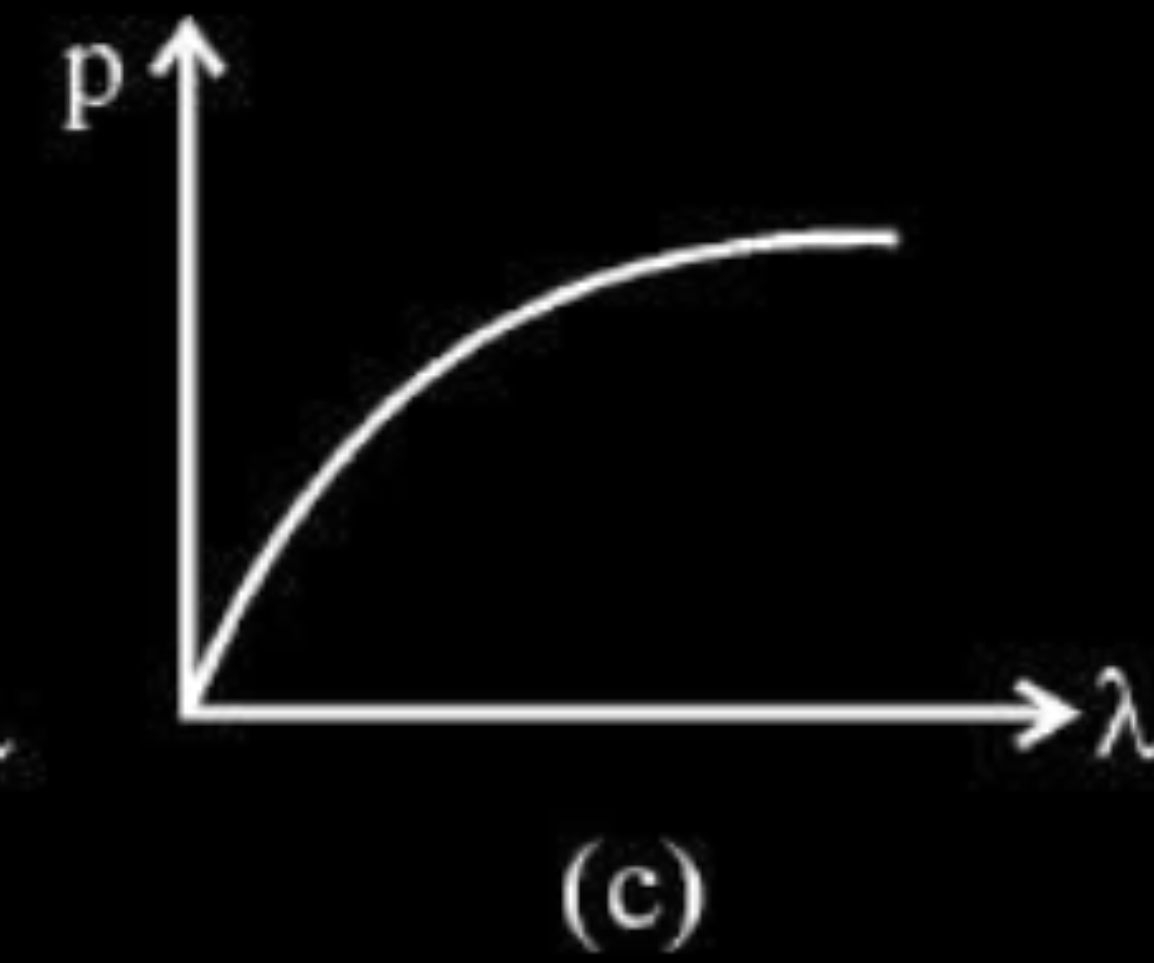
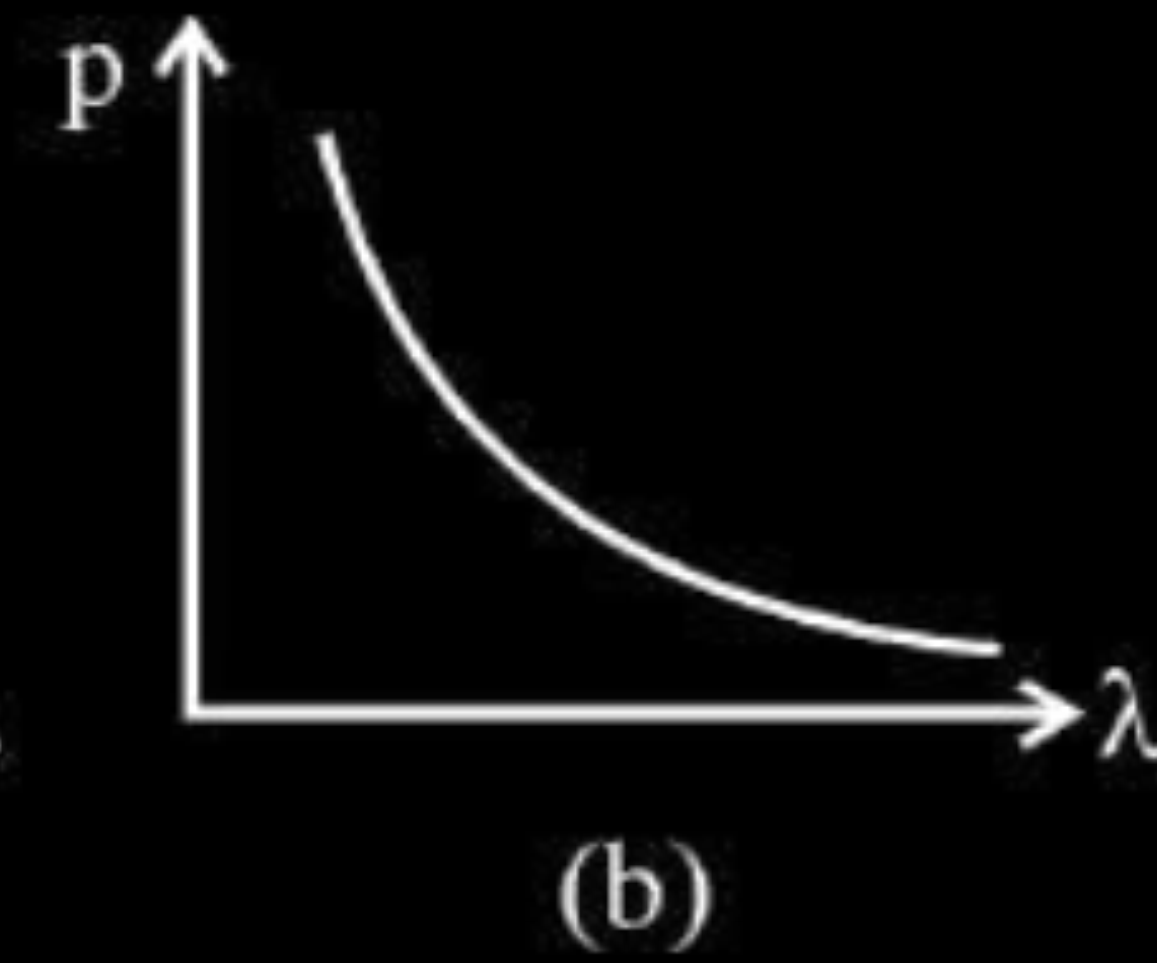
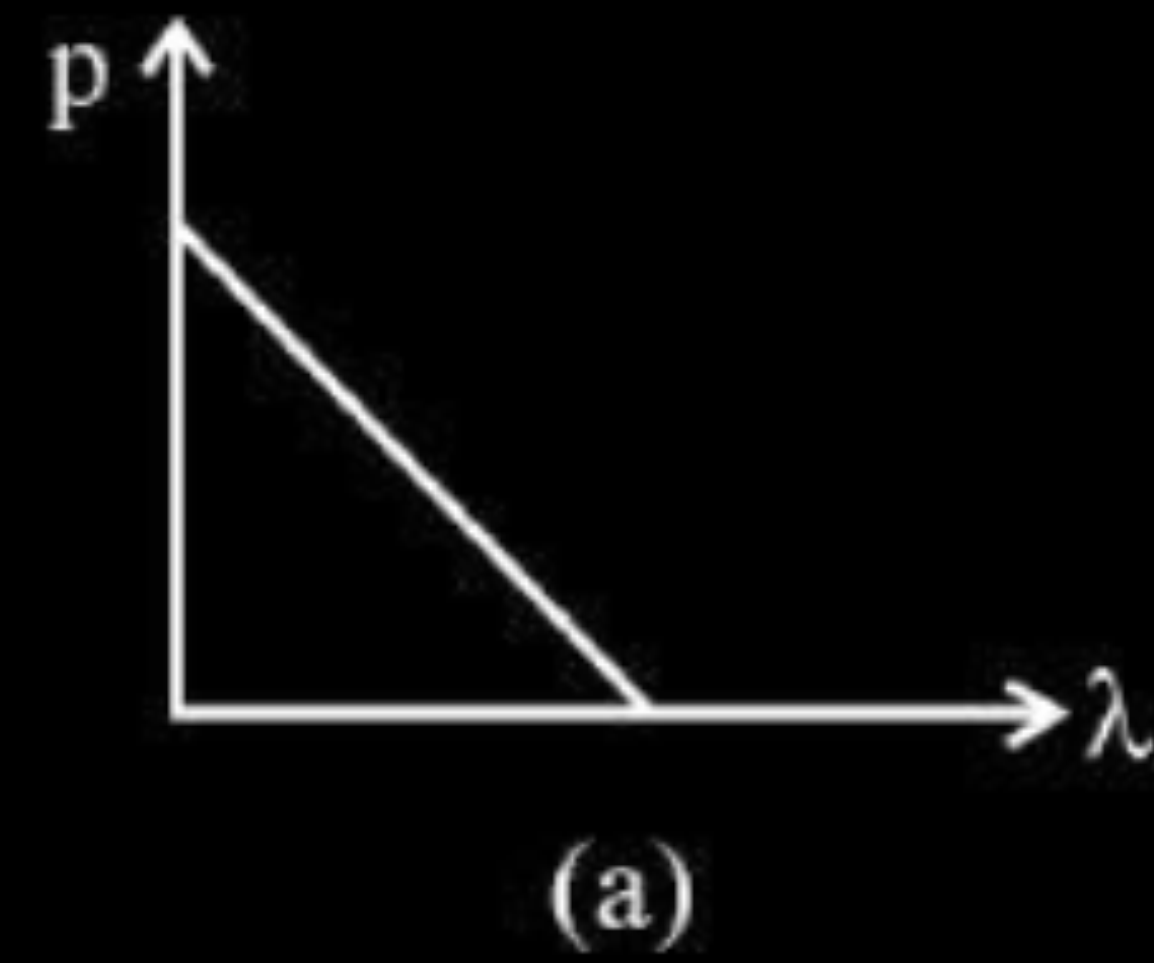
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- (a) have the same speeds,
- (b) have the same kinetic energy,
- (c) are accelerated through the same potential difference.

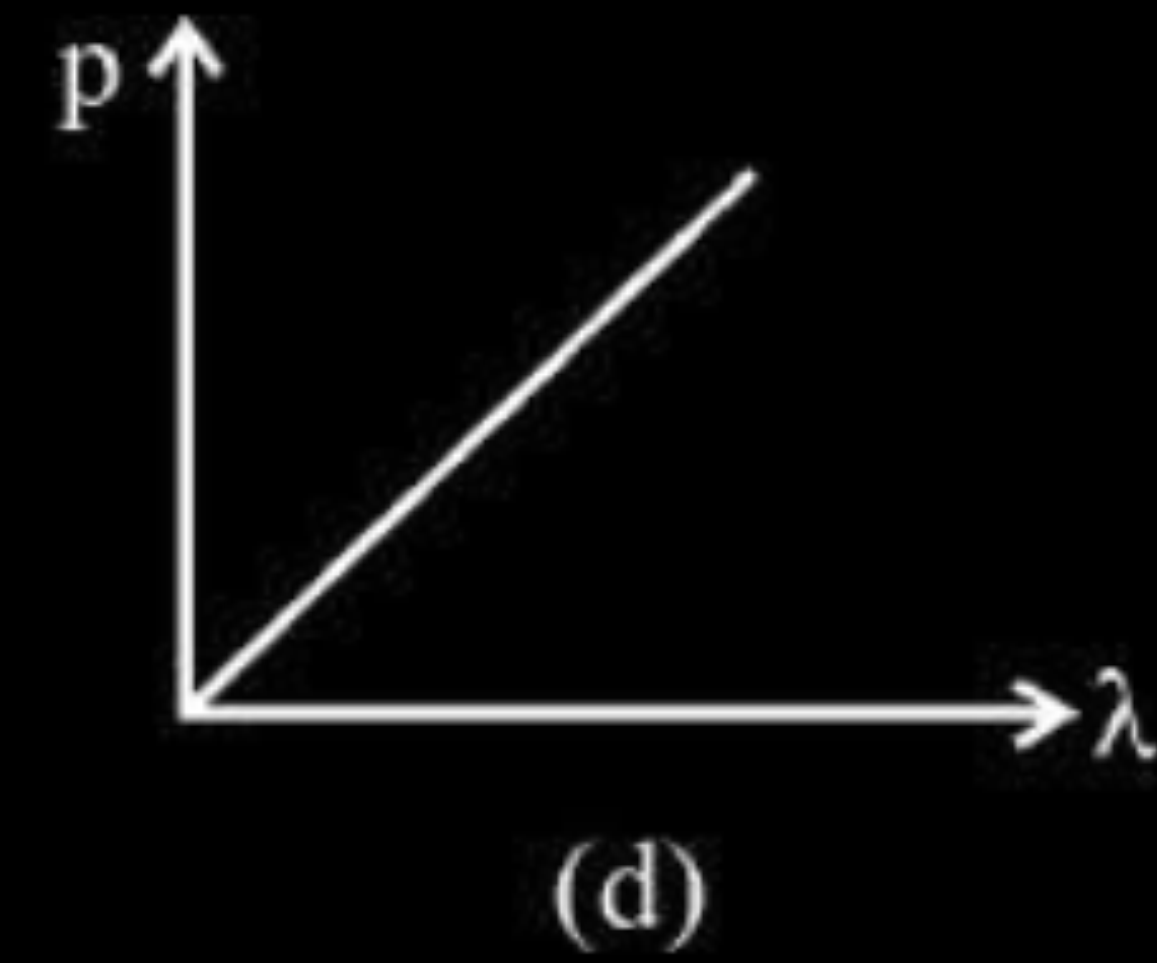
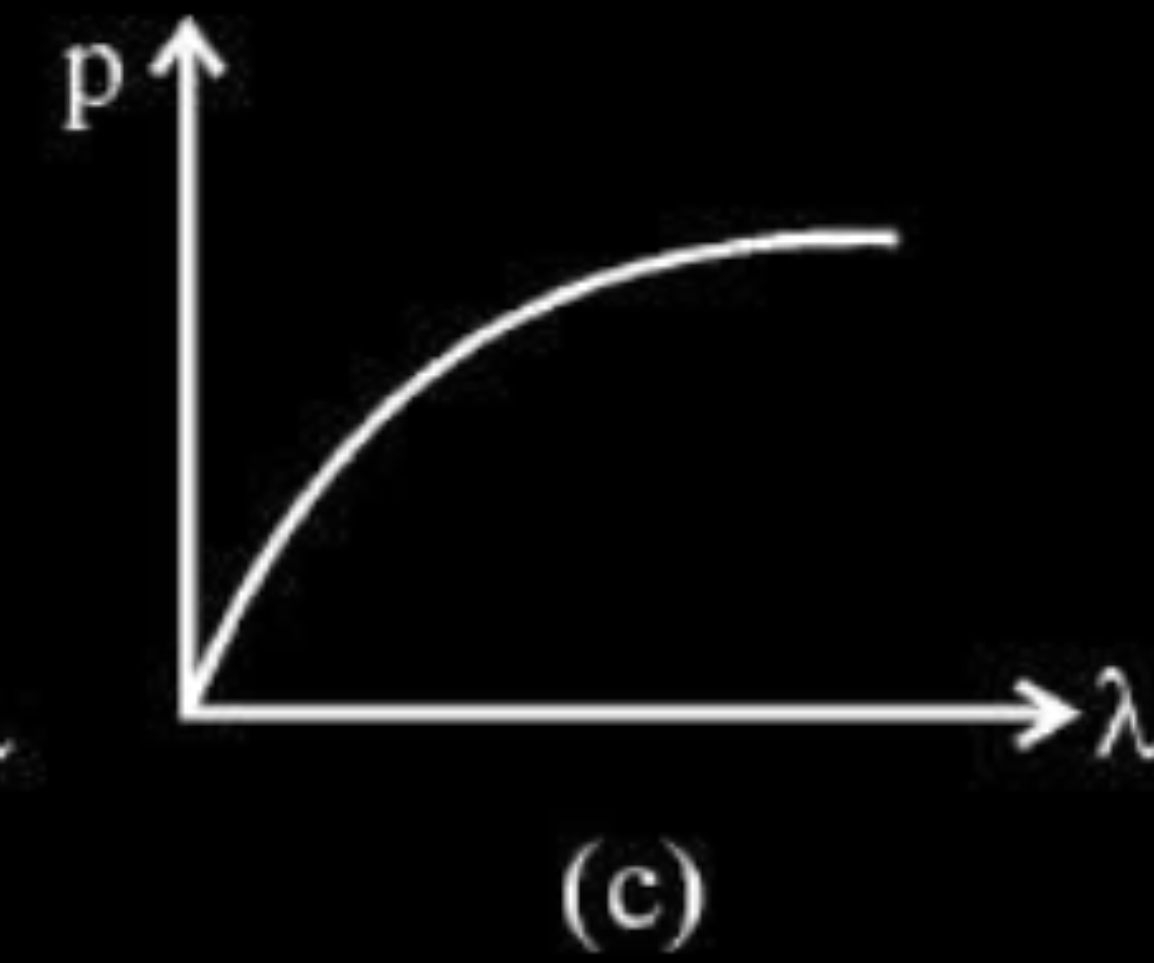
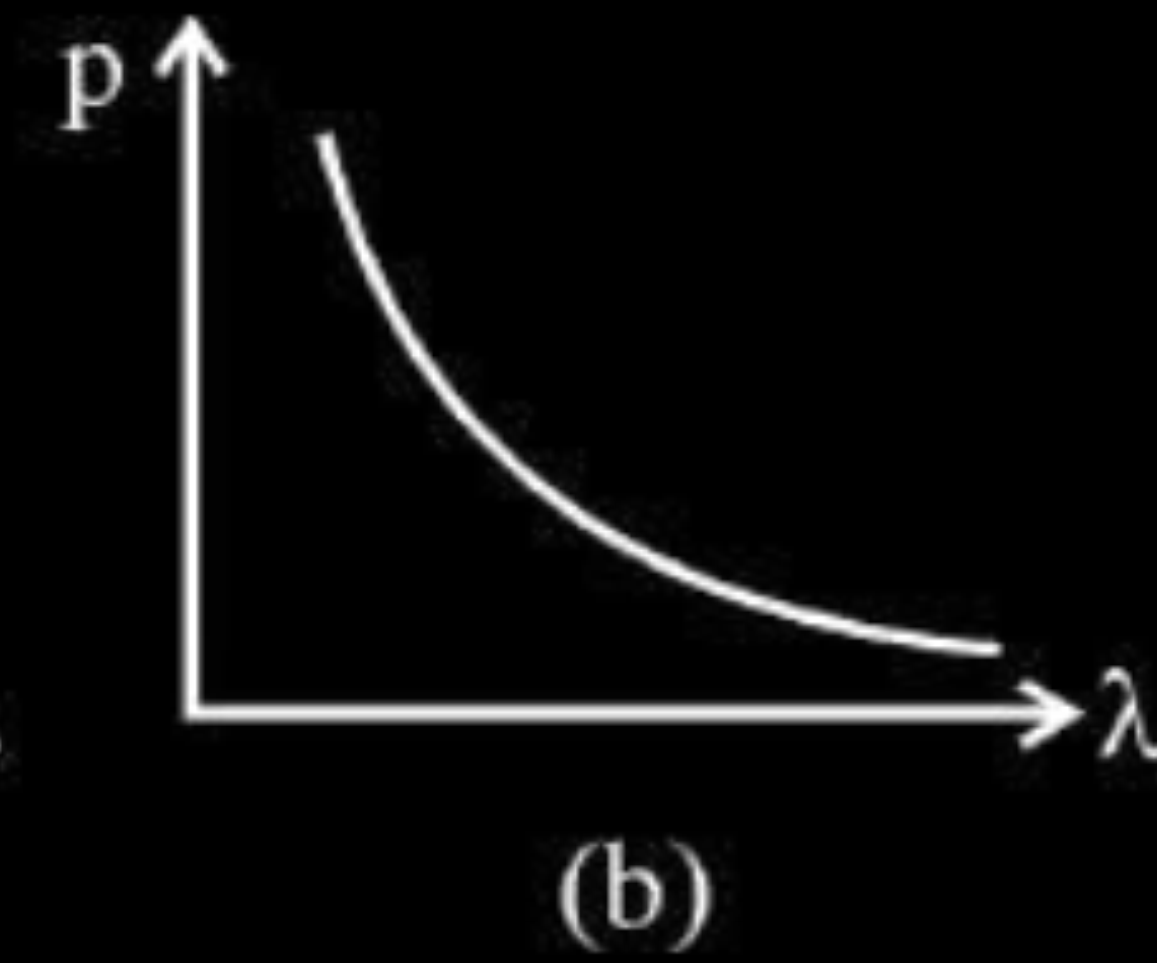
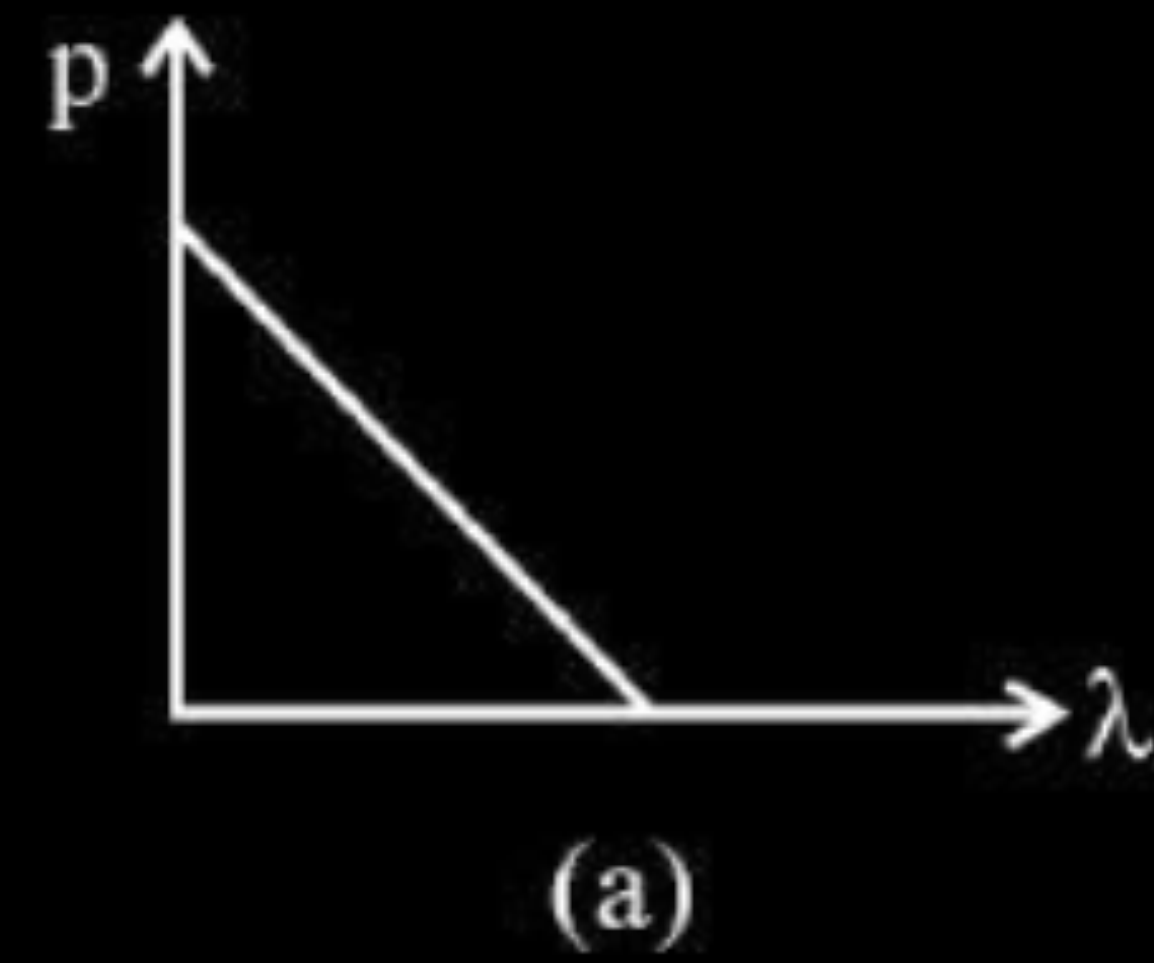
3

(a) $\lambda = \frac{h}{p}$	$\frac{1}{2}$
$\frac{\lambda_{\alpha}}{\lambda_p} = \frac{h}{m_{\alpha} v_{\alpha}} \times \frac{m_p v_p}{h} = \frac{1}{4}$	$\frac{1}{2}$
(b)	$\frac{1}{2}$
$p = \sqrt{2mK.E}$	$\frac{1}{2}$
$\frac{\lambda_{\alpha}}{\lambda_p} = \frac{h}{\sqrt{2m_{\alpha}(K.E.)_{\alpha}}} \times \frac{\sqrt{2m_p(K.E.)_p}}{h} = \sqrt{\frac{m_p}{m_{\alpha}}} = \frac{1}{2}$	$\frac{1}{2}$
(c)	$\frac{1}{2}$
$v = \sqrt{\frac{2qV}{m}}$	$\frac{1}{2}$
$\frac{\lambda_{\alpha}}{\lambda_p} = \frac{h}{m_{\alpha} v_{\alpha}} \times \frac{m_p v_p}{h} = \frac{m_p}{m_{\alpha}} \sqrt{\frac{2q_p V}{m_p}} \times \sqrt{\frac{m_{\alpha}}{2q_{\alpha} V}}$	
$= \frac{m_p}{m_{\alpha}} \times \sqrt{\frac{m_{\alpha}}{m_p}} \times \sqrt{\frac{q_p}{q_{\alpha}}} = \frac{1}{2\sqrt{2}}$	$\frac{1}{2}$

4. The graph showing the correct variation of linear momentum (p) of a charge particle with its de-Broglie wavelength (λ) is –

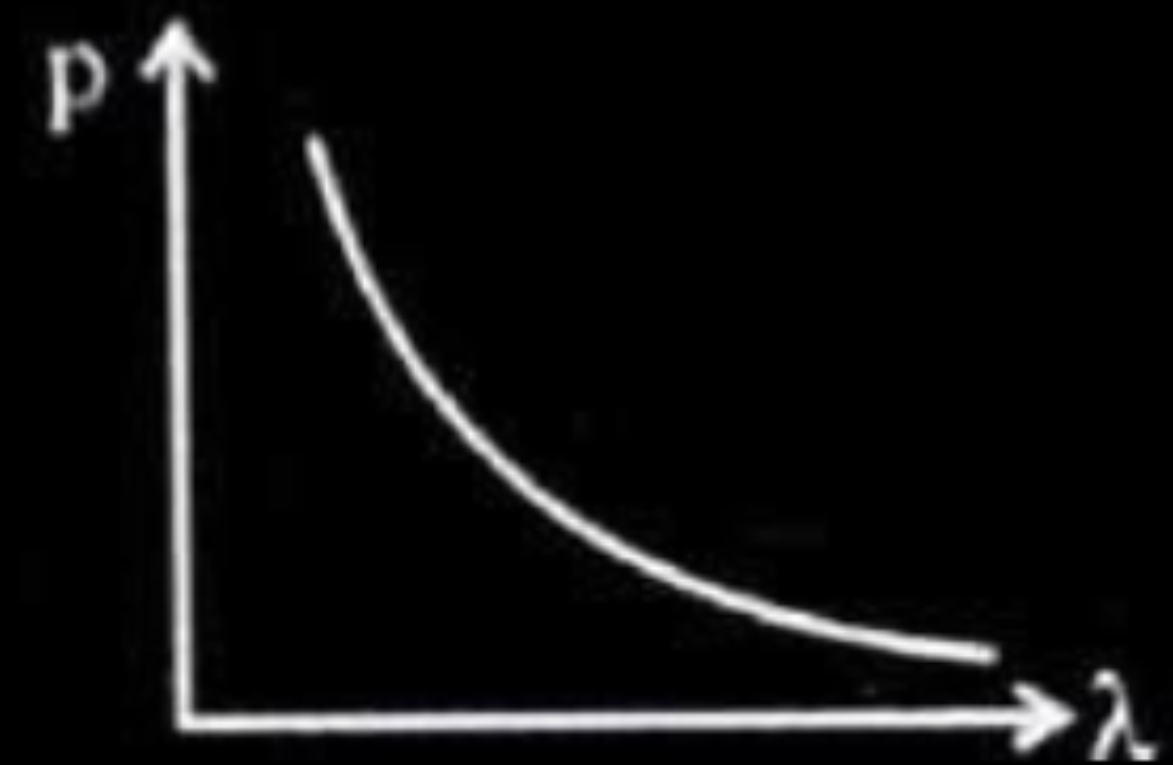


4. The graph showing the correct variation of linear momentum (p) of a charge particle with its de-Broglie wavelength (λ) is –



1

(b)



8. If photons of frequency ν are incident on the surfaces of metals A & B of threshold frequencies $\nu/2$ and $\nu/3$ respectively, the ratio of the maximum kinetic energy of electrons emitted from A to that from B is

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(a) $2 : 3$

(b) $3 : 4$

(c) $1 : 3$

(d) $\sqrt{3} : \sqrt{2}$

8. If photons of frequency ν are incident on the surfaces of metals A & B of threshold frequencies $\nu/2$ and $\nu/3$ respectively, the ratio of the maximum kinetic energy of electrons emitted from A to that from B is

CBSE 2020

(a) $2 : 3$

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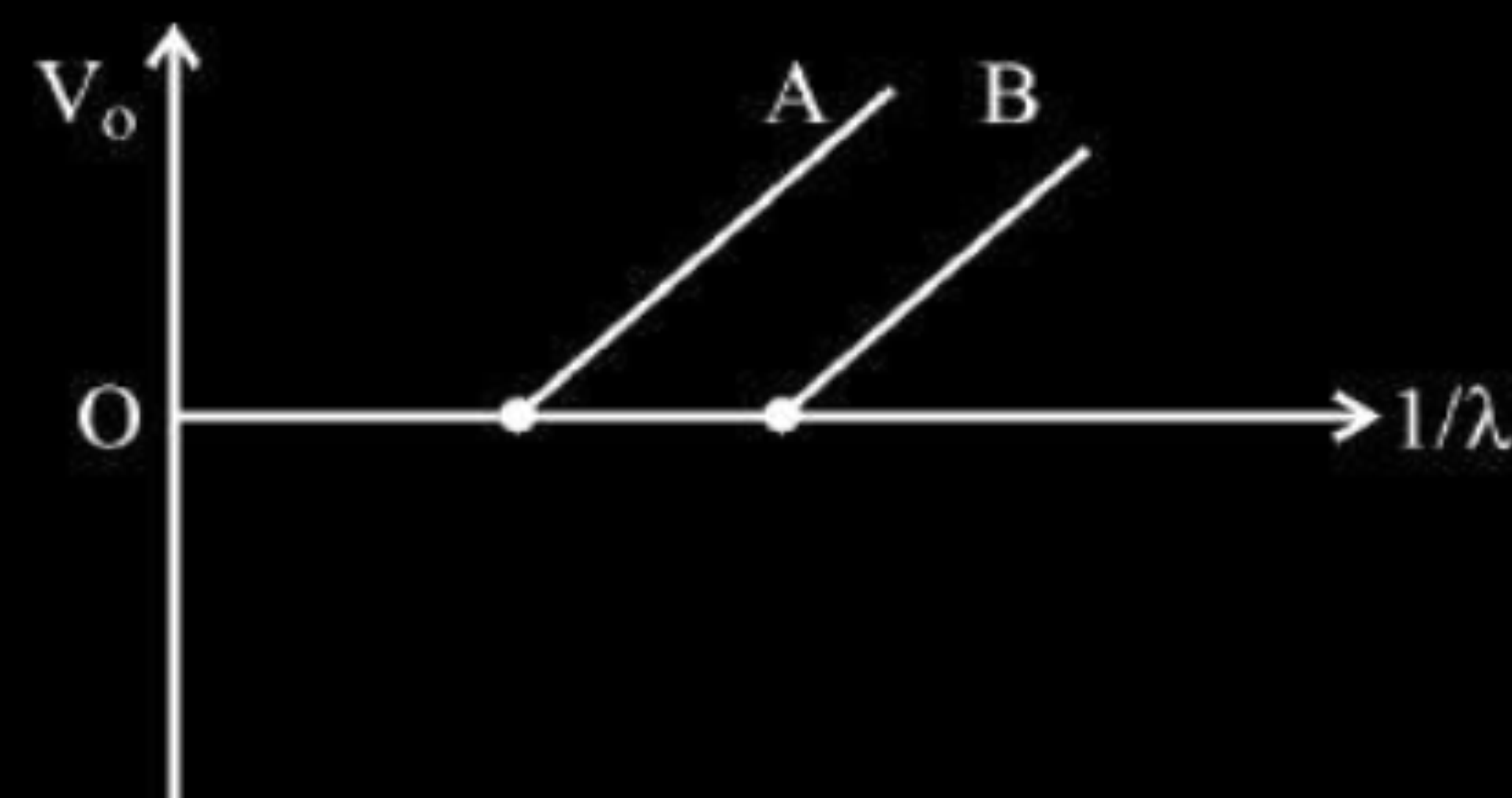
(c) $1 : 3$

(d) $\sqrt{3} : \sqrt{2}$

1

(b) 3:4

25. Figure shows the stopping potential (V_0) for the photo electron versus ($1/\lambda$) graph, for two metals A and B, λ being the wavelength of incident light.



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- (a) How is the value of Planck's constant determined from the graph ?
- (b) If the distance between the light source and the surface of metal A is increased, how will the stopping potential for the electrons emitted from it be effected ? Justify your answer.

According to Einstein's Photo electric equation

$$h\nu = \phi_0 + eV_s$$

$$eV_s = h\nu - \phi_0$$

$$V_s = \frac{h\nu}{e} - \frac{\phi_0}{e}$$

$$\text{since } \nu = c / \lambda$$

$$\begin{aligned}\therefore V_s &= \frac{hc}{e\lambda} - \frac{\phi_0}{e} \\ &= \left(\frac{hc}{e} \right) \frac{1}{\lambda} + \left[\frac{-\phi_0}{e} \right]\end{aligned}$$

Comparing with the equation of straight line $y=mx +c$

(a) The slope of the line $m = \frac{hc}{e}$. Hence, Planck's constant $h = \frac{me}{c}$

(b) Stopping potential will remain same

Justification

Variation of distance of light source from the metal surface will alter the intensity while the stopping potential however depends only on the frequency and not on the intensity of the incident light.

1/2

1/2

1/2

1/2

5. The kinetic energy of a proton and that of an α -particle are 4 eV and 1 eV, respectively. The ratio of the de-Broglie wavelengths associated with them, will be

(a) 2:1

(b) 1:1

(c) 1:2

(d) 4:1

1

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5. The kinetic energy of a proton and that of an α -particle are 4 eV and 1 eV, respectively. The ratio of the de-Broglie wavelengths associated with them, will be

(a) 2:1

(b) 1:1

(c) 1:2

(d) 4:1

1

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(b) 1:1

24. Light of same wavelength is incident on three photo-sensitive surfaces A, B and C. The following observations are recorded.
- (i) From surface A, photo electrons are not emitted.
 - (ii) From surface B, photo electrons are just emitted.
 - (iii) From surface C, photo electrons with some kinetic energy are emitted.
- Compare the threshold frequencies of the three surfaces and justify your answer.

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24. Light of same wavelength is incident on three photo-sensitive surfaces A, B and C. The following observations are recorded.
- (i) From surface A, photo electrons are not emitted.
 - (ii) From surface B, photo electrons are just emitted.
 - (iii) From surface C, photo electrons with some kinetic energy are emitted.
- Compare the threshold frequencies of the three surfaces and justify your answer.

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2

(a) Let ν_{0A} , ν_{0B} and ν_{0C} be their threshold frequencies for the surfaces A, B and C

$$\text{Therefore } \nu_{0A} > \nu_{0B} > \nu_{0C}$$

(b) Justification :-

If the frequency of incident light/photon is ν

$$h\nu = h\nu_0 + E_k$$

$$\text{Therefore } \nu_{0A} > \nu, \nu_{0B} = \nu \text{ and } \nu_{0C} < \nu$$

1

$\frac{1}{2}$

$\frac{1}{2}$

If the frequency of light incident on the cathode of a photo-cell is increased, how will the following be affected ? Justify your answer.

- (i) Energy of the photo electrons.
- (ii) Photo current.

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2

If the frequency of light incident on the cathode of a photo-cell is increased, how will the following be affected ? Justify your answer.

- (i) Energy of the photo electrons.
- (ii) Photo current.

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2

(i)	The energy of the emitted photoelectrons increases As $E_k = h\nu - \phi_0$ As ν increases, E_k also increases	$\frac{1}{2}$
(ii)	Photo current will not be affected As, increase of ν , E_k will increase but not the number of photoelectrons [Alternatively photocurrent depends upon intensity of light and not on frequency]	$\frac{1}{2}$

15. In photoelectric effect, the number of emitted photoelectrons is proportional to _____ of incident light. *1*

Light of frequency ν is incident on a photosensitive surface of threshold frequency ν_0 ($\nu > \nu_0$). The value of kinetic energy of the emitted photoelectrons will be _____ . *1*

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15. In photoelectric effect, the number of emitted photoelectrons is proportional to Intensity of incident light. 1

Light of frequency ν is incident on a photosensitive surface of threshold frequency ν_0 ($\nu > \nu_0$). The value of kinetic energy of the emitted photoelectrons will be $h(\nu - \nu_0)$. 1

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- 26.** Define the terms (a) threshold frequency, and (b) stopping potential. How were these terms incorporated in Einstein's photoelectric equation ? 2

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26. Define the terms (a) threshold frequency, and (b) stopping potential. How were these terms incorporated in Einstein's photoelectric equation ?

2

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(a) Threshold Frequency: The minimum cut off frequency ν_0 below which no photoelectric emission is possible, even if the intensity is large

$\frac{1}{2}$

(b) Stopping Potential: The minimum negative (retarding) potential V_0 given to the plate for which the photocurrent stops or becomes zero is called the cut off or stopping potential.

$\frac{1}{2}$

$$h\nu = \phi_0 + \frac{1}{2}mV_{max}^2$$

$\frac{1}{2}$

$$h\nu = h\nu_0 + eV_0$$

$\frac{1}{2}$

19. What is the wavelength of a photon of energy $3.3 \times 10^{-19} \text{ J}$?

1

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19. What is the wavelength of a photon of energy $3.3 \times 10^{-19} \text{ J}$? *1*

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$$6.03 \times 10^{-7} \text{ m}$$

[Award full 1 mark even if a student writes $6 \times 10^{-7} \text{ m}$]

1

20. Define the term ‘threshold frequency’ in photoelectric emission.

1

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20. Define the term 'threshold frequency' in photoelectric emission.

1

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For a given photosensitive material, there exists a certain minimum cut-off frequency of the incident radiation, called the **threshold frequency**, below which no emission of photo electrons takes place, no matter how intense the incident light is.

1

- 32.** The maximum kinetic energy of the photoelectrons emitted is doubled when the wavelength of light incident on the photosensitive surface changes from λ_1 to λ_2 . Deduce expressions for the threshold wavelength and work function for the metal surface in terms of λ_1 and λ_2 .

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32. The maximum kinetic energy of the photoelectrons emitted is doubled when the wavelength of light incident on the photosensitive surface changes from λ_1 to λ_2 . Deduce expressions for the threshold wavelength and work function for the metal surface in terms of λ_1 and λ_2 .

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3

$$K_{\max} = \frac{hc}{\lambda_1} - \phi_0 = hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_0} \right)$$

when $\lambda = \lambda_2$

$$2K_{\max} = hc \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_0} \right)$$

$$\frac{K_{\max}}{2K_{\max}} = \frac{1}{2} = \frac{\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_0} \right)}{\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_0} \right)}$$

$$\lambda_0 = \frac{\lambda_1 \lambda_2}{2\lambda_2 - \lambda_1} = \text{Threshold wavelength}$$

$$\phi_0 = \frac{hc}{\lambda_0} = \frac{hc(2\lambda_2 - \lambda_1)}{(\lambda_1 \lambda_2)}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

10. Photons of energies 1 eV and 2 eV are successively incident on a metallic surface of work function 0.5 eV. The ratio of kinetic energy of most energetic photoelectrons in the two cases will be

1

(A) 1 : 2

(B) 1 : 1

(C) 1 : 3

(D) 1 : 4

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10. Photons of energies 1 eV and 2 eV are successively incident on a metallic surface of work function 0.5 eV. The ratio of kinetic energy of most energetic photoelectrons in the two cases will be

1

(A) 1 : 2

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(C) 1 : 3

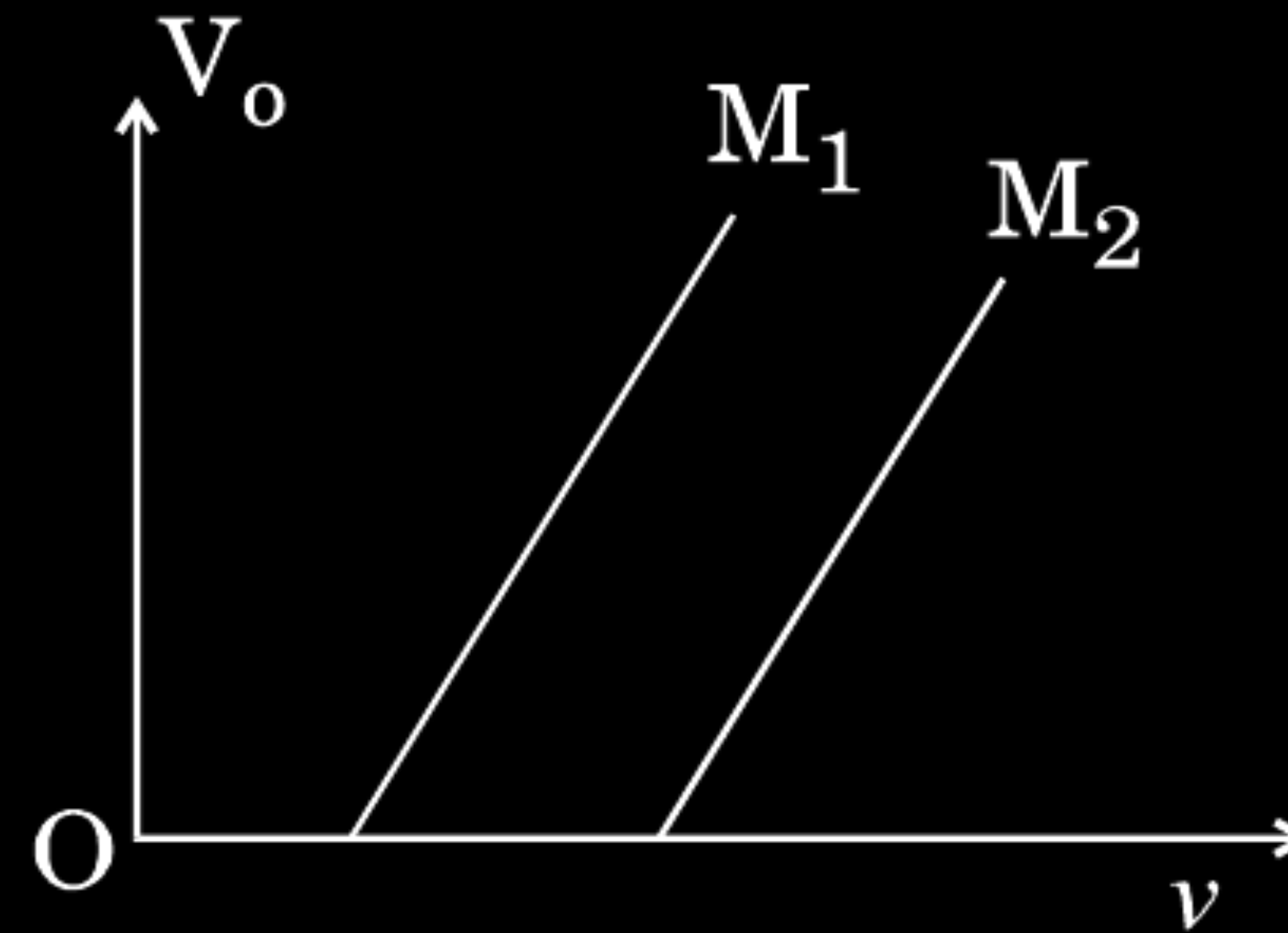
(D) 1 : 4

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(C)

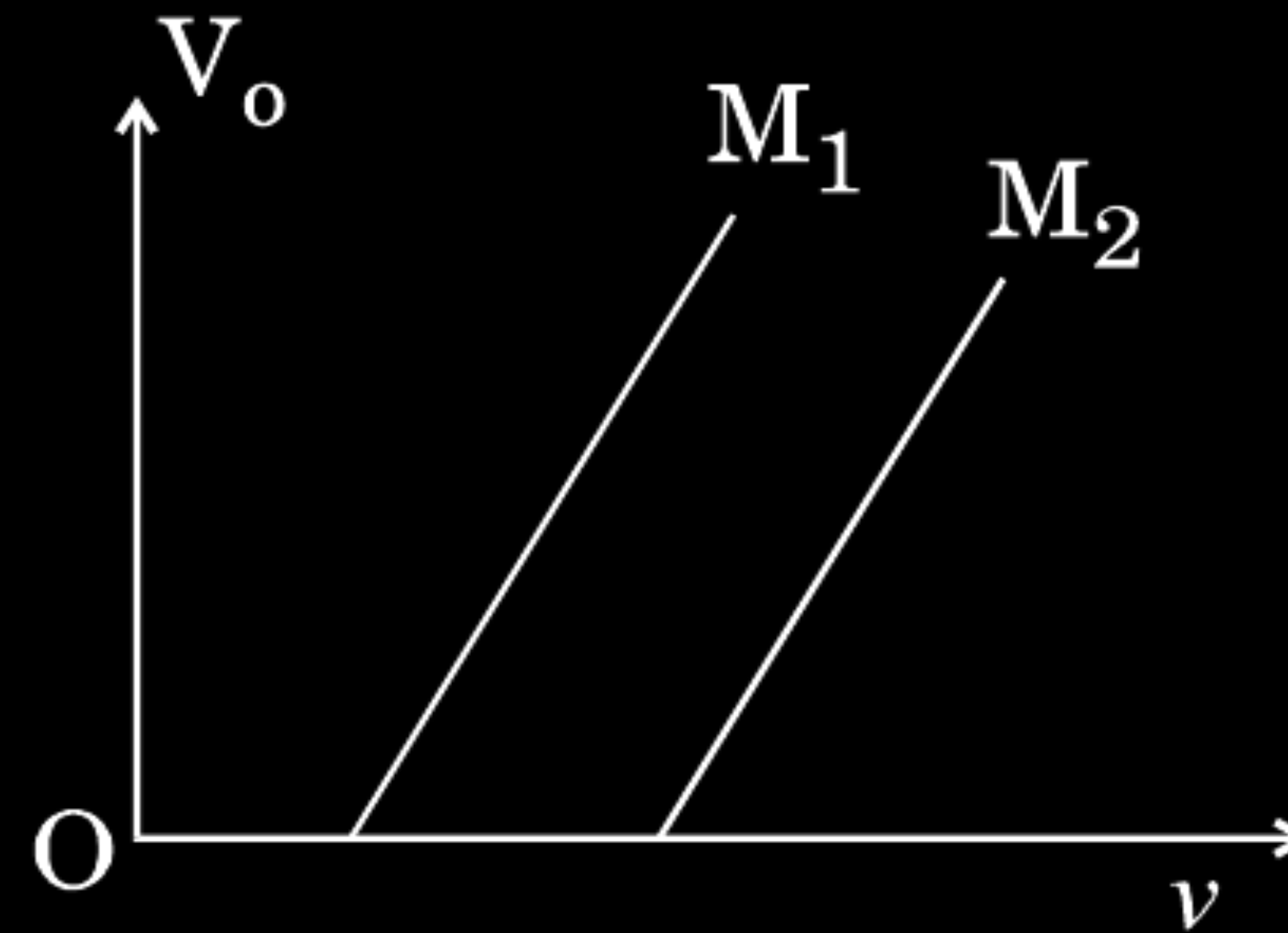
1:3

- 19.** The variation of the stopping potential (V_0) with the frequency (ν) of the light incident on two different photosensitive surfaces M_1 and M_2 is shown in the figure. Identify the surface which has greater value of the work function.



CBSE 2020

19. The variation of the stopping potential (V_0) with the frequency (ν) of the light incident on two different photosensitive surfaces M_1 and M_2 is shown in the figure. Identify the surface which has greater value of the work function.

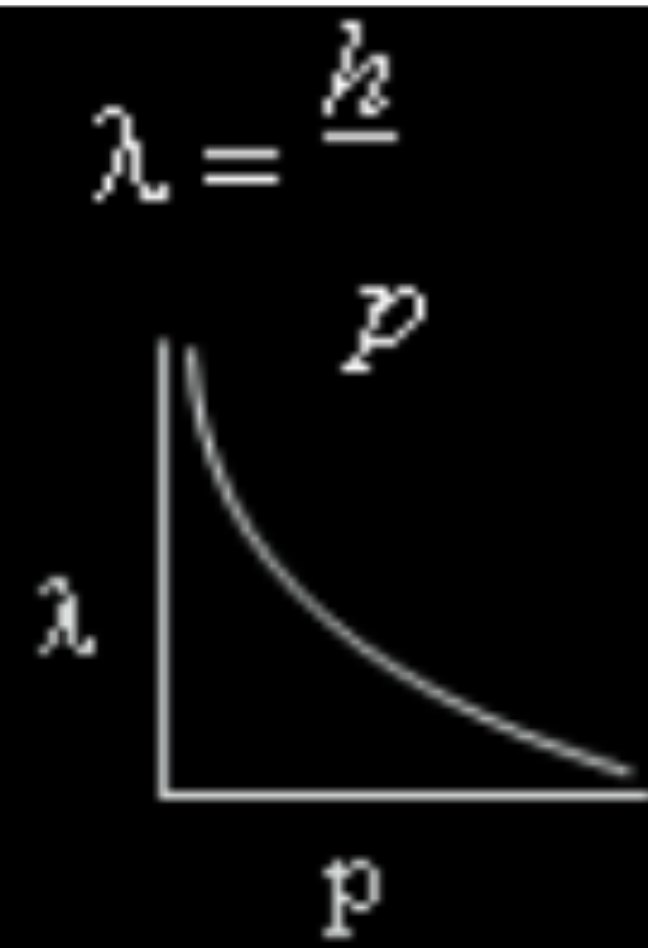


CBSE 2020

M_2

Plot a graph of the de-Broglie wavelength associated with a proton versus its momentum.

Plot a graph of the de-Broglie wavelength associated with a proton versus its momentum.

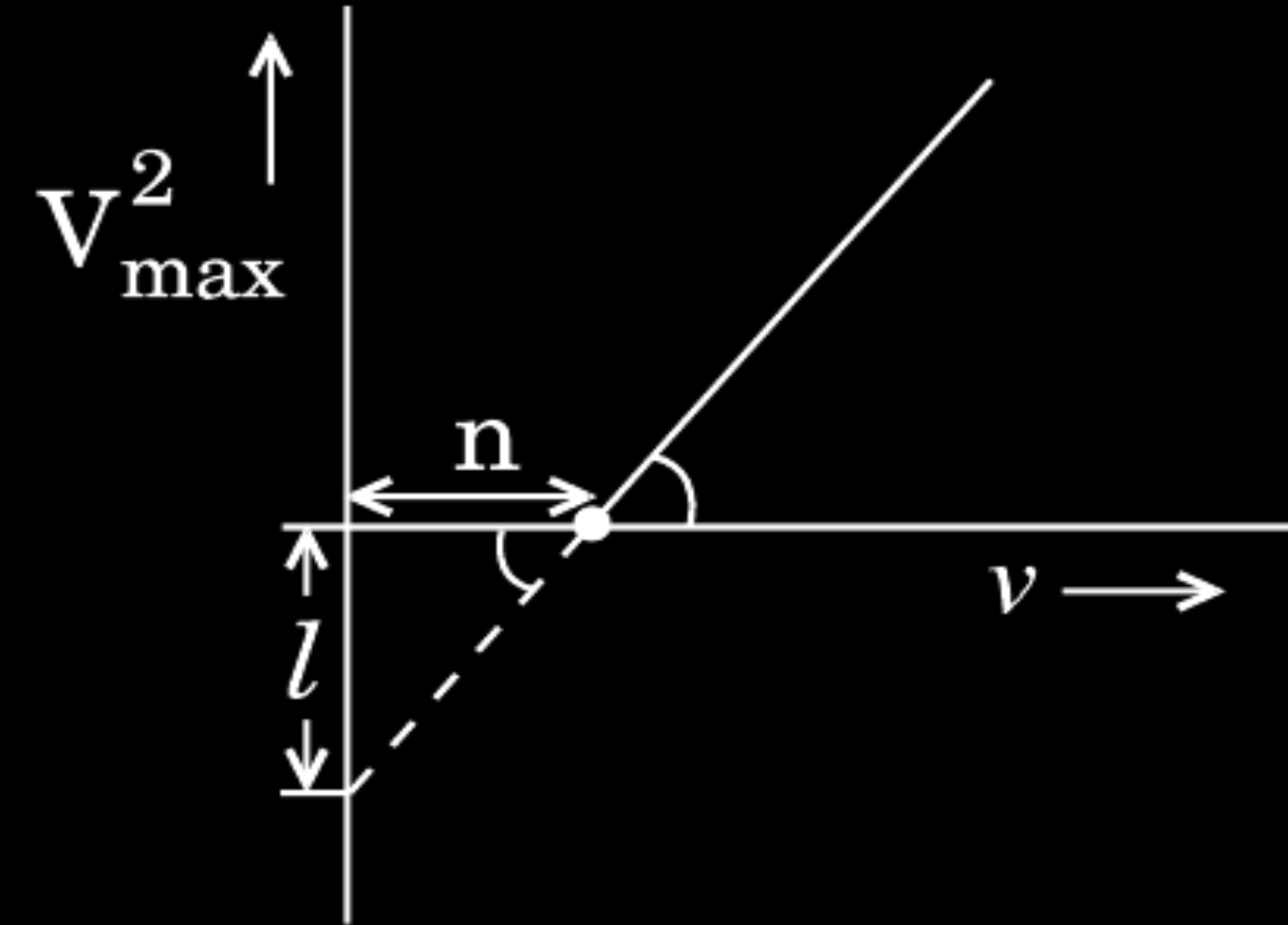


[If only the graph is drawn by the student, award full one mark]

$\frac{1}{2}$

$\frac{1}{2}$

When a given photosensitive material is irradiated with light of frequency ν , the maximum speed of the emitted photoelectrons equals $V_{\max}$. The graph shown in the figure gives a plot of $V_{\max}^2$ varying with frequency ν .



Obtain an expression for

- Planck's constant, and
- The work function of the given photosensitive material in terms of the parameters ' l ', ' n ' and the mass ' m ' of the electron.
- How is threshold frequency determined from the plot ?

(a) Einstein, Photoelectric equation

$$K_{\max} = \frac{1}{2} m v_{\max}^2 = h\nu - \phi_0$$

$$V_{\max}^2 = \left(\frac{2h}{m} \right) \nu - \frac{2\phi_0}{m}$$

$$\text{Slope of the graph is } \frac{2h}{m} = \frac{l}{n}$$

$$\text{Planck constant } h = \frac{lm}{2n}$$

(b) Intercept on $V_{\max}^2$, axis, $\frac{2\phi_0}{m} = l$

$$\text{Work function } \phi_0 = \frac{ml}{2}$$

(c) Threshold frequency is the intercept on ν axis i.e. $\nu_0 = n$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$

$$\frac{1}{2}$$

- (a) Write two main observations of photoelectric effect experiment which could only be explained by Einstein's photoelectric equation.
- (b) Draw a graph showing variation of photocurrent with the anode potential of a photocell.

- (a) Write two main observations of photoelectric effect experiment which could only be explained by Einstein's photoelectric equation.
- (b) Draw a graph showing variation of photocurrent with the anode potential of a photocell.

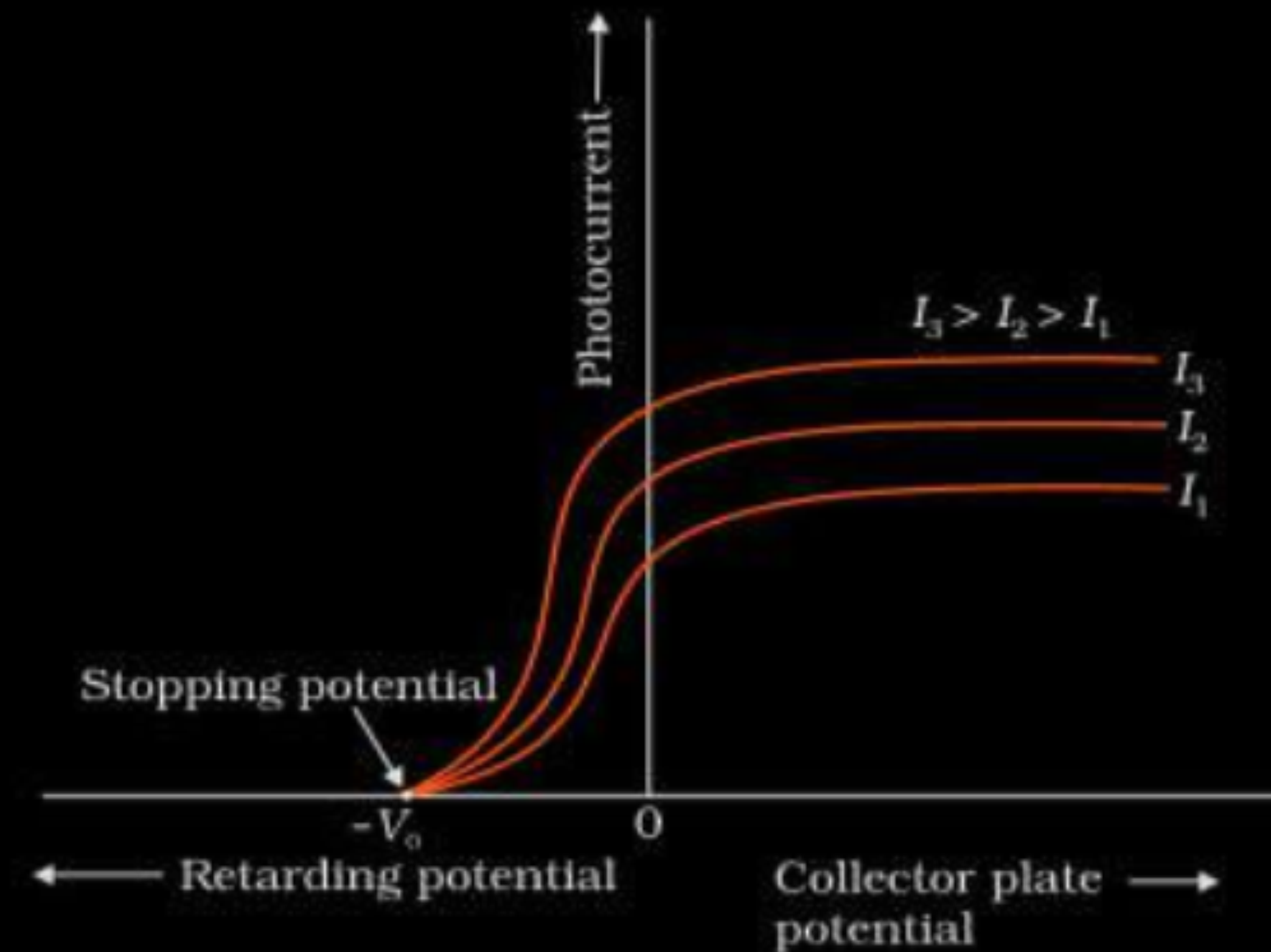
CBSE 2020

2

a)

- (i) There exists a threshold frequency below which no photoelectron is ejected.
- (ii) KE of electron depends linearly on frequency and is independent of intensity of radiation.
- [or any other correct observation]

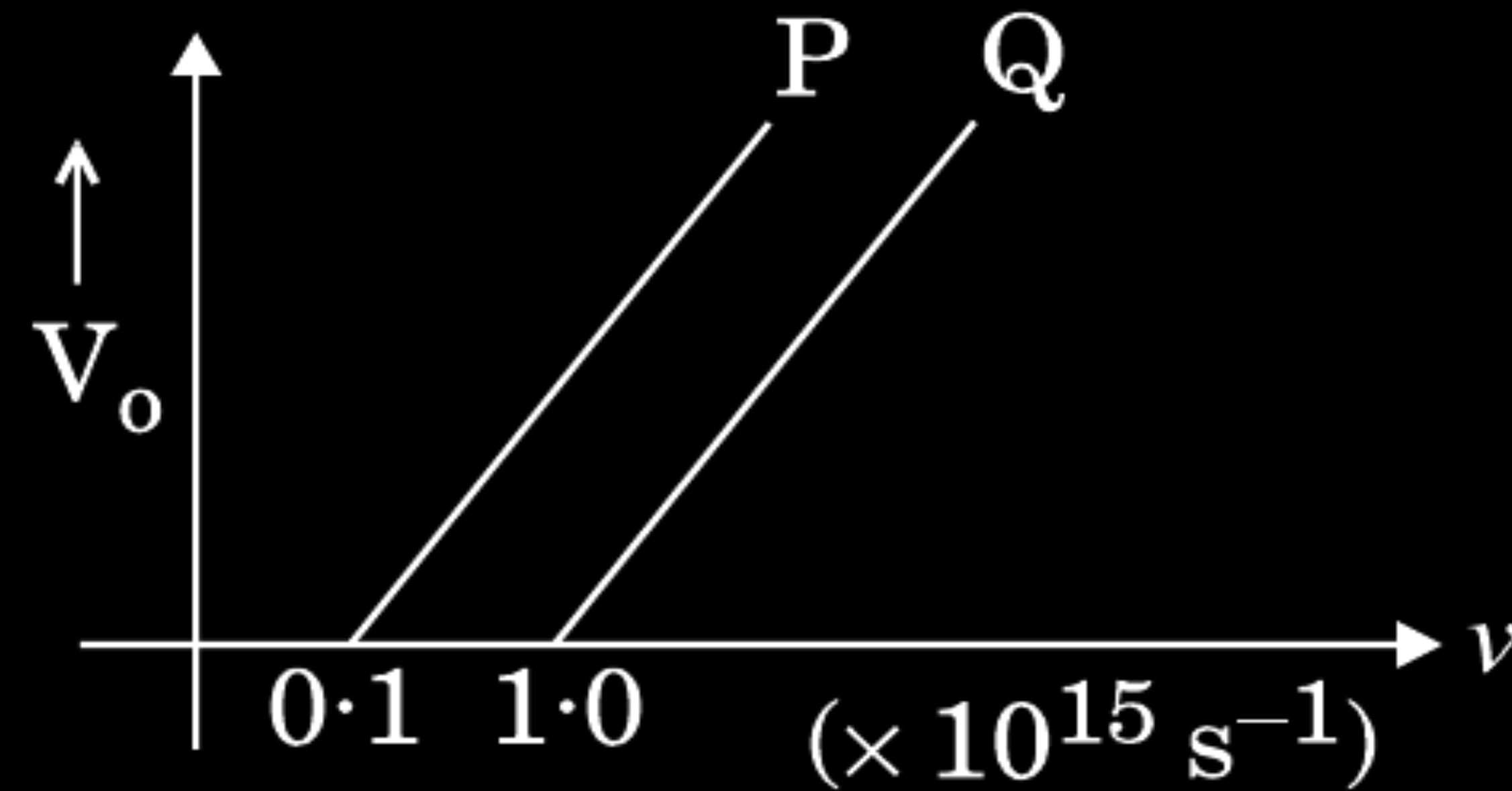
b)



[only curve is essential to draw]

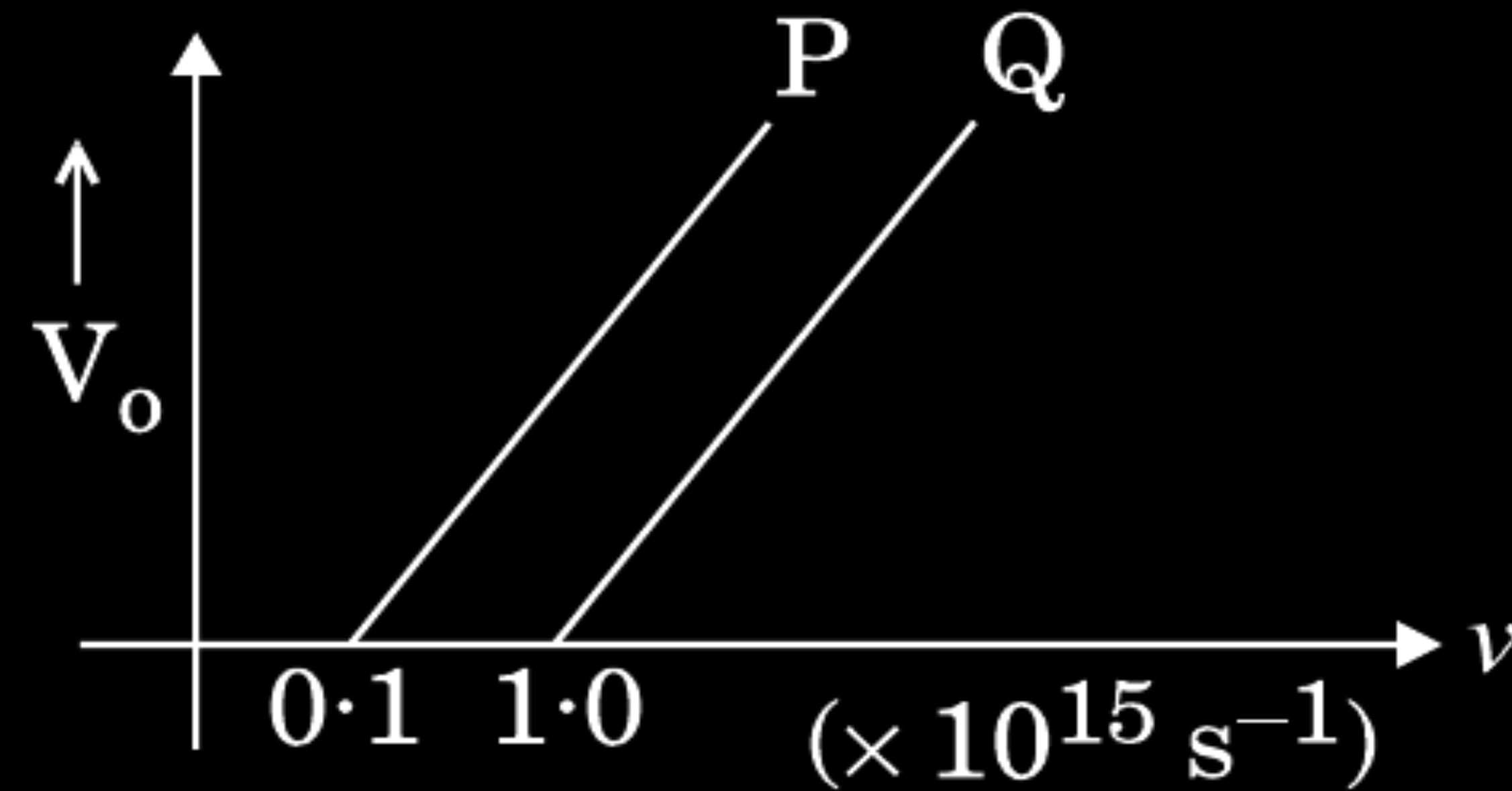
1

The figure shows the variation of stopping potential V_0 with the frequency ν of the incident radiations for two photosensitive metals P and Q. Which metal has smaller threshold wavelength ? Justify your answer.



CBSE 2019

The figure shows the variation of stopping potential V_0 with the frequency ν of the incident radiations for two photosensitive metals P and Q. Which metal has smaller threshold wavelength ? Justify your answer.



CBSE 2019

Metal Q has smaller threshold wavelength	$\frac{1}{2}$
Since $\lambda_0 = \frac{c}{\nu_0}$	$\frac{1}{2}$

- (b) Obtain an expression for the ratio of the accelerating potentials required to accelerate a proton and an α -particle to have the same de-Broglie wavelength associated with them.

CBSE 2019

- (b) Obtain an expression for the ratio of the accelerating potentials required to accelerate a proton and an α -particle to have the same de-Broglie wavelength associated with them.

CBSE 2019

b) de Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

$\frac{1}{2}$

$$\therefore \frac{h}{\sqrt{2m_p e V_p}} = \frac{h}{\sqrt{2m_\alpha e V_\alpha}}$$

$\frac{1}{2}$

$$\therefore \frac{V_p}{V_\alpha} = \frac{8}{1}$$

$\frac{1}{2}$

- (a) An electron and a proton are accelerated through the same potential. Which one of the two has
- (i) greater value of de-Broglie wavelength associated with it, and
 - (ii) lesser momentum ?

Justify your answer in each case.

- (b) How is the momentum of a particle related with its de-Broglie wavelength ? Show the variation on a graph.

CBSE 2019

a) i) Since $\lambda = \frac{h}{\sqrt{2mqV}}$

$$\lambda \propto \frac{1}{\sqrt{m}} \quad (\text{For other variables constant})$$

$$\therefore m_p > m_e$$

$$\text{Therefore } \lambda_{\text{electron}} > \lambda_{\text{proton}}$$

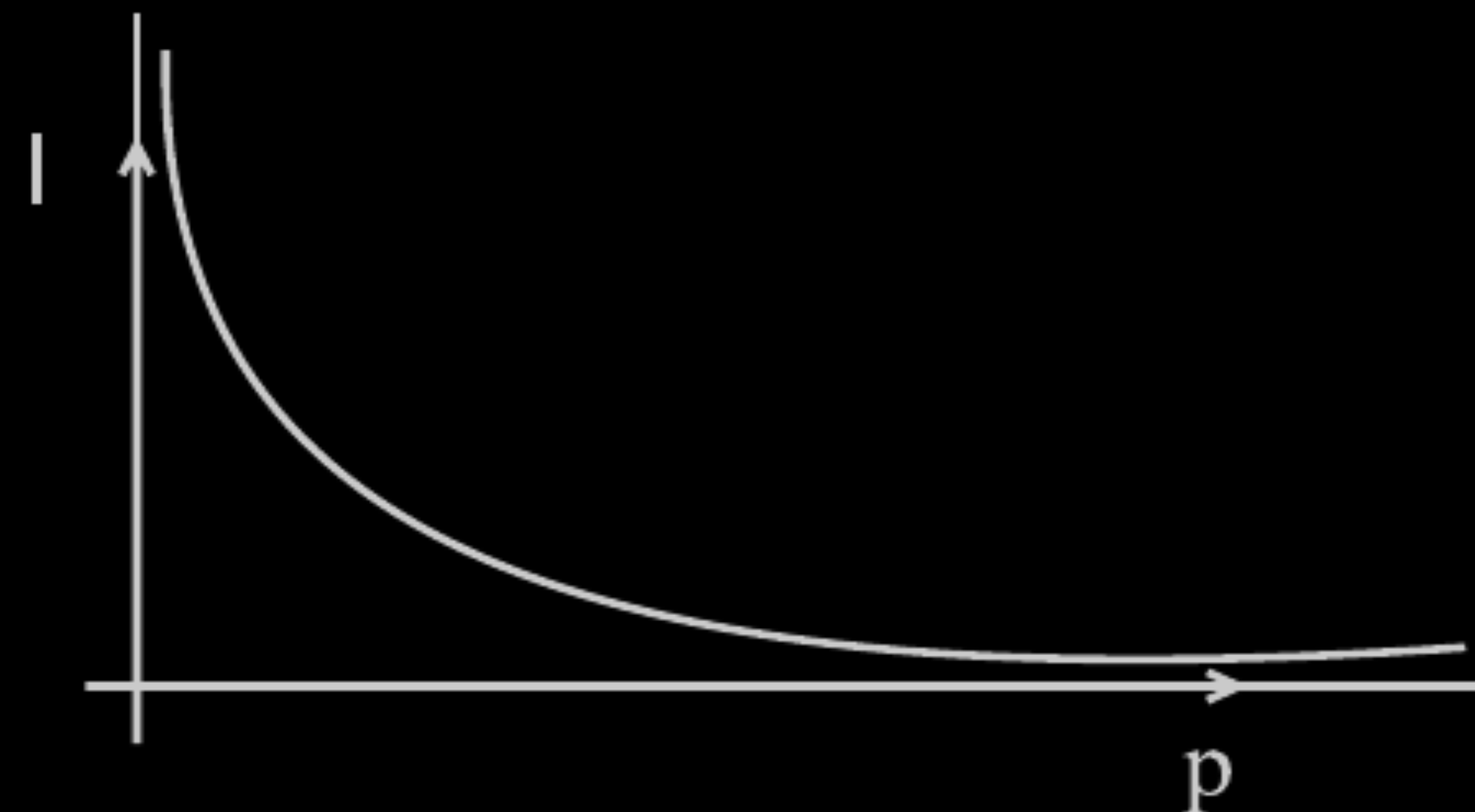
ii) momentum $p = \frac{h}{\lambda}$

$$\therefore \lambda_{\text{electron}} > \lambda_{\text{proton}}$$

$\therefore$ momentum of electron is lesser.

b) $\lambda = \frac{h}{p}$

Graph between p & λ



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

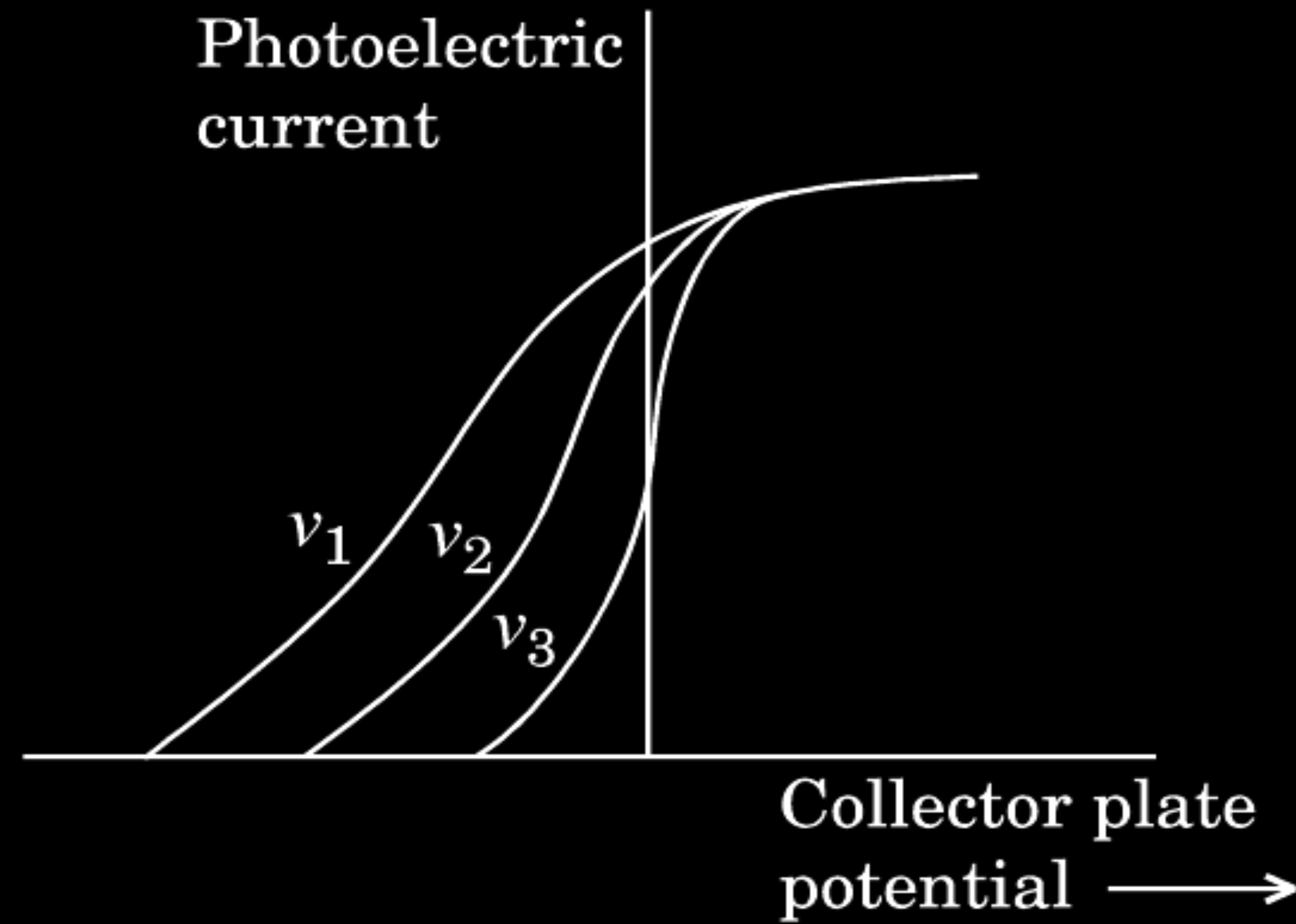
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

On the basis of the graphs shown in the figure, answer the following questions :

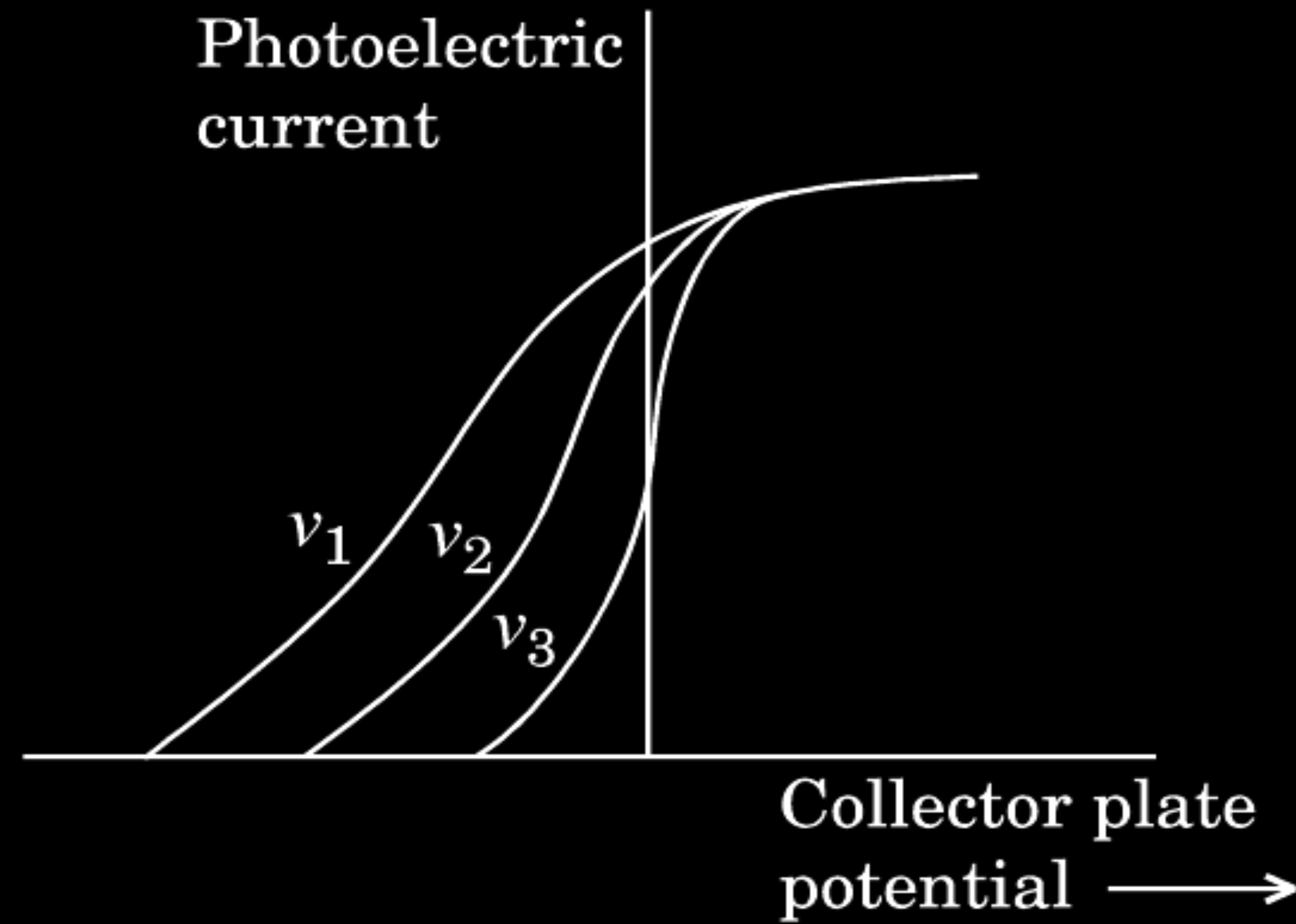
- (a) Which physical parameter is kept constant for the three curves ?
- (b) Which is the highest frequency among ν_1 , ν_2 and ν_3 ?



CBSE 2019

On the basis of the graphs shown in the figure, answer the following questions :

- (a) Which physical parameter is kept constant for the three curves ?
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CBSE 2019

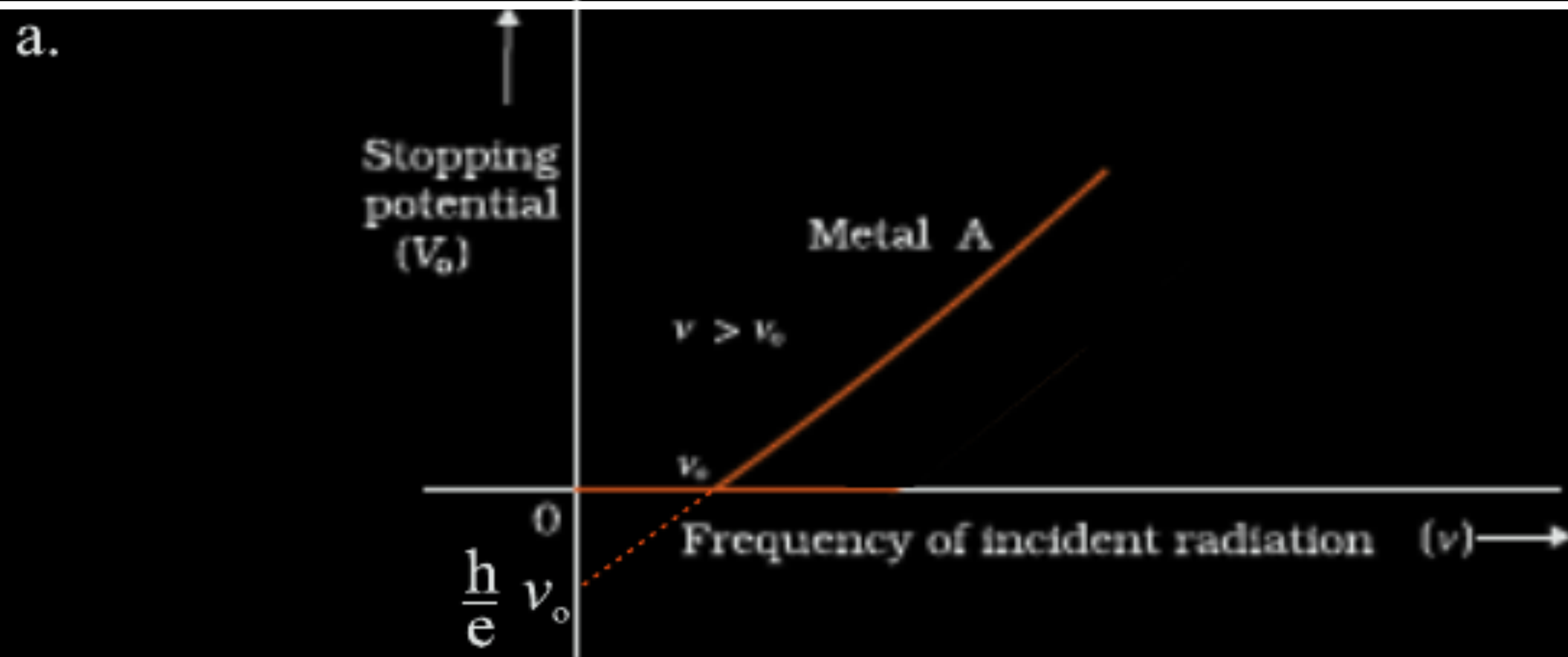
- (a) Intensity of incident radiation is constant
(b) ν_1 is the highest frequency.

$\frac{1}{2}$

$\frac{1}{2}$

- (a) Plot a graph to show the variation of stopping potential with frequency of incident radiation in relation to photoelectric effect.
- (b) Use Einstein's photoelectric equation to show how from this graph,
 - (i) Threshold frequency, and
 - (ii) Planck's constant can be determined.

CBSE 2019



1

b. From Einstein's Equation

$$eV_0 = h\nu - h\nu_0$$

$$V_0 = \frac{h}{e} \nu - \frac{h}{e} \nu_0$$

comparing

$$y = mx + c$$

$\frac{1}{2}$

(i) Threshold frequency ν_0 is the intercept along ν axis.

$\frac{1}{2}$

(Alternatively, intercept on V_0 axis, $c = \frac{h}{e} \nu_0$
 $\nu_0 = \frac{ec}{h}$)

(ii) Planck's constant $h = e \times \text{slope}$

1

- (a) How does one explain the emission of electrons from a photosensitive surface with the help of Einstein's photoelectric equation ?
- (b) Work function of aluminium is 4.2 eV . If two photons each of energy 2.5 eV are incident on its surface, will the emission of electrons take place ? Justify your answer.
- (c) The stopping potential in an experiment on photoelectric effect is 1.5 V . What is the maximum kinetic energy of the photoelectrons emitted ? Calculate in Joules.

CBSE 2019

- a. When a photon of the energy $h\nu$ is absorbed by an electron in the photosensitive material, a part of the energy absorbed is used up in liberating it from the surface (the work function ϕ_0). The remaining energy appears as KE of the photoelectron.

1

Alternatively:

$$K_{\max} = h\nu - \phi_0$$

if $h\nu \geq \phi_0$, $k_{\max}$ is positive and electron is emitted

- b. Emission of electron will not take place.

Energy $h\nu$, of a single photon, is less than the work function ϕ_0 .

(Alternatively - $k_{\max} = h\nu - \phi_0$

$h\nu < \phi_0$ so $k_{\max}$ is negative; Hence no emission will take place.)

- c. $V_0 = 1.5 \text{ V}$

$$k_{\max} = eV_0 = 1.6 \times 10^{-19} \times 1.5 = 2.4 \times 10^{-19} \text{ J}$$

[If a student just writes, $k_{\max} = 1.5 \text{ eV}$ award $\frac{1}{2}$ mark]

$\frac{1}{2}$
 $\frac{1}{2}$

1

A photon and a proton have the same de-Broglie wavelength λ . Prove that the energy of the photon is $(2m\lambda c/h)$ times the kinetic energy of the proton.

CBSE 2019

A photon and a proton have the same de-Broglie wavelength λ . Prove that the energy of the photon is $(2m\lambda c/h)$ times the kinetic energy of the proton.

CBSE 2019

$$\text{Energy of photon } E_p = \frac{hc}{\lambda}$$

$$\text{For proton } \lambda = \frac{h}{mv}$$

$$mv = \frac{h}{\lambda}$$

$$\text{Kinetic energy of proton } E_k = \frac{1}{2}mv^2$$

$$E_k = \frac{1}{2} \frac{h^2}{m\lambda^2}$$

$$E_p = \left(\frac{2m\lambda c}{h} \right) E_k$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

Define the term “threshold frequency”, in the context of photoelectric emission.

OR

Define the term “Intensity” in photon picture of electromagnetic radiation.

CBSE 2019

Define the term “threshold frequency”, in the context of photoelectric emission.

OR

Define the term “Intensity” in photon picture of electromagnetic radiation.

CBSE 2019

Threshold frequency equals the minimum frequency of incident radiation (light) that can cause photoemission from a given photosensitive surface.

(Alternatively)

The frequency below which the incident radiations cannot cause the photoemission from photosensitive surface.

OR

Intensity of radiation is proportional to (/ equal to) the number of energy quanta (photons) per unit area per unit time.

1

Why is wave theory of electromagnetic radiation not able to explain photo electric effect ? How does photon picture resolve this problem ?

CBSE 2019

Why is wave theory of electromagnetic radiation not able to explain photo electric effect ? How does photon picture resolve this problem ?

CBSE 2019

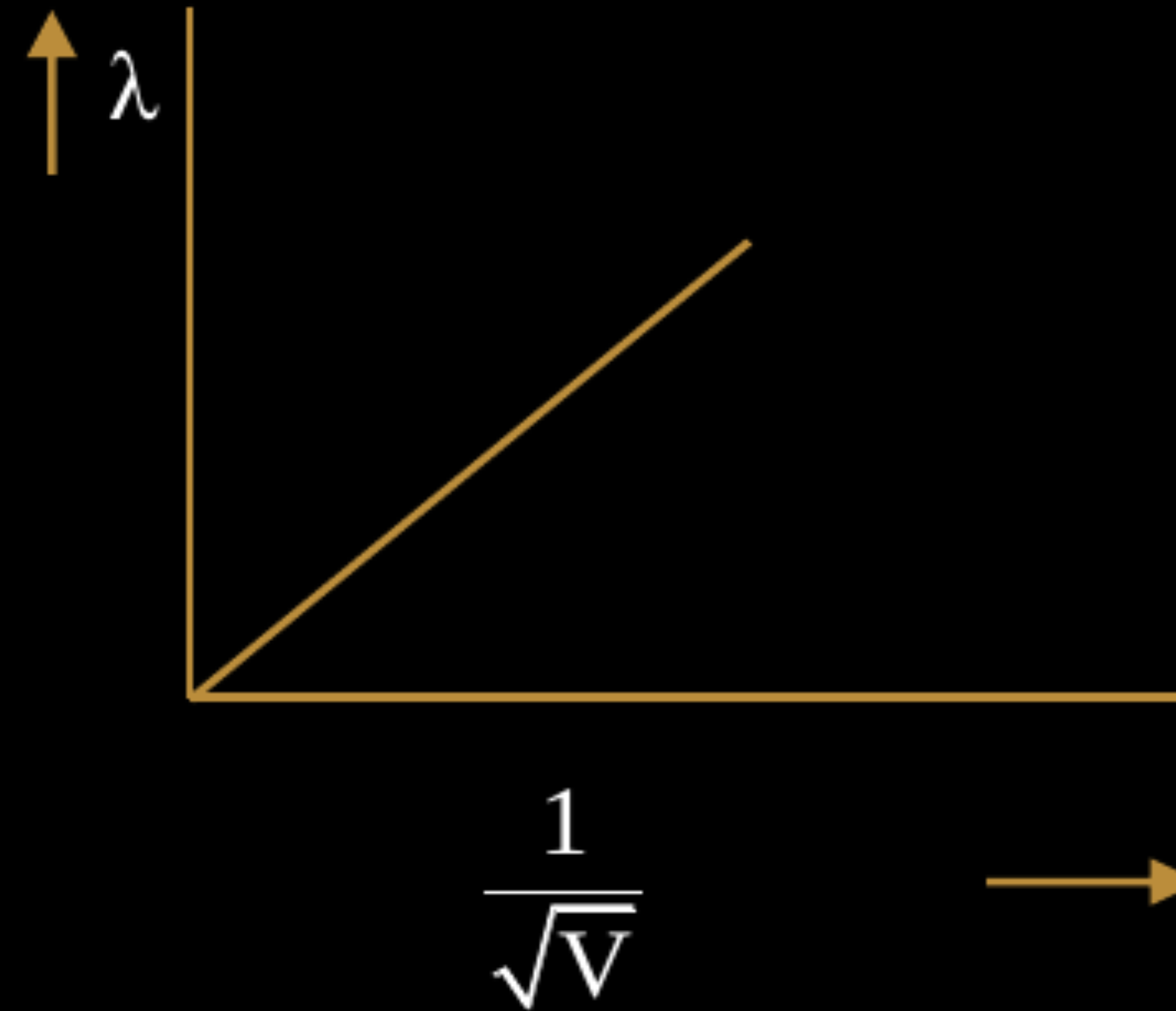
The explanation based on e.m theory does not agree with the experimental observations (instantaneous nature , max K.E of emitted photoelectron is independent of intensity, existence of threshold frequency) on the photoelectric effect.	1
[Note: Do not deduct any mark if the student does not mention the relevant experimental observation or mentions any one or any two of these observation.] The photon picture resolves this problem by saying that light, in interaction with matter behaves as if it is made of quanta or packets of energy, each of energy $h \nu$. This picture enables us to get a correct explanation of all the observed experimental features of photoelectric effect.	1
[NOTE: Award the first mark if the student just writes “As per E.M. theory the free electrons at the surface of the metal absorb the radiant energy continuously, this leads us to conclusions which do not match with the experimental observations”] Also award the second mark if the student just writes “The photon picture give us the Einstein photoelectric equation $K_{\max} (= eV_o) = h \nu - \phi_o$ which provides a correct explanation of the observed features of the photoelectric effect.	

Plot a graph showing variation of de Broglie wavelength (λ) associated with a charged particle of mass m , versus $1/\sqrt{V}$, where V is the potential difference through which the particle is accelerated. How does this graph give us the information regarding the magnitude of the charge of the particle ?

CBSE 2019

Plot a graph showing variation of de Broglie wavelength (λ) associated with a charged particle of mass m , versus $1/\sqrt{V}$, where V is the potential difference through which the particle is accelerated. How does this graph give us the information regarding the magnitude of the charge of the particle ?

CBSE 2019



$$\because \lambda = \frac{h}{\sqrt{2mqV}}$$

$$\frac{\lambda}{\left(\frac{1}{\sqrt{V}}\right)} = \frac{h}{\sqrt{2mq}} = \text{slope}$$

$$q = \frac{h^2}{2m (\text{slope})^2}$$

1

$\frac{1}{2}$

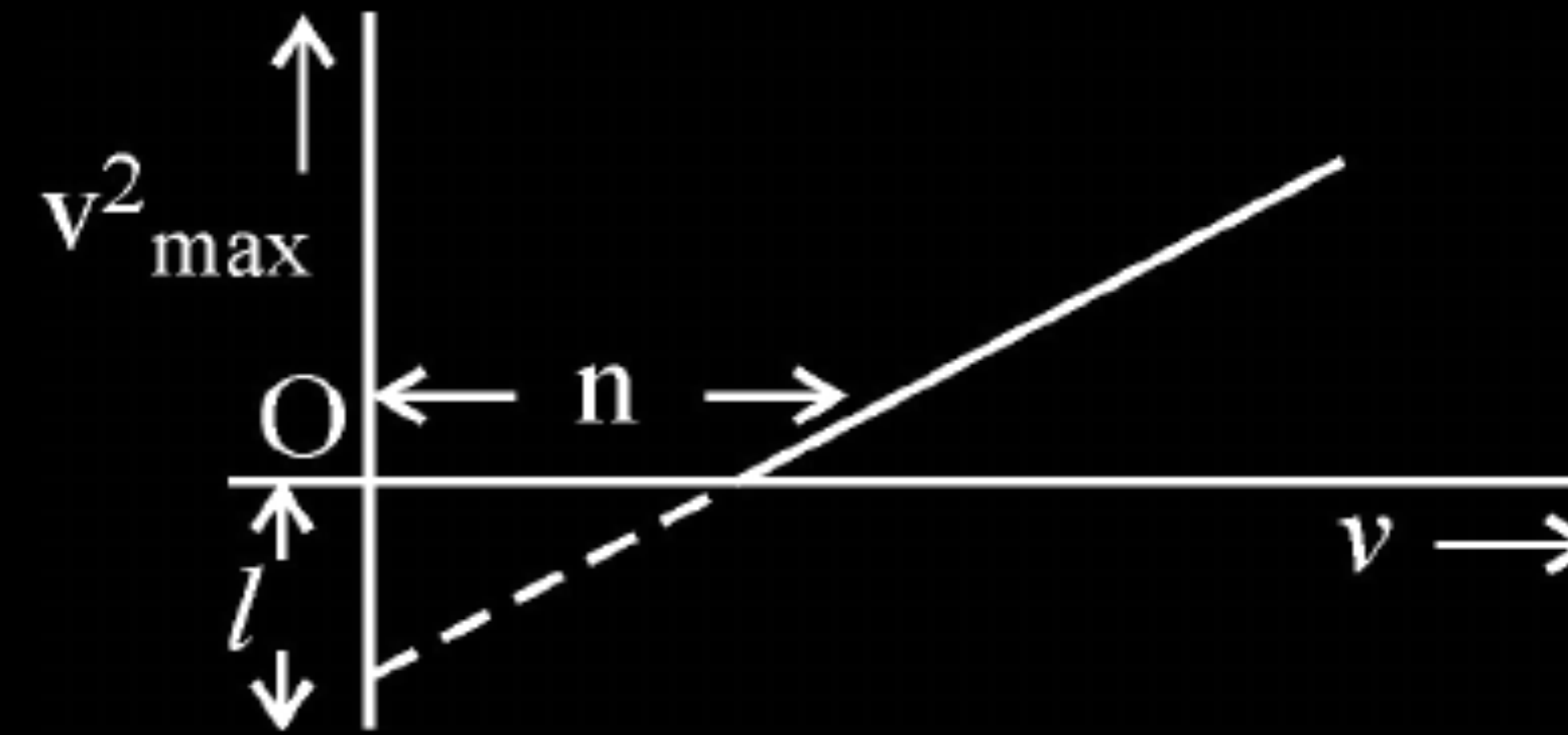
Find the frequency of light which ejects electrons from a metal surface, fully stopped by a retarding potential of 3.3 V. If photo electric emission begins in this metal at a frequency of 8×10^{14} Hz, calculate the work function (in eV) for this metal.

CBSE 2018

Monochromatic light of frequency 6.0×10^{14} Hz is produced by a laser. The power emitted is 2.0×10^{-3} W. Calculate the (i) energy of a photon in the light beam and (ii) number of photons emitted on an average by the source.

CBSE 2018

State Einstein's photoelectric equation explaining the symbols used.



CBSE 2018

Light of frequency ν incident is on a photosensitive surface. A graph of the square of the maximum speed of the electrons (v_{max}^2) vs. ν is obtained as shown in the figure.

Using Einstein's photoelectric equation, obtain expressions for (i) Planck's constant
(ii) work function of the given photosensitive material in terms of parameters l , n and mass of the electron m .

- (i) How does one explain the emission of electrons from a photosensitive surface with the help of Einstein's photoelectric equation ?
- (ii) The work function of the following metals is given : Na = 2.75 eV, K = 2.3 eV, Mo = 4.17 eV and Ni = 5.15 eV. Which of these metals will not cause photoelectric emission for radiation of wavelength $3300 \text{ \AA}$ from a laser source placed 1 m away from these metals ? What happens if the laser source is brought nearer and placed 50 cm away ?

CBSE 2017

- (i) How does one explain the emission of electrons from a photosensitive surface with the help of Einstein's photoelectric equation ?
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CBSE 2017

- (i) Einstein's Photoelectric equation is

$\frac{1}{2}$

When a photon of energy ' $h\nu$ ' is incident on the metal, some part of this energy is utilized as work function to eject the electron and remaining energy appears as the kinetic energy of the emitted electron.

1

- (ii) $\frac{hc}{\lambda} - \phi$
 $= 3.77 \text{ eV}$

$\frac{1}{2}$

The work function of Mo and Ni is more than the energy of the incident photons; so photoelectric emission will not take place from these metals.

$\frac{1}{2}$

Kinetic energy of photo electrons will not change, only photoelectric current will change.

$\frac{1}{2}$

Using photon picture of light, show how Einstein's photoelectric equation can be established. Write two features of photoelectric effect which cannot be explained by wave theory.

CBSE 2017

Using photon picture of light, show how Einstein's photoelectric equation can be established. Write two features of photoelectric effect which cannot be explained by wave theory.

CBSE 2017

In the photon picture, energy of the light is assumed to be in the form of photons, each carrying an energy $h\nu$.	1/2
Einstein assumed that photoelectric emission occurs because of a single collision of a photon with a free electron.	1/2
The energy of the photon is used to	
(i) free the electrons from the metal. [For this, a minimum energy, called the work function (=W) is needed].	
And	
(ii) provide kinetic energy to the emitted electrons.	1/2

Hence	
$(K.E.)_{\max} = h\nu - W$	
$/ \left(\frac{1}{2} m v_{\max}^2 = h\nu - W \right)$	
This is Einstein's photoelectric equation	1/2
Two features (which cannot be explained by wave theory):	
i) 'Instantaneous' emission of photoelectrons	
ii) Existence of a threshold frequency	
iii) 'Maximum kinetic energy' of the emitted photoelectrons, is independent of the intensity of incident light	
(Any two)	1/2 + 1/2

Name the phenomenon which shows the quantum nature of electromagnetic radiation.

CBSE 2017

Name the phenomenon which shows the quantum nature of electromagnetic radiation.

CBSE 2017

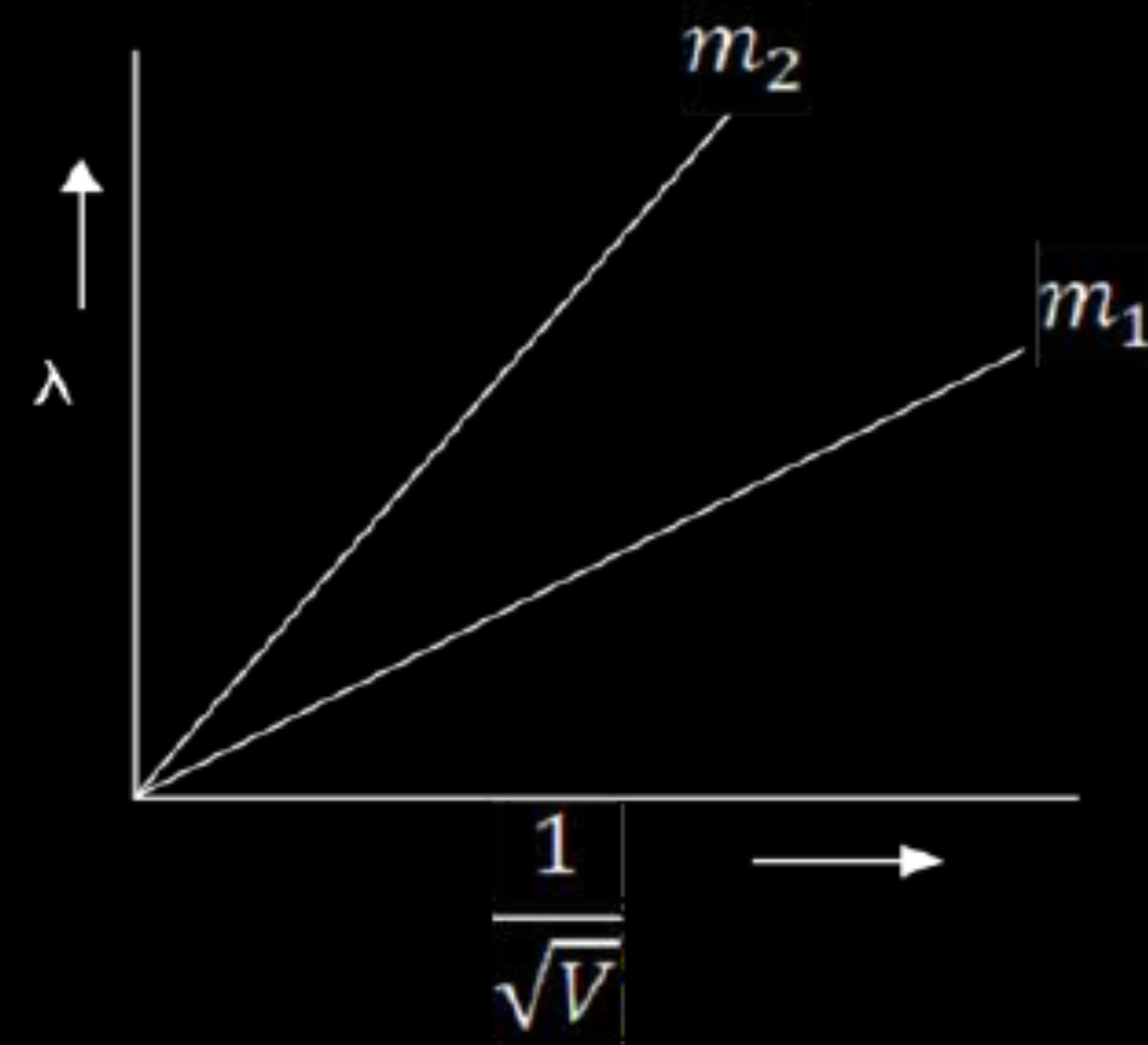
Photoelectric Effect (/Raman Effect/ Compton Effect)	1
--	----------

Plot a graph showing variation of de-Broglie wavelength λ versus $\frac{1}{\sqrt{V}}$, where V is accelerating potential for two particles A and B carrying same charge but of masses m_1, m_2 ($m_1 > m_2$). Which one of the two represents a particle of smaller mass and why ?

CBSE 2016

Plot a graph showing variation of de-Broglie wavelength λ versus $\frac{1}{\sqrt{V}}$, where V is accelerating potential for two particles A and B carrying same charge but of masses m_1, m_2 ($m_1 > m_2$). Which one of the two represents a particle of smaller mass and why ?

CBSE 2016



$\frac{1}{2} + \frac{1}{2}$

$$\text{As } \lambda = \frac{h}{\sqrt{2mqV}}$$

$\frac{1}{2}$

As the charge of two particles is same, therefore

$$\frac{\lambda}{\left(\frac{1}{\sqrt{V}}\right)} \propto \frac{1}{\sqrt{m}} \quad \text{i.e.} \quad \text{Slope} \propto \frac{1}{\sqrt{m}}$$

Hence, particle with lower mass (m_2) will have greater slope.

$\frac{1}{2}$

Write three characteristic features in photoelectric effect which cannot be explained on the basis of wave theory of light, but can be explained only using Einstein's equation.

CBSE 2016

Write three characteristic features in photoelectric effect which cannot be explained on the basis of wave theory of light, but can be explained only using Einstein's equation.

CBSE 2016

The three characteristic features which can't be explained by wave theory are:

- | | | |
|------|---|---|
| i. | Kinetic energy of emitted electrons are found to be independent of intensity of incident light. | 1 |
| ii. | Below a certain frequency (threshold) there is no photo-emission. | 1 |
| iii. | Spontaneous emission of photo-electrons. | 1 |

- (a) Define the term ‘intensity of radiation’ in photon picture.
- (b) Plot a graph showing the variation of photo current vs collector potential for three different intensities $I_1 > I_2 > I_3$, two of which (I_1 and I_2) have the same frequency ν and the third has frequency $\nu_1 > \nu$.
- (c) Explain the nature of the curves on the basis of Einstein’s equation.

CBSE 2016

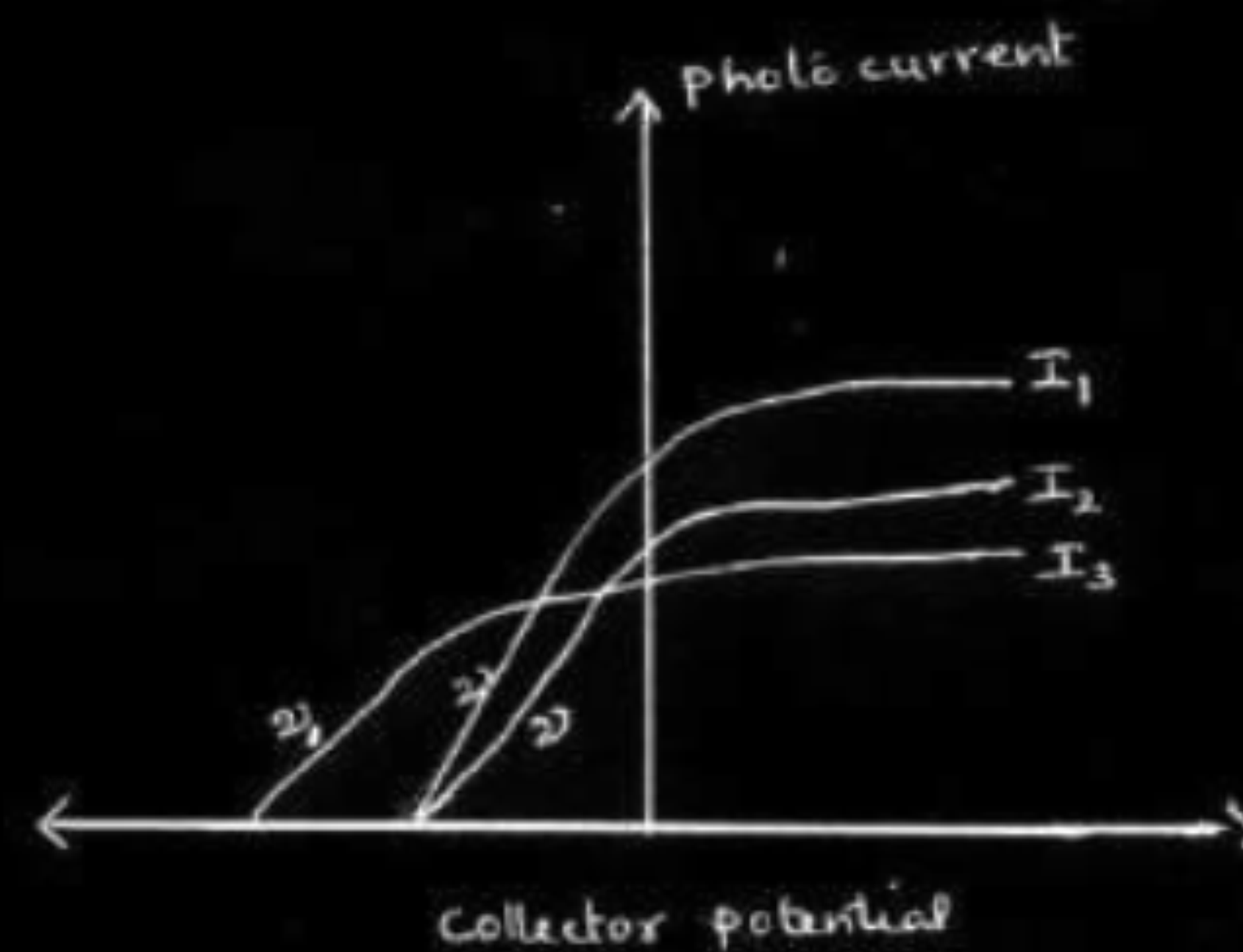
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CBSE 2016

- a) Intensity of radiation equals the energy of all the Photons incident normally per unit area per unit time.
Alternatively, The intensity of radiation is proportional to the number of photons emitted per unit area per unit time.

1

b)



1

- c) As per Einstein's equation,
- (i) The stopping potential is same for I_1 and I_2 as they have the same frequency.
- (ii) The saturation currents are as shown, because $I_1 > I_2 > I_3$

$\frac{1}{2}$

$\frac{1}{2}$

A proton and an α particle are accelerated through the same potential difference. Which one of the two has (i) greater de-Broglie wavelength, and (ii) less kinetic energy ? Justify your answer.

CBSE 2016

A proton and an α particle are accelerated through the same potential difference. Which one of the two has (i) greater de-Broglie wavelength, and (ii) less kinetic energy ? Justify your answer.

CBSE 2016

$$(i) \lambda = \frac{h}{p} = \frac{h}{\sqrt{2mqV}}$$

$\because (mq)$ is more for α – particle, we have

$$\lambda_{proton} > \lambda_{\alpha - particle}$$

(Also, accept if the student writes $\frac{\lambda_{proton}}{\lambda_{\alpha}} = 2\sqrt{2}$ (or $\sqrt{8}$)

$\frac{1}{2}$

$\frac{1}{2}$

$$(ii) K.E. = qV$$

$\because q$ is less for proton, we have

$$(K.E)_{proton} < (K.E)_{\alpha - particle}$$

(Also accept if the student writes $\frac{(K.E.)_{\alpha}}{(K.E.)_{\rho}} = 2$)

$\frac{1}{2}$

$\frac{1}{2}$

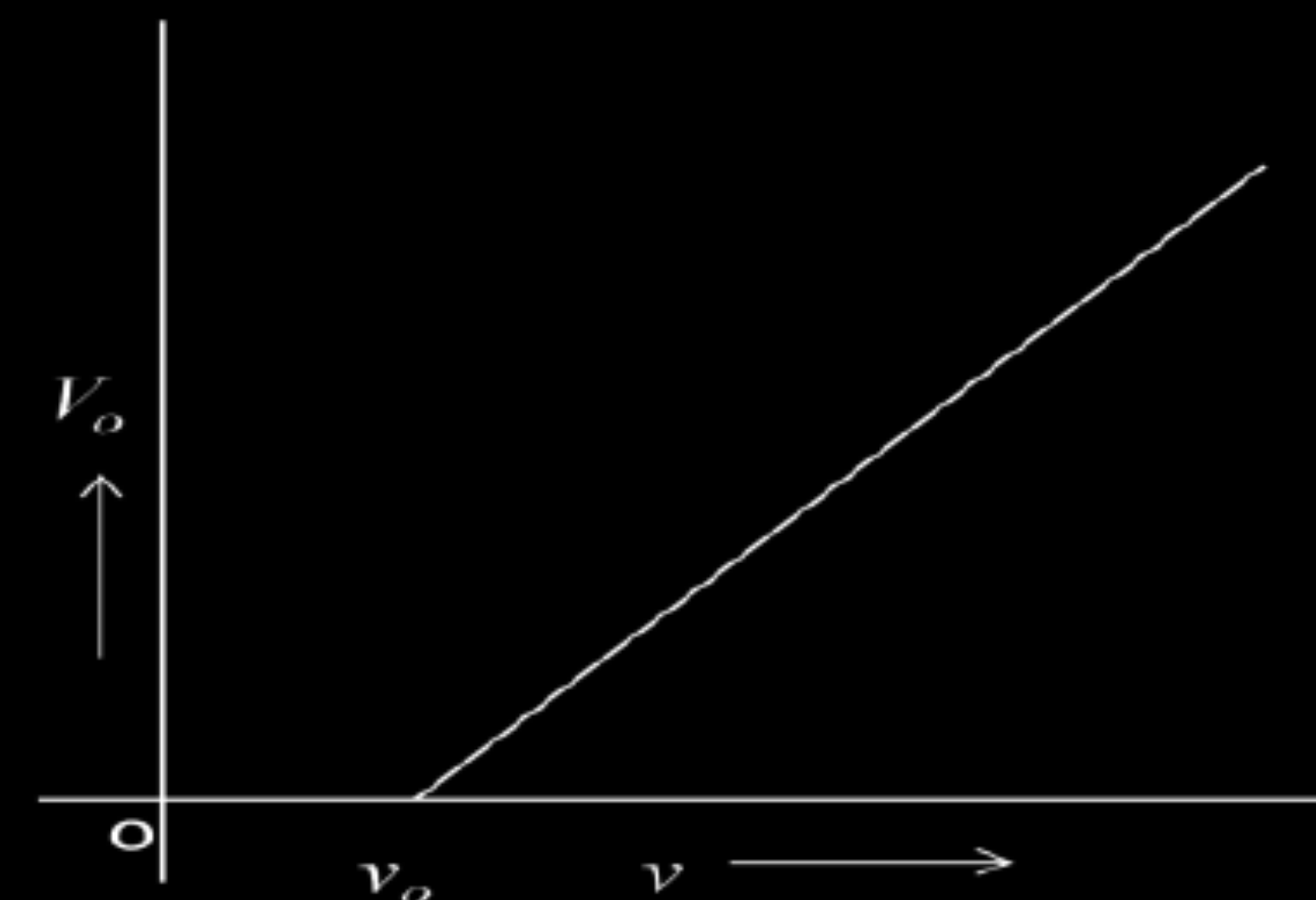
State two important properties of photon which are used to write Einstein's photoelectric equation. Define (i) stopping potential and (ii) threshold frequency, using Einstein's equation and drawing necessary plot between relevant quantities.

CBSE 2016

State two important properties of photon which are used to write Einstein's photoelectric equation. Define (i) stopping potential and (ii) threshold frequency, using Einstein's equation and drawing necessary plot between relevant quantities.

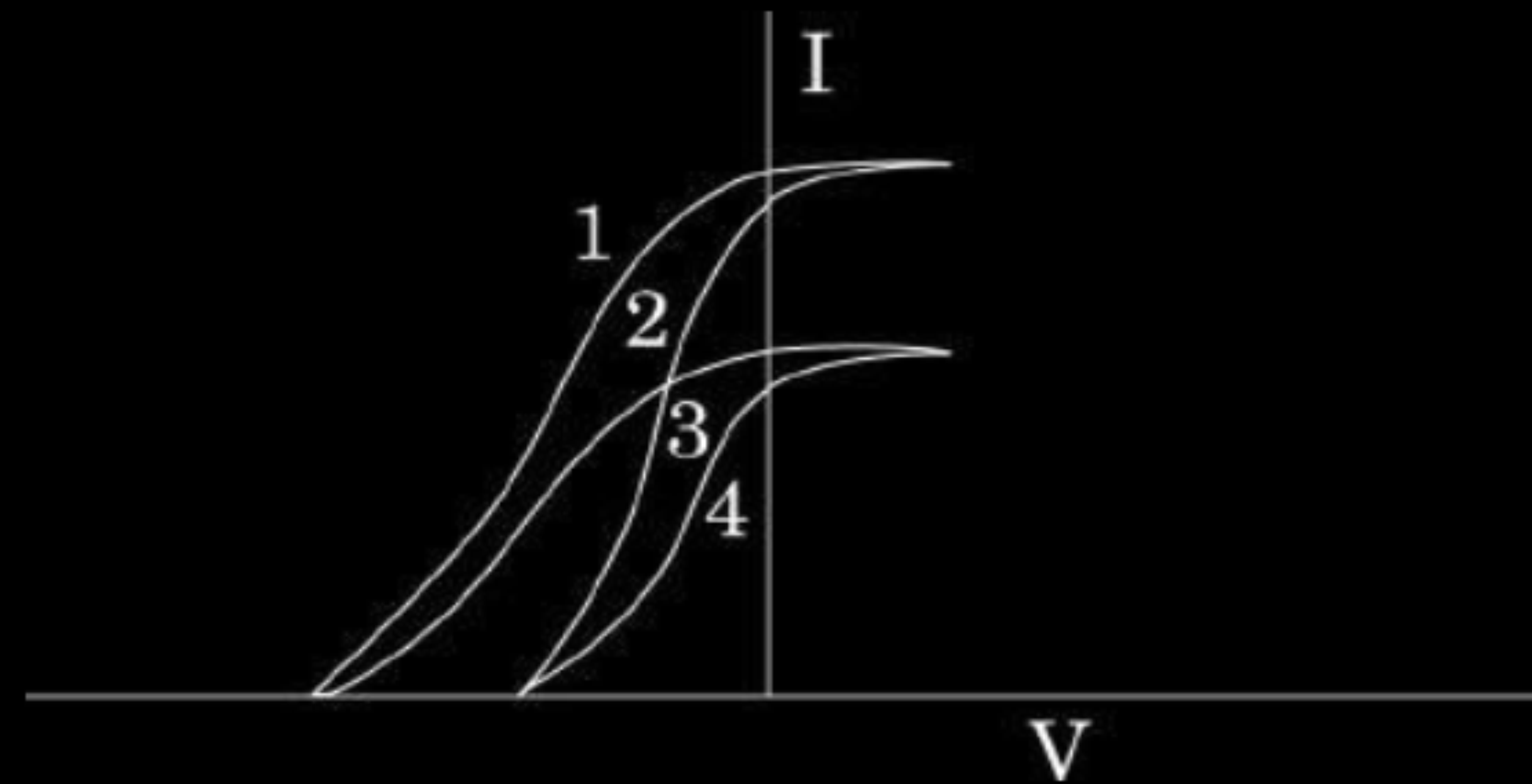
CBSE 2016

Properties of Photon (i) For a radiation of frequency ν, each photon has an energy, $E = h\nu$, associated with it (ii) The energy of a photon is independent of the intensity of incident radiation. (iii) During the collision of a photon, with an electron, the total energy of the photon gets absorbed by the electron. (Any two)	$\frac{1}{2} + \frac{1}{2}$		
Einstein's photoelectric equation is $K_{max} = h\nu - \phi_0$ or $eV_0 = h\nu - \phi_0$	$\frac{1}{2}$		
(a) Stopping potential, V_0, equals that value of the negative potential for which $eV_0 = K_{max}$	$\frac{1}{2}$		
		(Alternatively: The stopping potential (V_0) equals that (least) value of the (negative) plate potential that just stops the most energetic emitted photoelectrons from reaching the plate.) (b) Threshold frequency (ν_0) equals that value of the frequency of incident radiation for which $K_{max} = 0$. (Alternatively: For a given photosensitive surface, its threshold frequency is the minimum value of the frequency of incident radiation for which photoelectrons can be just emitted from that surface or that maximum frequency of incident radiation below which no photo emission takes place.) The plot, between V_0 and ν, has the form shown:	1



$\frac{1}{2}$

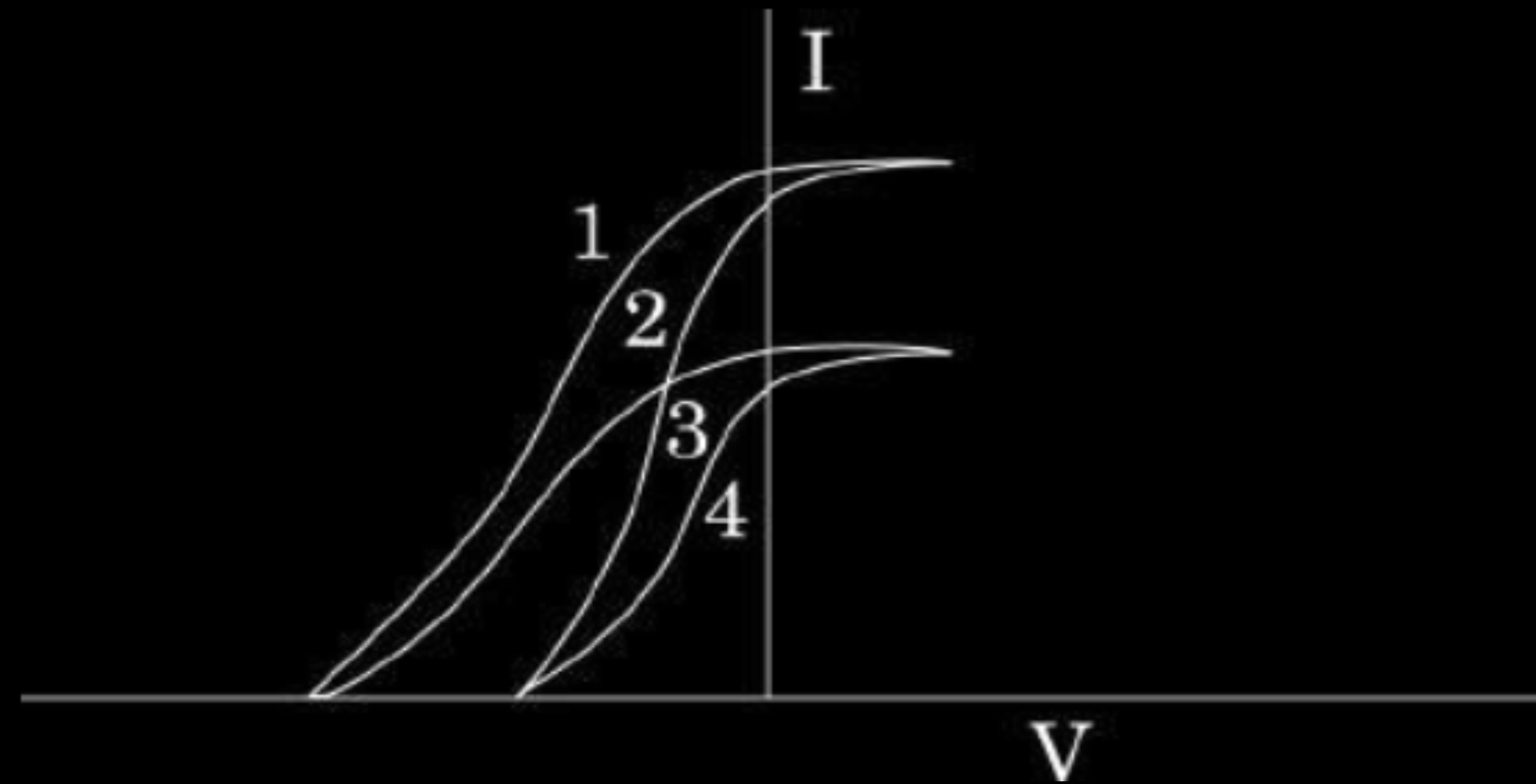
The given graph shows the variation of photo-electric current(I) with the applied voltage(V) for two different materials and for two different intensities of the incident radiations. Identify and explain using Einstein's photo electric equation the pair of curves that correspond to (i) different materials but same intensity of incident radiation, (ii) different intensities but same materials.



CBSE 2016

The given graph shows the variation of photo-electric current(I) with the applied voltage(V) for two different materials and for two different intensities of the incident radiations. Identify and explain using Einstein's photo electric equation the pair of curves that correspond to (i) different materials but same intensity of incident radiation, (ii) different intensities but same materials.

CBSE 2016



(1, 2) correspond to same intensity but different material.

$\frac{1}{2}$

(3, 4) correspond to same intensity but different material.

$\frac{1}{2}$

As saturation currents are same and stopping potentials are different.

$\frac{1}{2}$

(1, 3) correspond to different intensity but same material.

$\frac{1}{2}$

(2, 4) correspond to different intensity but same material.

$\frac{1}{2}$

As stopping potentials are same but saturation currents are different.

$\frac{1}{2}$

Sketch the graphs showing variation of stopping potential with frequency of incident radiations for two photosensitive materials A and B having threshold frequencies $\nu_A > \nu_B$.

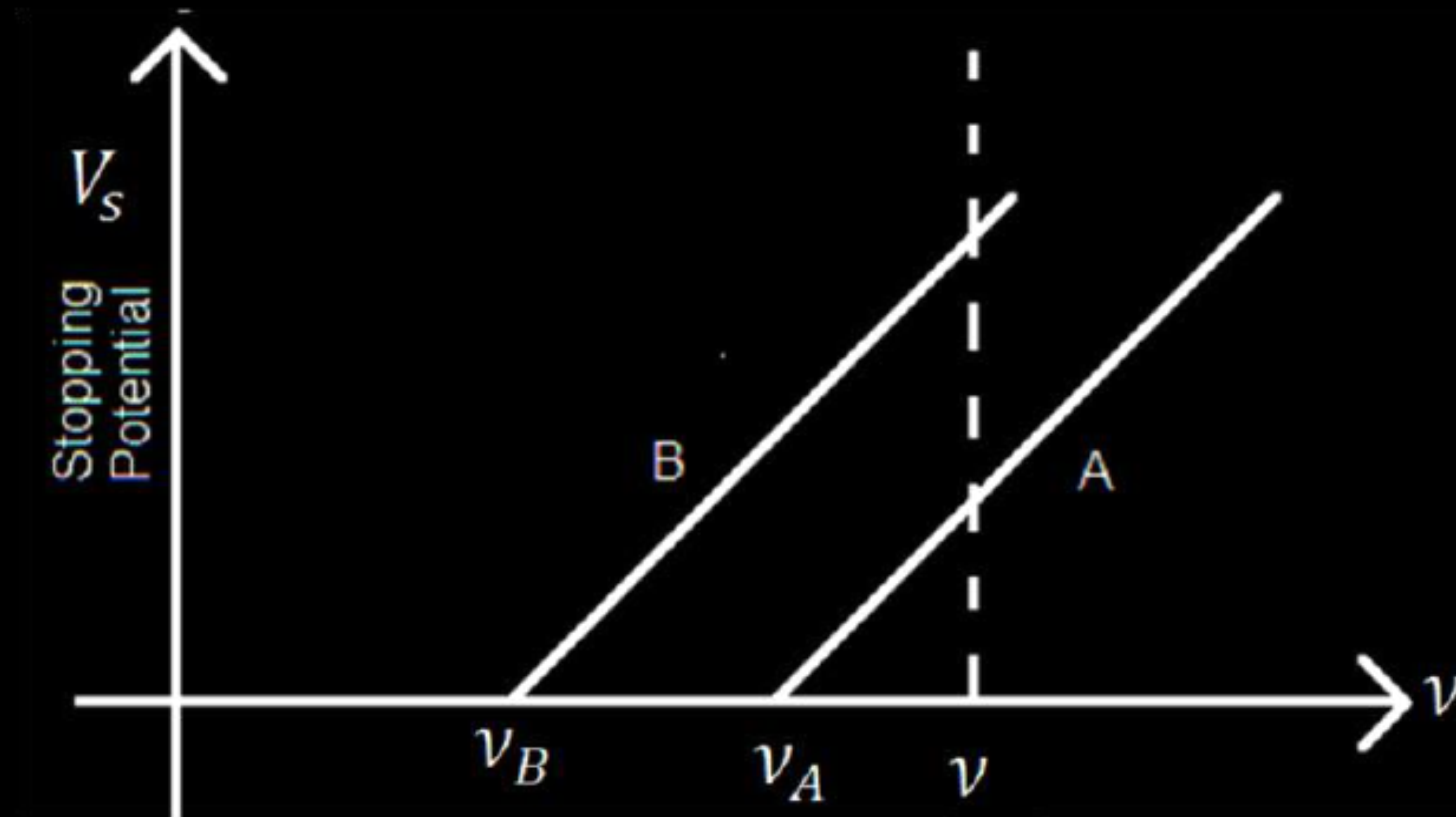
- (i) In which case is the stopping potential more and why ?
- (ii) Does the slope of the graph depend on the nature of the material used ? Explain.

CBSE 2016

Sketch the graphs showing variation of stopping potential with frequency of incident radiations for two photosensitive materials A and B having threshold frequencies $\nu_A > \nu_B$.

- In which case is the stopping potential more and why ?
- Does the slope of the graph depend on the nature of the material used ? Explain.

CBSE 2016



- For material B

From the graph for the same value of ' ν ', stopping potential is more for material 'B'

$$[V_0 = \frac{h}{e}(\nu - \nu_0)]$$

$\therefore, V_0$ is higher for lower value of ν_0

- No

As slope is given by $\frac{h}{e}$ which is constant.